

ROUGH DRAFT – OUTLINE

- This is a rough layout of the intended service class. \
- Not all note sections are complete and accurate.
- The concept is to review the installation and setup issues that cause issues in the field.
- Then get into the overview and operation of the electrical and refrigerant cycles.
- Giving a explanation as to the causes and effect within the refrigerant cycle.
- Then some basic electrical troubleshooting.

Hylex

Advanced Service Class

Why Attend Training? What Generation are you?



Are there differences in manufacturers? Are There Similarities?



Are there differences in operation, performance, service tools and techniques between manufacturers?

Just because you can do something with one manufacturer doesn't mean you can with the other.

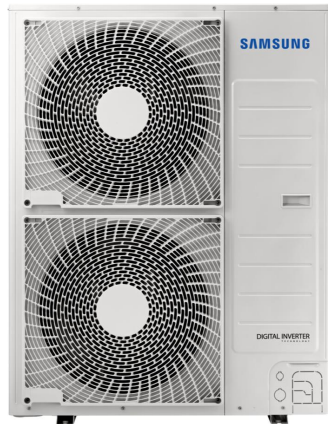
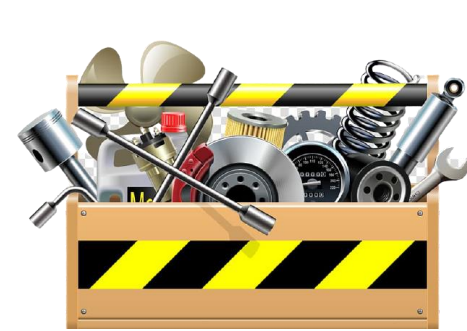
Do you have the right tools and knowledge to work on the equipment?



What is your expectation for service?



OR



What is your customers expectation for service?



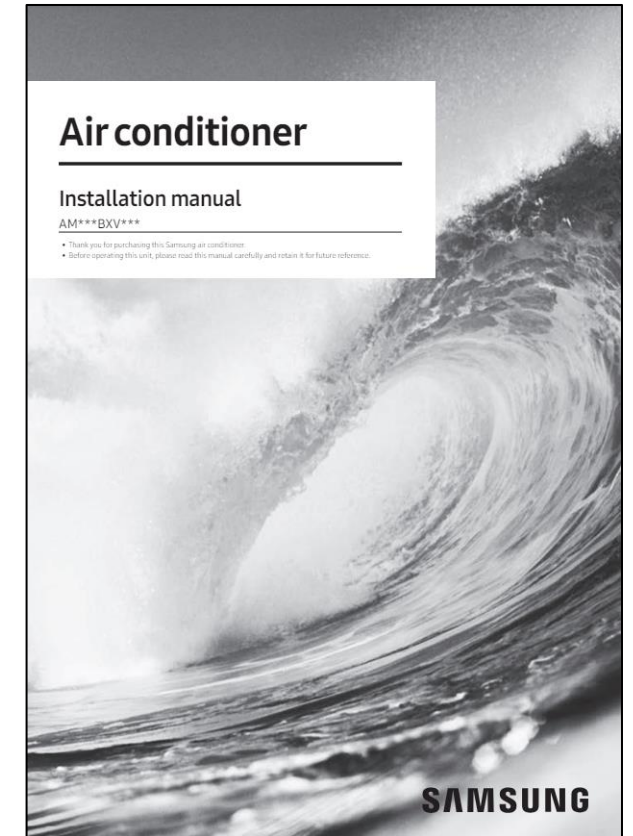
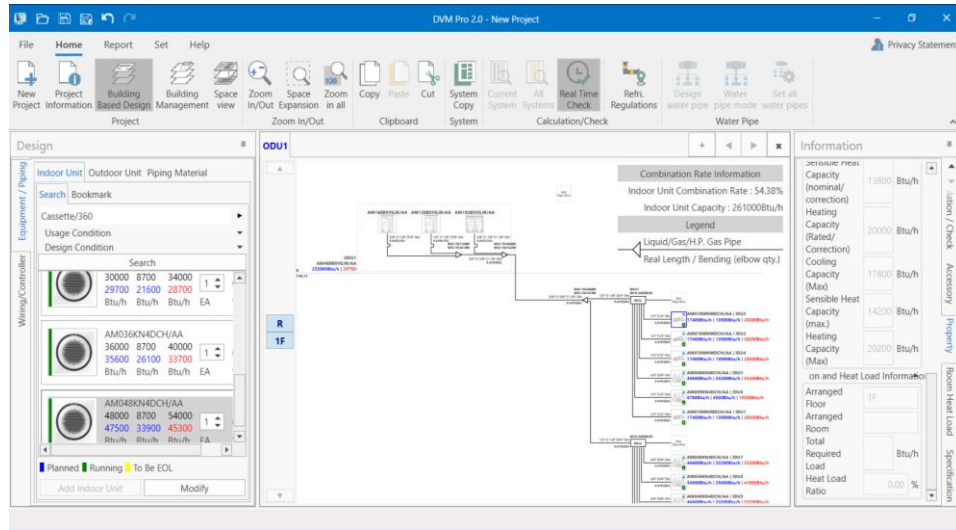
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And

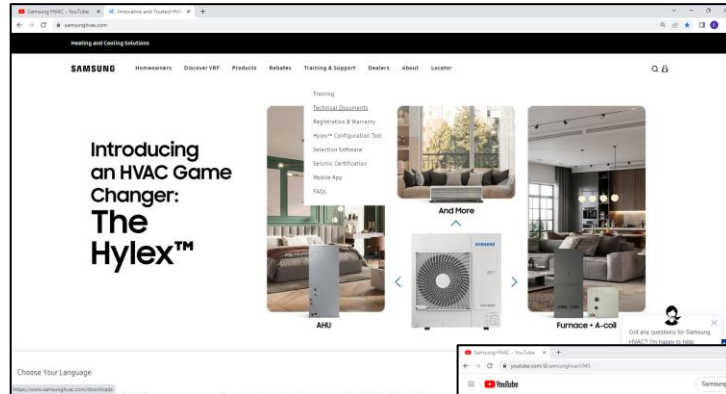


When things go bad?

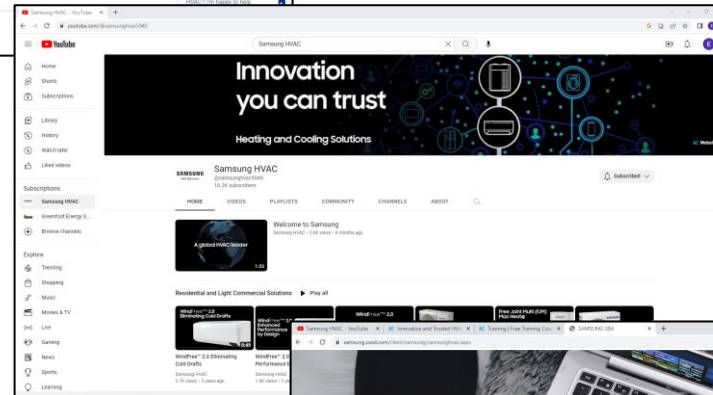
How to avoid being the source of the issue.



Tools - Technical Resources

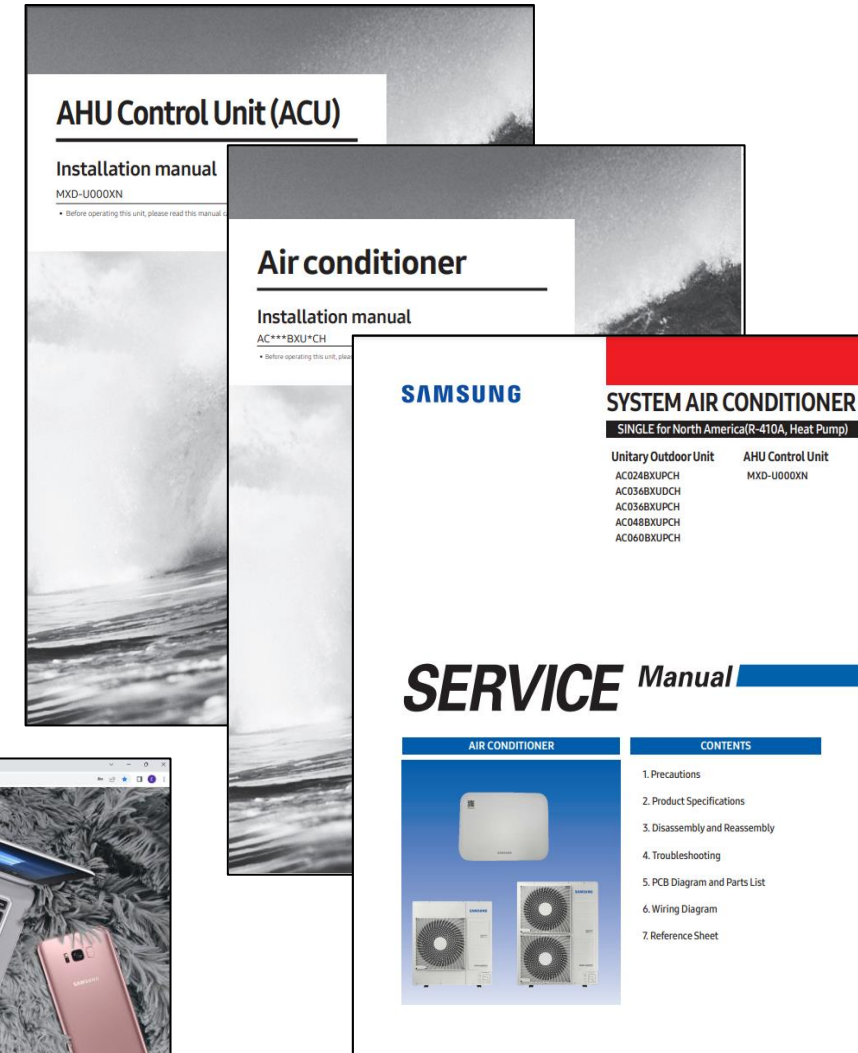


Web Sites



Mobile App.

Are you using the correct procedures and service information to diagnose the system?



Manuals

What is Hylex

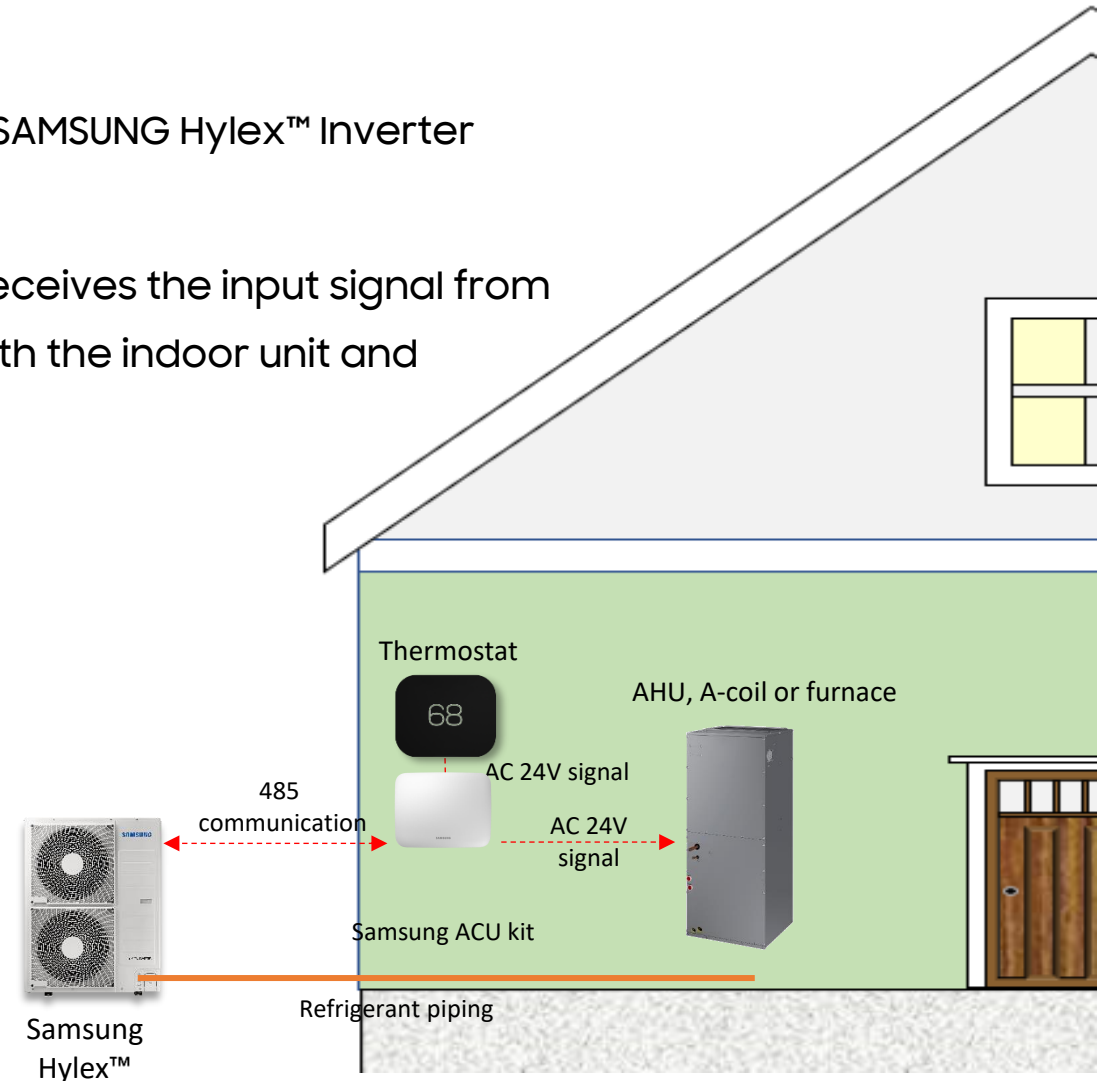
- Universal Connection
 - Universal, inverter driven heat pump
 - Connect to almost any coil with a TXV (conditions apply)
 - Allows the homeowner to use any thermostat they choose to operate their system
 - Provides the option for 3 fan speed control of third-party units (requires the URK-3A relay kit)



System Components

- Almost any type of indoor unit can be connected to the new SAMSUNG Hylex™ Inverter Outdoor Unit using the Air Handler Control Unit (ACU)
- The ACU may be installed either on or near the AHU where it receives the input signal from a standard 24 VAC thermostat and transmits that signal to both the indoor unit and outdoor unit, respectively
- Indoor unit coil requirements:
 - ✓ R410A rated
 - ✓ Contains a Thermostatic Expansion Valve (TXV)
 - ✓ Must be heat pump rated if used in a heat pump application

Not approved for use in microchannel evaporator coil applications

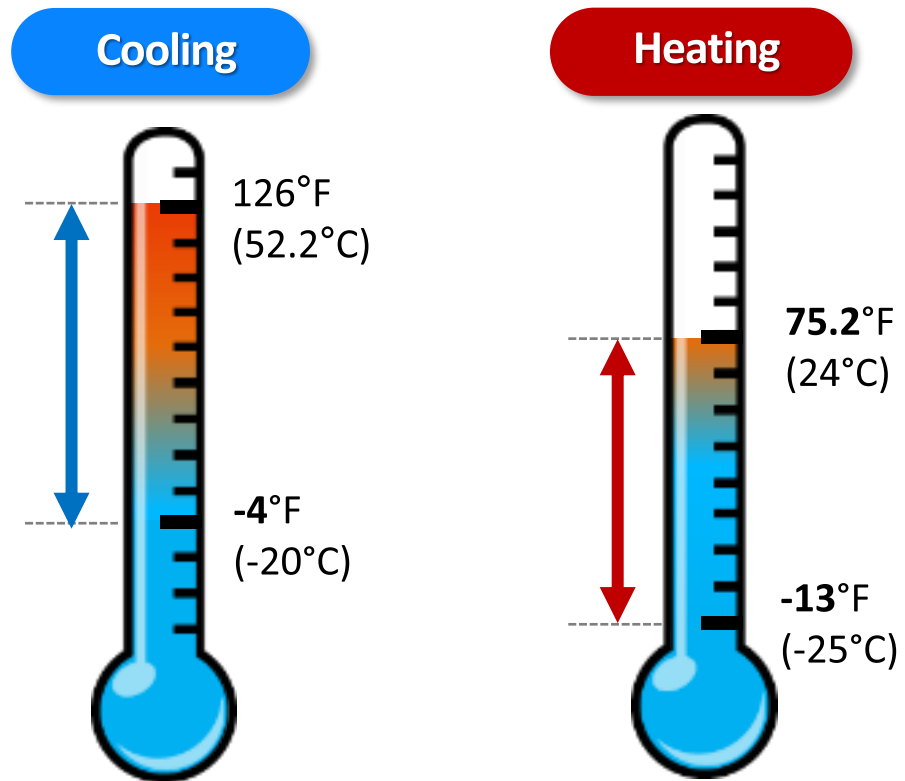


Features

- Cooling to 125°F (51.6°C)
- Heating to -13°F (-25°C)

- Operating Range

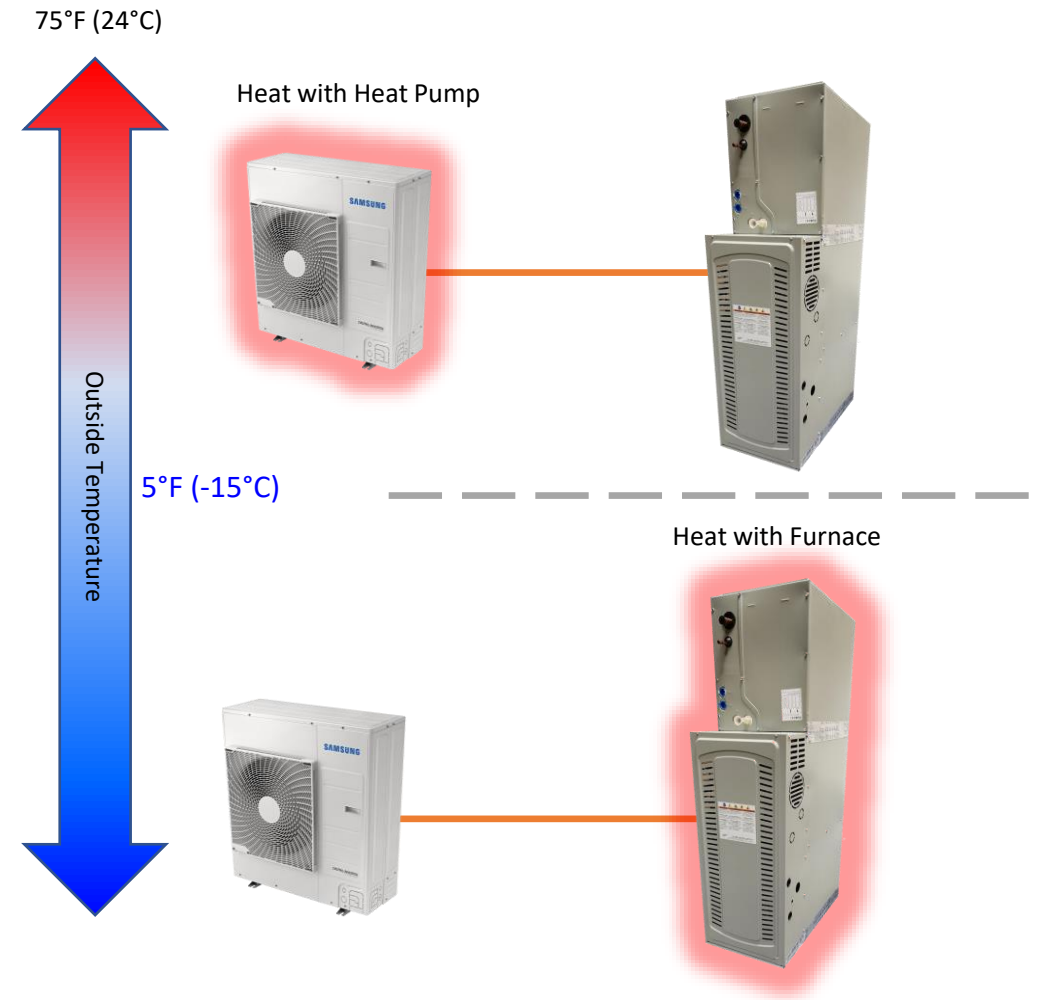
100% Rated Heating
Capacity at 5°F (-15°C)!



Features

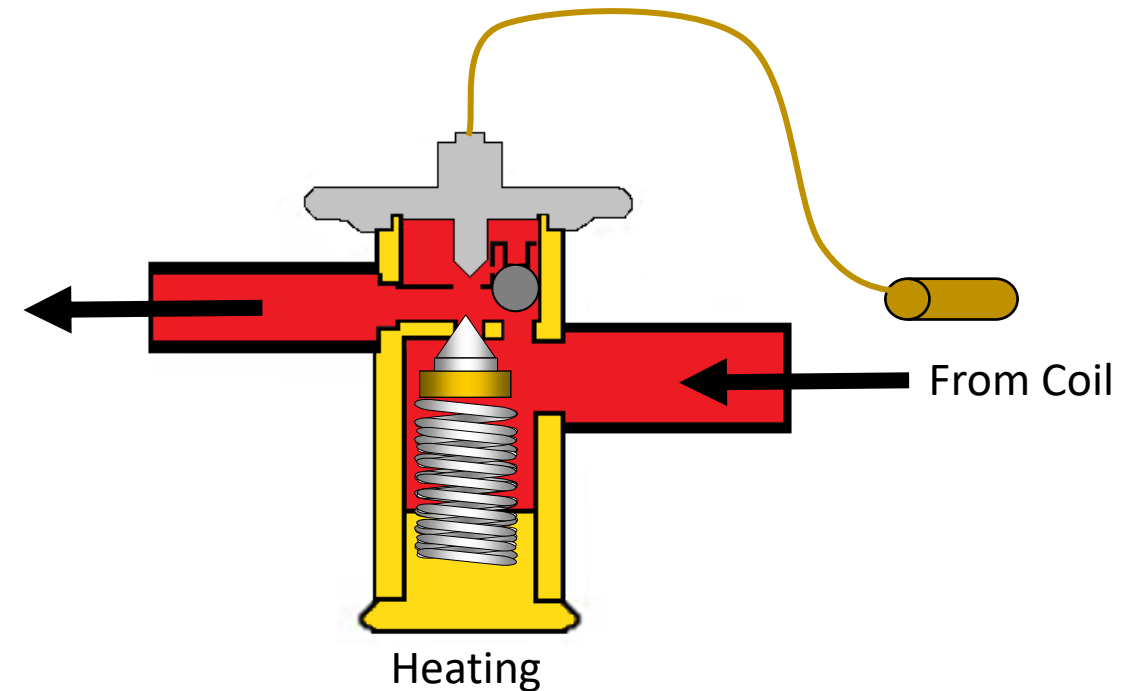
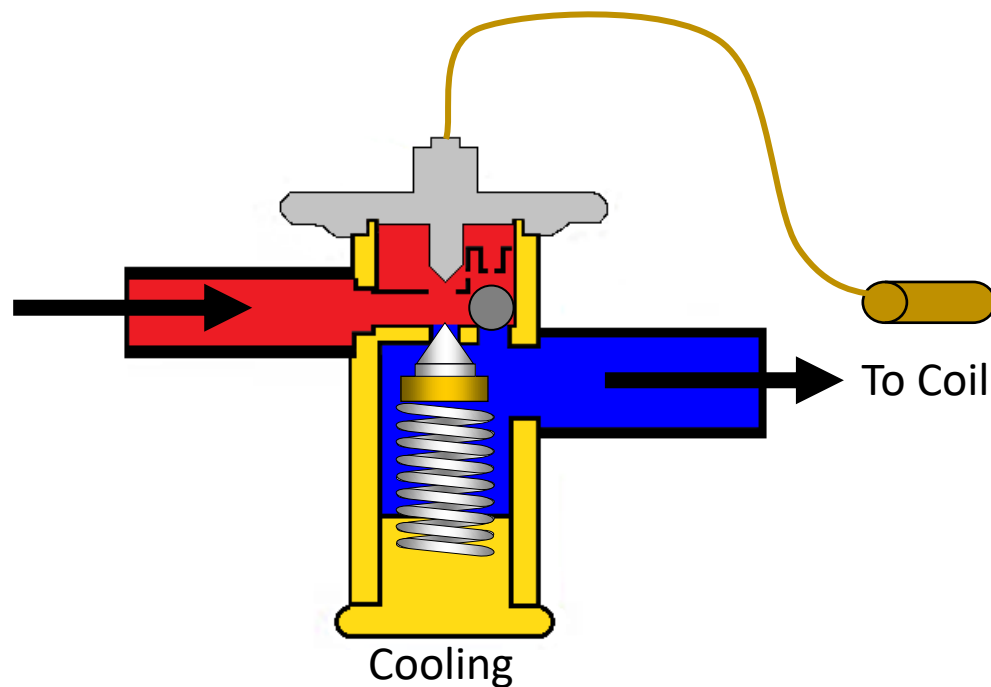
- Dual Fuel Option
 - The ACU can be set up to run in a dual-fuel operation to heat down to a desired setpoint or ambient temp then switch to another fuel source
 - If the Samsung unit is set for cooling-only, the system can be configured for manual changeover to heat pump heating by the occupant as “emergency heat” for an event where the fossil fuel system is not functioning.

Note: Requires a Heat Pump evap. and a thermostat that supports this capability

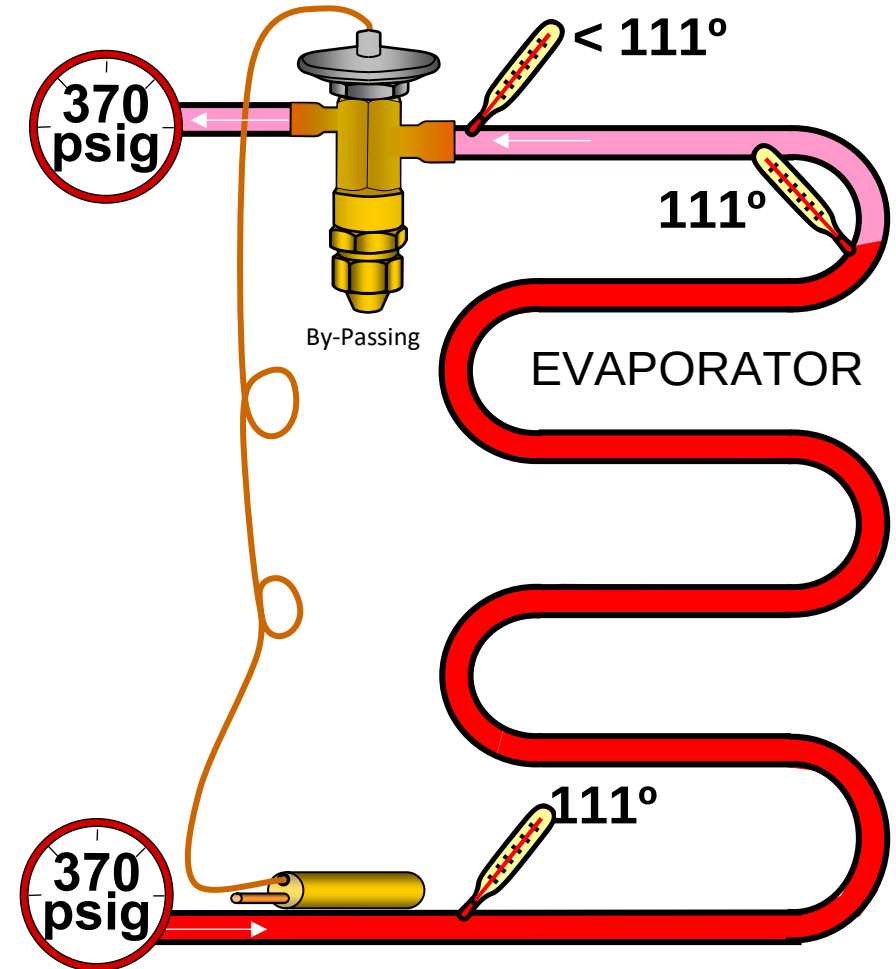
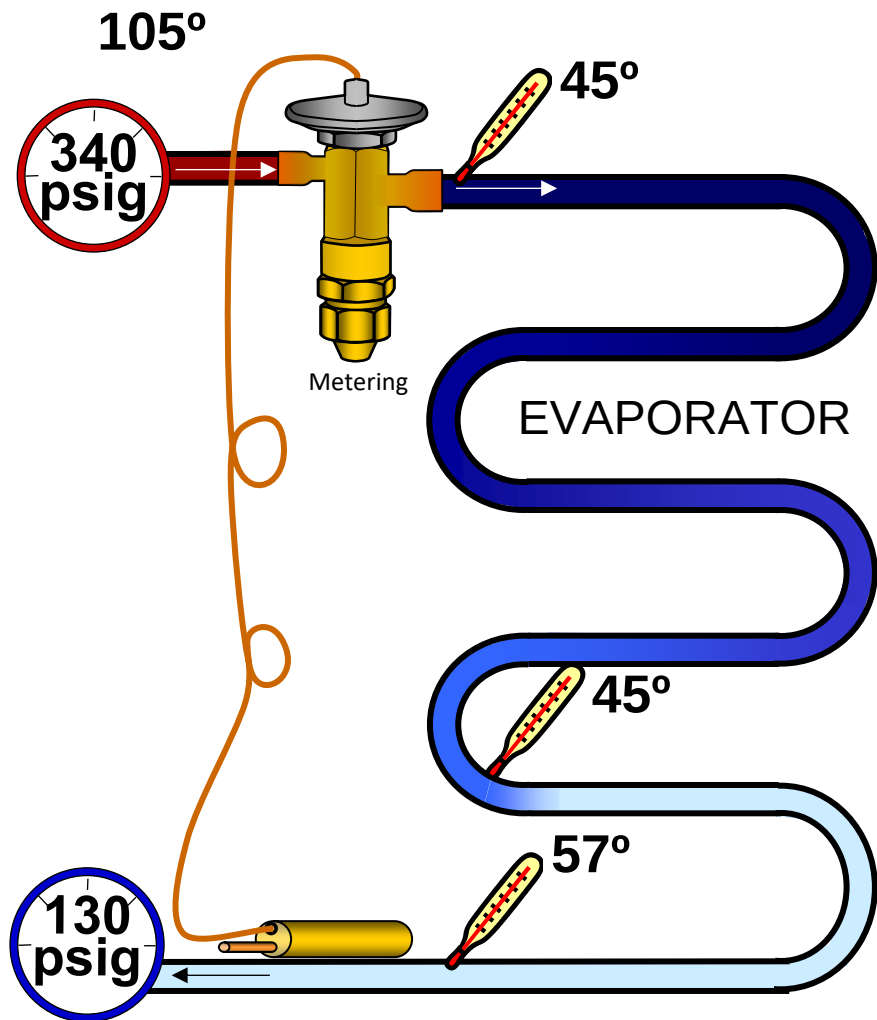


What is a Heat Pump TEV

- In the cooling mode the TEV is the metering device for the indoor coil
- In the heating mode the internal check valve opens allowing hot gas to by-pass the metering port and enter the indoor coil

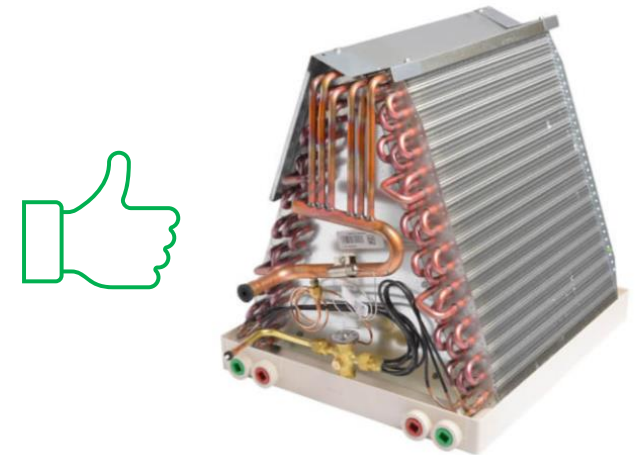
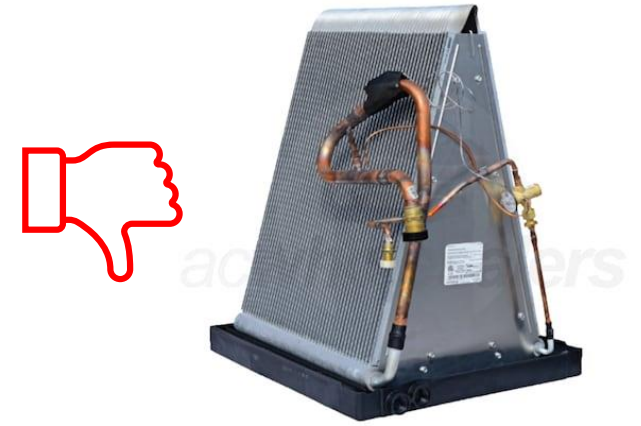


Indoor Heat Pump TEV Operation



Coils

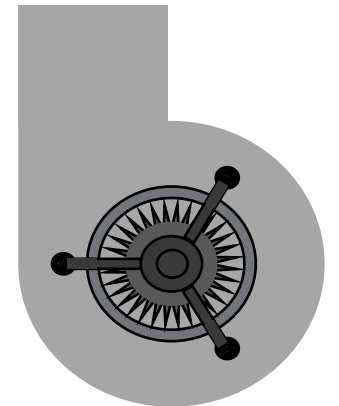
- Coils need to be rated for R410A refrigerant
- Micro Channel coils are not allowed
- Must have a Heat pump TEV if being used for heat.
- The IDU Coil is the evaporator in the cooling mode and the condenser in the heating mode



Air Flow

- For the system to operate correctly the indoor coil must be supplied with the correct amount of air flow (CFM)
- Failure to do so will result in performance issue.
- The following slide details the air flow requirements for each Hylex Outdoor unit.
- CFM must be set per the coil manufacturers recommendation.
- If CFM is not known use the charts on the following slide as guide.

IDU Coil



IDU Fan

Air Flow Requirements

Min / Max Fan Speeds

- A part of a properly installed system is understanding airflow
- Below is a table for the minimums and maximums allowed for proper system operation

Model Number	Low Speed (CFM)			Medium (CFM)			High (CFM)		
	Min.	Recommended	Max.	Min.	Recommended	Max.	Min.	Recommended	Max.
AC024BXUPCH/AA	360	500	500	518	700	700	740	1000	1000
AC036BXUDCH/AA	540	600	750	777	840	1050	1110	1200	1500
AC036BXUPCH/AA	540	600	750	777	840	1050	1110	1200	1500
AC048BXUPCH/AA	720	800	1000	1036	1120	1400	1480	1600	2000
AC060BXUPCH/AA	900	900	1250	1295	1295	1750	1850	1850	2500

Common Issues

- Installation
- Improper setup
- Air flow
- Charge

Installation

- What is the total length of the piping system?
- Is the right size piping being used.
- Are there any restrictions in the piping
- Does the piping travel through an uninsulated space?

Piping

Piping Reminder

Maximum Pipe Lengths

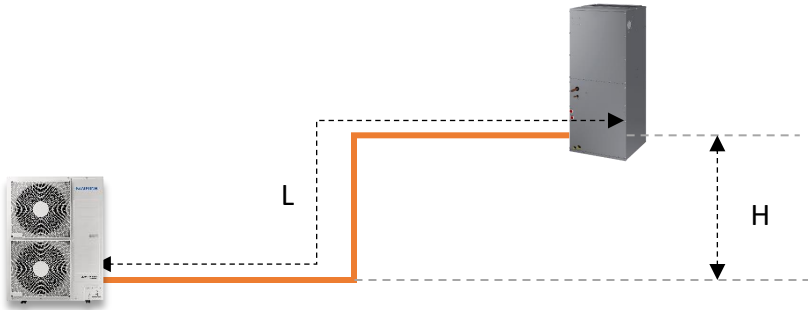
	Outdoor Unit Capacity (Tons)			
	2	3	4	5
Max. Length (feet)	100	164		100
Max. Vertical Separation	65	100		65

Flexible Piping Sizes

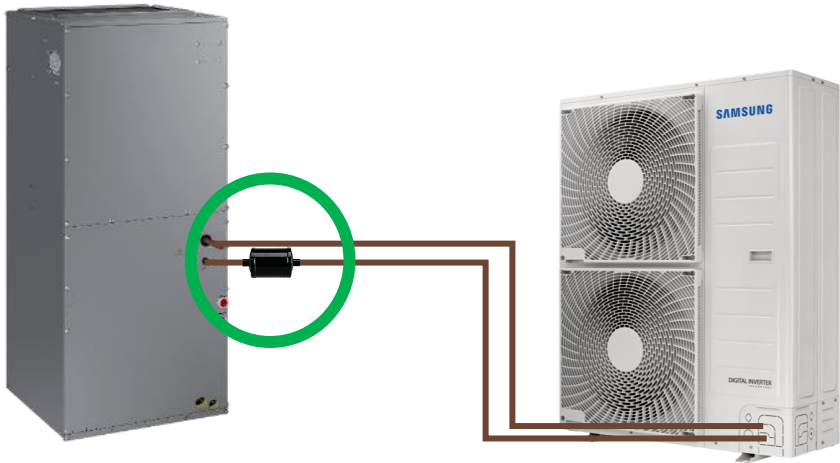
Allowed suction pipe diameters (3/8" liquid as standard)

Gas pipe	5/8"	3/4"	7/8"	1 1/8"
2 ton	O	Standard	O	
3 ton	O	Standard	O	
3 ton (high efficiency)		O	Standard	O
4 ton		O	Standard	O
5 ton		O	Standard	O

*10' Minimum Piping length



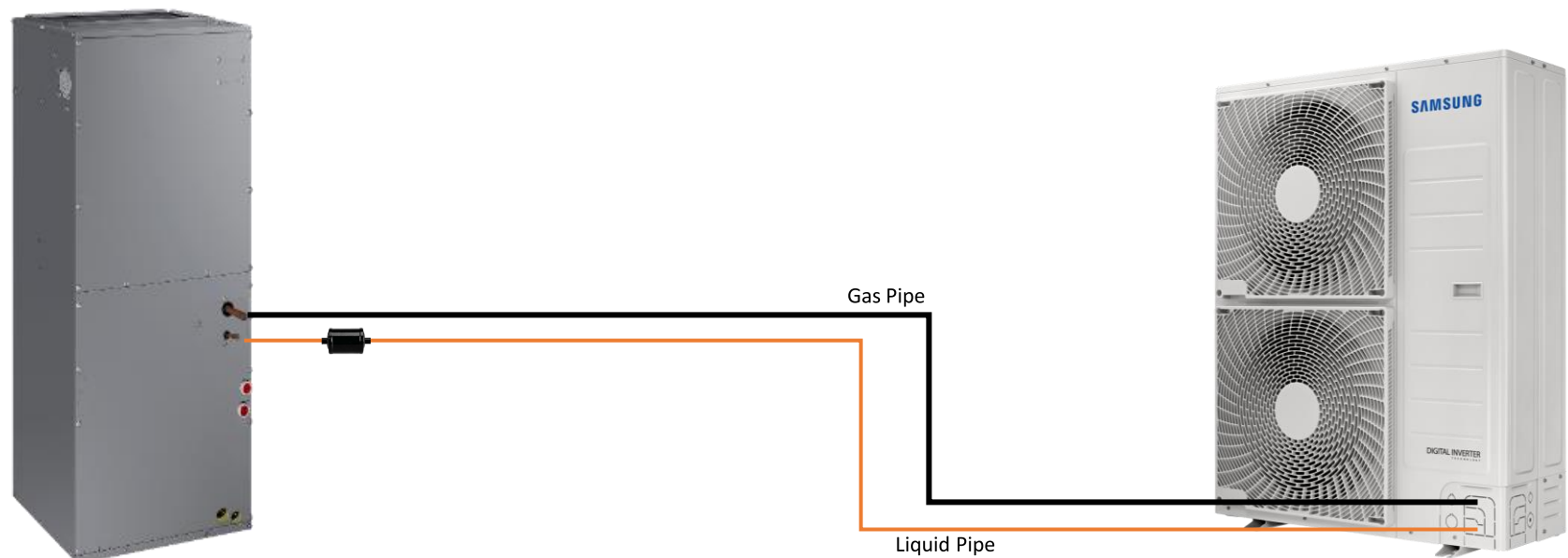
If using existing piping, a field-supplied bi-flow filter drier in the liquid line should be installed. It is recommended that service ports be installed at the IDU.



Installation

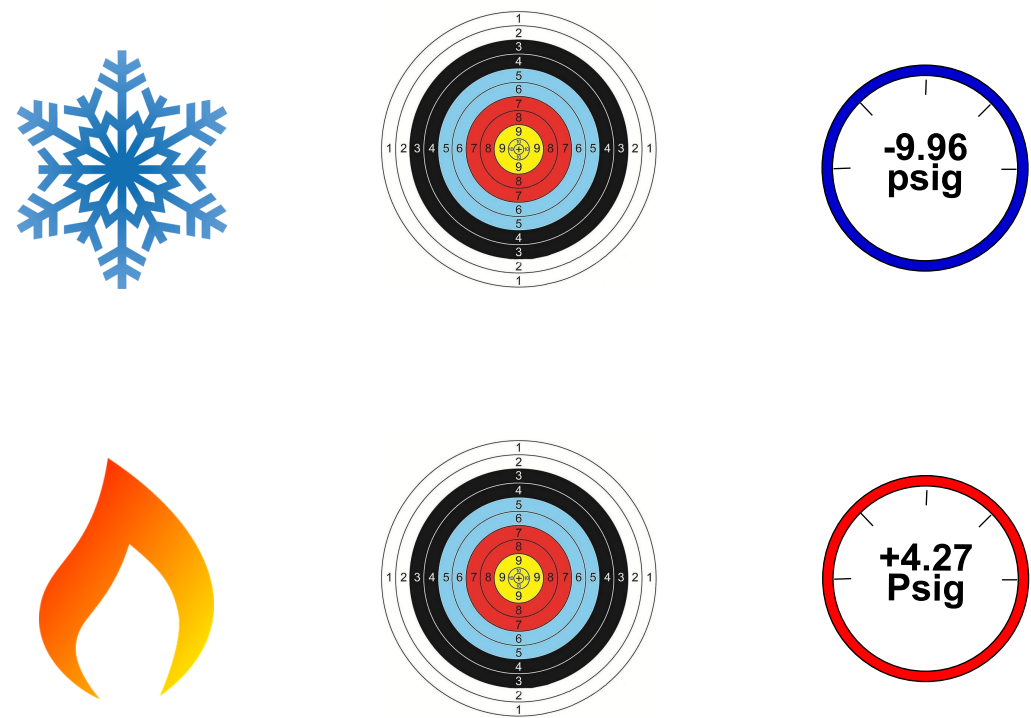
Install Flexibility

- There may be a situation where the existing gas piping isn't large enough
- Options settings on the Hylex ODU allows the installer to use the smaller existing gas piping
- 10' Minimum Piping length



Option	SEG1	SEG2	SEG3	SEG4	Function	Remark
Long-piping pressure calibration	0	8	0	0	Within 98.4 ft (30 m)	Set this value only when the smaller size of the gas pipe is used than the standard size.
			0	1	98.4 ~ 164 ft (30 ~ 50 m)	

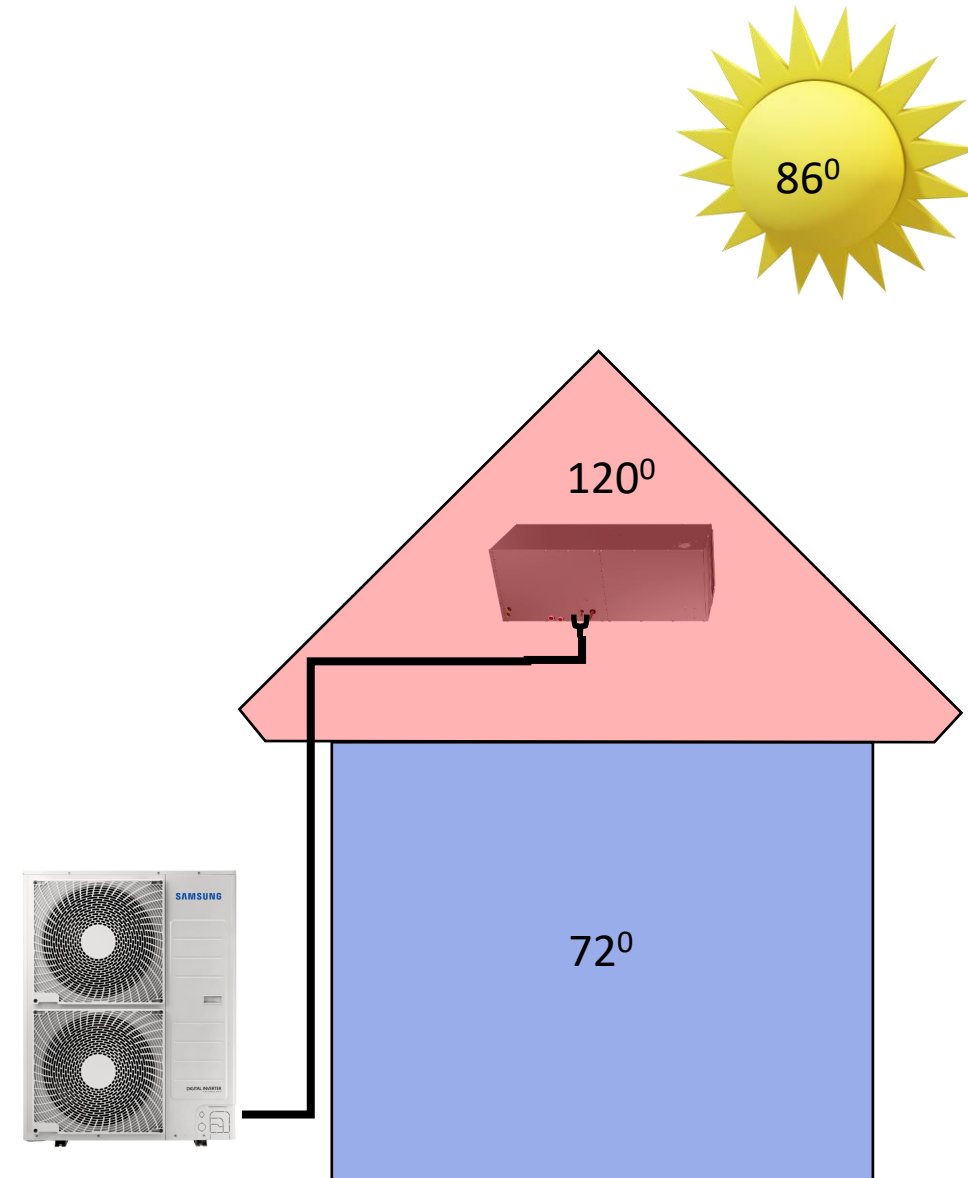
What Does Long Pipe do?



Long-piping pressure calibration ²⁾	0	8	0	0	Within 98.4 ft (30 m) (Factory default)
			0	1	98.4 ~ 164.0 ft (30 ~ 50 m) (Cooling:- 9.96psi, Heating:+4.27psi)

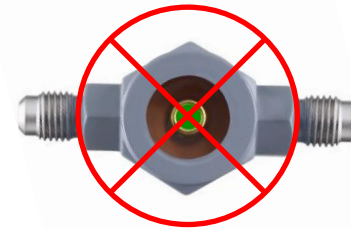
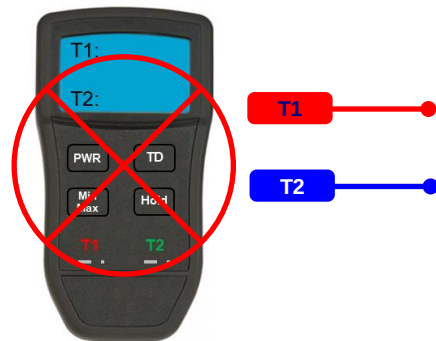
Loss of capacity

- Are the refrigerant pipes running through an unconditioned space.
 - This can cause a loss of capacity
- Is the liquid line uninsulated.
- Is there any sub-cooling at the IDU?
- What is the actual return air temperature?



System Charge

- Getting the correct charge in the system is critical for system operation
- You must calculate the required charge for the given system.
- The correct charge must be weighed in.
- Any rules of thumbs you may have applied in the past are not relevant here.
- System are not charged using superheat, sub-cooling or sight glasses



Refrigerant Charging

- These are critically charged systems. (Why)
- What affect does overcharging have on system pressures?
- What effect does low charge have on the system pressure?
- How do they affect discharge temperature?
- How does the compressor respond to these conditions?



Service Ports

Cooling mode:

Liquid line - Liquid pressure

Gas Line - Suction pressure

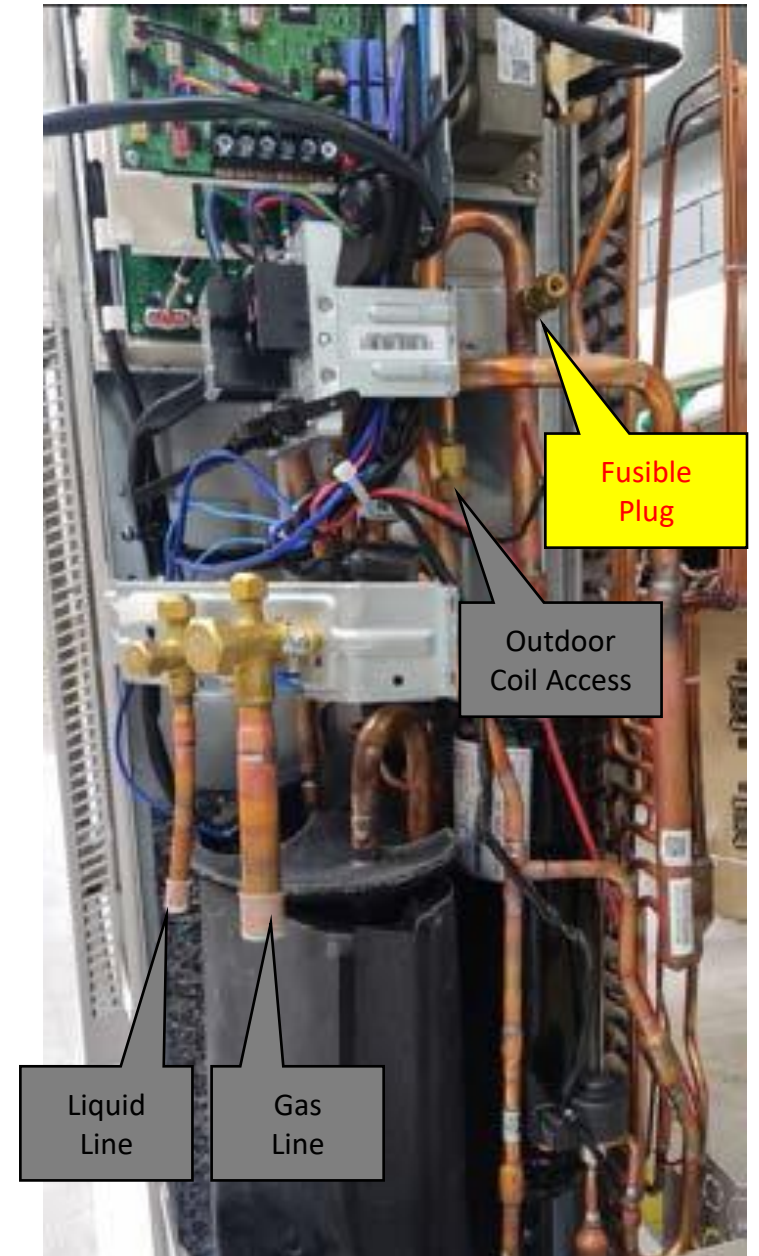
Outdoor Coil - High Side pressure

Heating mode:

Liquid line - Liquid pressure

Gas Line - High Side Pressure

Outdoor Coil - Evaporator Pressure



Calculating the refrigerant charge

- There are 3 ways to calculate the refrigerant Charge.
- Which formula you use will depend on what you know about the IDU coil
- The 3 calculations are as follows:
 - System is matched with a AHRI listed indoor unit
 - System Retrofit and the coil internal volume is known
 - System Retrofit and the coil's internal volume NOT known
- The system charge must not exceed the value listed below.

Maximum total allowable refrigerant charging amount

	Factory charging (lb)	**Maximum total refrigerant amount (lb)
AC024BXUPCH	7.06	9.36
AC036BXUDCH	7.94	12.29
AC036BXUPCH	9.26	13.62
AC048BXUPCH	9.26	13.62
AC060BXUPCH	10.58	12.89

Charging

When connecting to an AHRI listed indoor

- When connecting to an AHRI listed indoor unit used as part of the tested basic model group (tested combination).
- Total refrigerant amount = Amount charged at factory + Additional refrigerant amount to fit piping length¹⁾

Table 1 Additional refrigerant charge for piping length

Model	0 ~ 24.6 ft (7.5 m)	24.6 ft + (7.5 m) ~
All models (AC***BXU*CH)	0	0.5 oz / ft (46.5 g / m)

Table 4 Maximum total allowable refrigerant charging amount

	Factory charging (lb)	**Maximum total refrigerant amount (lb)
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AC036BXUDCH	7.94	12.29
AC036BXUPCH	9.26	13.62
AC048BXUPCH	9.26	13.62
AC060BXUPCH	10.58	12.89

Matched system charging example

- This is a matched system with 110ft of connected pipe.
- The additional refrigerant charge is 2.66 pounds.
- The total system charge is 10.6 pounds.
- For this system to operate correctly 2.66 pounds of refrigerant must be added to the system.



AC036BXUDCH/AA



110' total Piping



ADP Air Handler
Model#BVRMC3937

SAMSUNG		SUBMITTAL AC036BXUDCH/AA (CXH36UDB)		Page 1 of 3																																																																																																																																	
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Safety	Certifications		UL 60336-2-40																																																																																																																																		
	Devices		PCB fuses, indoor unit terminal block thermal fuse, current transformer, over-voltage protection, crankcase heating, temperature limit protection logic, compressor overload sensing																																																																																																																																		
Heat Pump Outdoor Unit General Information - Inverter driven heat pump condensing unit compatible with air handling units and A-coils with a thermostatic expansion valve (conditions apply, refer to supporting technical documents for details). 100% Rated heating capacity at 5°F (-15°C) outside temperature (refer to capacity tables in outdoor unit technical data book for full capacity details). - Soft-start compressor minimizing current inrush - Base pan heater equipped as standard - The condensing unit heat exchanger salt spray test method: ISO-9227 - The heat exchanger showed no unusual rust or corrosion development to 3,000 hours - The system (outdoor unit with MXD-U000XN AHU Control Unit) shall provide three (3) fan speed control of the indoor unit increasing performance and dehumidification. Construction - The unit shall be galvanized steel with a baked-on powder coated finish for durability. - Refrigerant pipe connections and service ports are located inside the heat pump cabinet. Refrigerant pipes can enter the cabinet on the front, side, back, or bottom. Controls - Requires MXD-U000XN AHU Control Unit (purchased separately) installed near the indoor unit for operation - Control wiring shall be 2 X 18 AWG (minimum) - Compatible with standard 24VAC thermostats - Requires B reversing valve signal (24VAC energized for heating) Optional Settings - The system (outdoor unit with AHU Control Unit) shall provide cold air prevention (optional) to delay indoor fan operation until the indoor coil is above a fixed temperature while in heat mode - Target low pressure calibration to increase or decrease indoor coil and discharge air temperature in cooling mode - Target high pressure calibration to increase or decrease indoor section coil and discharge air temperature in heating mode - Snow accumulation prevention option setting to prevent snow drifting against an idle outdoor unit - The outdoor unit shall feature optional night quiet modes to reduce outdoor unit sound (3 levels) with automatic activation or manual activation (with MM-B14 accessory). - System can be set up as heating/cooling, cooling only, or heating only via outdoor unit option setting. - Maximum Current Control configurable from 50% - 100% in 5% increments via outdoor unit Refrigerant System - The compressor shall be hermetically sealed, inverter-controlled BLDC rotary type. - The outdoor unit heat exchanger shall be mechanically bonded aluminum fin to copper tube - Uses traditional unitary line sets (liquid pipe does not require insulation) - Flexible pipe diameter compatibility (three suction pipe diameter options) - Refrigerant flow shall be controlled by an electronic expansion valve in the outdoor unit for heating operation. Refrigerant flow for cooling shall be controlled via Thermostatic Expansion Valve (TXV) at the indoor coil (bleed or non-bleed). Warranty 10 Years compressor, 10 years parts. Registration required. Conditions apply.																																																																																																																																					

Table 4 Maximum total allowable refrigerant charging amount

	Factory charging (lb)	**Maximum total refrigerant amount (lb)
AC024BXUPCH	7.06	9.36
AC036BXUDCH	7.94	12.29
AC036BXUPCH	9.26	13.62
AC048BXUPCH	9.26	13.62
AC060BXUPCH	10.58	12.89

Charging

Retrofit Coil Volume NOT Known

- When installing in a retrofit application and internal coil volume of indoor unit is not known.
- Total refrigerant amount = Standard refrigerant amount for retrofitting³⁾ + Additional refrigerant amount based on piping length¹⁾

Table3 Standard refrigerant amount for Retrofitting when internal coil volume is not know

Model	Factory charging (lb)	Charging amount for retrofit. (lb)
AC024BXUPCH	7.06	5.63
AC036BXUDCH	7.94	6.62
AC036BXUPCH	9.26	7.67
AC048BXUPCH	9.26	7.67
AC060BXUPCH	10.58	9.7

Table 1 Additional refrigerant charge for piping length

Model	0 ~ 24.6 ft (7.5 m)	24.6 ft + (7.5 m) ~
All models (AC***BXU*CH)	0	0.5 oz / ft (46.5 g / m)

Retrofit

Coil Volume NOT KNOWN

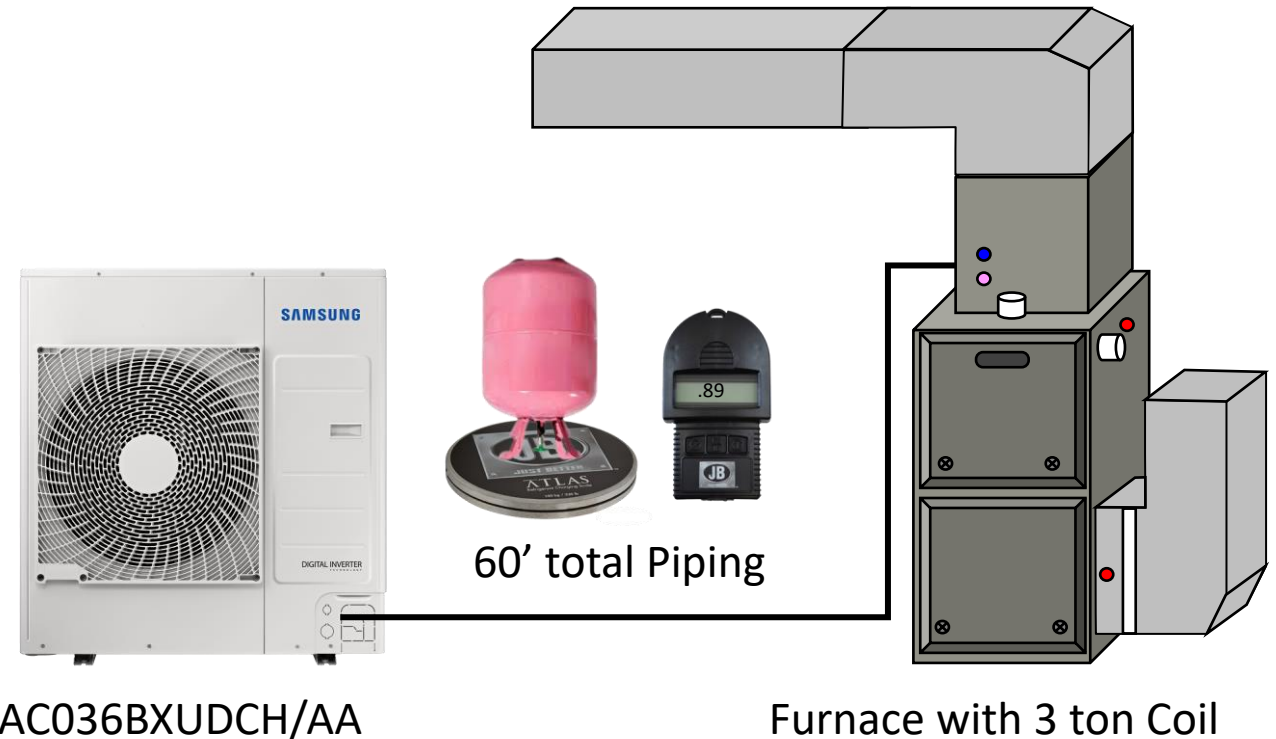
- The base charge for this retrofit system is 6.62 pounds.
- Calculate the amount needed for the additional line set.
- Add these 2 together to determine the total system charge.
- Subtract this from the system factory charge.
- This is the amount of refrigerant you add or remove from the system

Table3 Standard refrigerant amount for Retrofitting when internal coil volume is not known

Model	Factory charging (lb)	Charging amount for retrofit. (lb)
AC024BXUPCH	7.06	5.63
AC036BXUDCH	7.94	6.62
AC036BXUPCH	9.26	7.67
AC048BXUPCH	9.26	7.67
AC060BXUPCH	10.58	9.7

Table 1 Additional refrigerant charge for piping length

Model	0 ~ 24.6 ft (7.5 m)	24.6 ft + (7.5 m) ~
All models (AC***BXU*CH)	0	0.5 oz / ft (46.5 g / m)



Charging

Retrofit Coil Volume Known

- When installing in a retrofit application
- Total refrigerant amount = Amount charged at factory + Amount of refrigerant addition/reduction by internal volume of indoor unit²⁾ + Additional refrigerant amount to fit piping length¹⁾

Table2 : Amount of refrigerant adjusted by internal coil volume of indoor heat exchanger

	Model Name	Factory Charging	Required Charge Adjustment for Indoor Coil Internal Volume [in ³]										
			61	91.5	122	152.6	183.1	213.6	244.1	274.6	305.1	335.6	366.1
2Ton	AC024BXUPCH	7.06 lbs	/	-1.43 lbs	-1.04 lbs	-0.64 lbs	-0.24 lbs	0	/	/	/	/	/
3Ton	AC036BXUDCH	7.94 lbs	/	/	-1.76 lbs	-1.32 lbs	-0.44 lbs	-0.53 lbs	0	/	/	/	/
3Ton	AC036BXUPCH	9.26 lbs	/	/	/	-2.38 lbs	-1.98 lbs	-1.59 lbs	-1.19 lbs	-0.79 lbs	-0.4 lbs	0	/
4Ton	AC048BXUPCH	9.26 lbs	/	/	/	-2.38 lbs	-1.98 lbs	-1.59 lbs	-1.19 lbs	-0.79 lbs	-0.4 lbs	0	/
5Ton	AC060BXUPCH	10.58 lbs	/	/	/	/	-0.37 lbs	0					

/ : Not allowed

0 : No addition or reduction

Table 1 Additional refrigerant charge for piping length

Model	0 ~ 24.6 ft (7.5 m)	24.6 ft + (7.5 m) ~
All models (AC***BXU*CH)	0	0.5 oz / ft (46.5 g / m)

Retrofit

Coil Volume is KNOWN

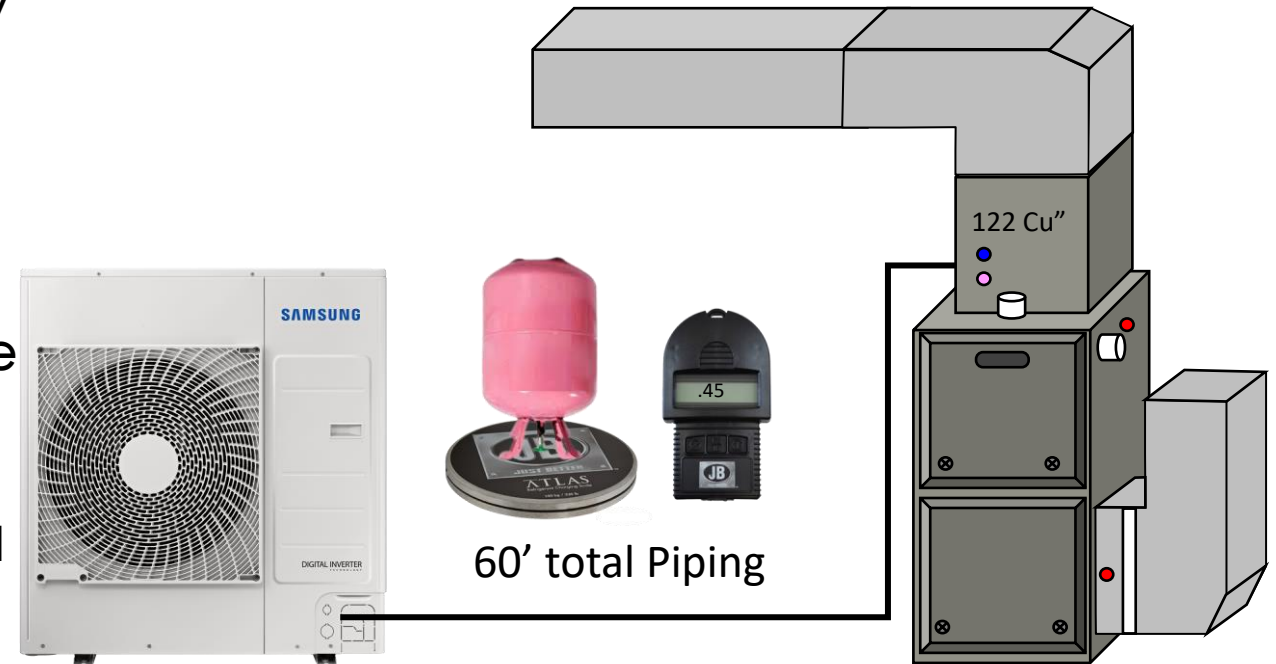
- The coil volume is 122 Cu"
- The factory charge is 7.94 pounds.
- Using table 2 locate the amount of refrigerant to remove from the factory charge for the smaller coil.
- Subtract this from the system factory charge. This is you new base charge.
- Calculate the amount of refrigerant required for the installed piping.
- Add this to the adjusted factory charge this is the total system charge.
- Subtract the original factory charge.
- This the amount of refrigerant you add or remove for the installed system.

Table2 : Amount of refrigerant adjusted by internal coil volume of indoor heat exchanger

	Model Name	Factory Charging	Required Charge Adjustment for Indoor Coil Internal Volume [in ³]										
			61	91.5	122	152.6	183.1	213.6	244.1	274.6	305.1	335.6	366.1
2Ton	AC024BXUPCH	7.06 lbs	/	-1.43 lbs	-1.04 lbs	-0.64 lbs	-0.24 lbs	0	/	/	/	/	/
3Ton	AC036BXUDCH	7.94 lbs	/	/	-1.76 lbs	-1.32 lbs	-0.44 lbs	-0.53 lbs	0	/	/	/	/
3Ton	AC036BXUPCH	9.26 lbs	/	/	/	-2.38 lbs	-1.98 lbs	-1.59 lbs	-1.19 lbs	-0.79 lbs	-0.4 lbs	0	/
4Ton	AC048BXUPCH	9.26 lbs	/	/	/	-2.38 lbs	-1.98 lbs	-1.59 lbs	-1.19 lbs	-0.79 lbs	-0.4 lbs	0	/
5Ton	AC060BXUPCH	10.58 lbs	/	/	/	/	-0.37 lbs	0					

Table 1 Additional refrigerant charge for piping length

Model	0 ~ 24.6 ft (7.5 m)	24.6 ft + (7.5 m) ~
All models (AC***BXU*CH)	0	0.5 oz / ft (46.5 g / m)



AC036BXUDCH/AA

Furnace with 3 ton Coil

ADP HE Coil Charging Recommendations

- TXV Charging – Use the charging method recommended by the outdoor unit instructions.
- Alternatively, ADP recommends charging to 12°F sub-cooling for AC units and 10°F sub-cooling for heat pump units.
- In addition, if equipped with an adjustable valve, adjust to 10°F superheat.
- For heat pump units initially charged in the cooling mode, final adjustments to charge in the heating mode are acceptable if necessary

System Set up

- What is the owner's intent usage for the system?
 - Heat pump or cooling only
 - Dual fuel application?
- Is the system wired for single stage or 2 stage operation?
- How are the ACU dip switches configured?
 - This will dictate how the system operates
- How is the ODU configured?
 - What setting in the ODU have been changed.

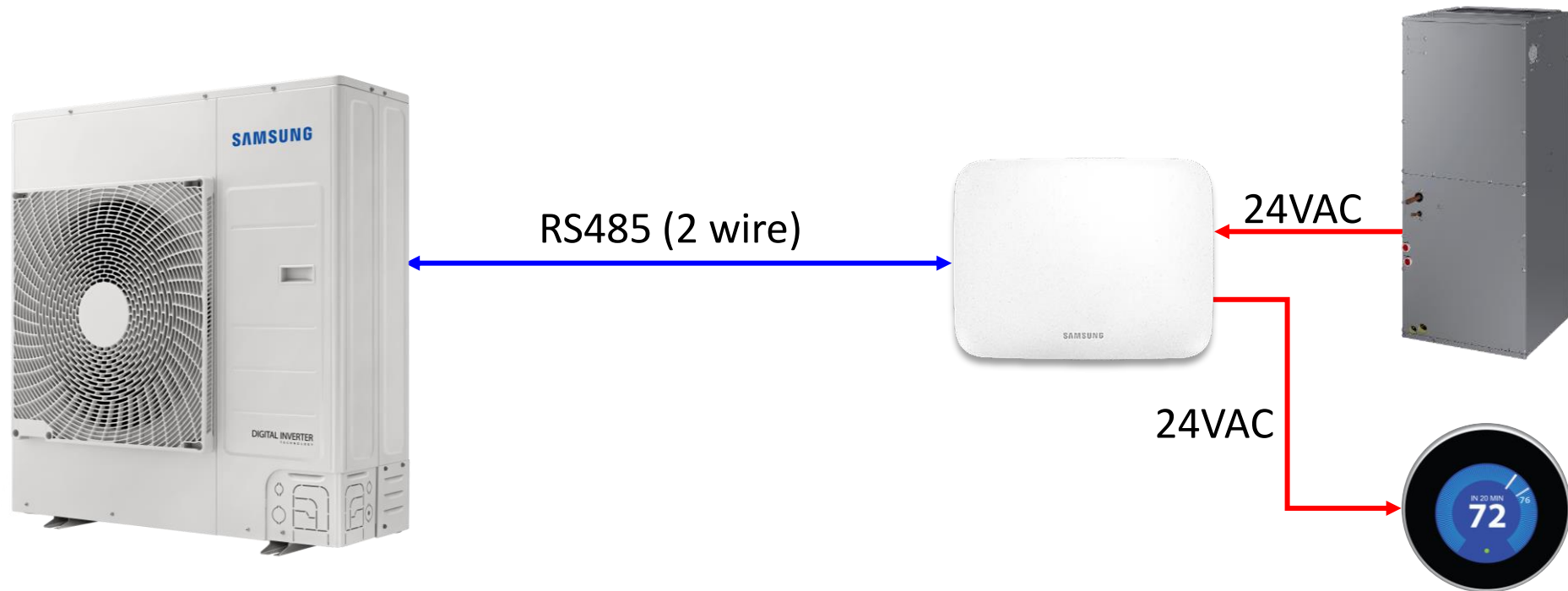
System Wiring

- The thermostat is wired to the ACU .
- The ACU is wired to the air handler/furnace
- The ODU is wired to the ACU.
- Conventional 18-gauge thermostat wire can be used for all connections
- 24 volts must be supplied from the indoor system to the ACU
- The G terminal Must be wired in for the system to operate.
- wiring guides are found in the ACU installation manual

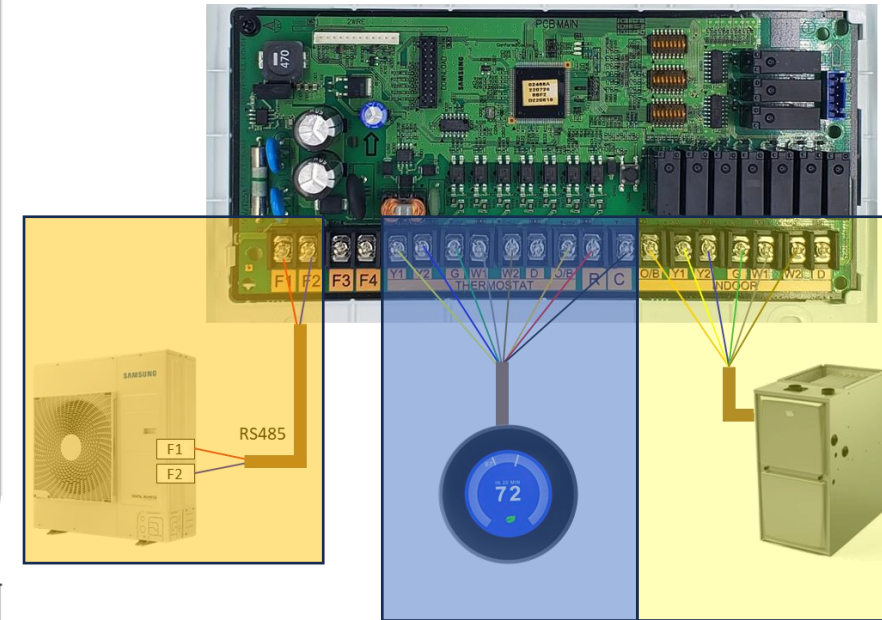
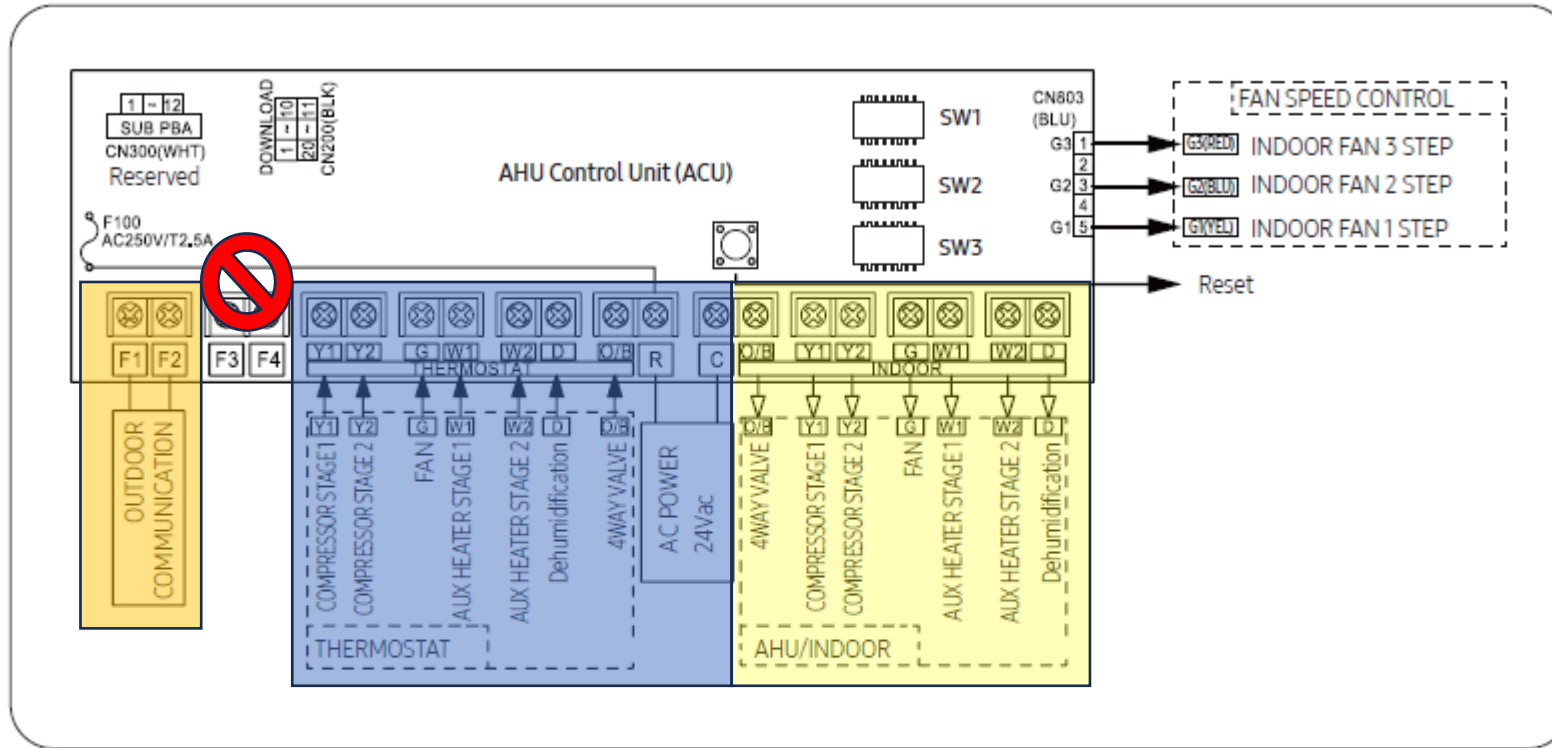
Integration Solutions

Advanced System Control with ACU (AHU Control Unit)

- The ACU will install between the thermostat, indoor unit, and the outdoor unit
- Installation should be on, or near the indoor unit



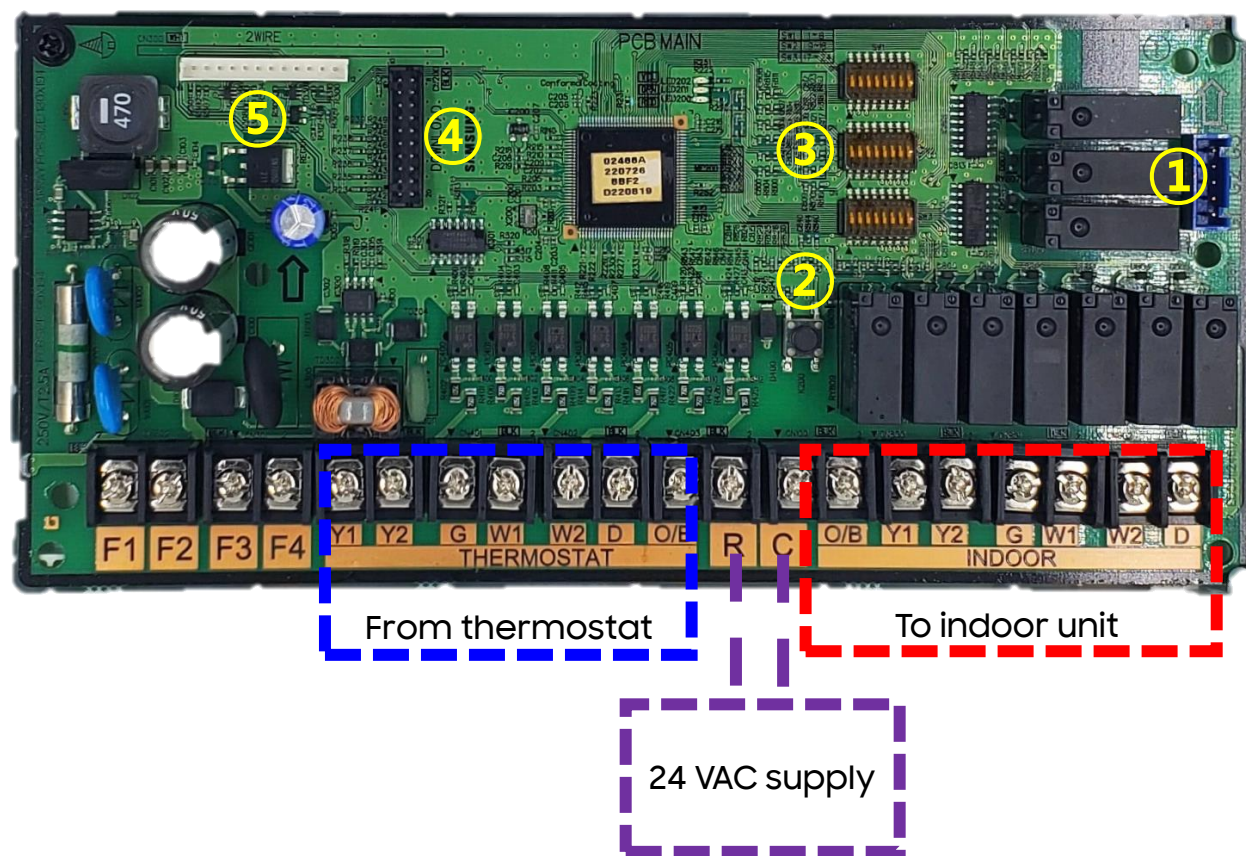
ACU Terminal Designations



Y1	1-stage operation	Y2	2-stage operation
G	Fan signal	D	Dehumidifier
W1	Auxiliary heater 1-stage(heat pump) Furnace 1-stage	W2	Auxiliary heater 2-stage(heat pump) Furnace 2-stage
O/D	4Way valve (24 VAC : Heating, 0 VAC : Cooling)	250V / T2.5A	FUSE

ACU PCB Overview

- ACU(AHU control unit) PBA



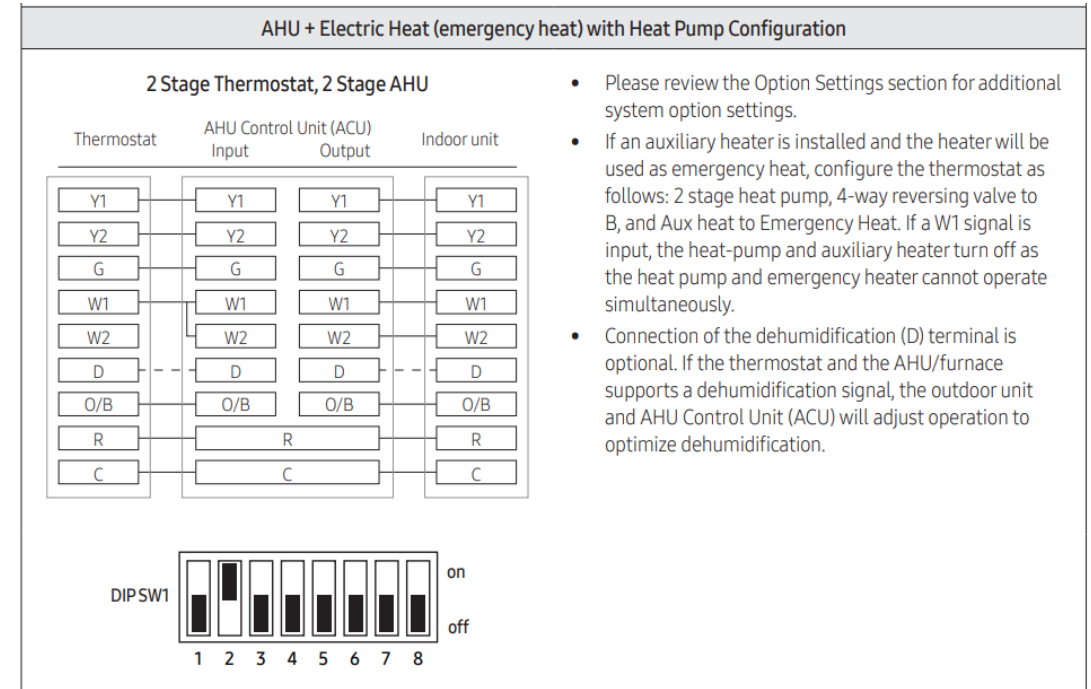
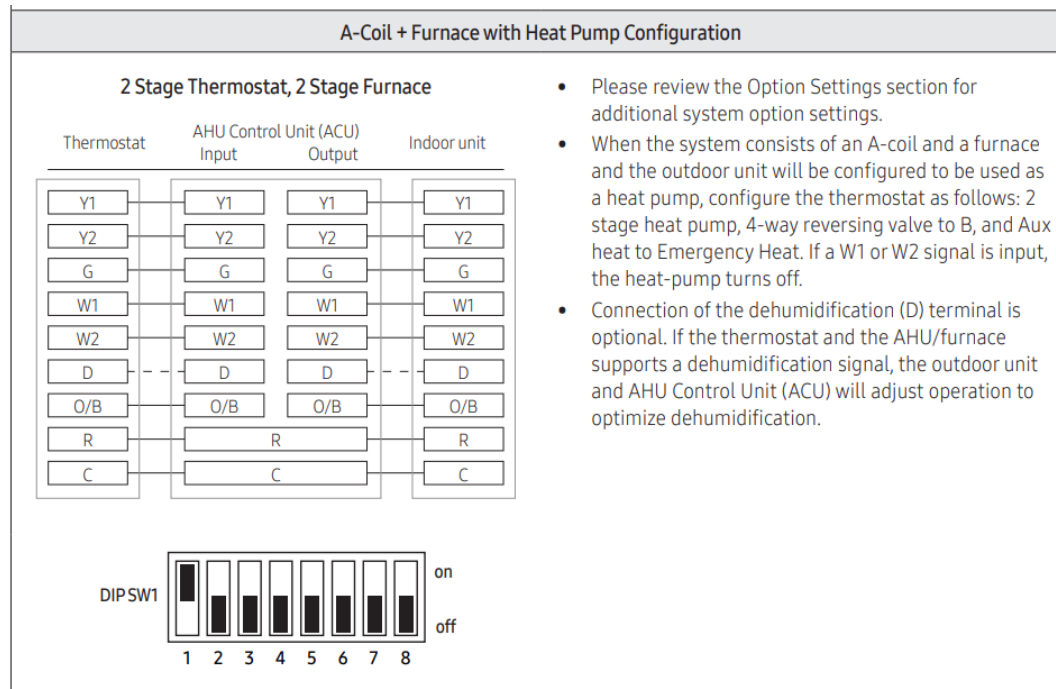
No.	Function
①	3 step fan wiring, G1(Low), G2(Middle), G3(High) fan step
②	K1 Reset button (After changing a setting of a Dip SW, press it)
③	Dip switches
④	Firmware updates
⑤	Not used

Terminal	Function
Y1	Compressor 1-stage
Y2	Compressor 2-stage
G	Indoor fan
W1	Auxiliary heater Stage1 / Furnace 1-stage
W2	Auxiliary heater Stage2 / Furnace 2-stage
D	Dehumidification
O/B	4-way reversing valve (24 VAC : Heating, 0 VAC : Cooling)

Program the thermostat to energize the reversing valve in the heating mode (B)

Wiring Configurations

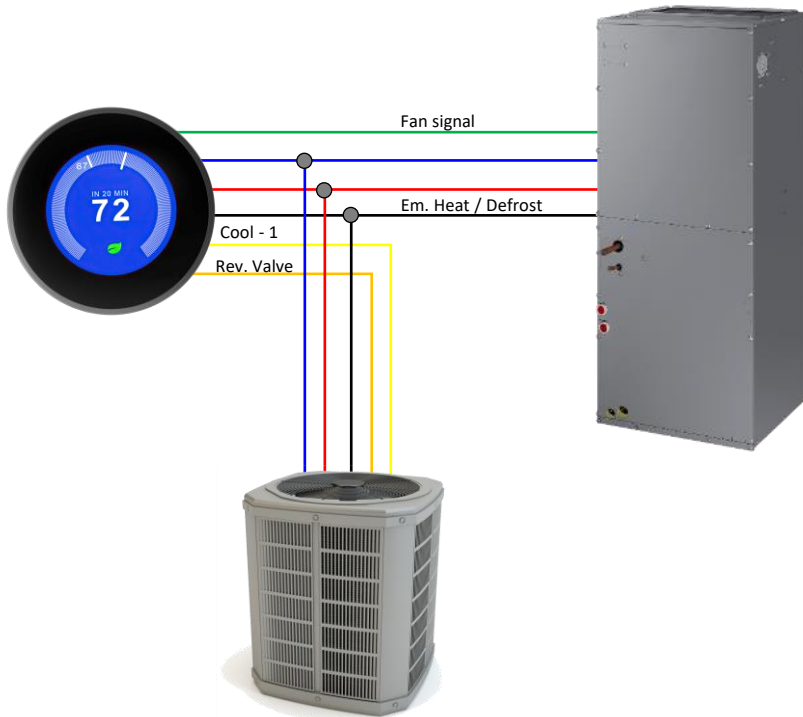
- Wiring Diagrams are found in the installation manual.
- Manuals can be found at Samsunghvac.com



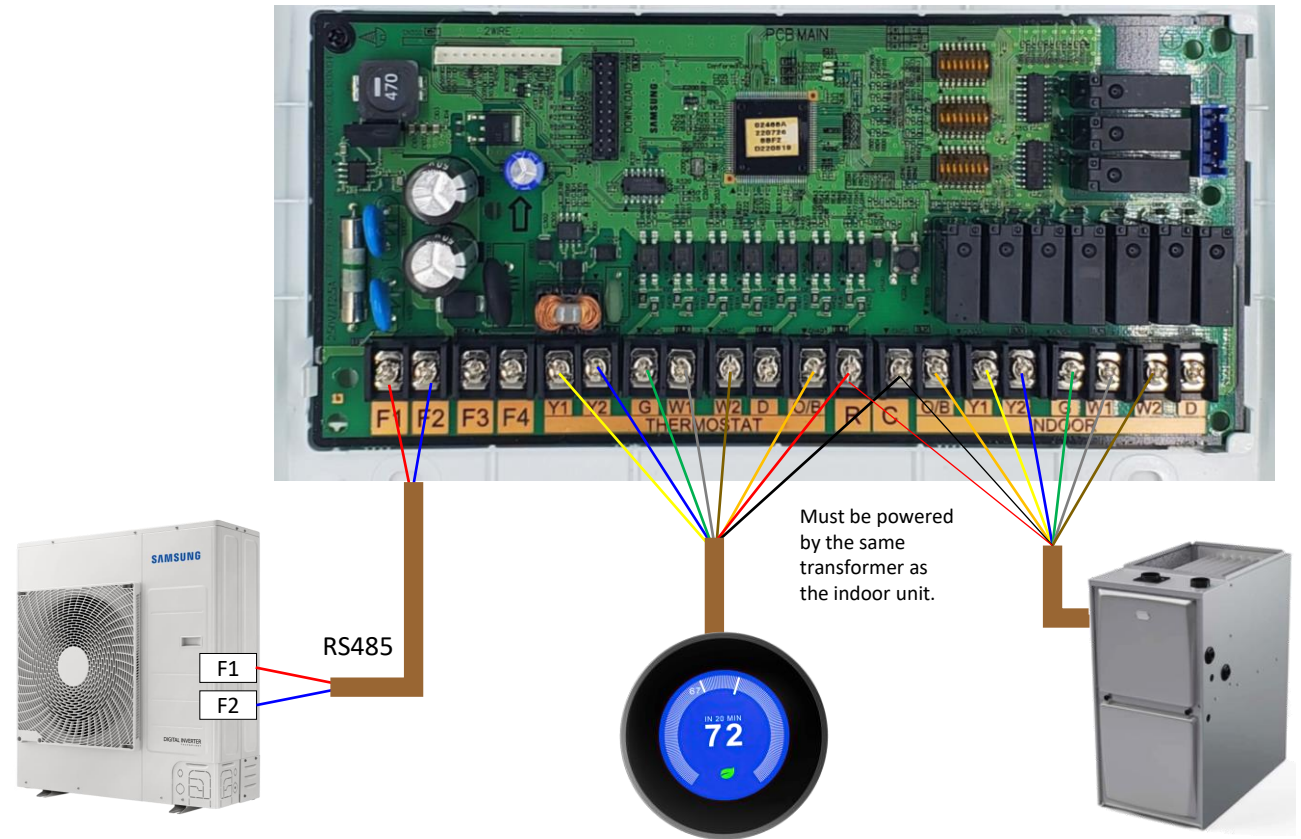
Wiring Example and Comparison

The ODU/ACU will have full control of the IDU, not the thermostat

Typical Unitary Example

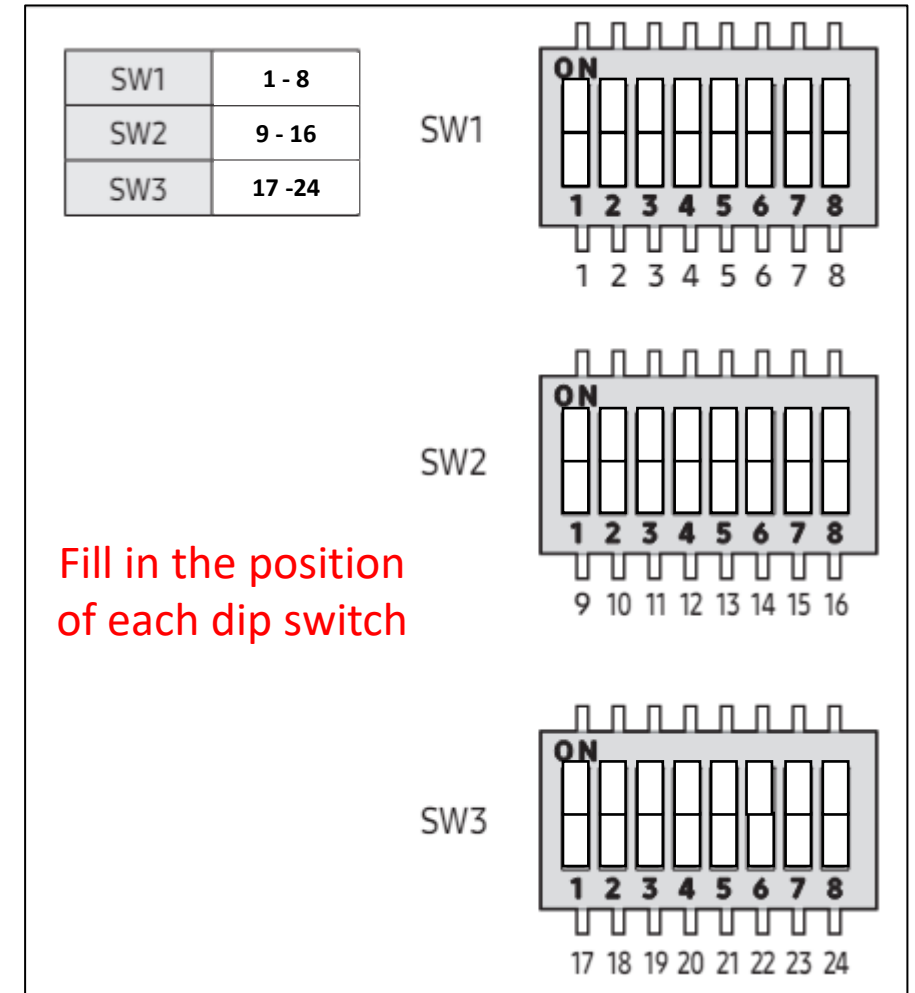


Configuration using ACU



System Configuration

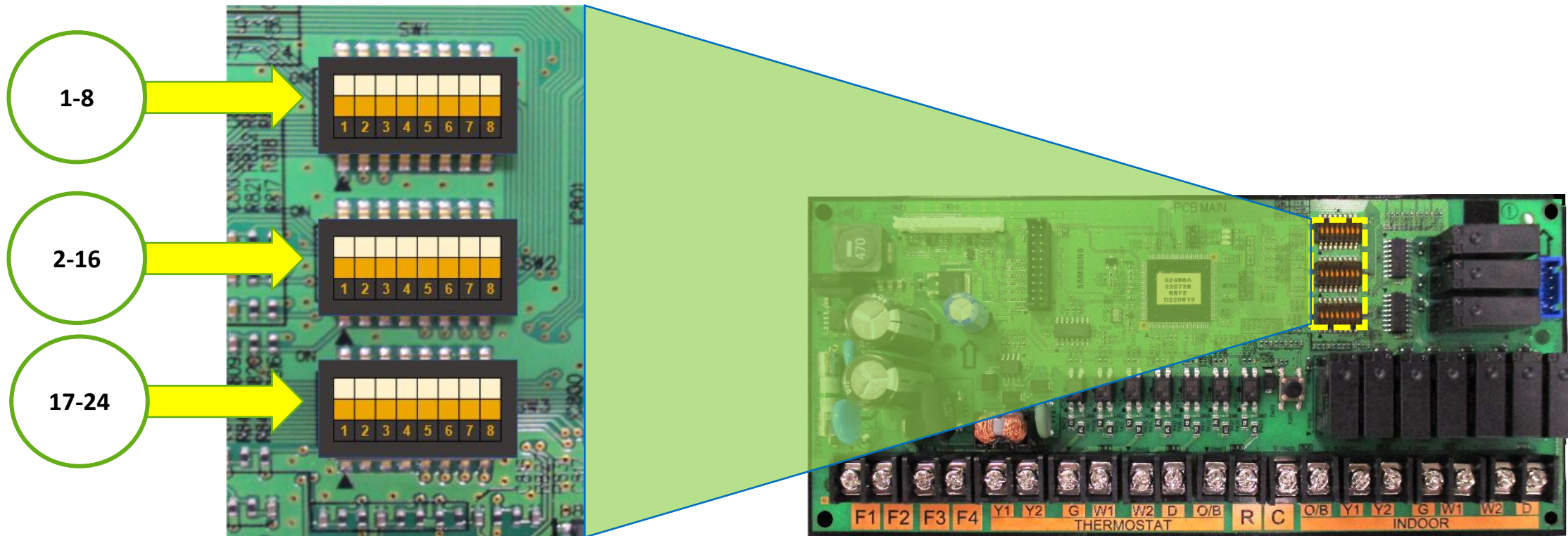
- Circle the setting for each option
- **What type of indoor unit is installed?**
 - Furnace – Air Handler
- **Is this a dual fuel application?**
 - Yes - No
- **Do you want to enable cold air prevention?**
 - Yes - No
- **Fan operation during defrost?**
 - Yes - No
- **Configure fan On/Off delay during defrost and cold air prevention?**
 - 5 – 10 – 15 seconds
- **Configure the desired fan step control?**
 - Disabled – 3 step (Y1 only) – 3 Step (Y1 & Y2)
- **Enable enhanced dehumidification?**
 - Yes - No
- **Configure the Fan Off delay timer when system turns Off?**
 - No Delay – 30 – 60 – 90 Seconds



Integration Solutions

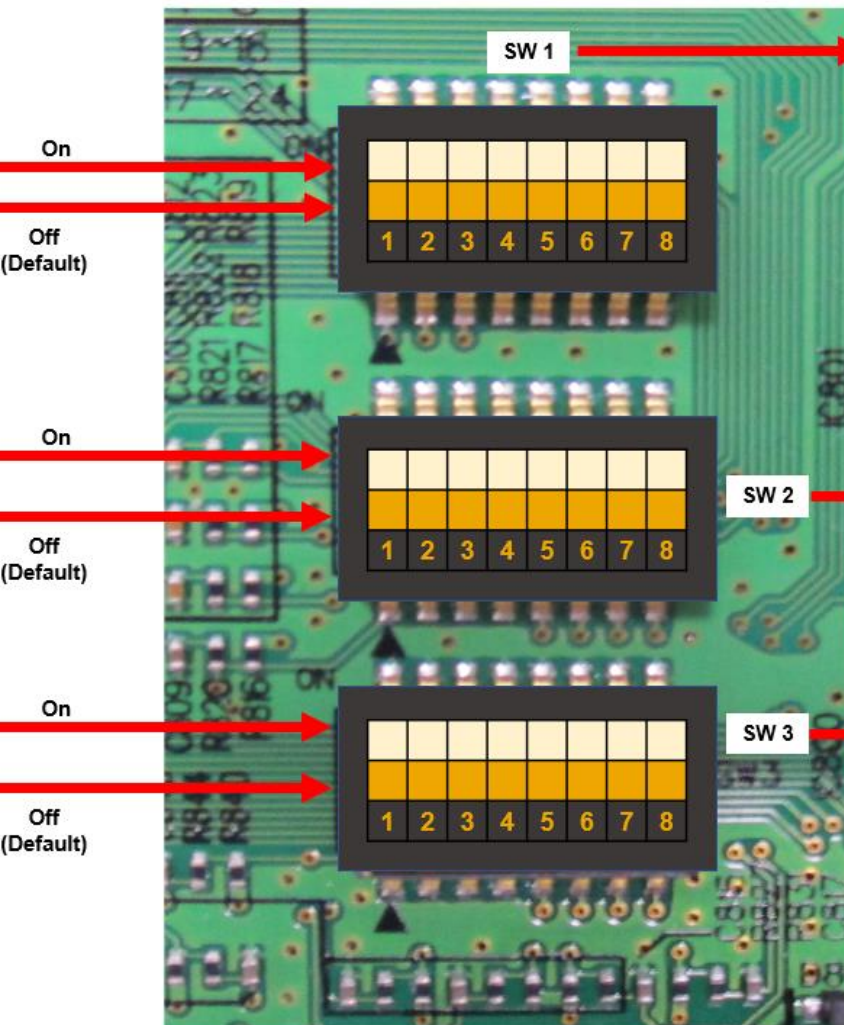
- System Configuration

- Part of the installation will be configuration
- Using industry standard DIP switches makes setting up the system quick and easy



Product information

- ACU DIP Switch Settings



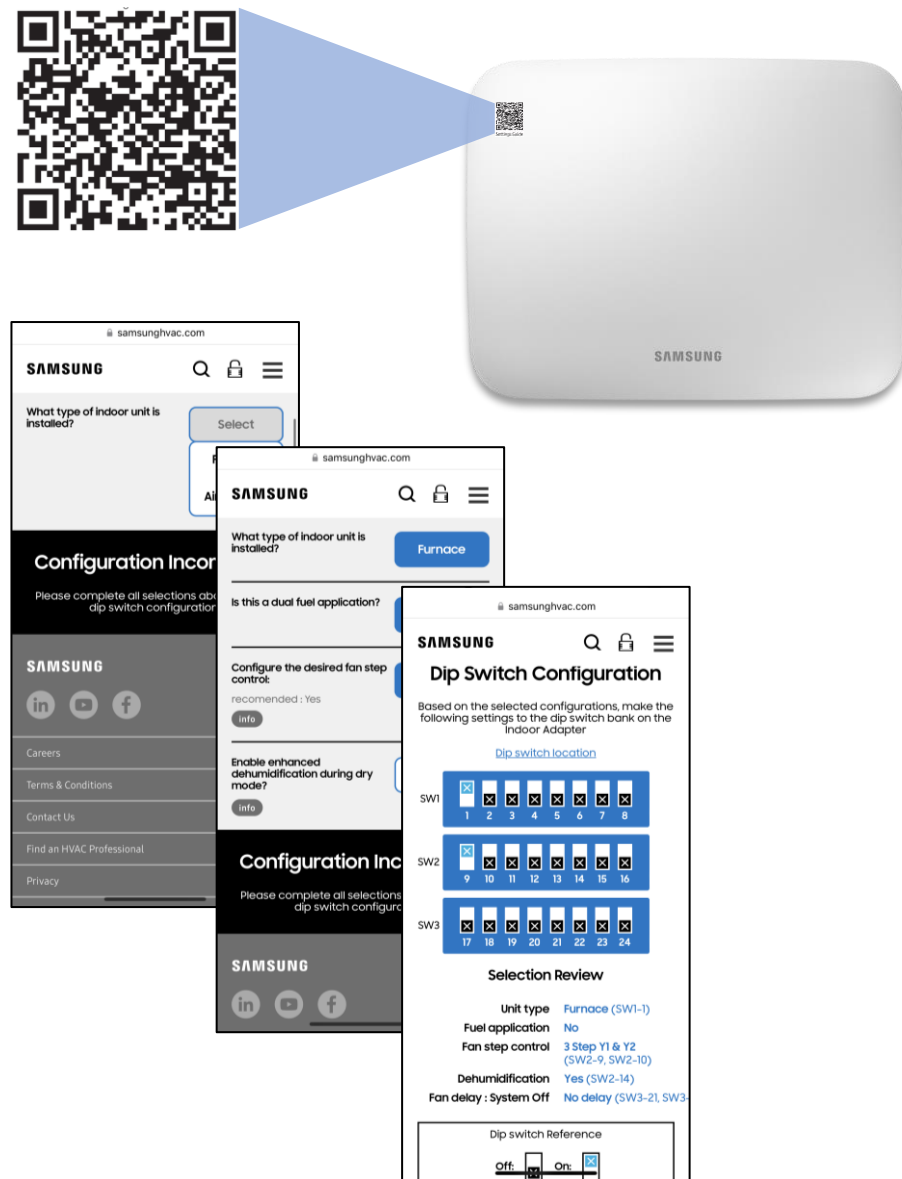
SW No.	Function	Descriptions
SW 1-1	System	- If an auxiliary heater is installed with a heat pump, the SW 1-2 should be set to the "On" status(SW 1-1 : Off). If not, while the W1 and W2 signals of the thermostat is activated, the heat pump turns off but the W1 and W2 signals of the ACU will not be activated. - If the SW 1-1 is set to "on" and SW 1-2 to off, the furnace output signal(Heating) of the thermostat should be connected to the W1 of the ACU. - If the SW 1-1 and SW 1-2 are set to "Off", when the W1 signal of the thermostat sends the ACU, the heat pump stops.
On	Furnace with A-coil (Dual fuel or cooling only)	
Off	Air handling unit (Default)	
SW 1-2	Auxiliary heater installed for heat pump (W1)	
On	Yes	- If there is the W2 of an auxiliary heater(2-stage), the SW 1-3 should be set to "On" status. - If the thermostat has the signal of "Emergency heat", set the SW 1-3 to "Off".
Off	No auxiliary heat / furnace (Default)	
SW 1-3	Heat sources option (W2)	
On	Auxiliary heat	
Off	Emergency heat (Default)	
SW 1-4 ~ SW 1-8	Reserved	

■ <https://www.samsunghvac.com/unitary-settings>

Integration Solutions

- System Configuration

- A QR code located at the top left of the ACU takes the installer to a dedicated web-based configuration tool
- A dynamic page will visually show dip switch option settings for simple configuration
- Just answer a few of questions and the configuration tool will make suggestions to walk you through it
- Direct Link:
<https://www.samsunghvac.com/unitary-settings>



ODU Configuration

- Have there been any changes made to the outdoor unit option settings?
- Is there a current restriction set within the ODU?
- Has the snow accumulation setting been changed?
- What mode of operation is the system set for?
- What is the low-pressure target settings
- What are the high-pressure targets if used as a heat pump?

ODU Setting Menu

Option item	SEG1	SEG2	SEG3	SEG4	Function
Channel address	0	0	A	U	Automatic setting (Factory default)
			00~15		Manual setting
Snow accumulation prevention control	0	1	0	0	Disabled (Factory default)
			0	1	Enabled
Step for Silence mode	0	2	0	0	Disabled (Factory default)
			0	1	Step1
			0	2	Step2
			0	3	Step3
Type of Silence mode	0	3	0	0	Automatic Silence mode (Factory default)
			0	1	Manual Silence mode
Temperature unit	0	4	0	0	Celsius
			0	1	Fahrenheit (default)

Option item	SEG1	SEG2	SEG3	SEG4	Function
Current restriction rate ¹⁾	0	6	0	0	100% (Factory default)
			0	1	95%
			0	2	90%
			0	3	85%
			0	4	80%
			0	5	75%
			0	6	70%
			0	7	65%
			0	8	60%
			0	9	55%
			1	0	50%
			1	1	100%
Dedicated mode for cooling/heating	0	7	0	0	Cooling / Heating operation (default)
			0	1	Cooling operation only
			0	2	Heating operation only
Long-piping pressure calibration ²⁾	0	8	0	0	Within 98.4 ft (30 m) (Factory default)
			0	1	98.4 ~ 164.0 ft (30 ~ 50 m) (Cooling:-9.96psi, Heating:+4.27psi)
Target low pressure calibration for cooling ³⁾	0	9	0	0	Initial set pressure for cooling (Factory default)
			0	1	-2.84 psi
			0	2	-5.69 psi
			0	3	-8.53 psi
			0	4	-11.38 psi
			0	5	-14.22 psi
			0	6	+2.84 psi
			0	7	+5.69 psi
			0	8	+8.53 psi
			0	9	+11.38 psi
			1	0	+14.22 psi

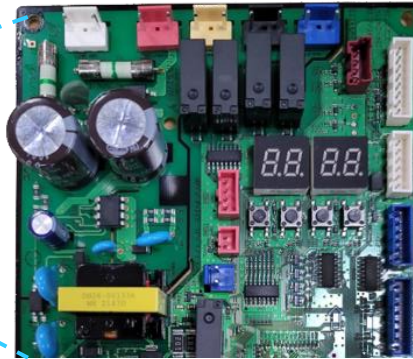
Option item	SEG1	SEG2	SEG3	SEG4	Function
Target high pressure calibration for heating	1	0	0	0	Initial set pressure for heating (Factory default)
			0	1	+2.84 psi
			0	2	+5.69 psi
			0	3	+8.53 psi
			0	4	+11.38 psi
			0	5	+14.22 psi
			0	6	-2.84 psi
			0	7	-5.69 psi
			0	8	-8.53 psi
			0	9	-11.38 psi
			1	0	-14.22 psi

New Defrost Settings

Cumulative defrost hour	Main	1	1	0	0	unused	default
				0	1	90 Mins	
				0	2	60 Mins	
				0	3	40 Mins	
Defrost Hz Offset	Main	1	2	0	0	unused	default
				0	1	-5Hz	
				0	2	-10Hz	
				0	3	-15Hz	
Exiting defrost Offset	Main	1	3	0	0	unused	default
				0	1	+3°C (5.4°F)	
				0	2	+6°C (10.8°F)	
				0	3	+9°C (16.2°F)	

How is the ODU Configured

- Some applications may require a cooling only or heating only installation and for that, the Hylex offers the mode lock function
- This setting is easily made by using the “K” buttons while following the table below.



Heating & Cooling
(Default)

or

Cooling only

or

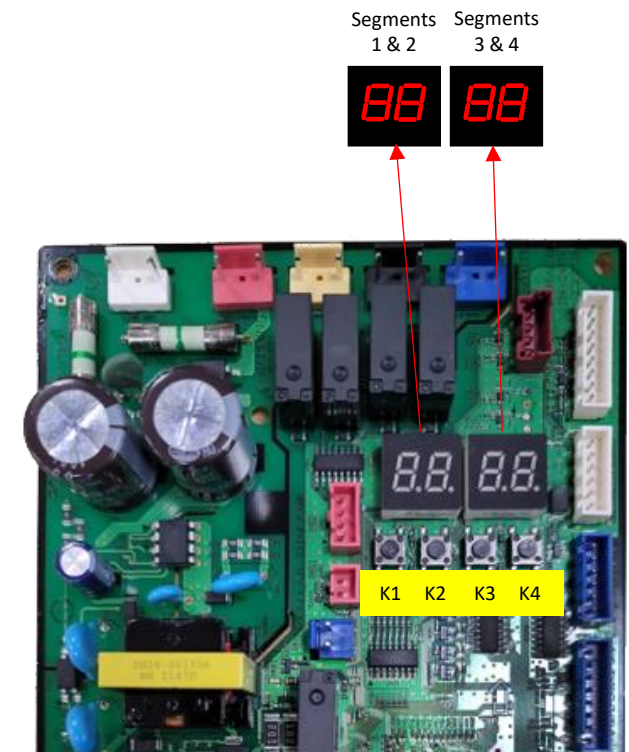
Heating only

- Mode Lock

SEG 1	SEG 2	SEG 3	SEG 4	Function
0	7	0	0	Cooling/Heating
		0	1	Cooling only
		0	2	Heating only

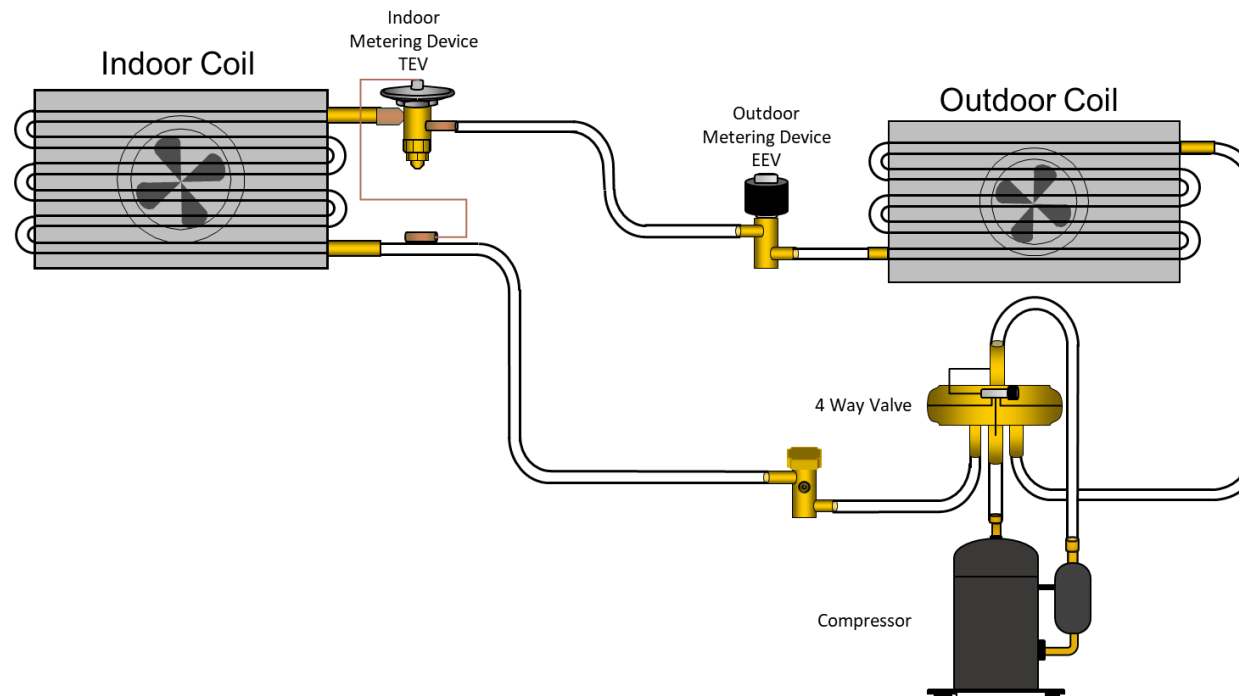
Entering the ODU option settings

- Press and hold the K2 button to enter ODU option menu
- Press K1 to change the value of the option
- Press K2 to transition through the options.
- Press and hold K2 to save the setting.

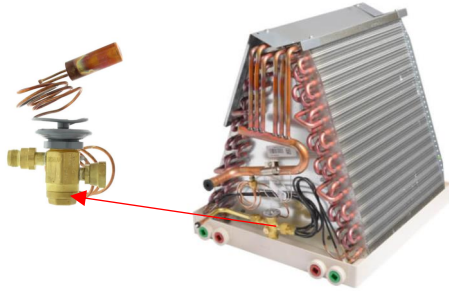


Operation

- In order to understand how the system operates it is important to understand the component that make up the refrigerant circuit.
- If you know how each component operates individually you will have a better understand on how the system operates in unison as a whole



Basic Refrigerant Components



Indoor TEV & Coil



ODU EEV



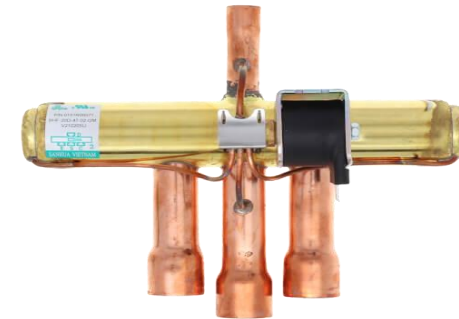
System Piping



outdoor Coil



Compressor



Reversing Valve

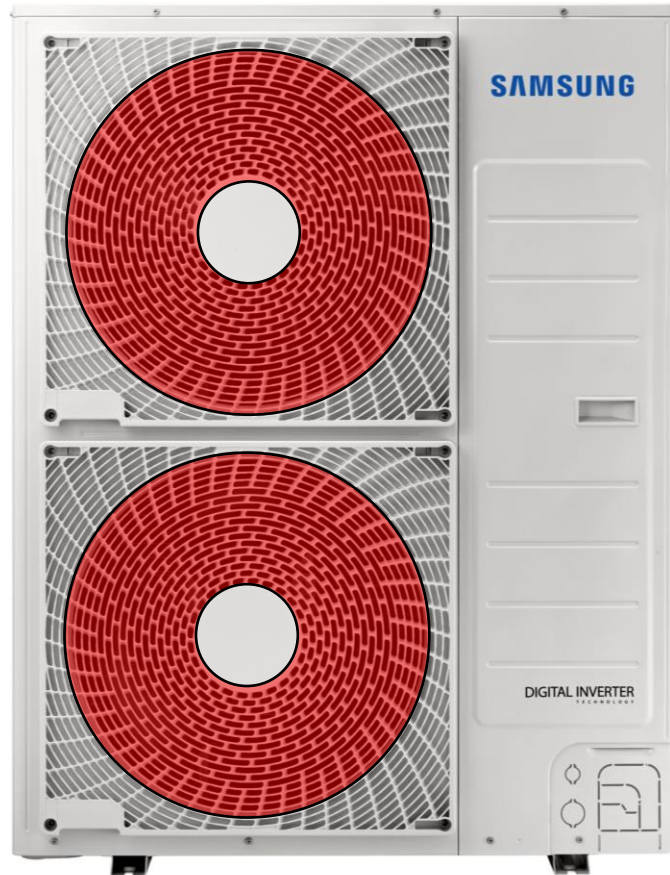
ODU Components Function in Cooling Mode



EEV



Pressure Transducer



Thermistor



4 Way Valve



Solenoid Valve

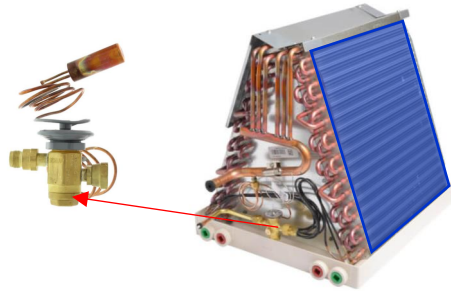


Fan Motor



Compressor

ODU Components Function in Cooling Mode



Indoor & TEV Coil



ODU EEV



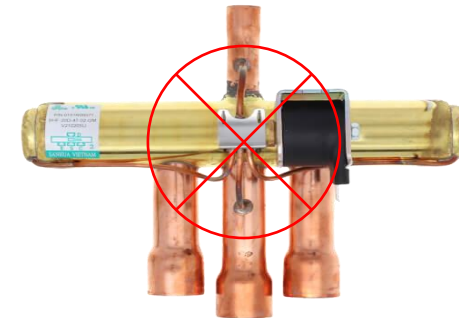
Piping



outdoor Coil



Compressor



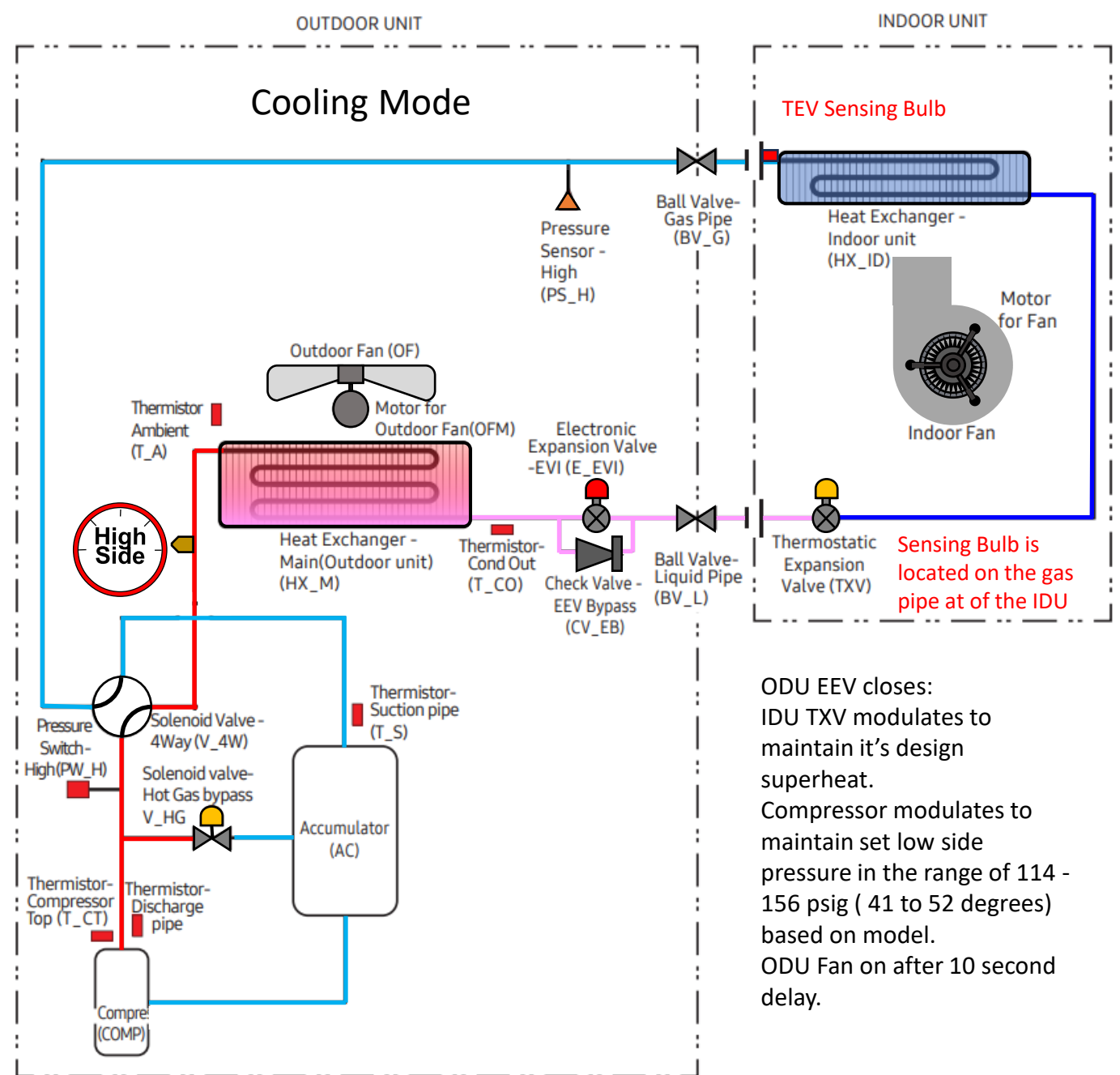
Reversing Valve

Cooling Cycle

Compressor runs to maintain target suction pressure measured by the sensor. The indoor TEV maintains superheat at the indoor coil of approximately 12 degrees. If indoor superheat is high, the discharge temperature will be high.

The system monitors discharge temperature and adjust the compressor accordingly.

The Main EEV in the DOU closes. Hot gas by-pass operates if suction pressure is low.



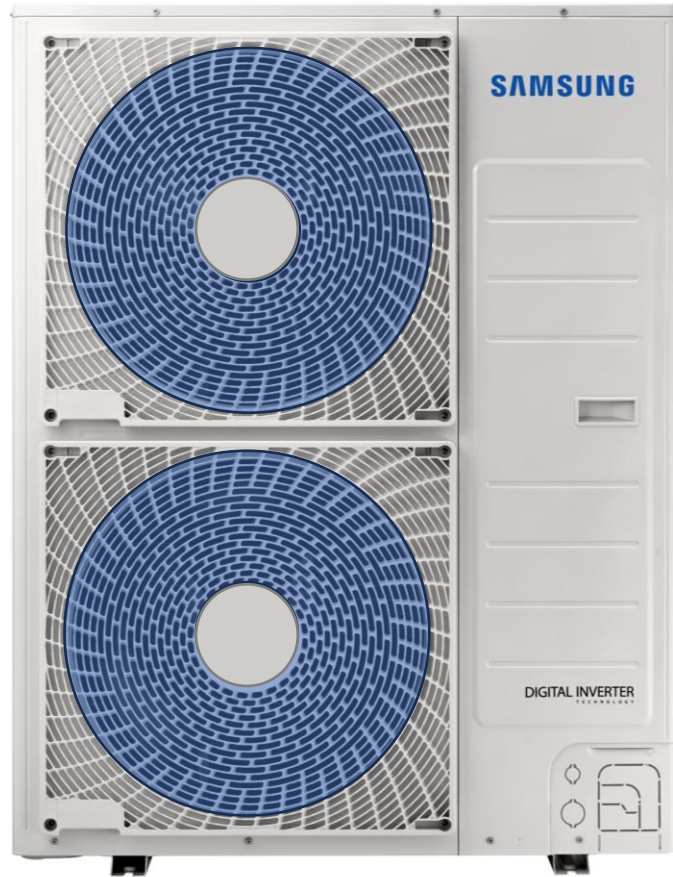
ODU Components Function in Heating Mode



EEV



Pressure Transducer



Thermistor



4 Way Valve



Solenoid Valve

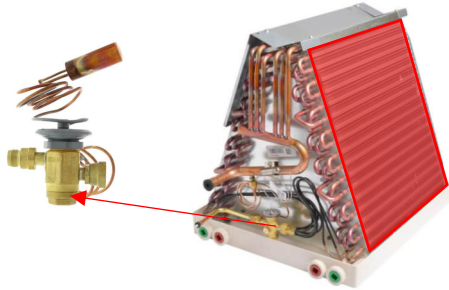


Fan Motor



Compressor

ODU Components Function in Heating Mode



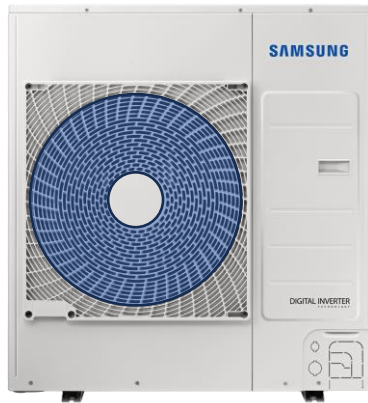
Indoor & TEV Coil



ODU EEV



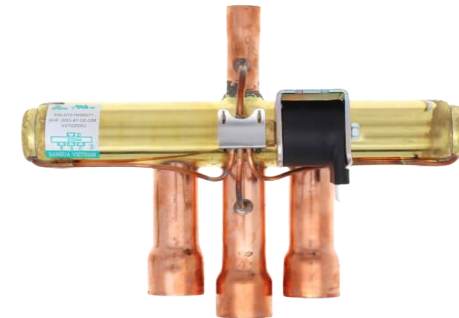
System Piping



outdoor Coil



Compressor



Reversing Valve

Reversing Valve

- Is energized in the heating mode.
- Coil Voltage is:
 - 208+ VAC

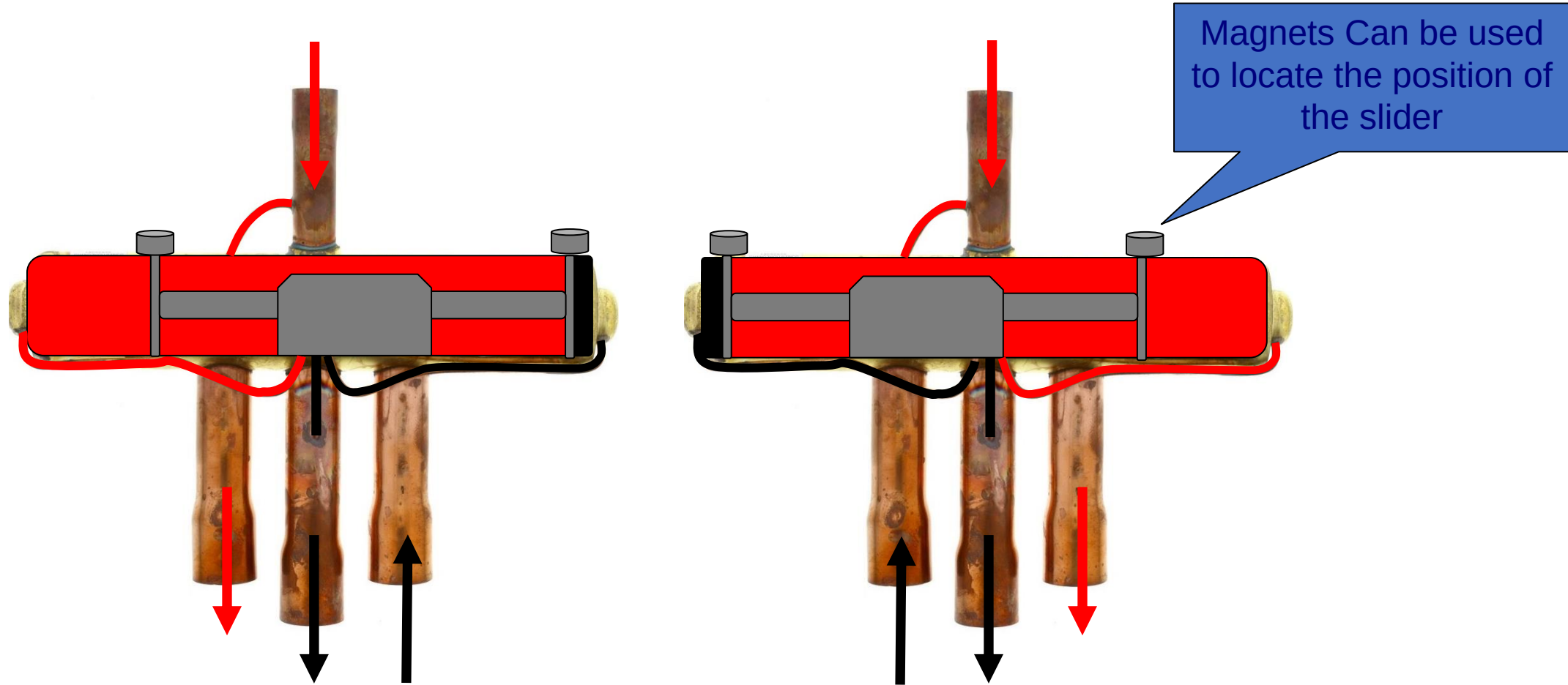
It's the difference in refrigerant pressure that causes the valve to shift



* Varies by model

Pressure differential between high side and low side causes the valve to shift!

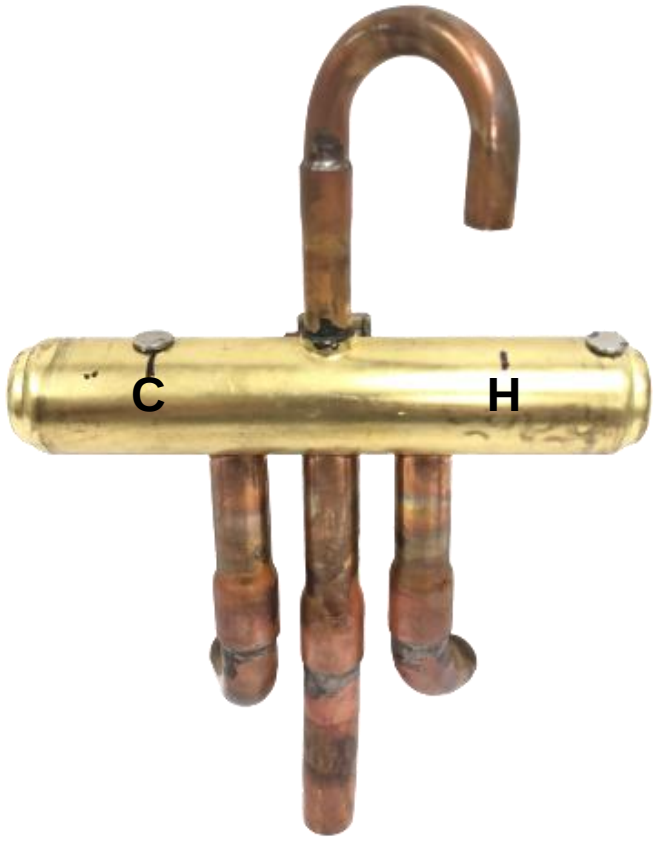
Refrigerant Flow



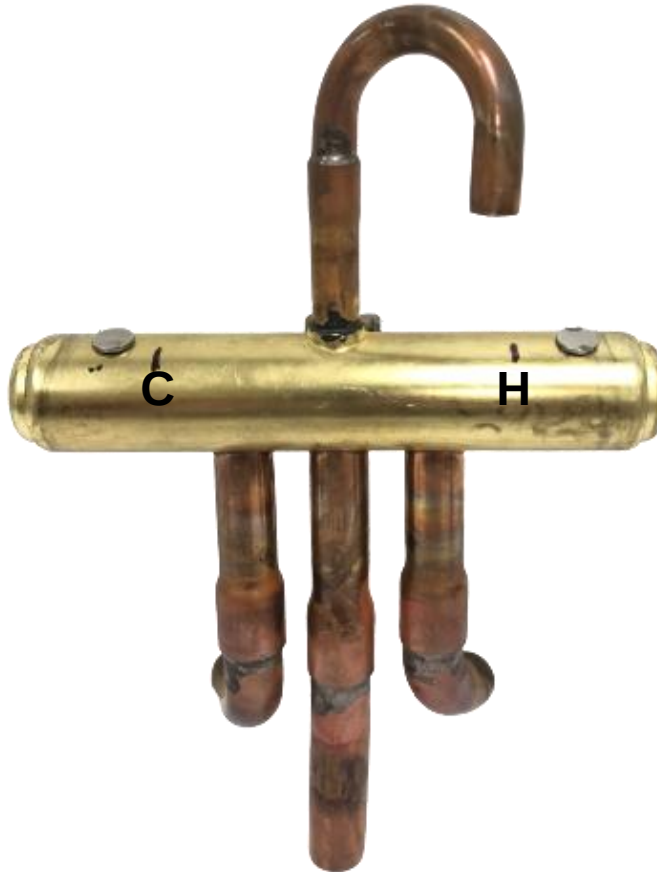
A 27 PSIG differential between high and low side pressure is required to move the slider. It may shift with as low as 18 PSIG differential.

Magnets Used as a Service Tool

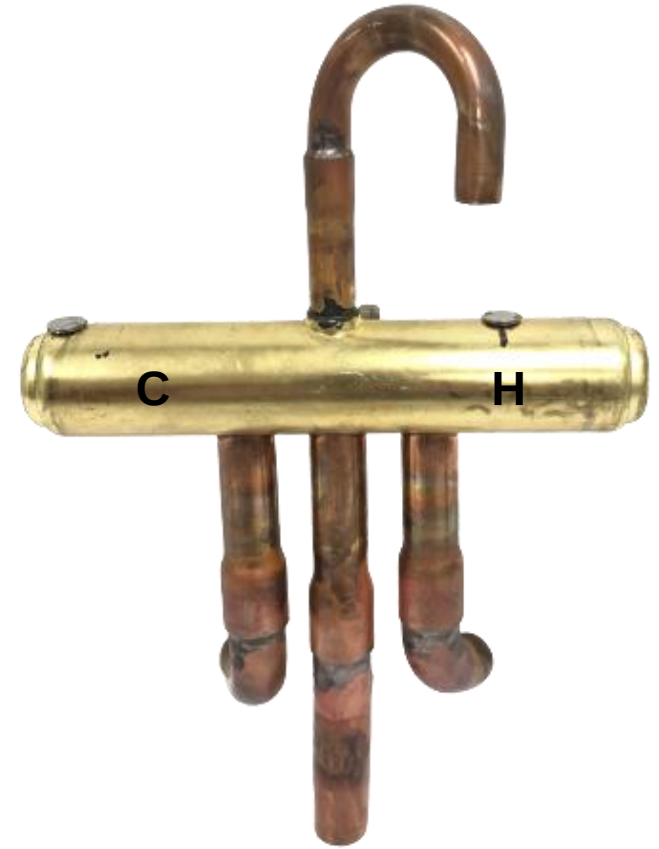
Normal Position Deenergized



Cooling Position

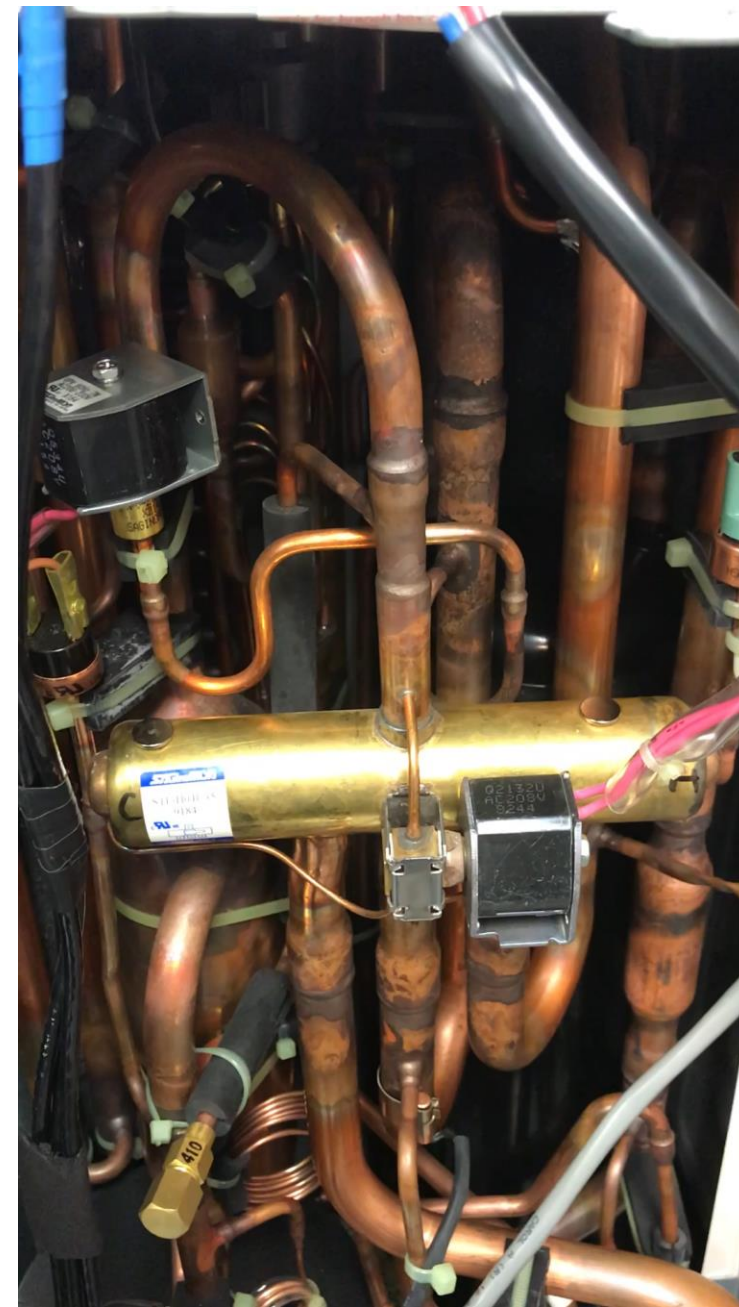


Mid Seated



Heating Position

Reversing Valve Operation



Reversing Valve

- This video show how you can use a magnet to find the position of the reversing valve slider and see its operation in the cooling mode.



Heating Cycle

The 4 way valve is energized in the heating mode
The compressor modulates to maintain target Head Pressure measured by the sensor.
The indoor TEV goes into By-pass mode.
The main EEV modulates to maintain ODU Coil superheat and discharge temperature.
IF the discharge temperature is high the adjust the compressor accordingly

ODU EEV set for an initial setting for 5 minutes then modulates to maintain 5.4 degrees of superheat.

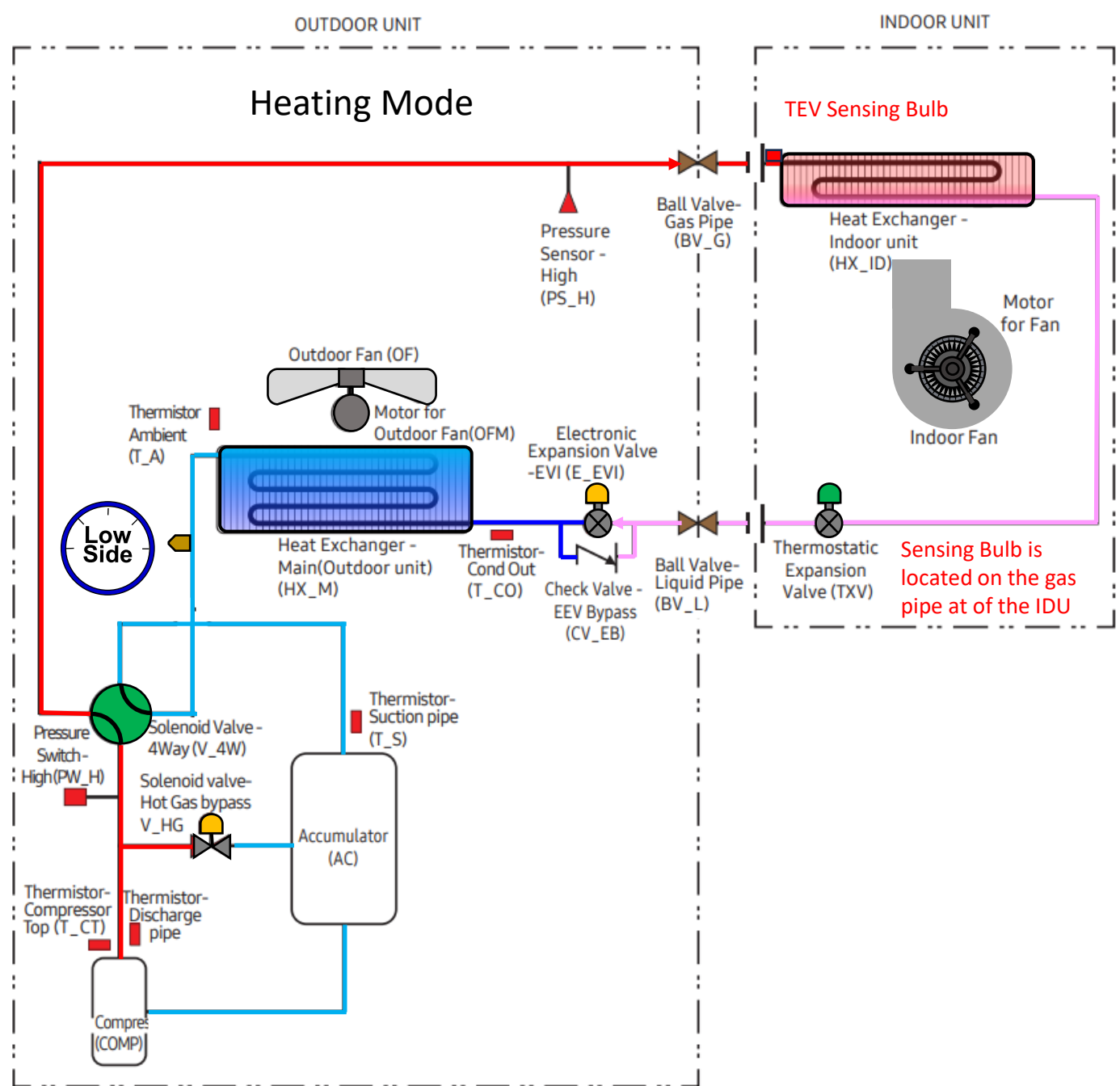
IDU TXV goes into by-pass mode.

Compressor modulates to maintain set high side pressure average between 355 psig (W1) – 370 psig (W2).

Reversing valve is energized after a 20 second delay.

Fan is delayed until High side saturation temperature reaches 93.2 degrees or after 5 minutes.

Snow Accumulation IF active in heating mode fan on for 1 minute when outdoor temperature is below 41 degrees from 30 minutes.



System Targets

How does the unit operate in each mode?



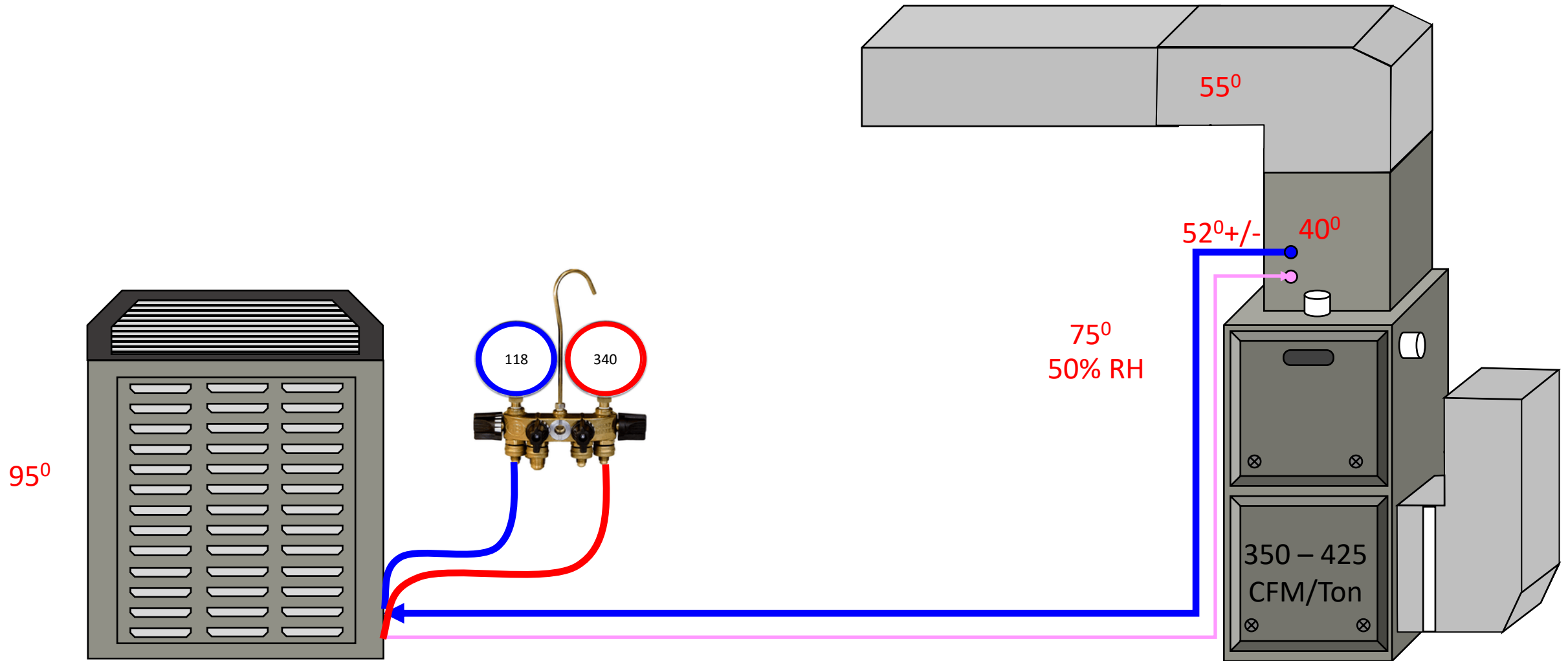
Low Side Pressure
Discharge Temperature



Head Pressure Control
Evaporator Superheat
Discharge Temperature

Design Conditions

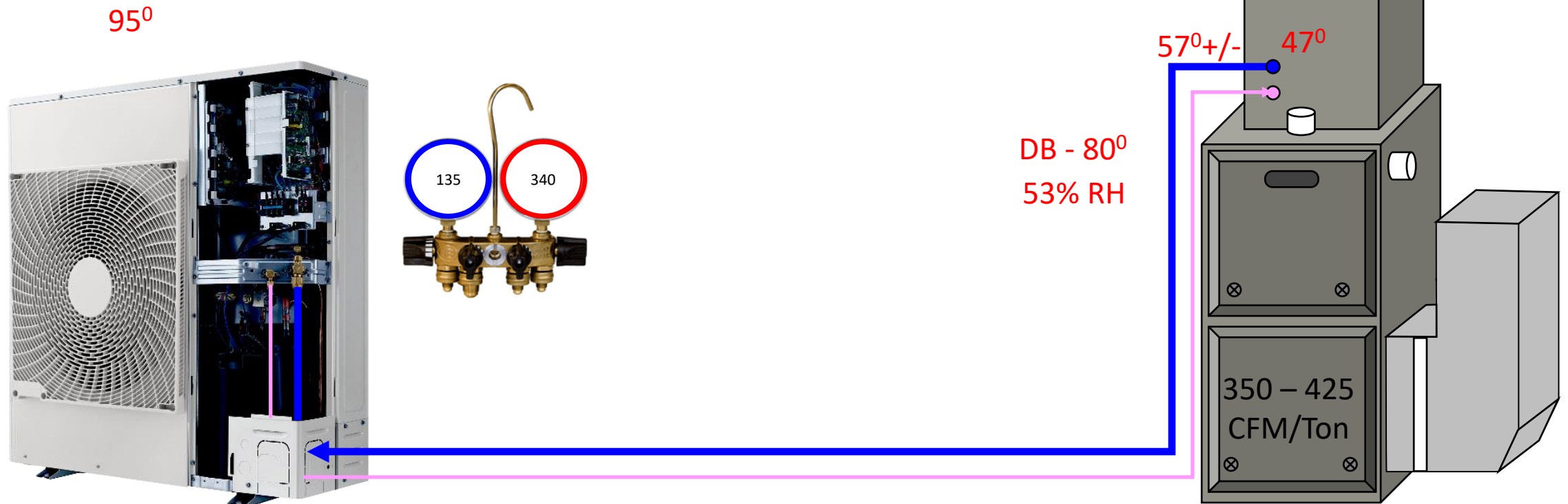
Unitary Air Conditioning



Design Conditions

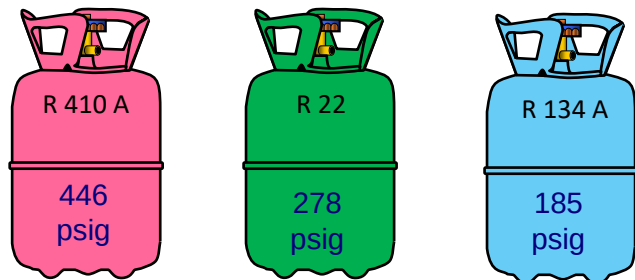
Hylex.

Cooling : Indoor temperature : 80°F(26.7°C) DB, 67°F(19.4°C) WB, Outdoor temperature : 95°F(35°C) DB, 75°F(23.9°C) WB
– Heating : Indoor temperature : 70°F(21.1°C) DB, 60°F(15.6°C) WB, Outdoor temperature : 47°F(8.3°C) DB, 43°F(6.1°C) WB
– Equivalent refrigerant pipe length 7.5m (24.6ft), Level differences 0m (0ft)



Refrigerants

- Refrigerants can exist in one of three states :
 - Saturated
 - Superheated
 - Sub-cooled
- Saturated refrigerants have a specific pressure / temperature relationship.
- If you know one of the values, you can determine the other using a PT Charts.



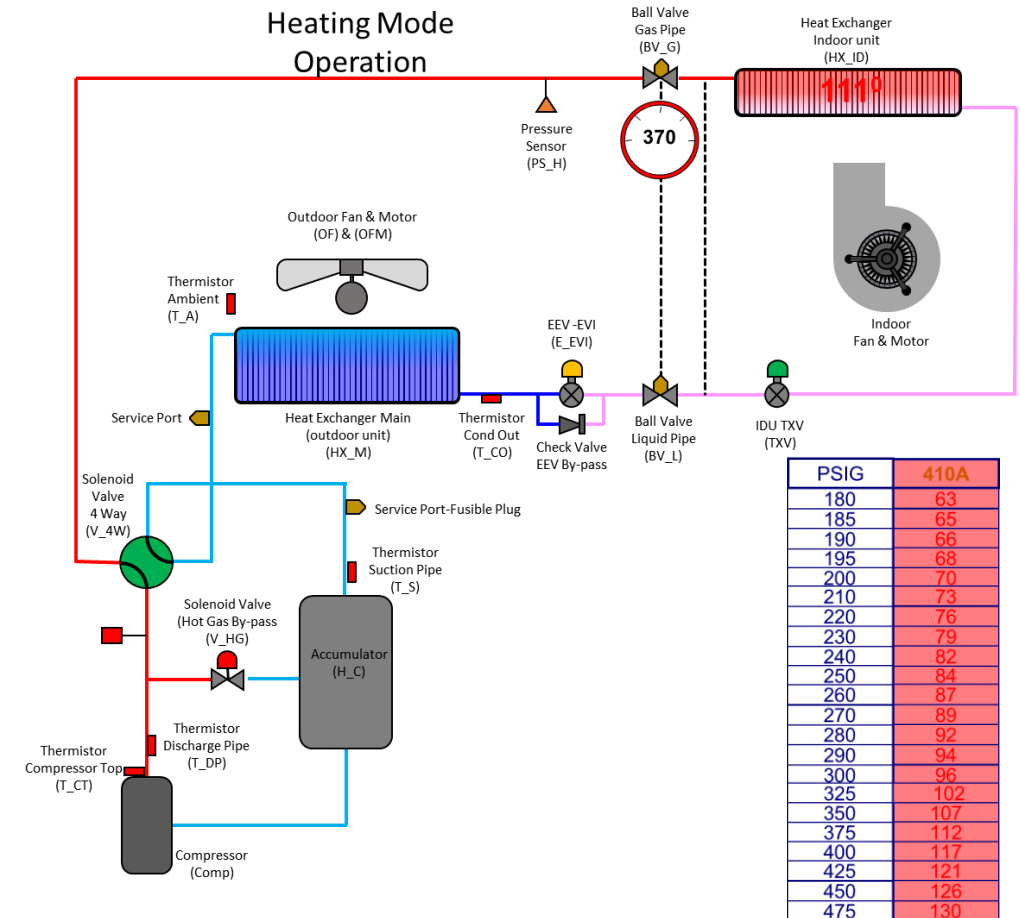
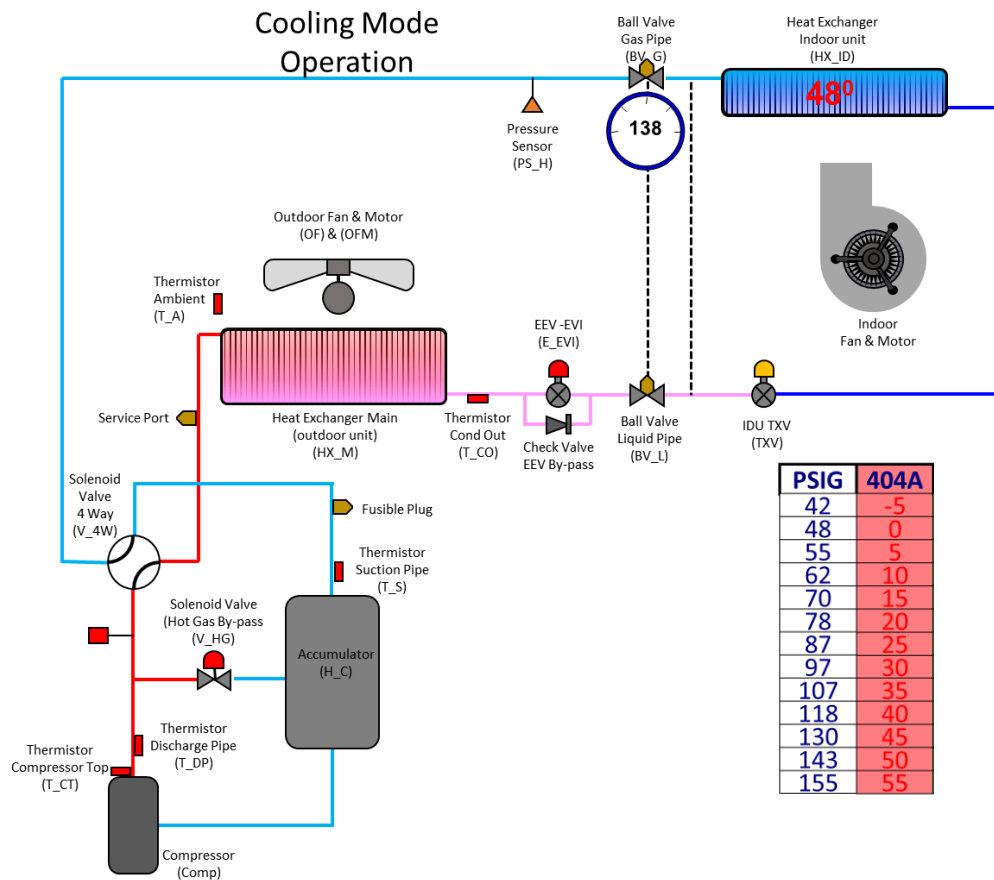
125°

PSIG	134a	22	410A
180	123	94	63
185	125	96	65
190	127	98	66
195	128	100	68
200	130	101	70
210	134	105	73
220	137	108	76
230	140	111	79
240	143	114	82
250	146	117	84
260	149	120	87
270	152	123	89
280	155	126	92
290		128	94
300		131	96
325		137	102
350		143	107
375		149	112
400		155	117
425			121
450			126
475			130

Saturation Temperatures

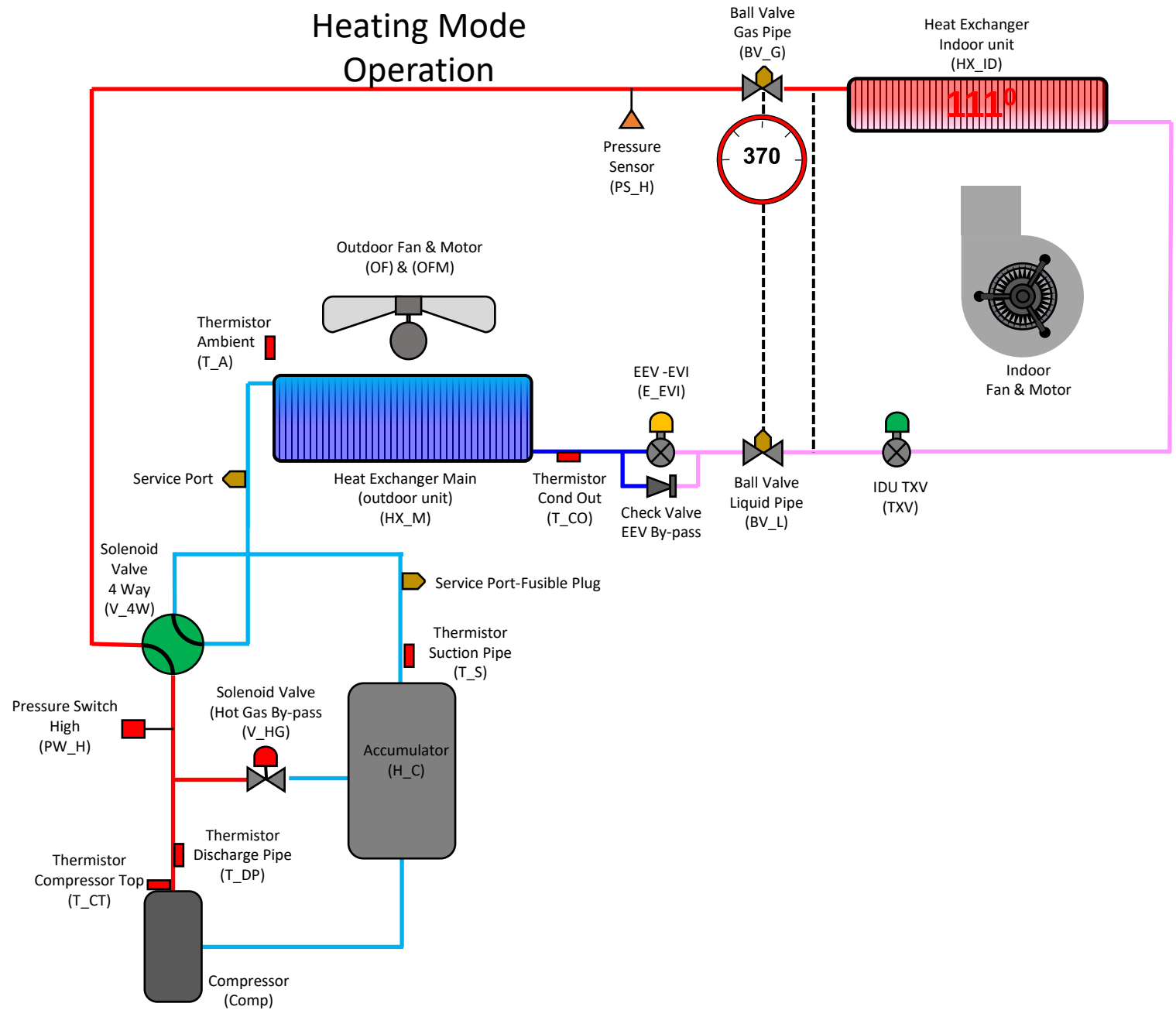
Pressure Temperature Relationship

The compressor modulates to maintain a specific pressure in the heating and cooling modes.
This provides the correct indoor coil temperatures for each mode.



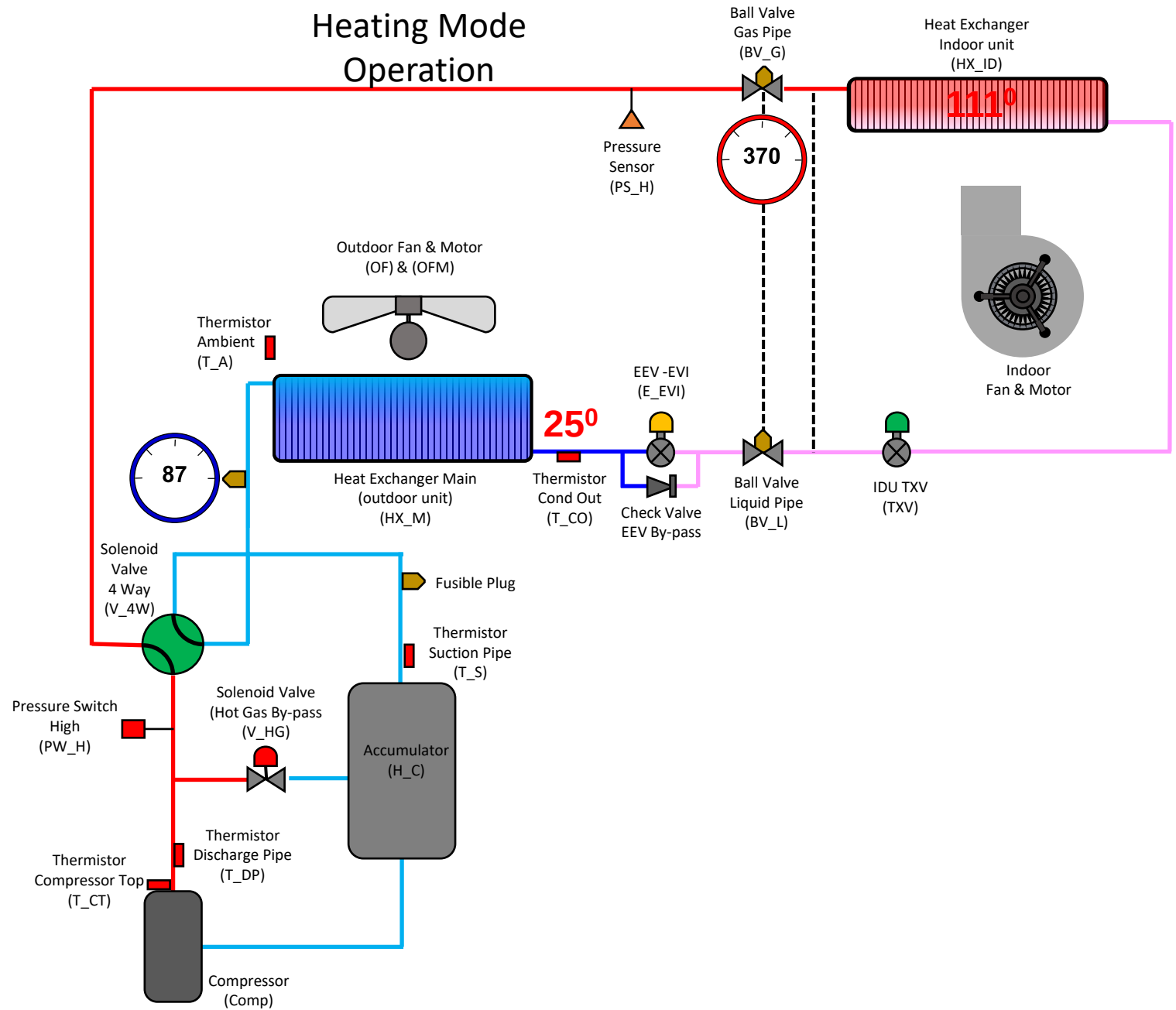
Example

- If the high side pressure is 370 psig.
- The saturation temperature of the indoor coil (condenser in the heating mode) would be 111 degrees.
- Its this pressure temperature relationship that allows the outdoor unit to know the temperature of the indoor coil.



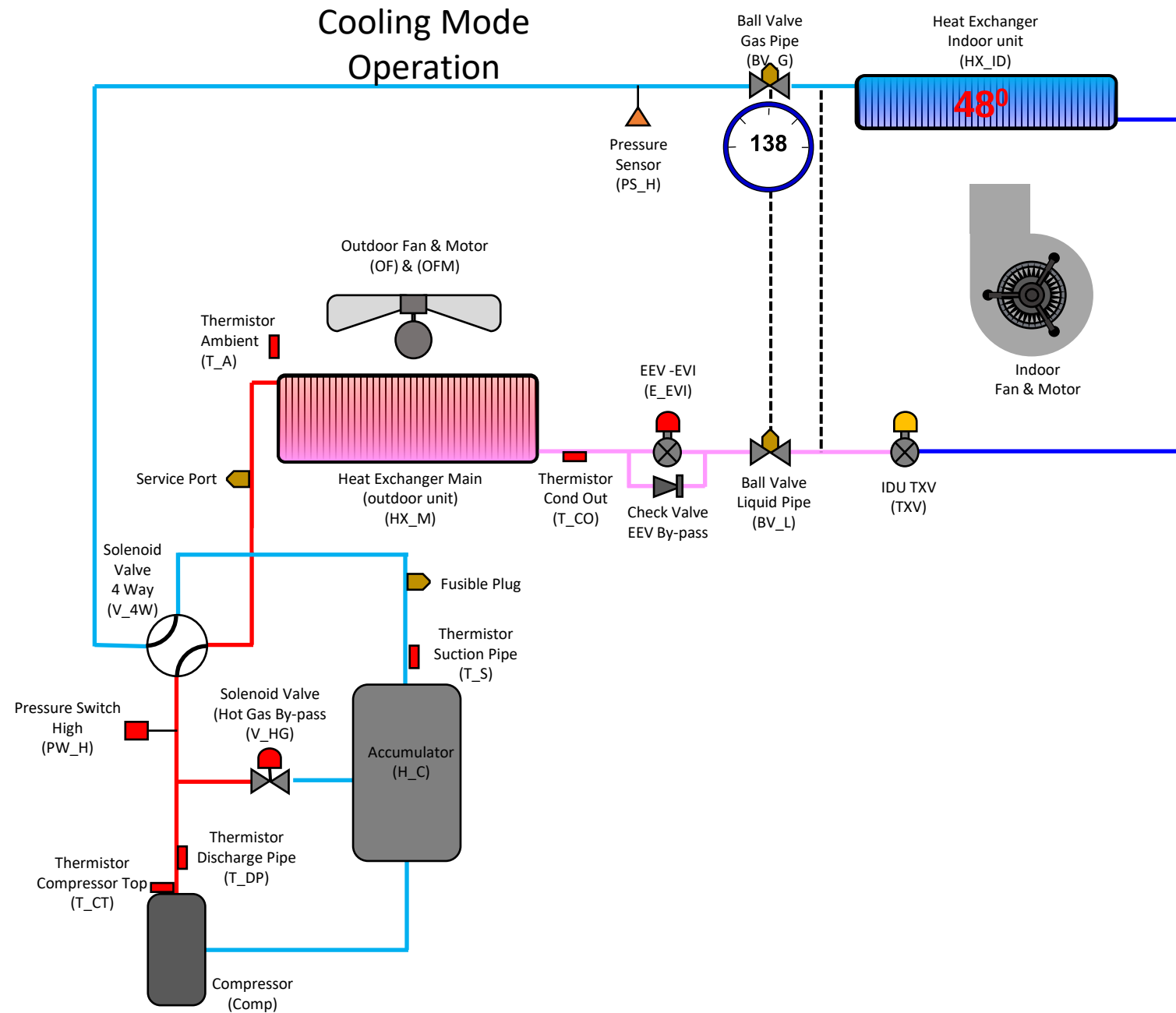
Example

- If the condenser out thermistor is 25 degrees.
- This is the low side saturation temperature of the outdoor coil (evaporator in the heating mode)
- Its corresponding pressure would be 87 psig
- Its this pressure temperature relationship that allows the outdoor unit to know the low side pressure.



Example

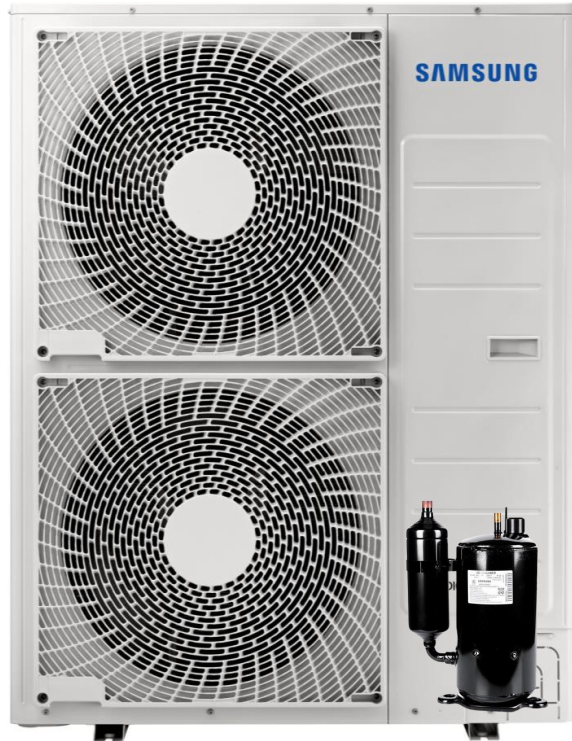
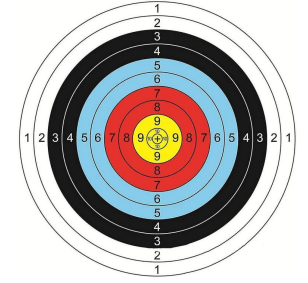
- If the low side pressure is 138 psig.
- The saturation temperature of the indoor coil (evaporator in the cooling mode) would be 48 degrees.
- Its this pressure temperature relationship that allows the outdoor unit to know the temperature of the indoor coil.



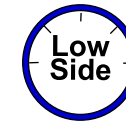
Compressor Targets

Example:

AC036BXUPCH/AA



Target Pressure		Saturation Temperature	
Stage2	Stage1	Stage2	Stage1
9	10	43.9	49.6
128	140		



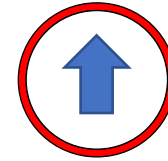
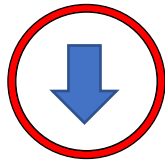
Target Pressure		Saturation Temperature	
Stage2	Stage1	Stage2	Stage1
26	25	110.9	85.4
370	256		



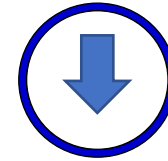
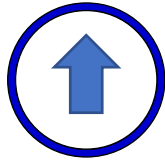
Compressor Speed – Pressure Relationship

(With no other changes in the system)

Heating
Mode



Cooling
Mode



Compressor Slow Down



Target Operation



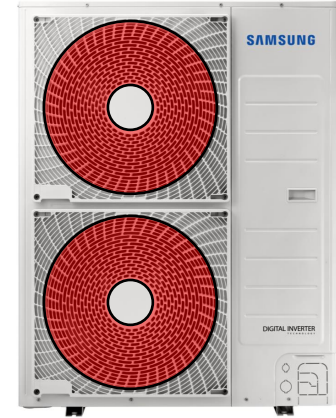
Compressor Speeds Up

System Default Targets

Cooling ❄️

AC024BXUPCH/AA		AC036BXUDCH/AA		AC036BXUPCH/AA		AC048BXUPCH/AA		AC060BXUPCH/AA	
Stage2	Stage1	Stage2	Stage1	Stage2	Stage1	Stage2	Stage1	Stage2	Stage1
9.5	10.5	9	10	9	10	8.5	9.5	8.5	9.5
135	150	128	140	128	140	121	135	121	135

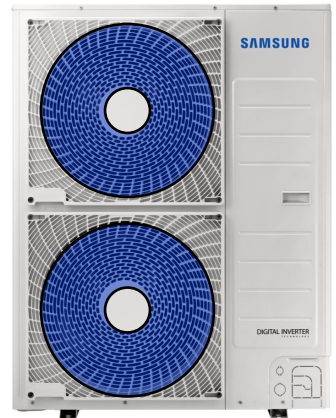
x14.223



Heating 🔥

AC024BXUPCH/AA		AC036BXUDCH/AA		AC036BXUPCH/AA		AC048BXUPCH/AA		AC060BXUPCH/AA	
Stage2	Stage1	Stage2	Stage1	Stage2	Stage1	Stage2	Stage1	Stage2	Stage1
26	25	26	25	26	25	28	27	25	24
370	256	370	256	370	256	398	384	256	341

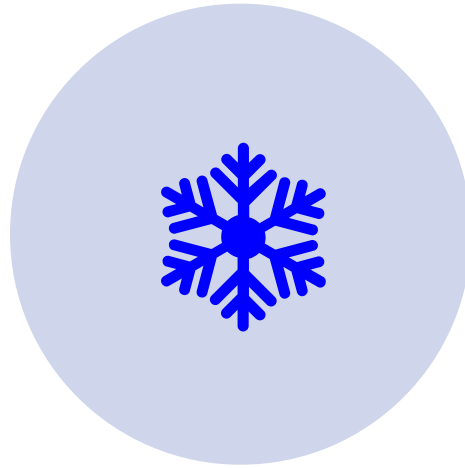
x14.223



Compressor Regulation



IN HEATING MODE, THE COMPRESSOR RUNS TO
MAINTAIN A TARGET HIGH SIDE PRESSURE
THE EEV MODULATES TO MAINTAIN THE CORRECT
AMOUNT OF SUPERHEAT



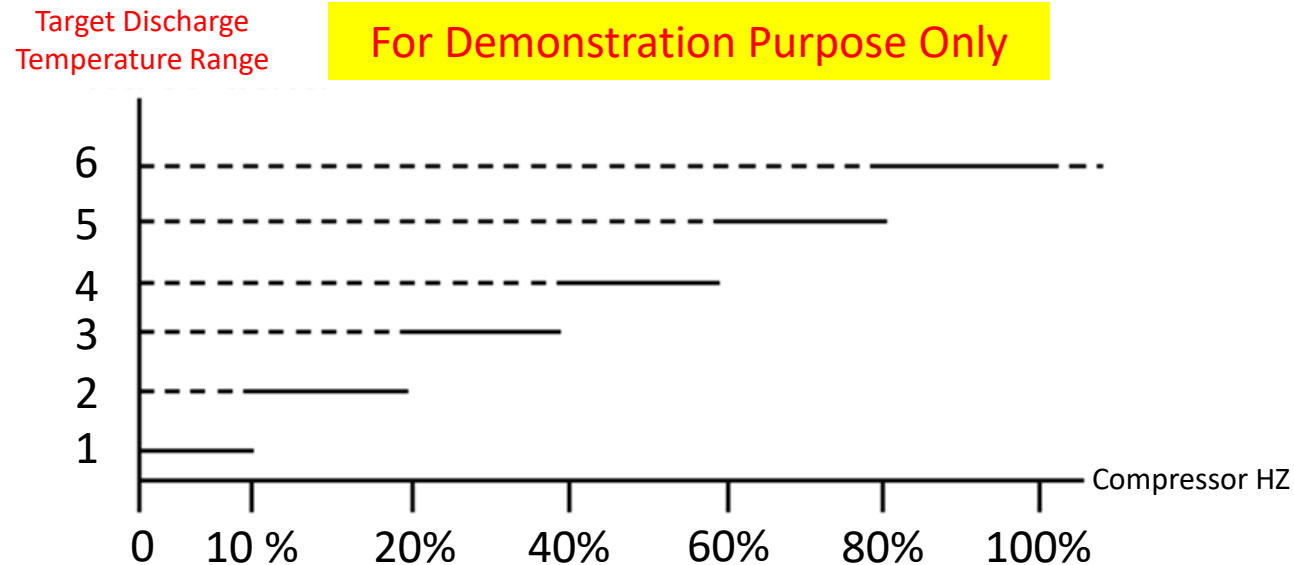
IN COOLING MODE, THE COMPRESSOR MODULATES TO
MAINTAIN A TARGET LOW SIDE PRESSURE
AND TARGET DISCHARGE



THE ARE SAFETY CONDITIONS THAT WILL OVERRIDE
COMPRESSOR OPERATION
FLOOD BACK CONDITION
(LOW DISCHARGE TEMPERATURE}
HIGH DISCHARGE TEMPERATURE

Target Discharge

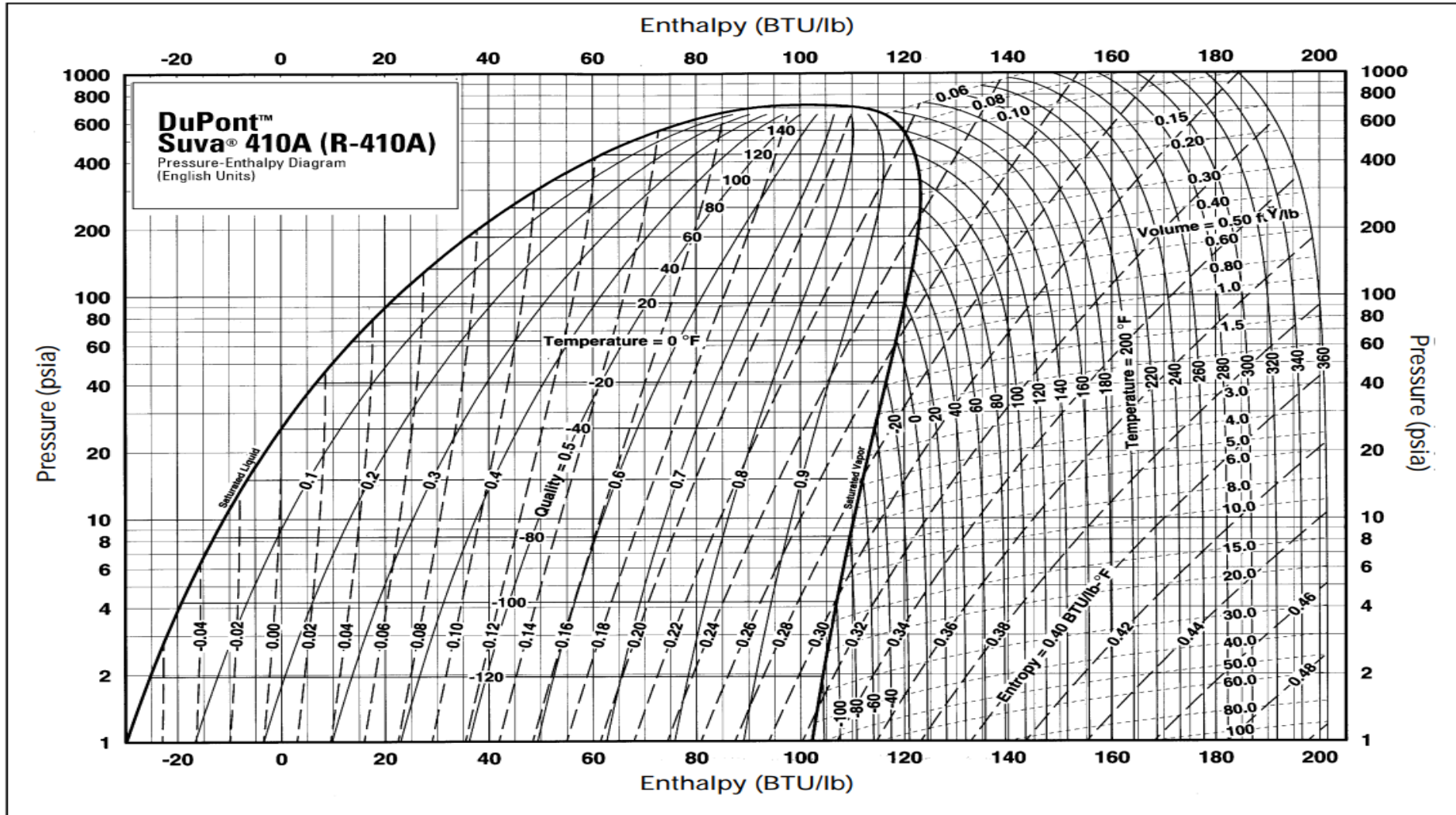
- The target discharge temperature is set based on the demand of the system.
- The system looks at compressor Hz and outside air temperature to determine the target discharge temperature.
- Target Discharge temperature can be viewed on the CAC, Hylex and FJM Products.
- Target discharge temperatures are different in the heating and cooling modes
- Where do these targets come from?



PE Diagrams

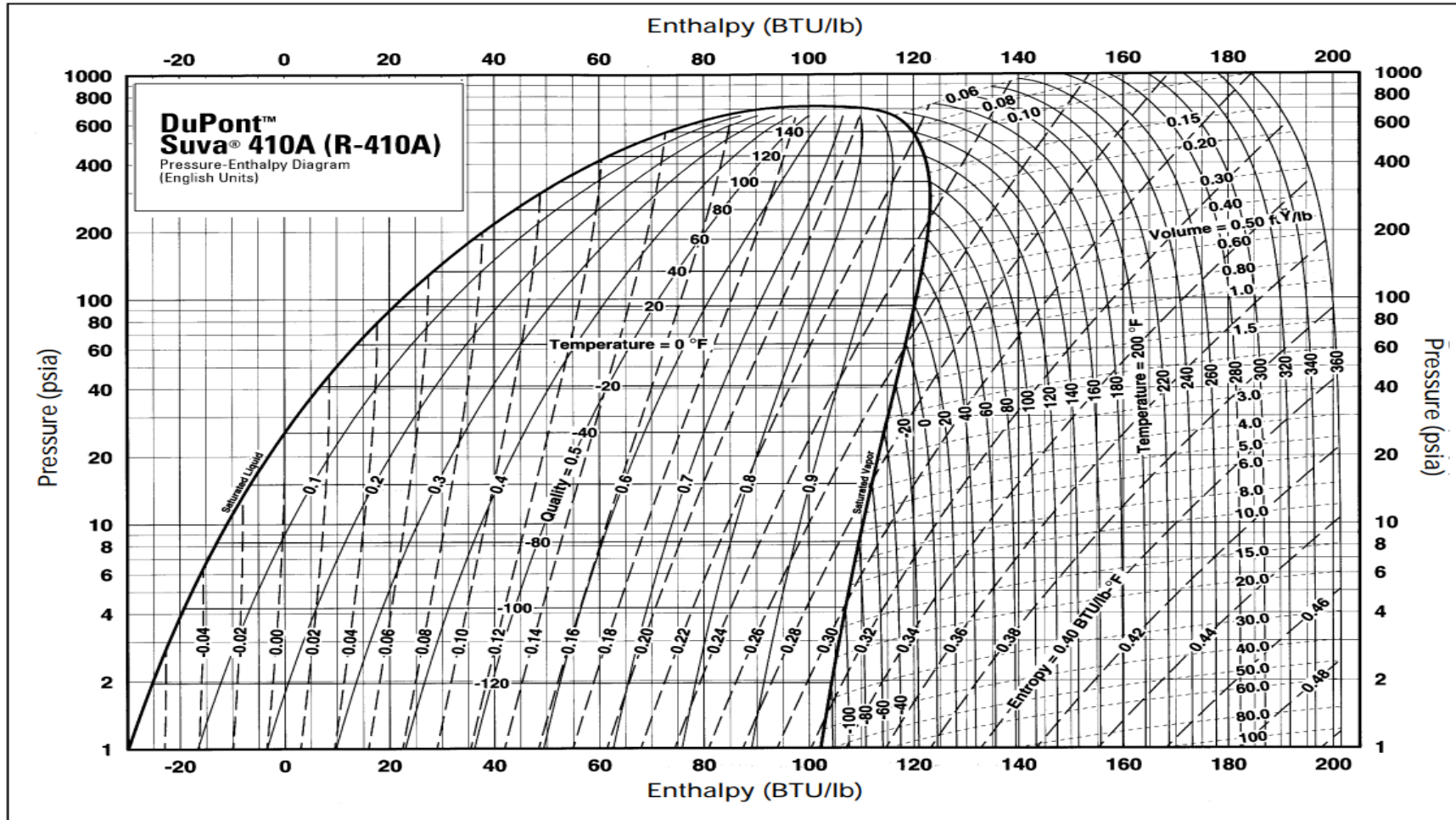
- Are charts that describe the relationship between pressure and enthalpy (total heat content)of a select refrigerant.
- They are the engineering charts for a refrigeration system.
- They show the total heat transfer within a system.
- A refrigeration's system operation is extremely predictable.
- PE diagrams show the operating characteristics of a system at various operating conditions.
- They are used to calculate:
 - Net Refrigeration Effect of the Evaporator
 - Heat of Compression of the compressor
 - Total heat of rejection from the condenser
 - Discharge temperature
 - The amount of refrigerant required for system operation

PE Diagrams

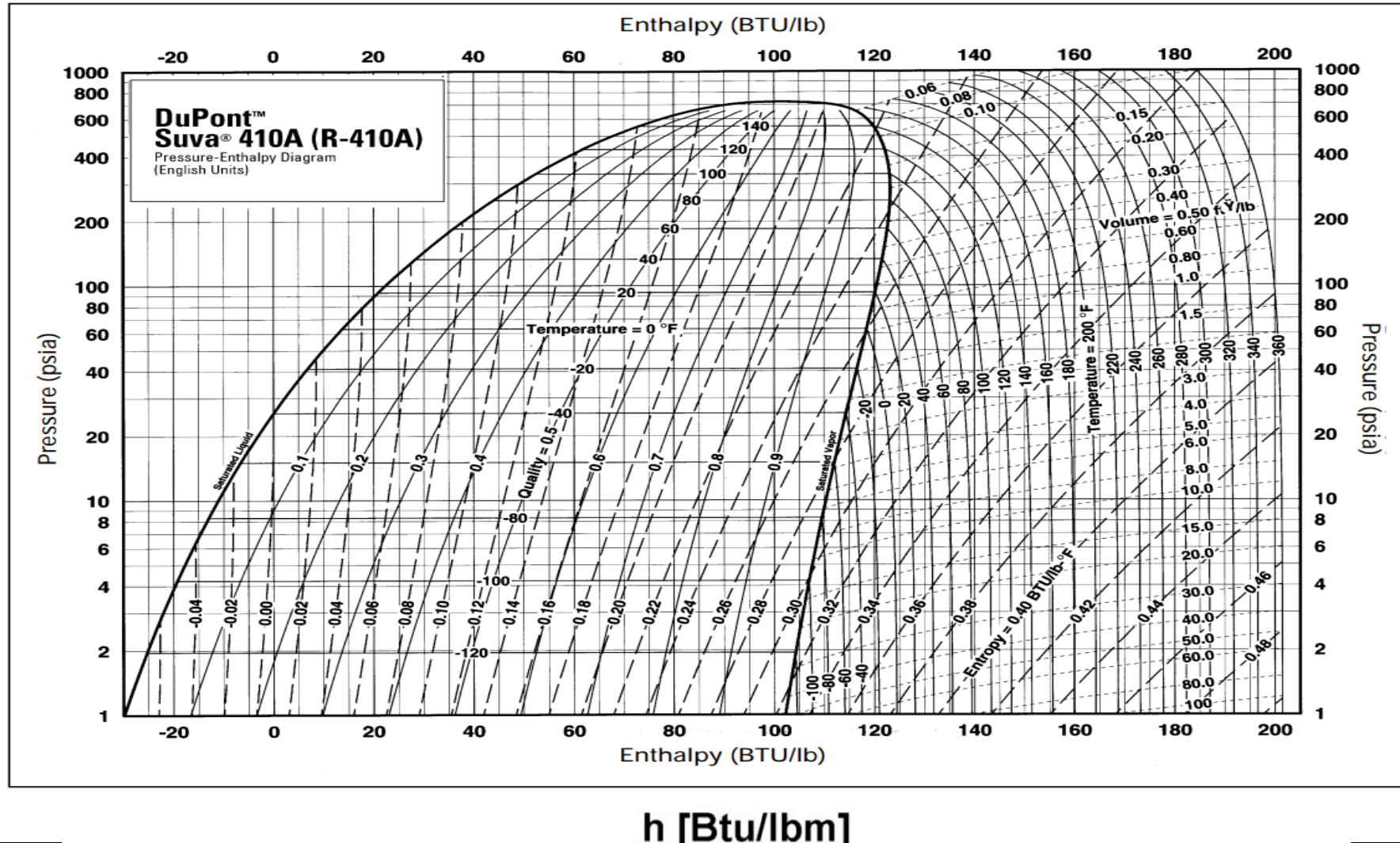


PE Diagrams

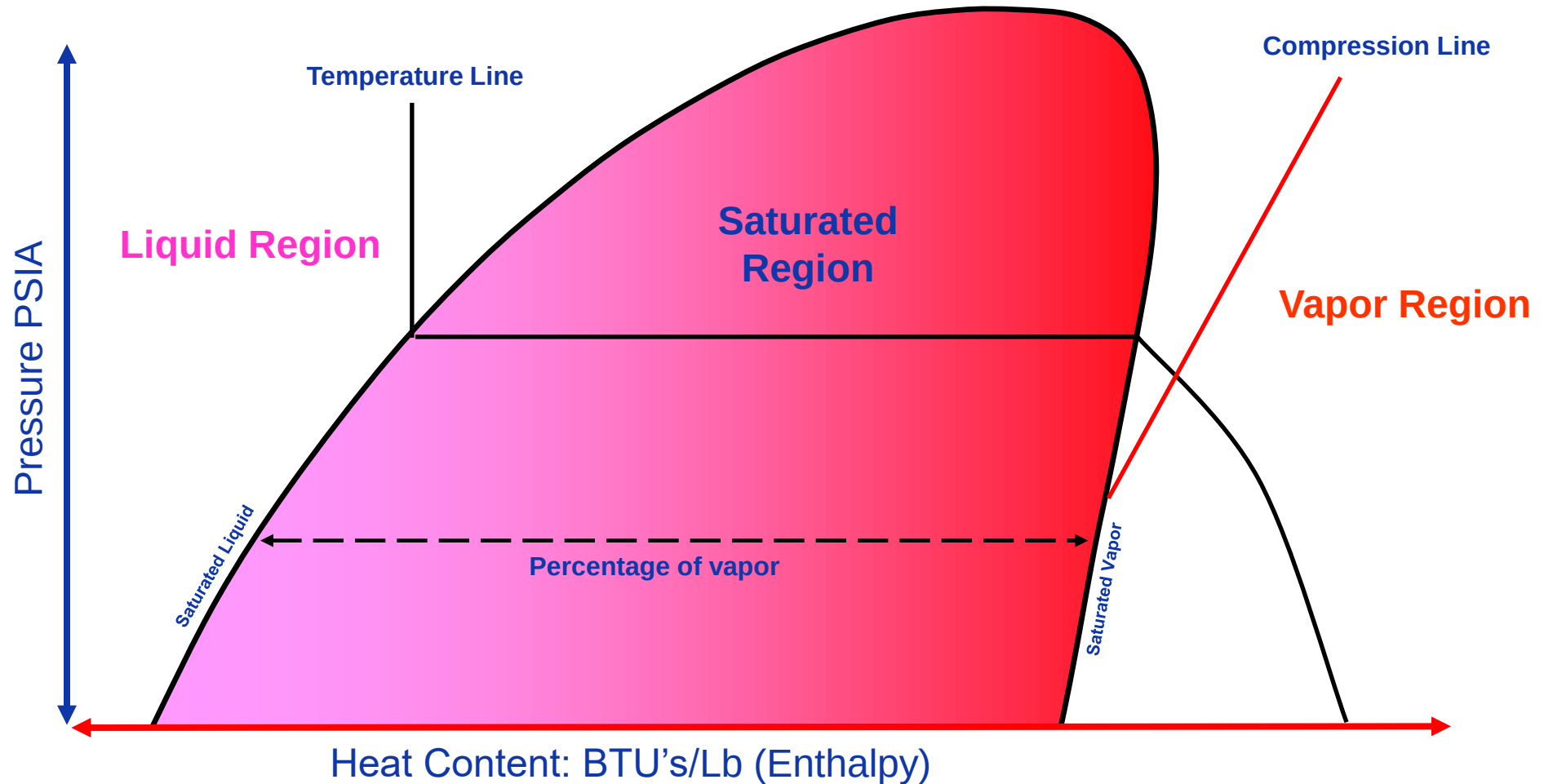
Combines slides 65&66



Understanding PE Diagrams



Understanding PE Diagrams

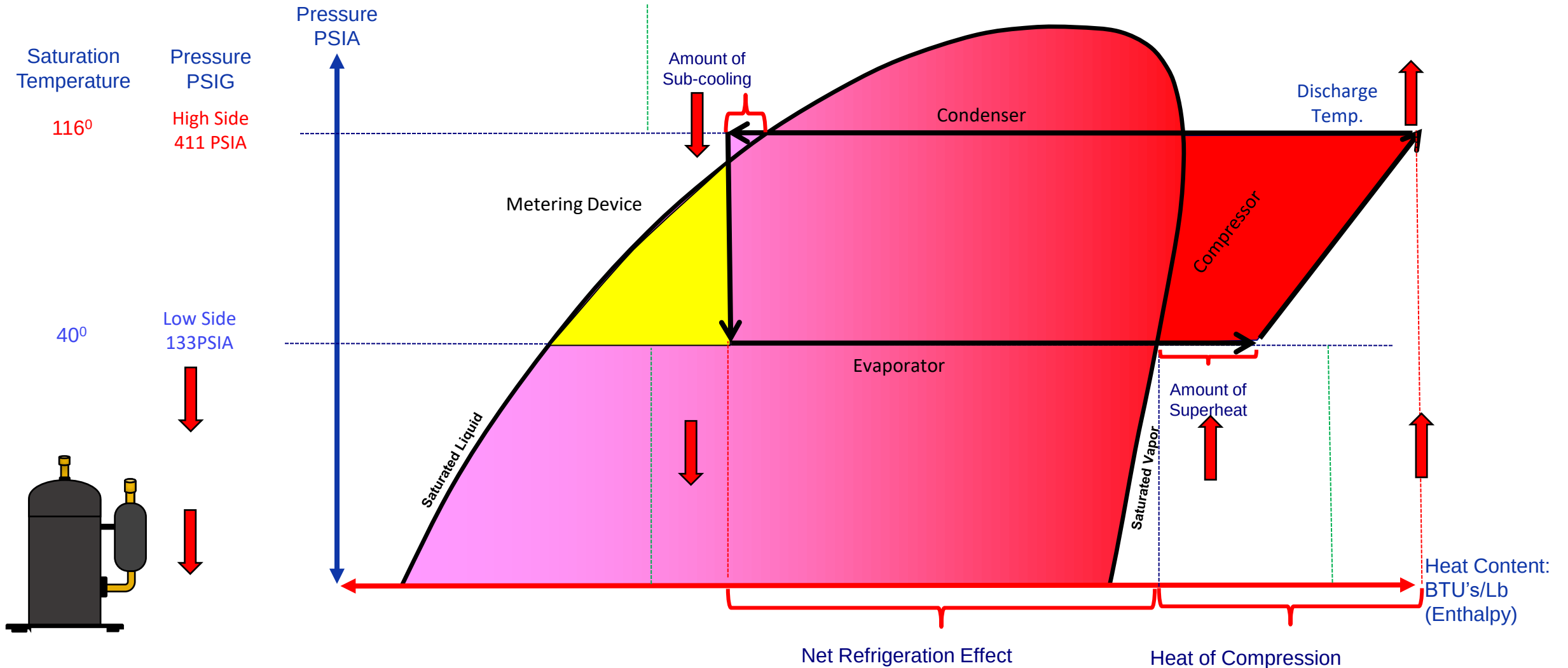


Putting it to use System Design

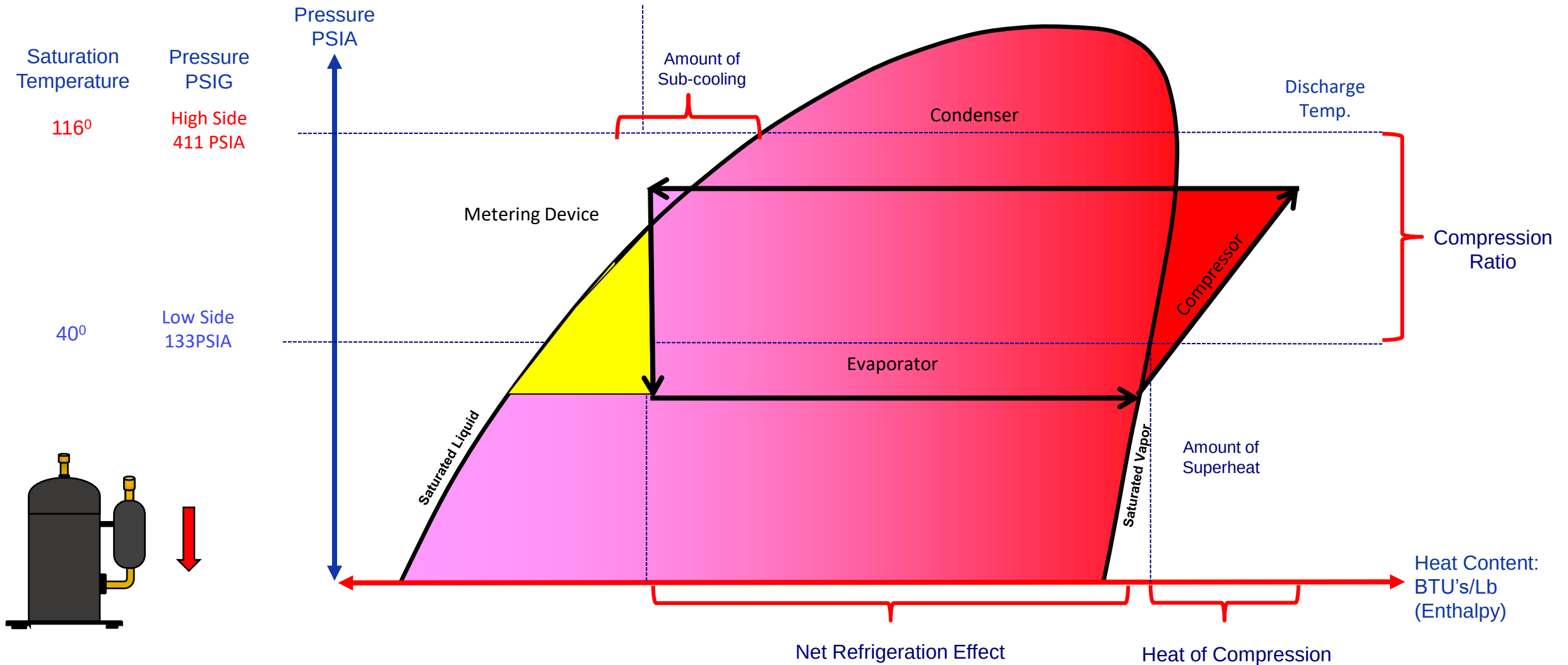


Why are they useful?

What Happens if a system is low on gas.

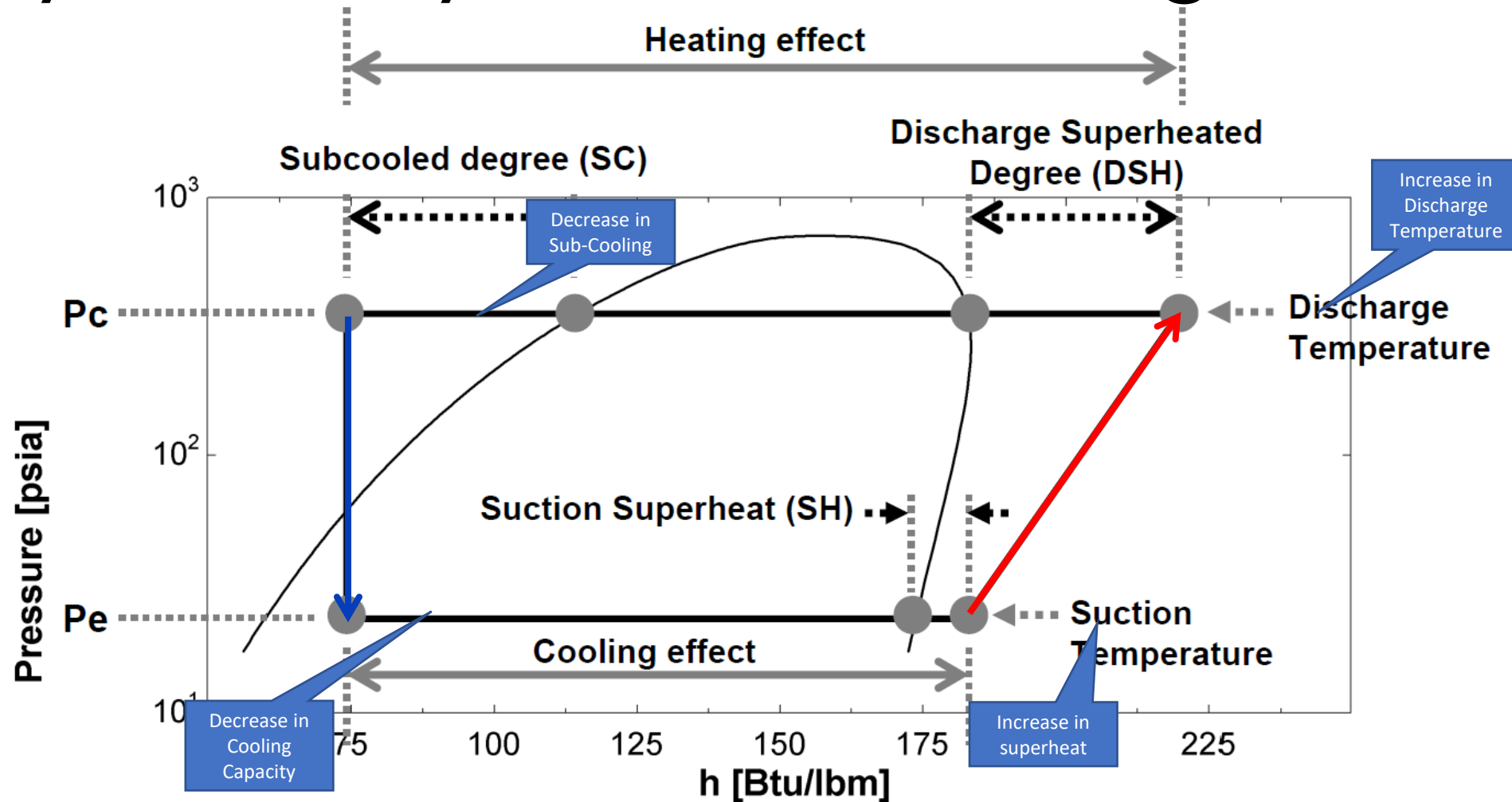


Low air flow



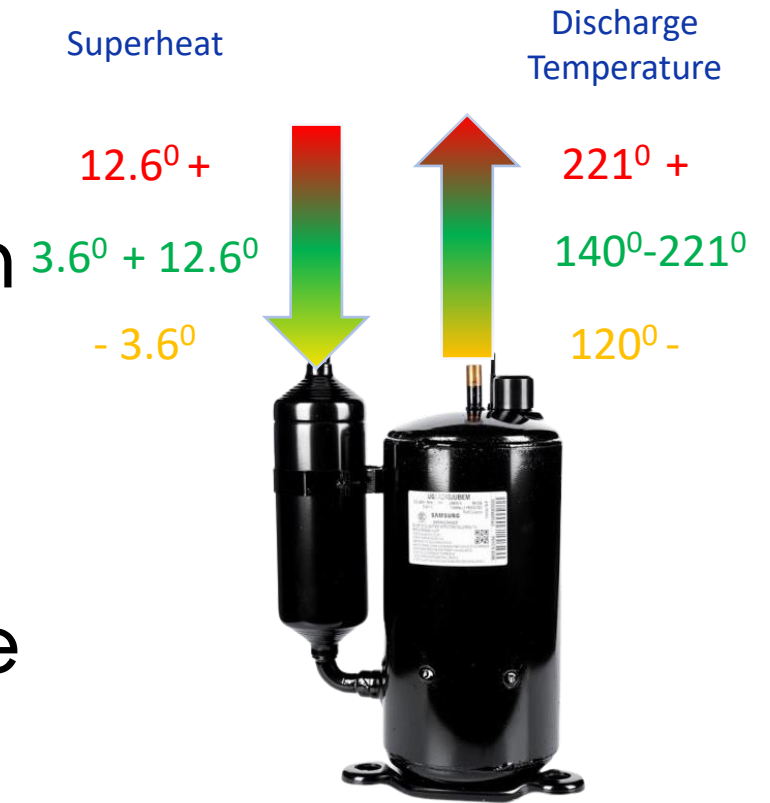
What do they tell us?

Why are they a useful training aide?



Discharge Temperature Range

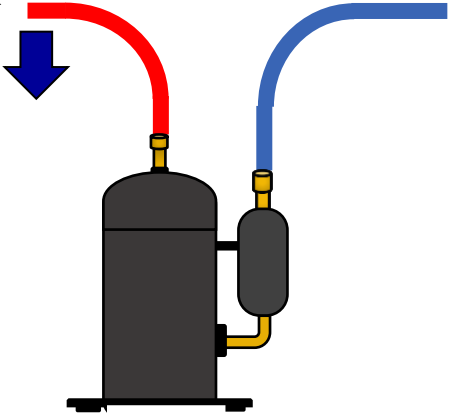
- The discharge temperature is affected by suction temperature coming into the compressor
- High Evaporator superheat results in high discharge temperatures.
- If the system charge, air flow and evaporator superheat are within acceptable ranges then the discharge temperature will be close to the target discharge temperature.



Discharge Temperature protections Vs Compressor Frequency

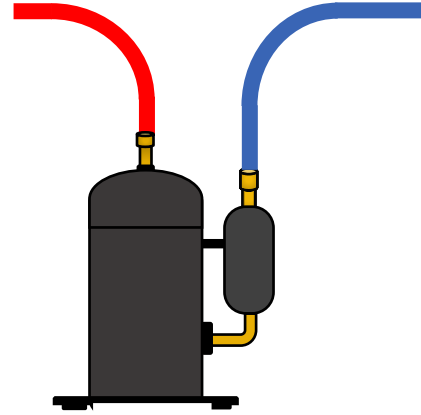
(With no other changes in the system)

Flood Back Condition



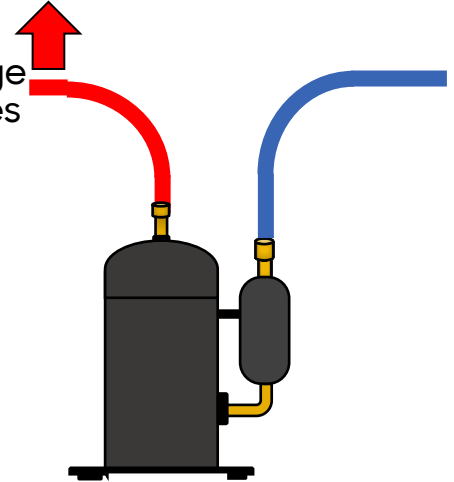
Compressor Slow Down

Target Temp



Stable Operation

High Discharge Temperatures



Compressor Slows Down

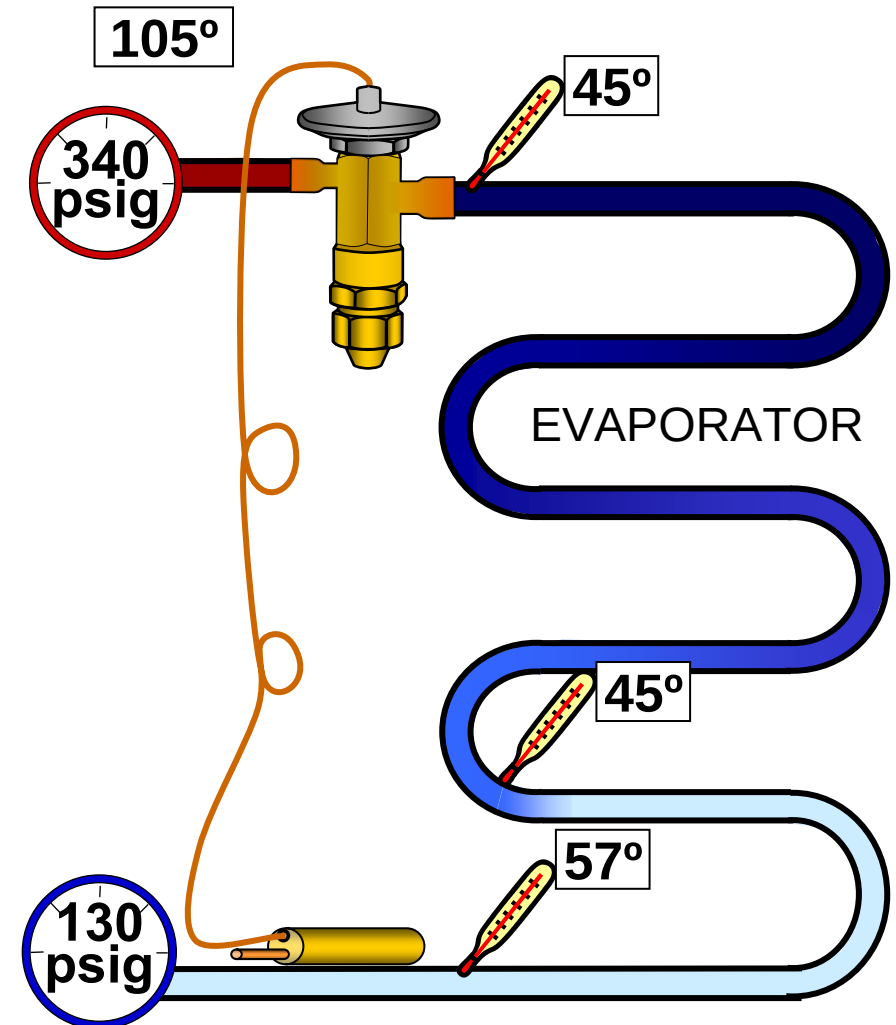
How to Calculate Super Heat

- Measure the refrigerant pressure.
 - Best practice are to measure the pressure at the location where you are calculating superheat. (this is not always possible).
- Use this pressure to determine the saturation temperature from the PT chart.
- Measure the temperature of the refrigerant pipe.
- Superheat = Measured line temperature – Saturation Temperature.

Calculating IDU Superheat

PSIG	410A
42	-5
48	0
55	5
62	10
70	15
78	20
87	25
97	30
107	35
118	40
130	45
143	50
135	55

- TEV manufacturer recommend a minimum of 5 degrees of sub-cooling at the inlet of the TEV for proper operation.
- This is not meant to be used as a gauge of the system charge.
- If the TEV can not maintain the correct amount of superheat (10 to 12 degrees) the sub-cooling must be verified prior to replacing the valve.
- This is why service ports should be installed at the indoor coil.
- Always check with the TEV manufacturer for specific troubleshooting guidelines.



Cooling Cycle Operation System Targets

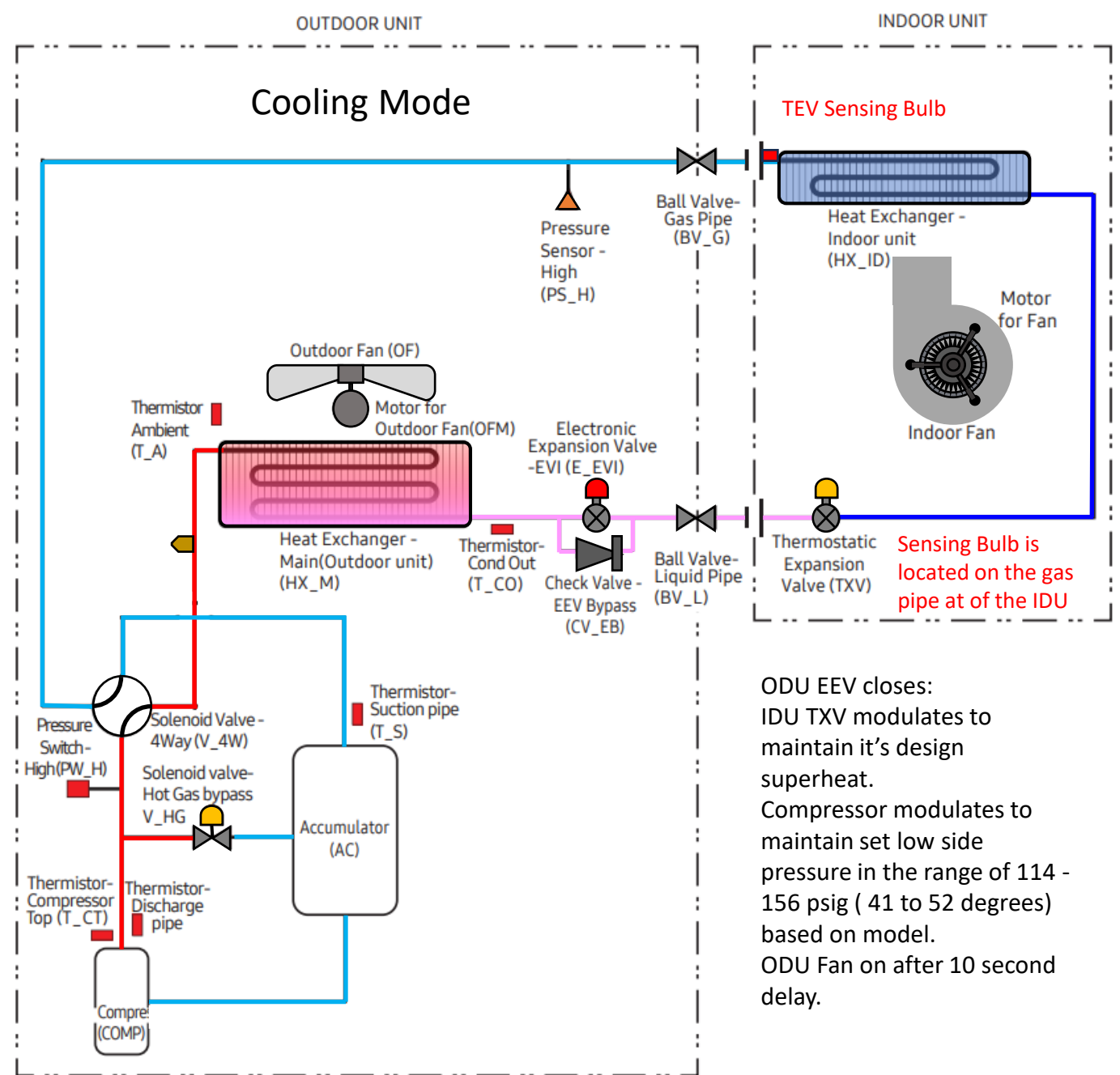
Compressor runs to maintain target suction pressure measured by the sensor. System monitors discharge temperature and adjust the compressor accordingly. The indoor TEV will modulate to maintain the correct amount of superheat set by the manufacturer.

TYPICALL unitary TEV's are set for approximately 10 - 12 degrees of superheat.

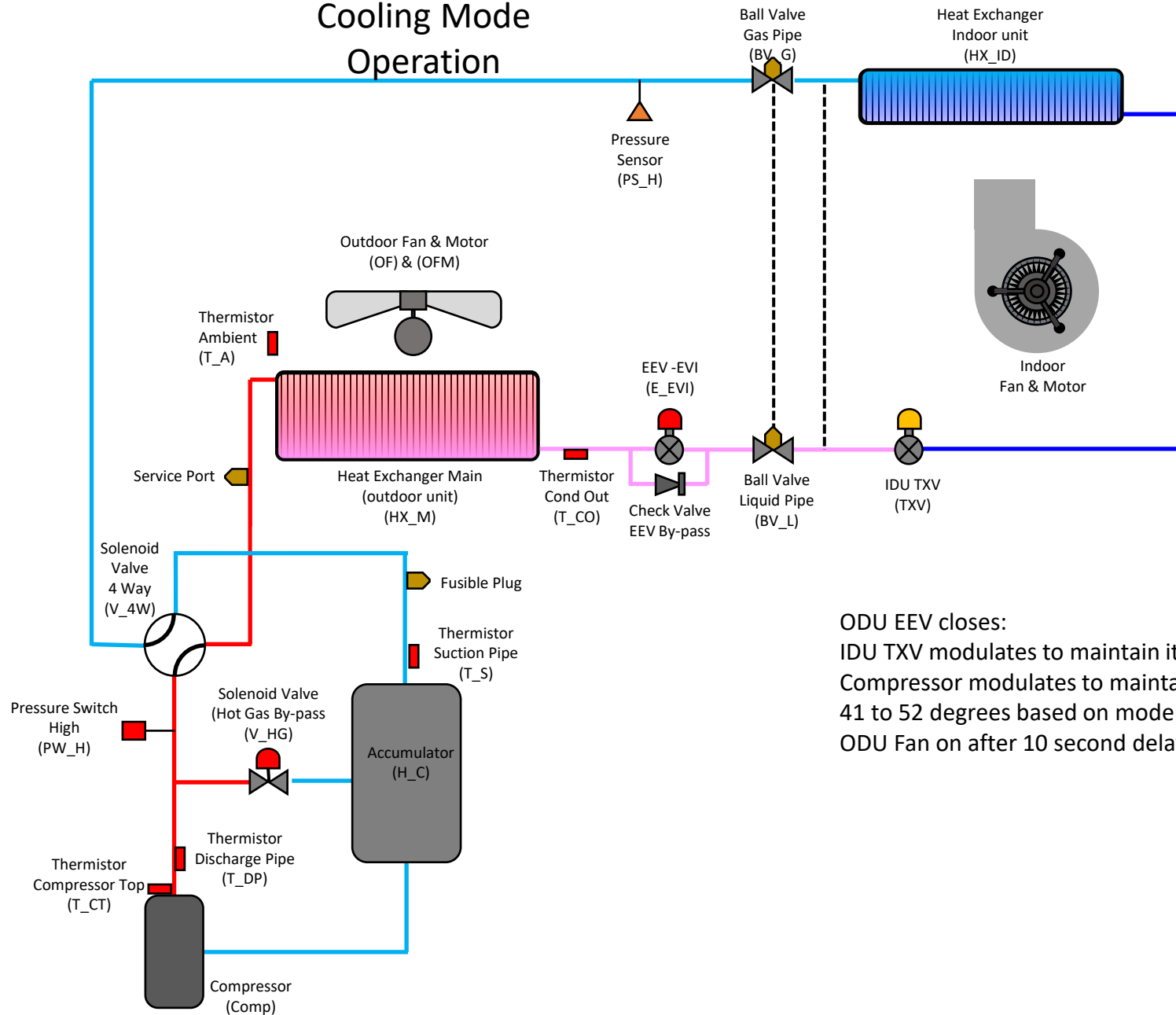
(its is recommended to check with the coil manufacturer for the specific superheat setting.

It is advisable to put service ports at the indoor coil.

It is good practice to measure Evaporator Superheat at the outlet of the evaporator and not at the inlet of the ODU.



Cooling Mode Operation



ODU EEV closes:

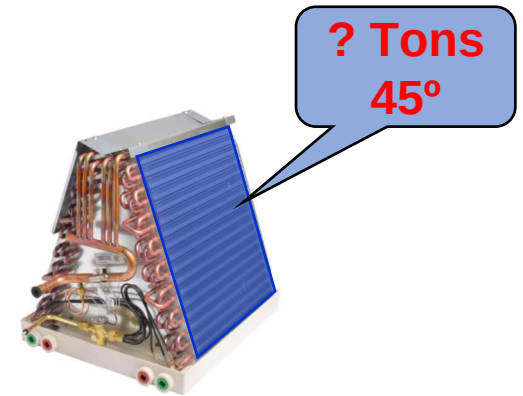
IDU TXV modulates to maintain it's design superheat.

Compressor modulates to maintain set low side pressure in the range of 41 to 52 degrees based on model.

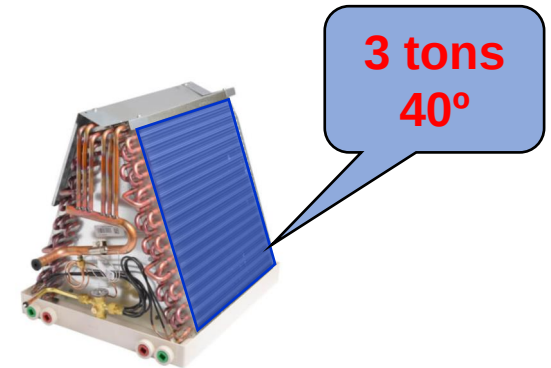
ODU Fan on after 10 second delay.

Indoor Coil Temperature

- The capacity of the coil (tonnage) is based on a given temperature (typically 40 degrees) and CFM
- If the temperature of the coil is increased the capacity in tons will decrease.
- The target evaporator temperature can be used to match the capacity of the indoor coil to the cooling requirements.



Indoor Coil
Saturation Temperature



Indoor Coil
Saturation Temperature

Cooling Targets

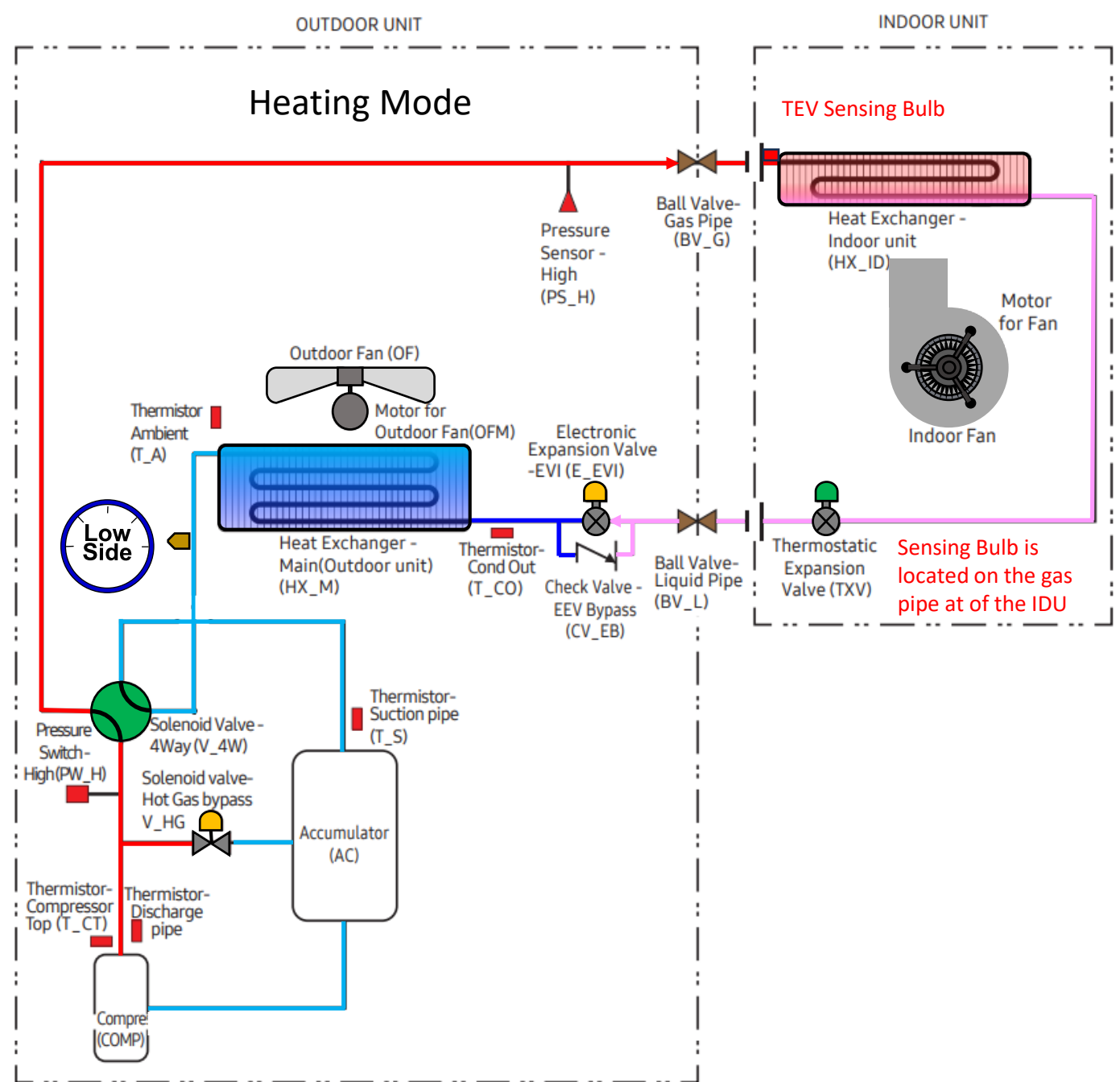
			Target pressure (psi) based on low pressure correction setpoint									
			AC024BXUPCH/AA		AC036BXUDCH/AA		AC036BXUPCH/AA		AC048BXUPCH/AA		AC060BXUPCH/AA	
			Stage2	Stage1	Stage2	Stage1	Stage2	Stage1	Stage2	Stage1	Stage2	Stage1
Target low pressure calibration for cooling	0(Default)	-	135	149	128	142	128	142	121	135	121	135
	1	- 2.84 psi (- 0.2 kg/cm2)	132	147	125	139	125	139	118	132	118	132
	2	- 5.69 psi (- 0.4 kg/cm2)	129	144	122	137	122	137	115	129	115	129
	3	- 8.53 psi (- 0.6 kg/cm2)	127	141	119	134	119	134	114	127	114	127
	4	- 11.38 psi (- 0.8 kg/cm2)	124	138	117	131	117	131	114	124	114	124
	5	- 14.22 psi (- 1.0 kg/cm2)	121	135	114	128	114	128	114	121	114	121
	6	+ 2.84 psi (+ 0.2 kg/cm2)	138	152	131	145	131	145	124	138	124	138
	7	+ 5.69 psi (+ 0.4 kg/cm2)	141	155	134	148	134	148	127	141	127	141
	8	+ 8.53 psi (+ 0.6 kg/cm2)	144	158	137	151	137	151	129	144	129	144
	9	+ 11.38 psi (+ 0.8 kg/cm2)	147	161	139	154	139	154	132	147	132	147
	10	+ 14.22 psi (+ 1.0 kg/cm2)	149	164	142	156	142	156	135	149	135	149

			Target Temp (°F)									
			AC024BXUPCH/AA		AC036BXUDCH/AA		AC036BXUPCH/AA		AC048BXUPCH/AA		AC060BXUPCH/AA	
			Stage2	Stage1	Stage2	Stage1	Stage2	Stage1	Stage2	Stage1	Stage2	Stage1
Target low pressure calibration for cooling	0(Default)	-	46.8	52.4	43.9	49.6	43.9	49.6	40.9	46.8	40.9	46.8
	1	- 2.84 psi (- 0.2 kg/cm2)	45.7	51.3	42.7	48.5	42.7	48.5	39.7	45.7	39.7	45.7
	2	- 5.69 psi (- 0.4 kg/cm2)	44.5	50.2	41.5	47.4	41.5	47.4	38.4	44.5	38.4	44.5
	3	- 8.53 psi (- 0.6 kg/cm2)	43.3	49.1	40.3	46.3	40.3	46.3	37.8	43.3	37.8	43.3
	4	- 11.38 psi (- 0.8 kg/cm2)	42.1	48.0	39.0	45.1	39.0	45.1	37.8	42.1	37.8	42.1
	5	- 14.22 psi (- 1.0 kg/cm2)	40.9	46.8	37.8	43.9	37.8	43.9	37.8	40.9	37.8	40.9
	6	+ 2.84 psi (+ 0.2 kg/cm2)	48.0	53.4	45.1	50.7	45.1	50.7	42.1	48.0	42.1	48.0
	7	+ 5.69 psi (+ 0.4 kg/cm2)	49.1	54.5	46.3	51.8	46.3	51.8	43.3	49.1	43.3	49.1
	8	+ 8.53 psi (+ 0.6 kg/cm2)	50.2	55.5	47.4	52.9	47.4	52.9	44.5	50.2	44.5	50.2
	9	+ 11.38 psi (+ 0.8 kg/cm2)	51.3	56.5	48.5	53.9	48.5	53.9	45.7	51.3	45.7	51.3
	10	+ 14.22 psi (+ 1.0 kg/cm2)	52.4	57.5	49.6	55.0	49.6	55.0	46.8	52.4	46.8	52.4

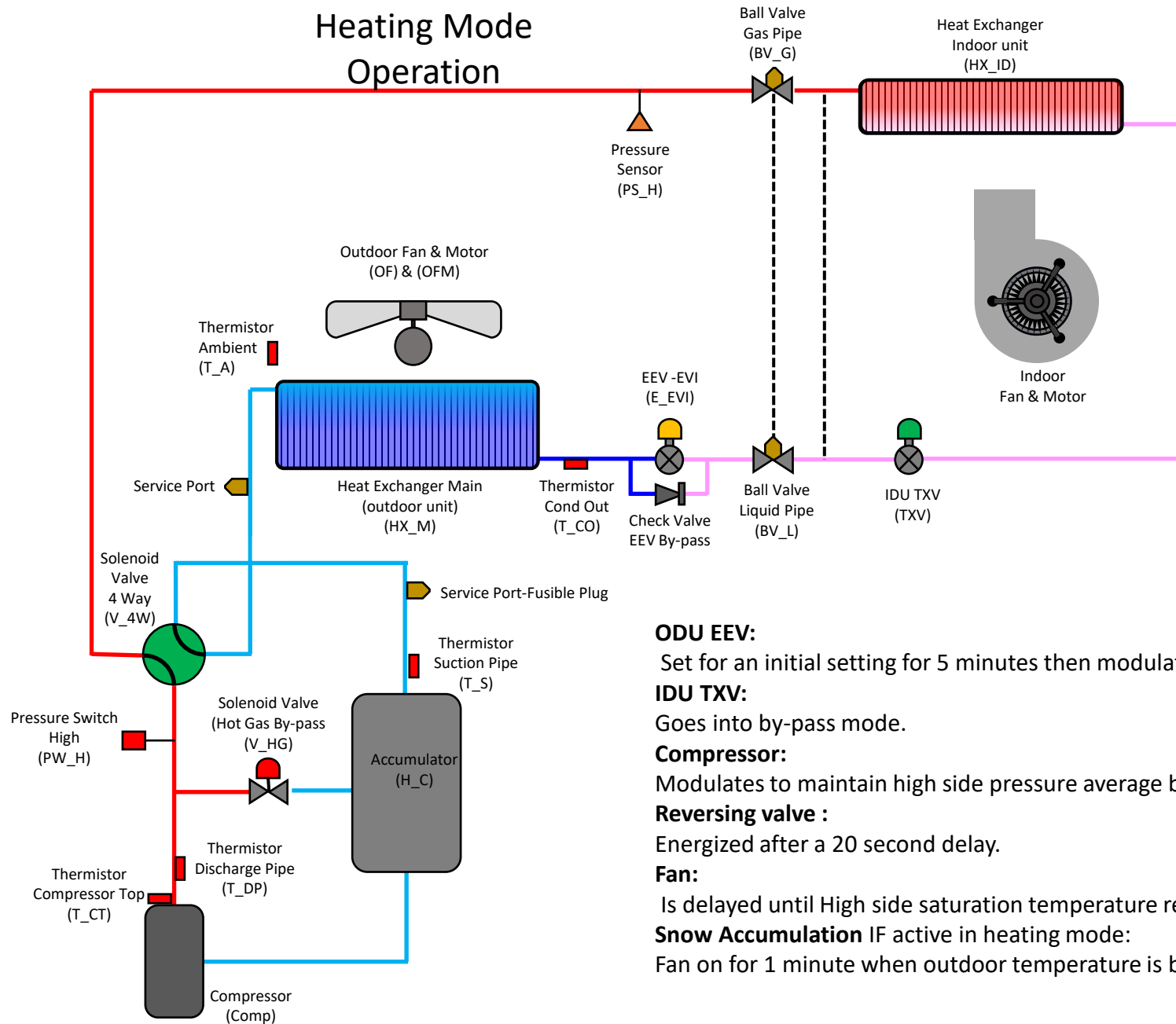
Heating Cycle

Compressor runs to maintain target Head Pressure measured by the sensor.
System monitors discharge temperature and adjust the compressor accordingly

ODU EEV set for an initial setting for 5 minutes then modulates to maintain 5.4 degrees of superheat.
IDU TXV goes into by-pass mode.
Compressor modulates to maintain set high side pressure average between 355 psig (W1) – 370 psig (W2).
Reversing valve is energized after a 20 second delay.
Fan is delayed until High side saturation temperature reaches 93.2 degrees or after 5 minutes.
Snow Accumulation **IF** active in heating mode fan on for 1 minute when outdoor temperature is below 41 degrees from 30 minutes.



Heating Mode Operation



ODU EEV:

Set for an initial setting for 5 minutes then modulates to maintain 5.4 degrees of superheat.

IDU TXV:

Goes into by-pass mode.

Compressor:

Modulates to maintain high side pressure average between 355 psig (W1) – 370 psig (W2).

Reversing valve :

Energized after a 20 second delay.

Fan:

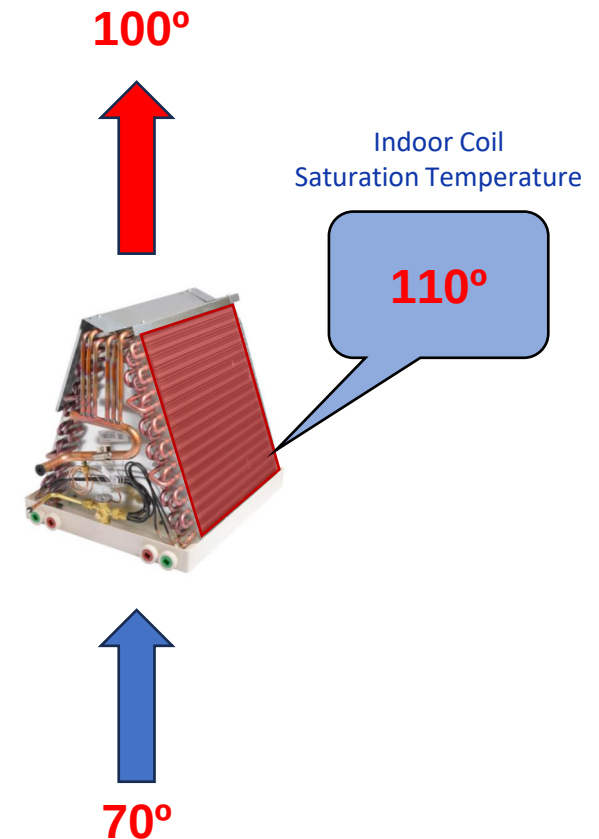
Is delayed until High side saturation temperature reaches 93.2 degrees or after 5 minutes.

Snow Accumulation IF active in heating mode:

Fan on for 1 minute when outdoor temperature is below 41 degrees from 30 minutes.

Indoor Coil Temperature

- The indoor coil should produce a minimum of a 30-degree Delta T across the coil..
- The target high side pressure can be adjusted to slightly change the coil saturation temperature.



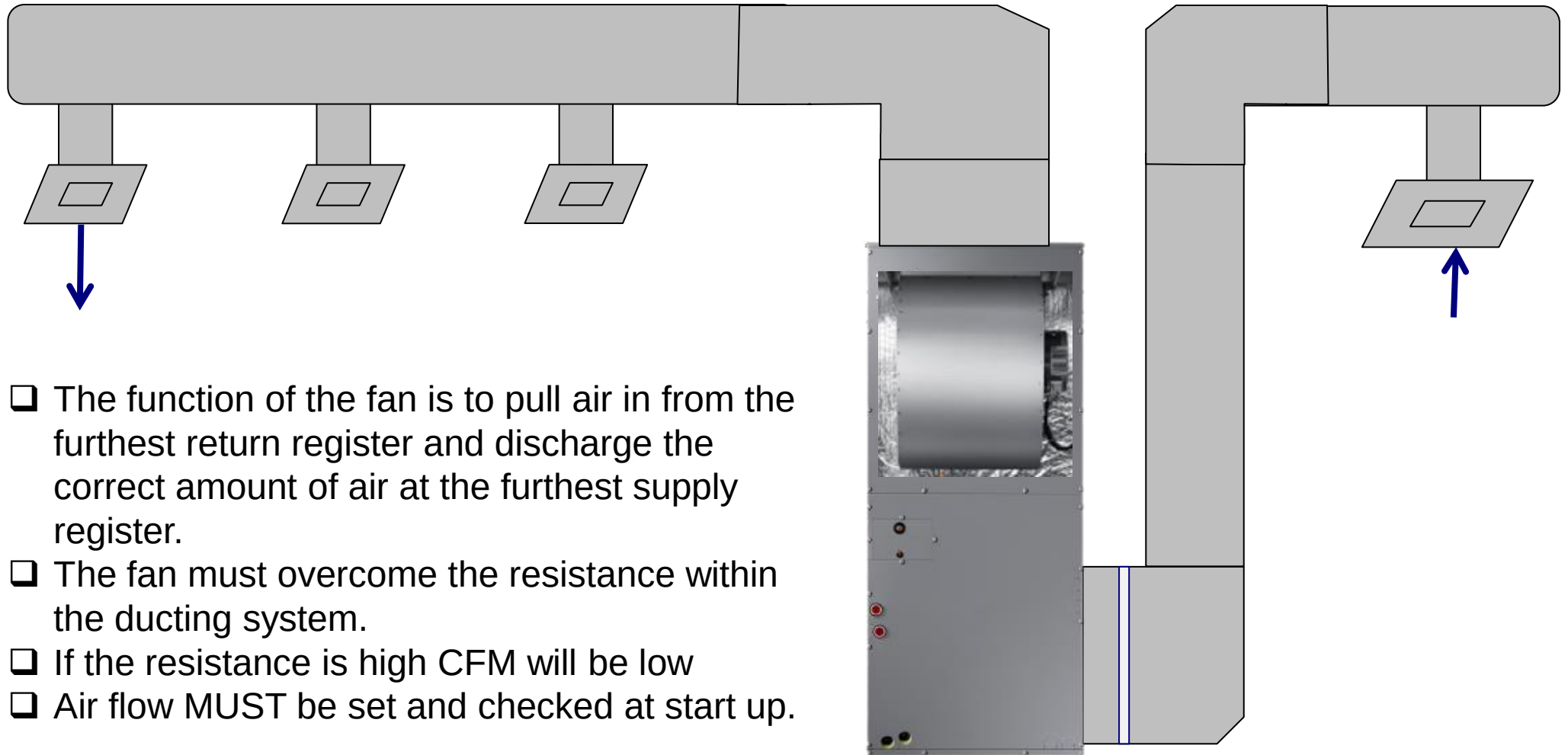
Heating Target Operation

			Target High Side Pressure (psig)									
			AC024BXUPCH/AA		AC036BXUDCH/AA		AC036BXUPCH/AA		AC048BXUPCH/AA		AC060BXUPCH/AA	
			Stage2	Stage1	Stage2	Stage1	Stage2	Stage1	Stage2	Stage1	Stage2	Stage1
Target high pressure calibration for heating	0(Default)	-	370.0	256.0	370.0	256.0	370.0	256.0	398.0	256.0	341.0	256.0
	1	+ 2.84 psi (- 0.2 kg/cm2)	372.8	253.8	372.8	253.8	372.8	253.8	400.8	253.8	343.8	253.8
	2	+ 5.69 psi (- 0.4 kg/cm2)	375.7	261.7	375.7	261.7	375.7	261.7	403.7	261.7	346.7	261.7
	3	+ 8.53 psi (- 0.6 kg/cm2)	378.5	264.5	378.5	264.5	378.5	264.5	406.5	264.5	349.5	264.5
	4	+ 11.38 psi (- 0.8 kg/cm2)	381.4	267.4	381.4	267.4	381.4	267.4	409.4	267.4	352.4	267.4
	5	+ 14.22 psi (- 1.0 kg/cm2)	384.2	270.2	384.2	270.2	384.2	270.2	412.2	270.2	355.2	270.2
	6	- 2.84 psi (+ 0.2 kg/cm2)	367.2	253.2	367.2	253.2	367.2	253.2	395.2	253.2	338.2	253.2
	7	- 5.69 psi (+ 0.4 kg/cm2)	364.3	250.3	364.3	250.3	364.3	250.3	392.3	250.3	335.3	250.3
	8	- 8.53 psi (+ 0.6 kg/cm2)	361.5	247.5	361.5	247.5	361.5	247.5	398.5	247.5	332.5	247.5
	9	- 11.38 psi (+ 0.8 kg/cm2)	358.6	244.6	358.6	244.6	358.6	244.6	386.6	244.6	329.6	244.6
	10	- 14.22 psi (+ 1.0 kg/cm2)	355.8	241.8	355.8	241.8	355.8	241.8	383.8	241.8	326.8	241.8

			High side saturation temperature (°F)									
			AC024BXUPCH/AA		AC036BXUDCH/AA		AC036BXUPCH/AA		AC048BXUPCH/AA		AC060BXUPCH/AA	
			Stage2	Stage1	Stage2	Stage1	Stage2	Stage1	Stage2	Stage1	Stage2	Stage1
Target high pressure calibration for heating	0(Default)	-	110.7	107.8	110.7	107.8	110.7	107.8	116.1	113.4	107.8	104.9
	1	+ 2.84 psi (+ 0.2 kg/cm2)	111.2	108.4	111.2	108.4	111.2	108.4	116.7	114.0	108.4	105.5
	2	+ 5.69 psi (+ 0.4 kg/cm2)	111.8	109.0	111.8	109.0	111.8	109.0	117.2	114.5	109.0	106.1
	3	+ 8.53 psi (+ 0.6 kg/cm2)	112.3	109.5	112.3	109.5	112.3	109.5	117.7	115.1	109.5	106.7
	4	+ 11.38 psi (+ 0.8 kg/cm2)	112.9	110.1	112.9	110.1	112.9	110.1	118.2	115.6	110.1	107.2
	5	+ 14.22 psi (+ 1.0 kg/cm2)	113.4	110.7	113.4	110.7	113.4	110.7	118.7	116.1	110.7	107.8
	6	- 2.84 psi (- 0.2 kg/cm2)	110.1	107.2	110.1	107.2	110.1	107.2	115.6	112.9	107.2	104.3
	7	- 5.69 psi (- 0.4 kg/cm2)	109.5	106.7	109.5	106.7	109.5	106.7	115.1	112.3	106.7	103.7
	8	- 8.53 psi (- 0.6 kg/cm2)	109.0	106.1	109.0	106.1	109.0	106.1	114.5	111.8	106.1	103.1
	9	- 11.38 psi (- 0.8 kg/cm2)	108.4	105.5	108.4	105.5	108.4	105.5	114.0	111.2	105.5	102.5
	10	- 14.22 psi (- 1.0 kg/cm2)	107.8	104.9	107.8	104.9	107.8	104.9	113.4	110.7	104.9	101.9

Air Flow

What's the Purpose of the fan ?



- ☐ The function of the fan is to pull air in from the furthest return register and discharge the correct amount of air at the furthest supply register.
- ☐ The fan must overcome the resistance within the ducting system.
- ☐ If the resistance is high CFM will be low
- ☐ Air flow **MUST** be set and checked at start up.

Air Flow

- Having the correct air flow is critical to system operation.
- The air flow **MUST** be set to meet the recommendations of the outdoor unit. See the following slide for recommendations.
- Knowing how to set air flow is a staple of the HVAC business

Installation

- Min / Max Fan Speeds
 - A part of a properly installed system is understanding airflow
 - Below is a table for the minimums and maximums allowed for proper system operation

Model Number	Low Speed (CFM)			Medium (CFM)			High (CFM)		
	Min.	Recommended	Max.	Min.	Recommended	Max.	Min.	Recommended	Max.
AC024BXUPCH/AA	360	500	500	518	700	700	740	1000	1000
AC036BXUDCH/AA	540	600	750	777	840	1050	1110	1200	1500
AC036BXUPCH/AA	540	600	750	777	840	1050	1110	1200	1500
AC048BXUPCH/AA	720	800	1000	1036	1120	1400	1480	1600	2000
AC060BXUPCH/AA	900	900	1250	1295	1295	1750	1850	1850	2500

Understanding ESP

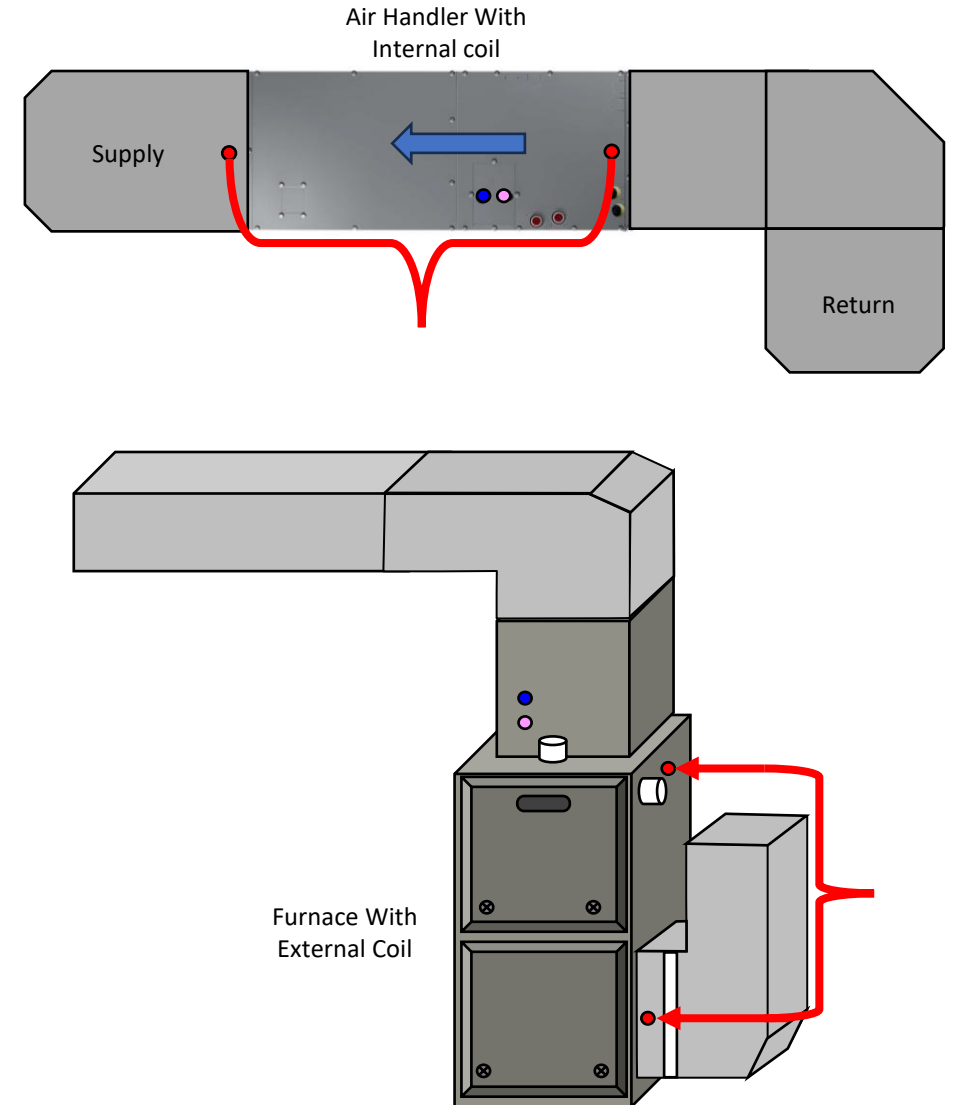
- External Static Pressure is the measurement of all the resistance in the duct system that the fan has to work against.
- The Static Capability of motor depends on the type of motor being used.
- If you have the fan performance chart of the motor and know the ESP, you can determine the air flow in CFM.
- If the measured ESP is greater than the maximum allowable of the fan, then you will not move the correct amount of air.

Measuring ESP

- ESP is the sum of the suction pressure (negative) and discharge pressure (positive) created by the equipment blower.
- External Static pressure is measured using a manometer and is expressed in inches of water column (i.e., #” WC).

Where to Measure ESP

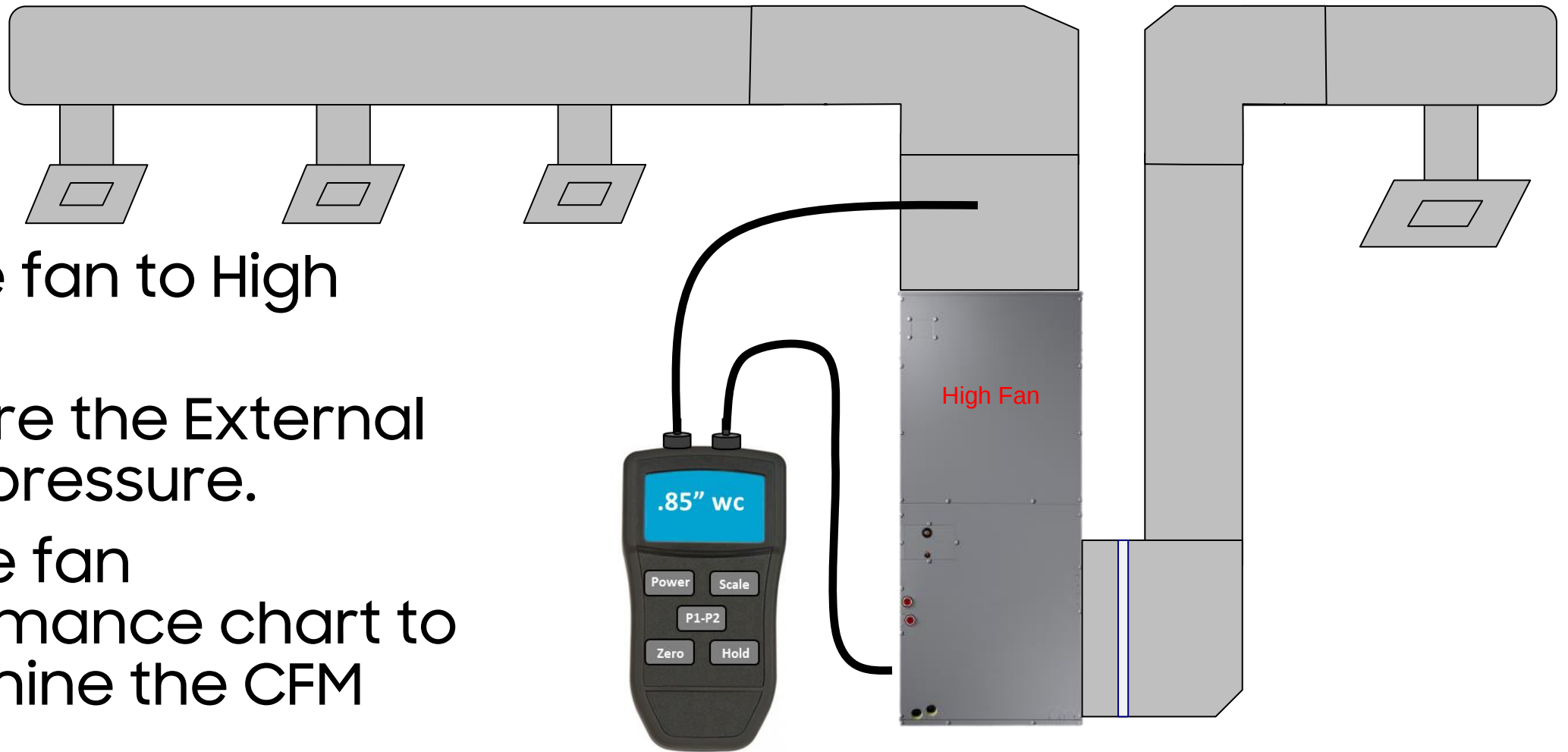
- Always check the air handlers or furnace manufacturers recommendations for measuring ESP.
- Readings are taken on a forced air furnace at the inlet of the furnace blower after the filter and exiting the discharge of the furnace blower before the evaporator coil.
- For air handlers, measurements are taken at the inlet of the blower after the filter and the discharge or outlet of the RTU or air handler.



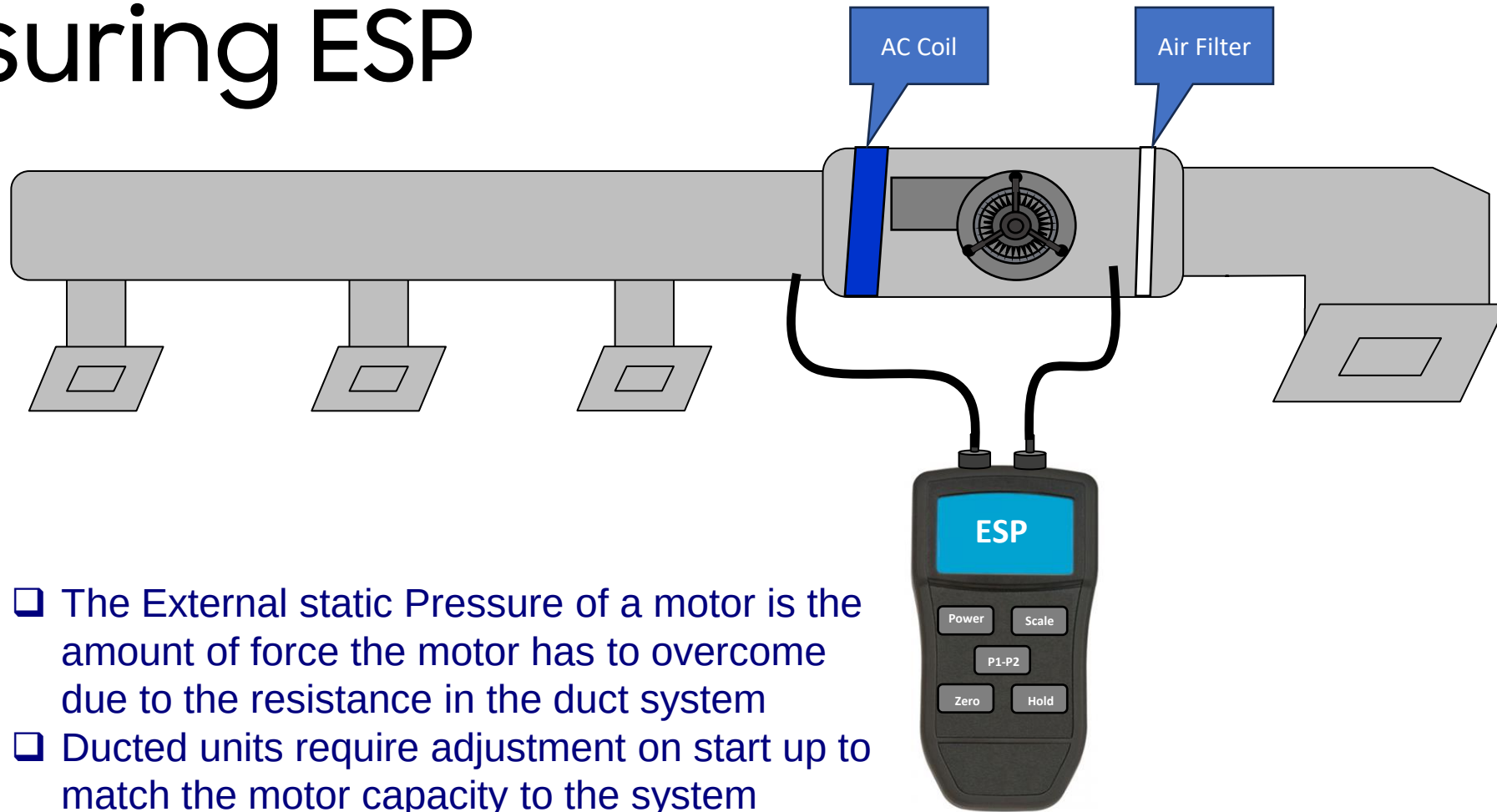
Air Handler Measurements

ESP

- Set the fan to High Speed
- Measure the External Static pressure.
- Use the fan performance chart to determine the CFM

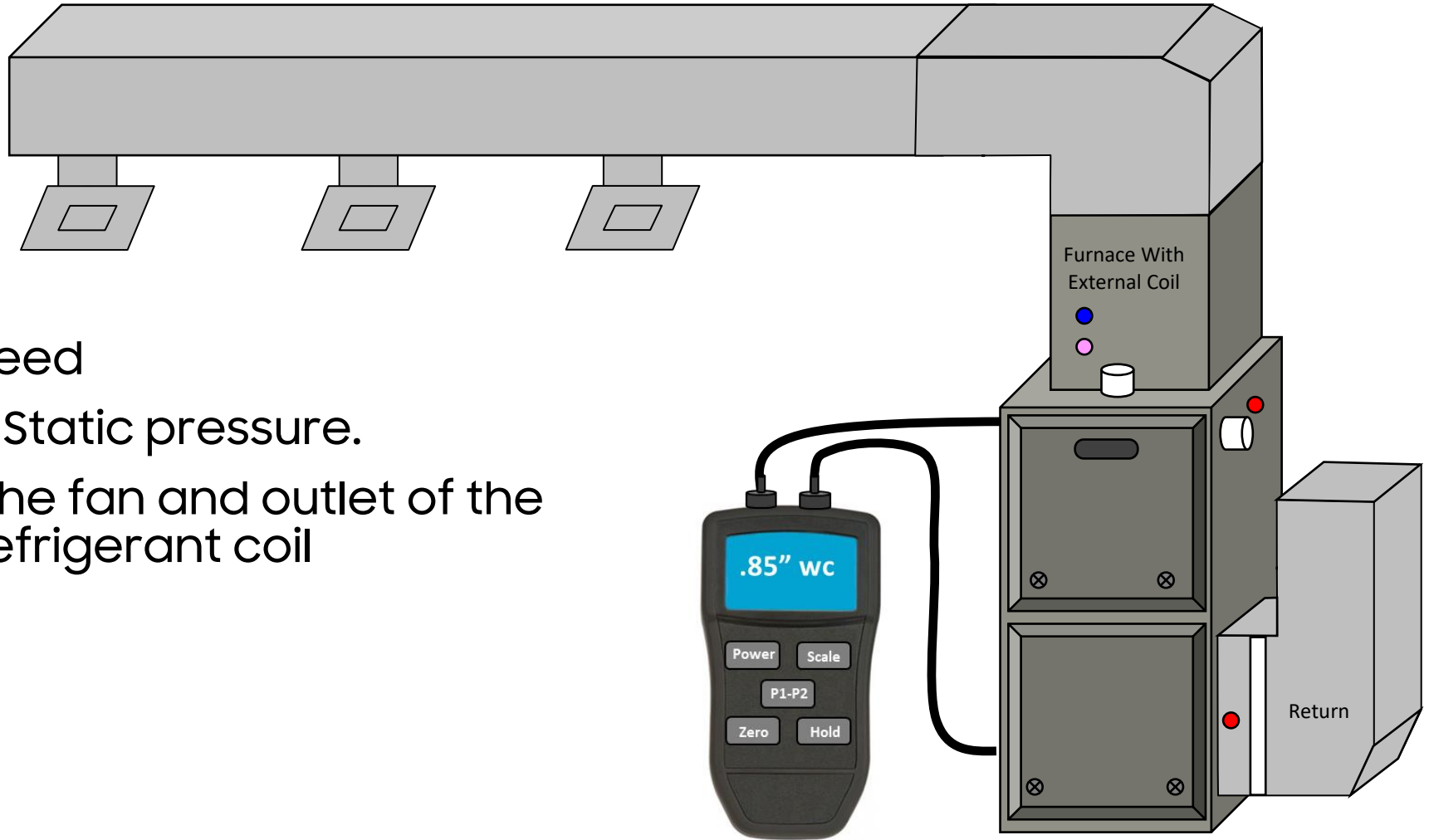


Measuring ESP



- ❑ The External static Pressure of a motor is the amount of force the motor has to overcome due to the resistance in the duct system
- ❑ Ducted units require adjustment on start up to match the motor capacity to the system resistance.

Furnace with a coil ESP Measurements



- Set the fan to High Speed
- Measure the External Static pressure.
- Between the inlet of the fan and outlet of the furnace before the Refrigerant coil

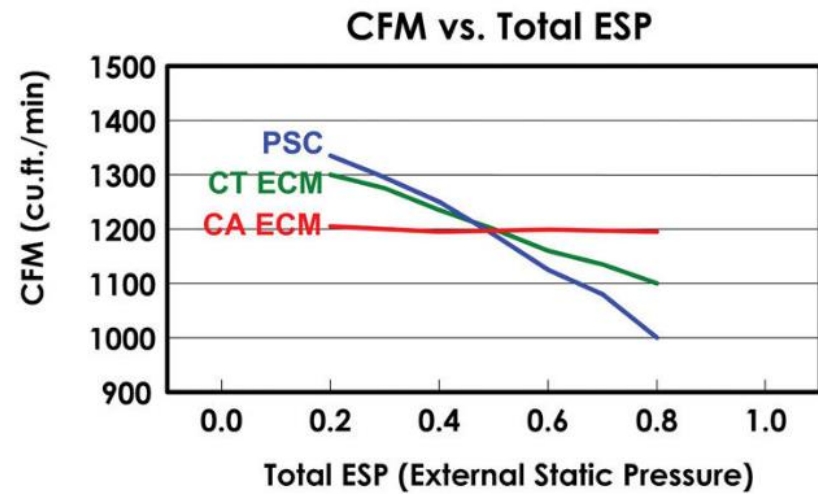
Air Flow

Types of motors and their characteristics

- PSC
 - Use fan speed taps to adjust air flow
 - These motors typical have low ESP ratings.
 - As ESP increases CFM decreases
- Constant torque ECM
 - Uses fan speed taps to adjust air flow.
 - These motors typical have Higher ESP ratings compared to PSC Motors
 - As ESP increases CFM decreases slightly less than a PSC motor.
 - Also, Energy usage and Velocity increases and ESP increases
- Variable Speed ECM
 - Uses fan speed taps to adjust air flow
 - Typically have high ESP rating (up to 1" wc)
 - These are communicating motors, and the Air Flow setting is specific to the equipment manufacturer.
 - They will deliver the programmed CFM as long as the motor operates within the required ESP range



Blower performance



PSC Motor

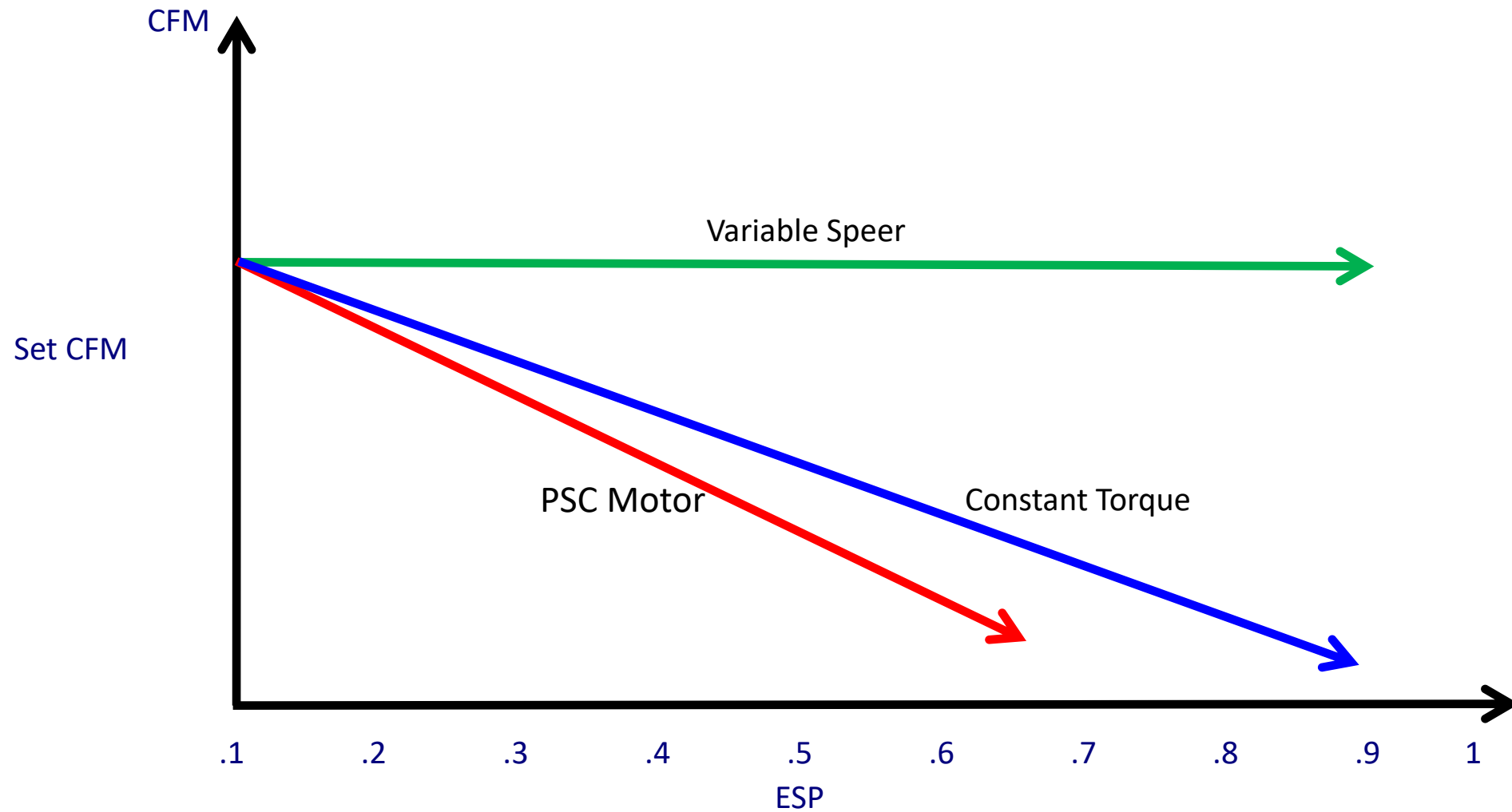


Constant
Torque

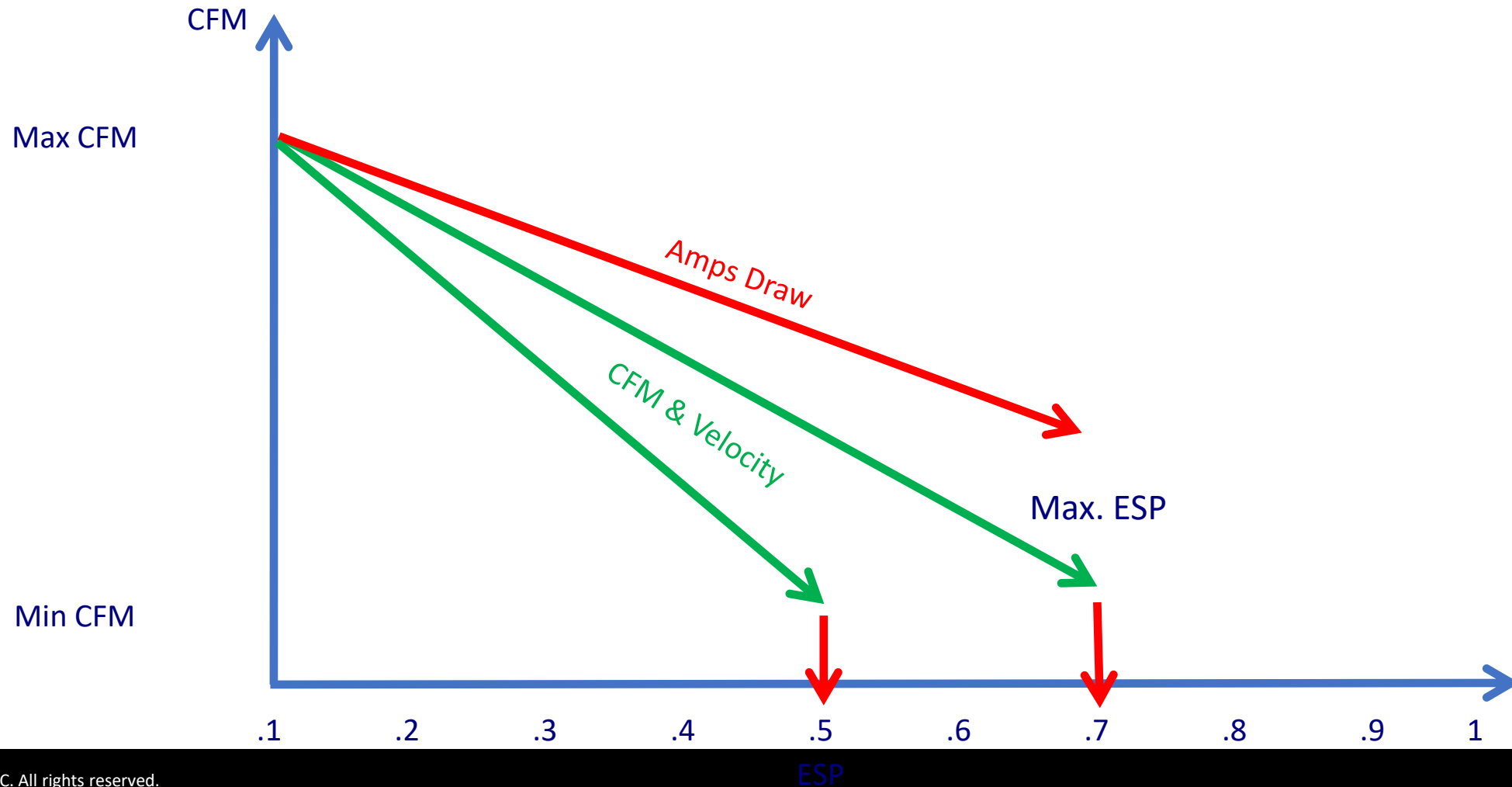


Variable
Speed

Blower Motor Performance



CFM to ESP Chart PSC Motor



PSC Fan Performance Chart

- In this Example the a 06M unit is running with at medium high fan speed operating against .4 inches of external static pressure will move 960 CFM

BLOWER PERFORMANCE DATA—

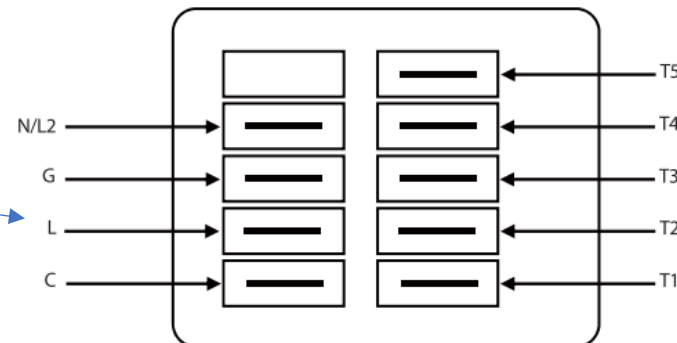
	BLOWER SIZE [mm]	MOTOR H.P. [W]	BLOWER SPEED	CFM [L/s] AIR DELIVERY EXTERNAL STATIC PRESSURE INCHES WATER COLUMN [kPa]						
				0.1 [.02]	0.2 [.05]	0.3 [.07]	0.4 [.10]	0.5 [.12]	0.6 [.15]	0.7 [.17]
04•M	11 x 7 [279 x 178]	1/2 [373]	LOW	805 [380]	780 [368]	760 [358]	740 [340]	685 [323]	645 [304]	605 [285]
			MED-LO	920 [434]	885 [417]	850 [401]	810 [382]	775 [365]	730 [344]	690 [325]
			MED-HI	1140 [538]	1110 [524]	1085 [512]	1055 [493]	1010 [476]	950 [448]	890 [420]
			HIGH	1360 [642]	1320 [623]	1280 [604]	1255 [583]	1195 [564]	1140 [538]	1080 [500]
06•M	11 x 7 [279 x 178]	1/2 [373]	LOW	770 [363]	740 [349]	710 [335]	675 [318]	645 [304]	605 [285]	570 [269]
			MED-LO	880 [415]	845 [398]	815 [384]	790 [373]	760 [358]	715 [337]	670 [316]
			MED-HI	1060 [500]	1025 [483]	990 [467]	960 [453]	925 [436]	880 [415]	835 [394]
			HIGH	1260 [594]	1215 [573]	1175 [554]	1135 [535]	1100 [519]	1040 [491]	985 [465]
07•Y	12 x 7 [305 x 178]	3/4 [559]	LOW	1105 [522]	1096 [517]	1080 [610]	1050 [498]	1030 [466]	1010 [477]	990 [467]
			MED-LO	1290 [609]	1275 [602]	1280 [605]	1220 [678]	1195 [564]	1170 [552]	1140 [538]
			MED-HI	1480 [698]	1435 [677]	1415 [668]	1390 [658]	1370 [647]	1300 [614]	1255 [592]
			HIGH	1705 [805]	1665 [788]	1615 [762]	1570 [741]	1540 [727]	1475 [696]	1400 [661]
07•M	11 x 7 [279 x 178]	1/2 [373]	LOW	780 [368]	745 [351]	710 [335]	675 [318]	640 [302]	595 [281]	555 [261]
			MED-LO	880 [415]	850 [401]	825 [389]	785 [370]	750 [354]	702 [331]	655 [309]
			MED-HI	1090 [514]	1050 [495]	1010 [477]	970 [458]	925 [436]	875 [413]	825 [389]
			HIGH	1300 [613]	1255 [592]	1210 [571]	1160 [547]	1110 [524]	1055 [498]	1005 [474]
09•Z ***	12 x 11 [305 x 279]	3/4 [559]	LOW	1235 [582]	1210 [571]	1185 [559]	1150 [543]	1120 [528]	1075 [507]	1035 [488]
			MED-LO	1490 [703]	1465 [691]	1440 [679]	1405 [663]	1375 [649]	1315 [620]	1255 [592]
			MED-HI	1720 [811]	1670 [788]	1620 [764]	1600 [755]	1580 [746]	1520 [717]	1460 [689]
			HIGH	2100 [991]	2050 [967]	2000 [944]	1955 [923]	1910 [901]	1825 [861]	1745 [823]
10•Z ***	12 x 11 [305 x 279]	3/4 [559]	LOW	1230 [680]	1205 [567]	1160 [557]	1155 [545]	1130 [583]	1090 [514]	1050 [495]
			MED-LO	1490 [703]	1445 [682]	1405 [663]	1375 [649]	1350 [637]	1295 [611]	1240 [585]
			MED-HI	1710 [807]	1665 [786]	1620 [764]	1570 [746]	1540 [727]	1475 [696]	1410 [665]
			HIGH	2010 [949]	1955 [923]	1900 [897]	1655 [875]	1810 [854]	1710 [607]	1610 [769]

Constant Torque Motor Terminal Designation



Terminal	Connection	Default speed tap settings					
		AC018KNZDCH/ AA	AC024KNZDCH/ AA	AC030KNZDCH/ AA	AC036/042KNZDCH/ AA	AC048KNZDCH/ AA	AC054KNZDCH/ AA
C	Speed tap common - 24 VAC common						
L	Supply voltage - 240 VAC Line 1						
G	Ground connection						
N/L2	Supply voltage - 240 VAC Line 2						
T1	Low speed tap - 24 VAC input		"Low" speed	"Low" speed	"Low" speed	"Low" speed	"Low" speed
T2	Medium-low speed tap - 24 VAC input	"Low" speed	"Mid" speed	"Mid" speed		"Mid" speed	"Mid" speed
T3	Medium speed tap - 24 VAC input	"Mid" speed		"High" speed			
T4	Medium-high speed tap - 24 VAC input	"High" speed "	"High" speed		High speed for electric heat	"High" speed	High speed for electric heat
T5	High speed tap - 24 VAC input	High speed for electric heat	High speed for electric heat	High speed for electric heat	"High" speed	High speed for electric heat	"High" speed

Motor control/voltage taps



Always refer to the specific unit's motor data!

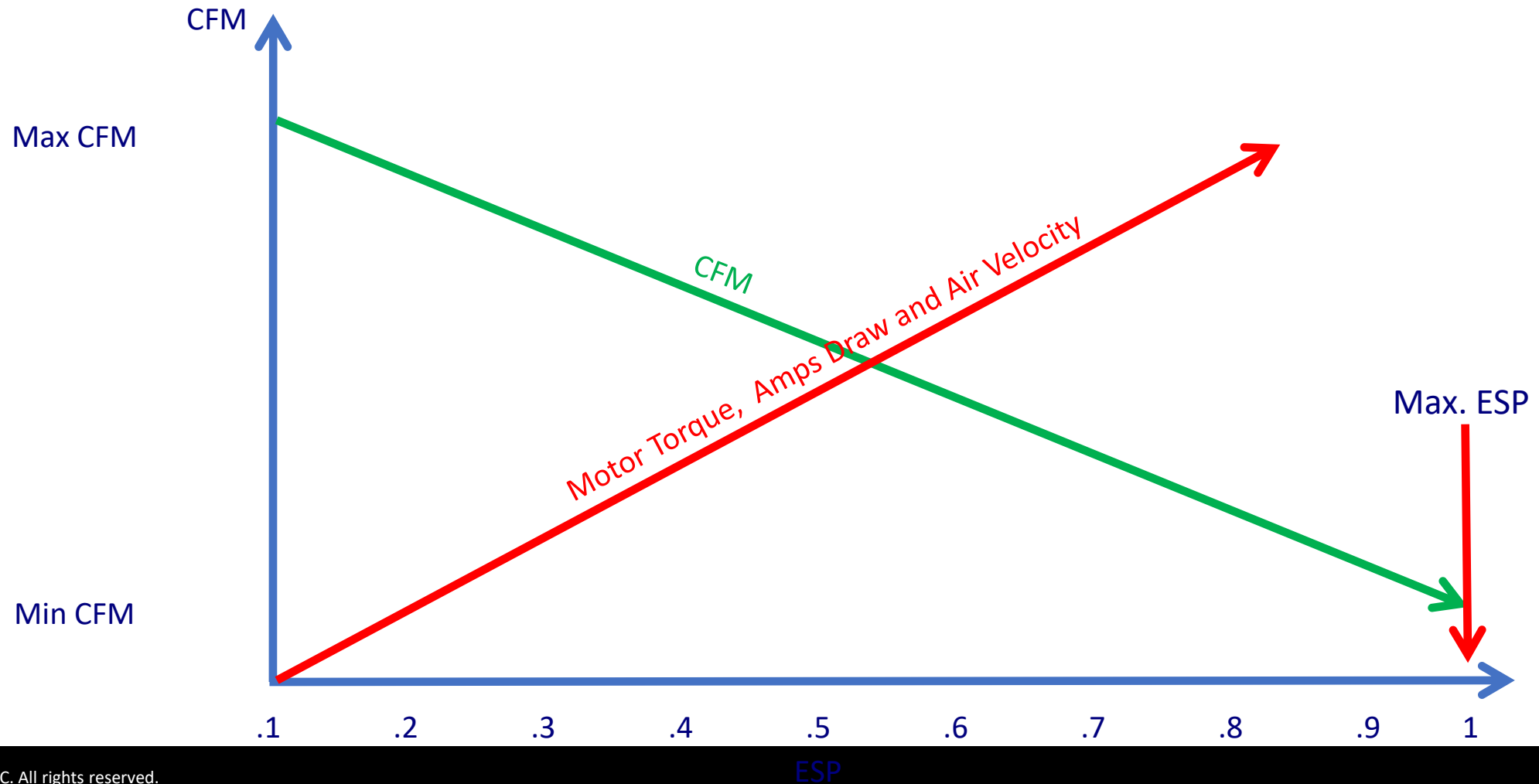
Constant Torque ECM

Fan Performance Charts



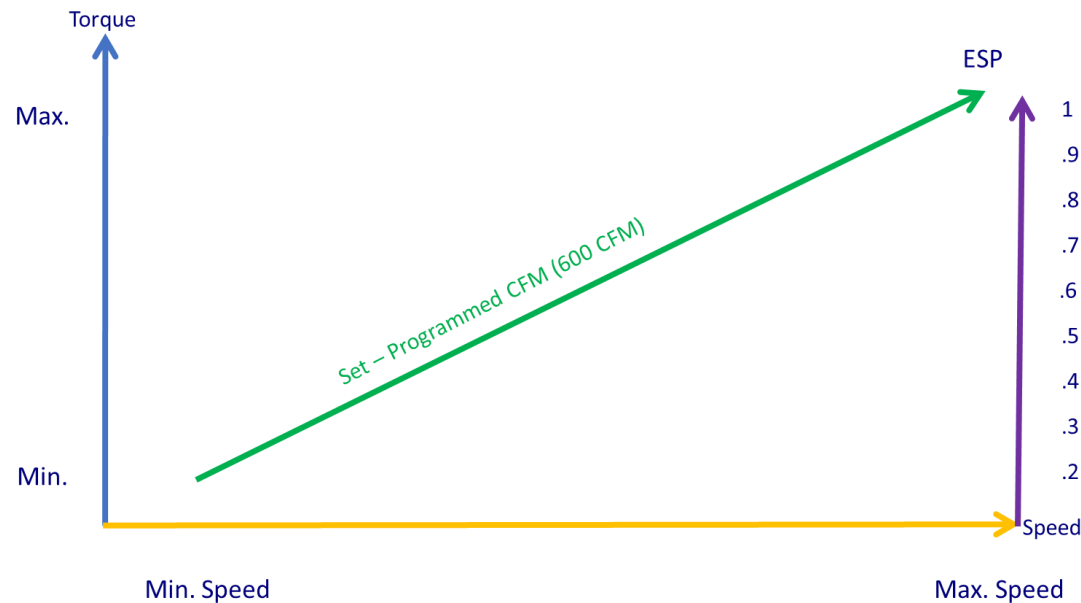
Motor Tap	P	(inwg)	CFMM	Motor Tap	P	(inwg)C	FM
1		0.1	1260	4		0.1	1538
		0.2	1167			0.2	1500
		0.3	1075			0.3	1462
		0.4	981			0.4	1422
		0.5	887			0.5	1379
		0.6	793			0.6	1331
		0.7	699			0.7	1280
		0.8	606			0.8	1224
		0.9	510			0.9	1158
		1	415			1	1078
2		0.1	1379	5		0.1	1739
		0.2	1293			0.2	1705
		0.3	1210			0.3	1670
		0.4	1130			0.4	1633
		0.5	1045			0.5	1597
		0.6	963			0.6	1559
		0.7	878			0.7	1522
		0.8	795			0.8	1485
		0.9	711			0.9	1446
		1	627			1	1406
3		0.1	1475	= Default Setting AC048KNZDCH/AA HP : 3/4 Default motor taps: High / Mid / Low = 4/2/1			
		0.2	1410				
		0.3	1348				
		0.4	1282				
		0.5	1216				
		0.6	1148				
		0.7	1076				
		0.8	1003				
		0.9	929				
		1	850				

CFM to ESP Chart Constant Torque Motor

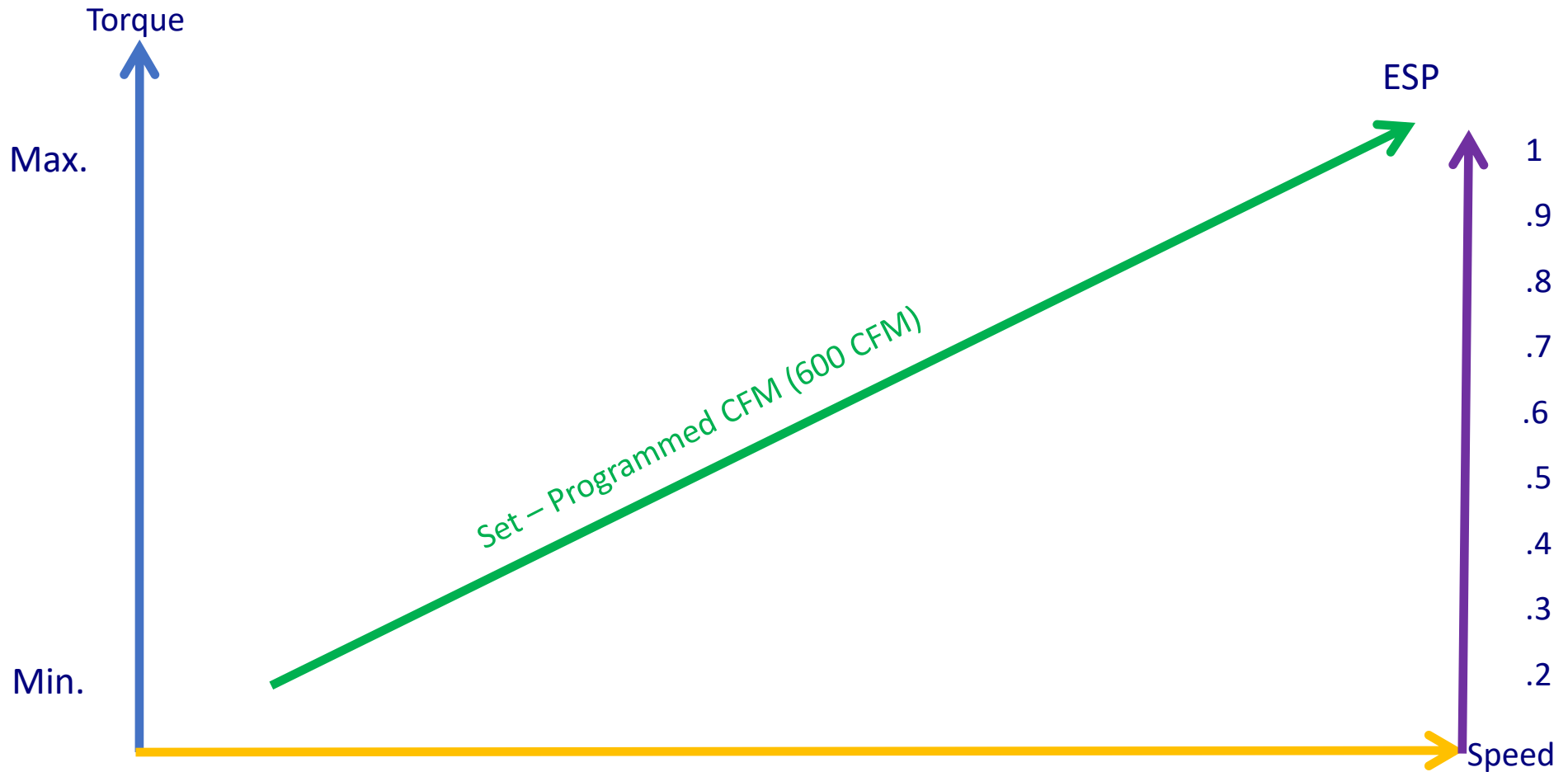


Variable Speed ECM

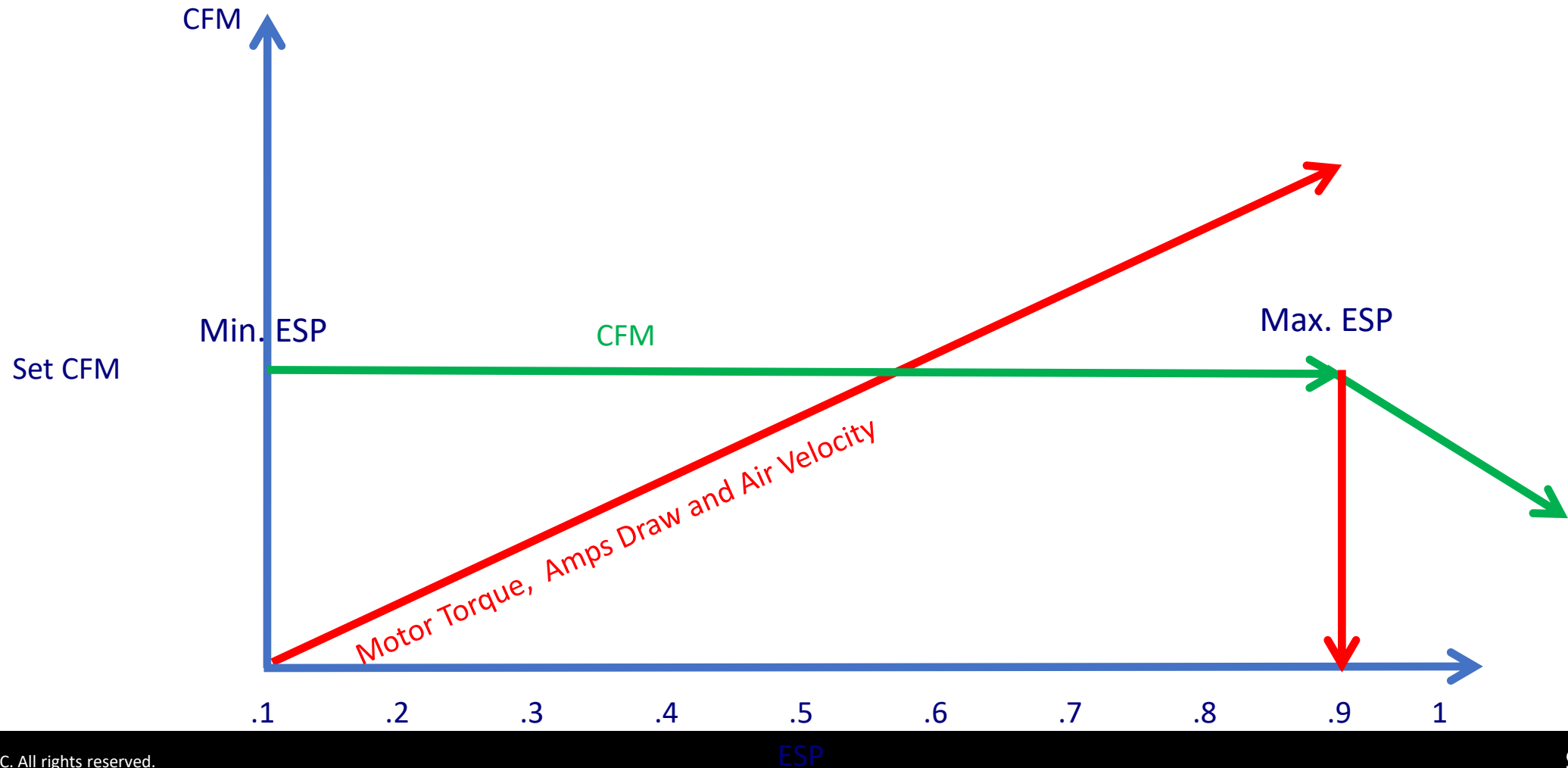
- The fan will deliver the Set – Programmed CFM as long as the ESP falls within the Min – Max Range.
- When Resistance (ESP) increases the fan increases the speed and torque until the speed and torque hits a spot on the fan performance chart for the given CFM



Variable Speed ECM Motor



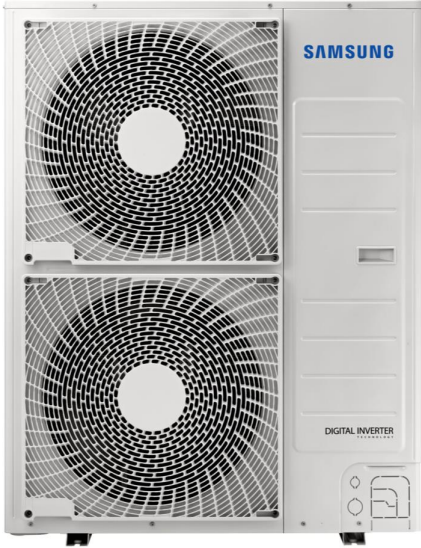
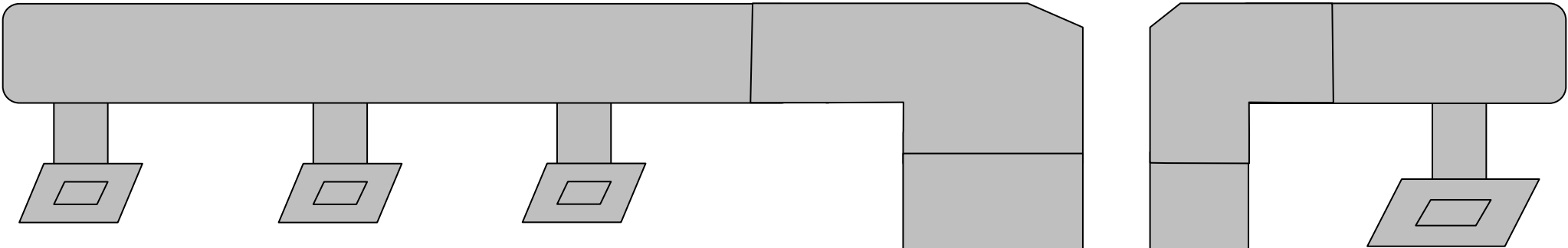
CFM to ESP Chart BLDC Motor



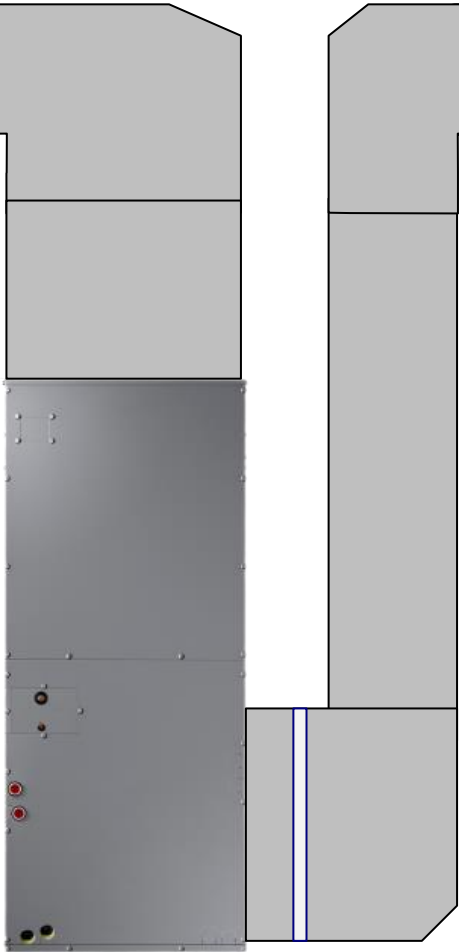
Determining CFM Example

- Determine the CFM of the air handler.
- Use the CFM value to calculate the heating and cooling capacity of the system.

Model	Model Number		AC036BXUPCH/AA
	US Code		CXH36UPB
Capacity	Nominal Capacity	Cooling (Btu/h)	36,000
		Heating (Btu/h)	34,600
	Capacity Range ¹	Cooling (Btu/h)	16,500 - 45,000
		Heating (Btu/h)	13,000 - 50,000



AC036BXUPCH/AA



3 Ton Air Handler

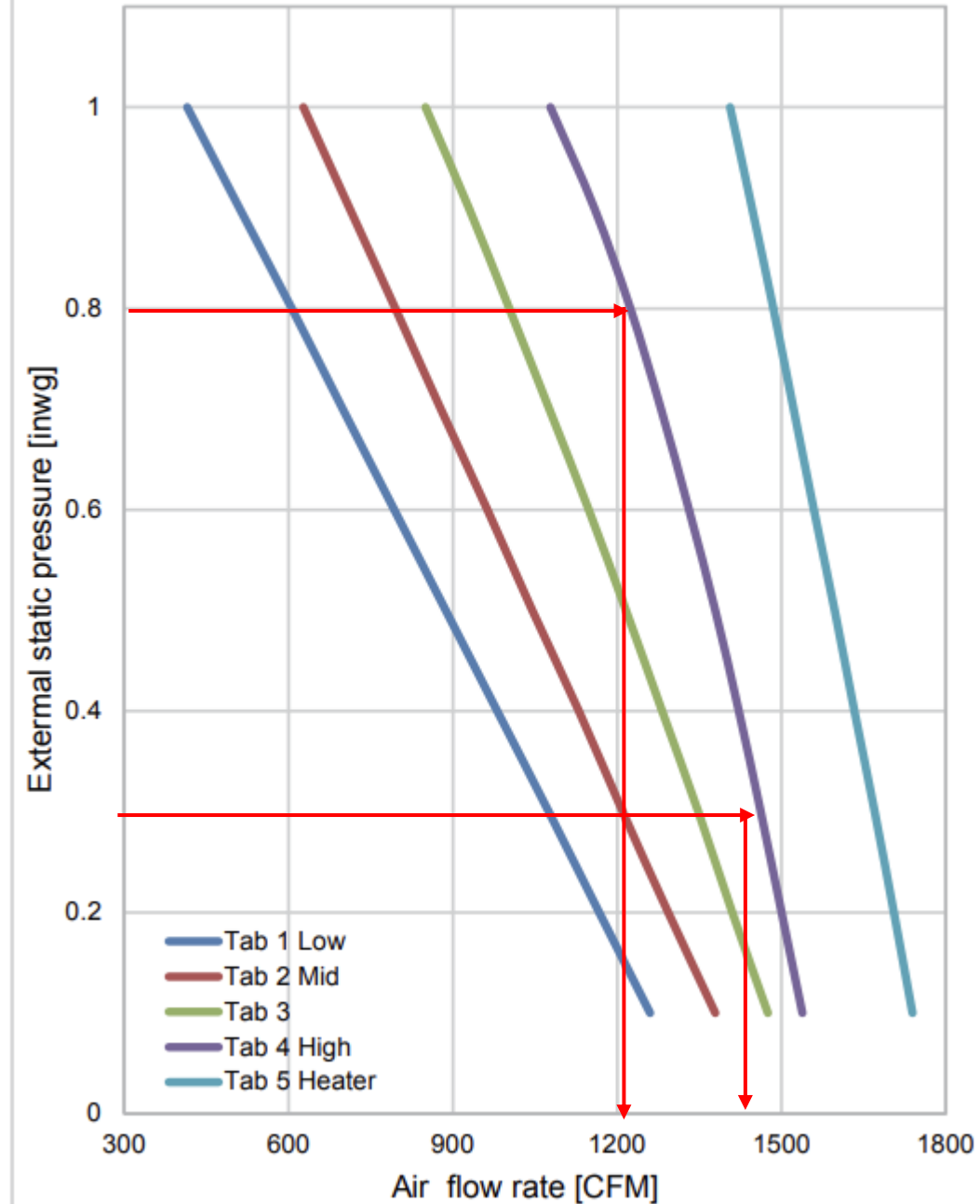
Fan Curves



- The air handler was rated with the fan on high speed with an external static pressure of .28"wc and 1,413 CFM
- If the ESP of the blower goes up CFM goes down.
- IF CFM goes down the delivered capacity goes down

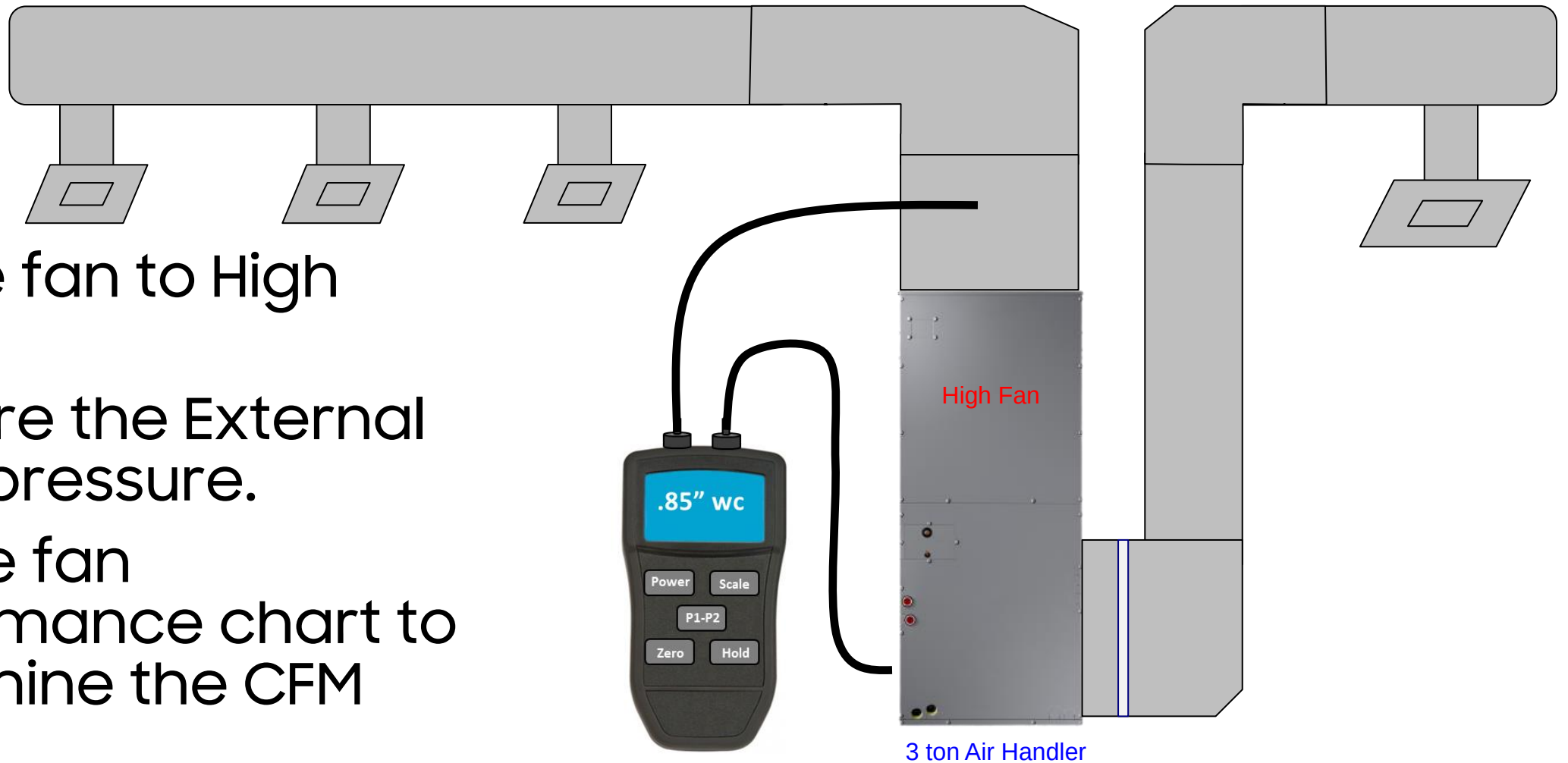
Evaporator Fan	Type		Double-inlet, forward curve, centrifugal (with ECM motor)
	Air Volume	CFM (L/M/H)	1,130 / 1,271 / 1,413 (at standard ESP)
	Output (W) / FLA (A)		410 W / 1.66 A
	Static Pressure	Standard ("WC)	0.28
		Min. / Max. ("WC)	0.01 - 0.98

3-ton Air Handler Fan chart



Determining CFM Example

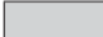
- Set the fan to High Speed
- Measure the External Static pressure.
- Use the fan performance chart to determine the CFM



Calculating CFM Example

- Set the fan to High Speed
- Measure the External Static pressure is .85" w.c.
- The delivered CFM is approximately 1,191 CFM
- Is this enough air flow
- What can you do now that you know the CFM?
- You can determine the delivered capacity of the indoor unit in heating and cooling modes.

Motor Tap	P (inwg)	CFM	Motor Tap	P (inwg)	CFM
1	0.1	1260	4	0.1	1538
	0.2	1167		0.2	1500
	0.3	1075		0.3	1462
	0.4	981		0.4	1422
	0.5	887		0.5	1379
	0.6	793		0.6	1331
	0.7	699		0.7	1280
	0.8	606		0.8	1224
	0.9	510		0.9	1158
	1	415		1	1078
2	0.1	1379	5	0.1	1739
	0.2	1293		0.2	1705
	0.3	1210		0.3	1670
	0.4	1130		0.4	1633
	0.5	1045		0.5	1597
	0.6	963		0.6	1559
	0.7	878		0.7	1522
	0.8	795		0.8	1485
	0.9	711		0.9	1446
	1	627		1	1406
3	0.1	1475			
	0.2	1410			
	0.3	1348			
	0.4	1282			
	0.5	1216			
	0.6	1148			
	0.7	1076			
	0.8	1003			
	0.9	929			
	1	850			

 = Default Setting

3-ton Air Handler Fan chart

HP : 3/4

Default motor taps:

High / Mid / Low = 4/2/1



Why is Air flow So important in variable speed equipment

- What affect does low air flow have on:
 - Suction pressure in the A/C mode
 - Head pressure in heating mode?
- How does the system respond to these changes?
- What Can you calculate it if you know the CFM of the system?

IDU Capacity

- The IDU capacity is directly related to the air flow setting, refrigerant temperature and space conditions.
- If the air flow setting is low the output capacity is low.
- If the air flow setting is high the output capacity will be high.

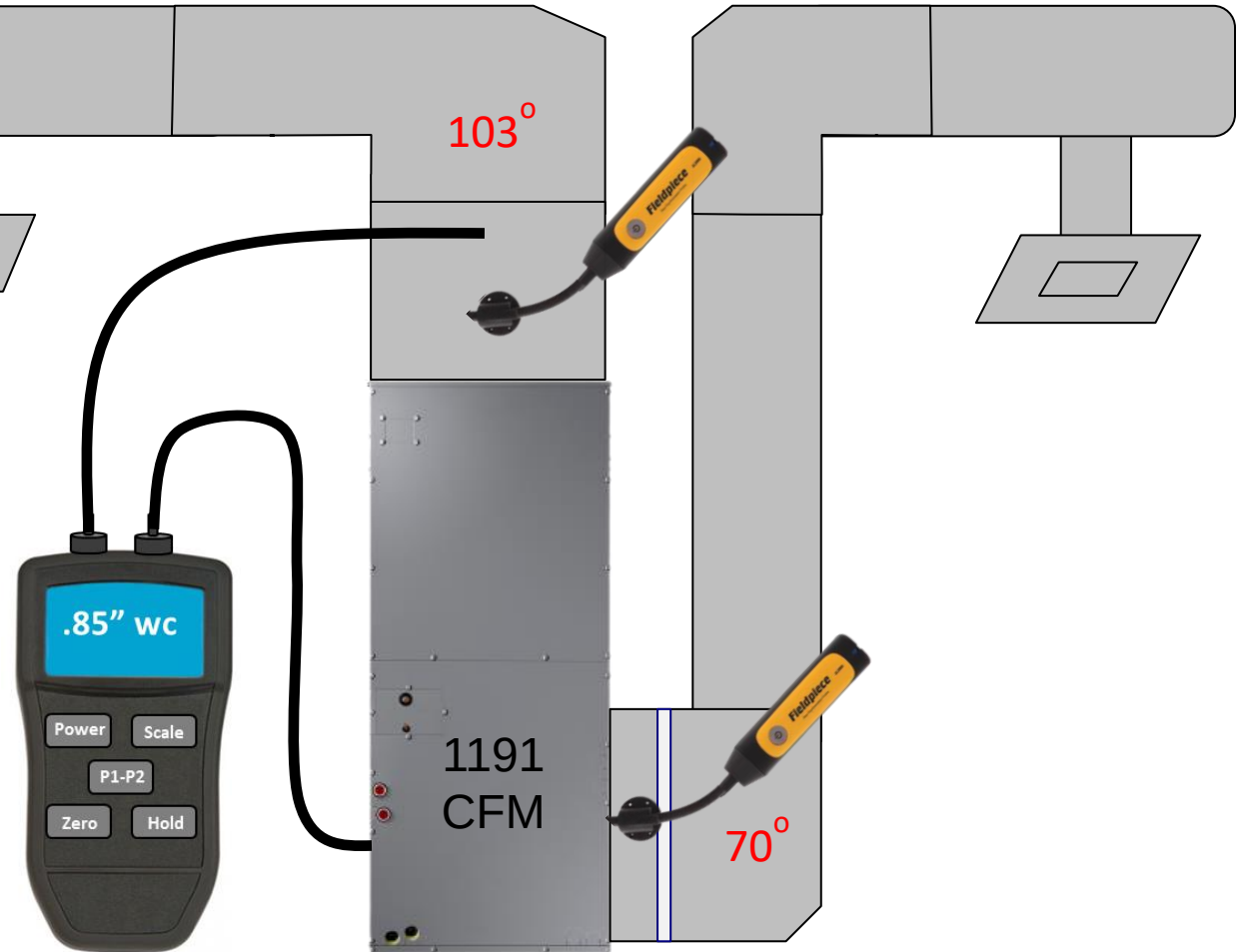
Calculating the IDU Heat Output

- Heating BTUs = CFM x Δt x (Sensible Heat Factor) 1.08
- CFM = the fan CFM plotted on the fan tables, or measured at several points in the system
- Δt = the dry bulb temperature change through the system
- 1.08 = the heating BTU multiplier at sea level, or .075-lbs. of air per CFM x .24 (the specific heat of air) x 60 (minutes in an hour.) This factor will vary at higher altitudes and temperatures

Calculating Heating Capacity

- Heating BTUs = CFM x Δt x (Sensible Heat Factor) 1.08
- BTU's = 1191 x 33 x 1.08
- Delivered BTU = 42,447

Model	Model Number	AC036BXUPCH/AA	
	US Code	CXH36UPB	
Capacity	Nominal Capacity	Cooling (Btu/h)	36,000
		Heating (Btu/h)	34,600
	Capacity Range ¹	Cooling (Btu/h)	16,500 - 45,000
		Heating (Btu/h)	13,000 - 50,000

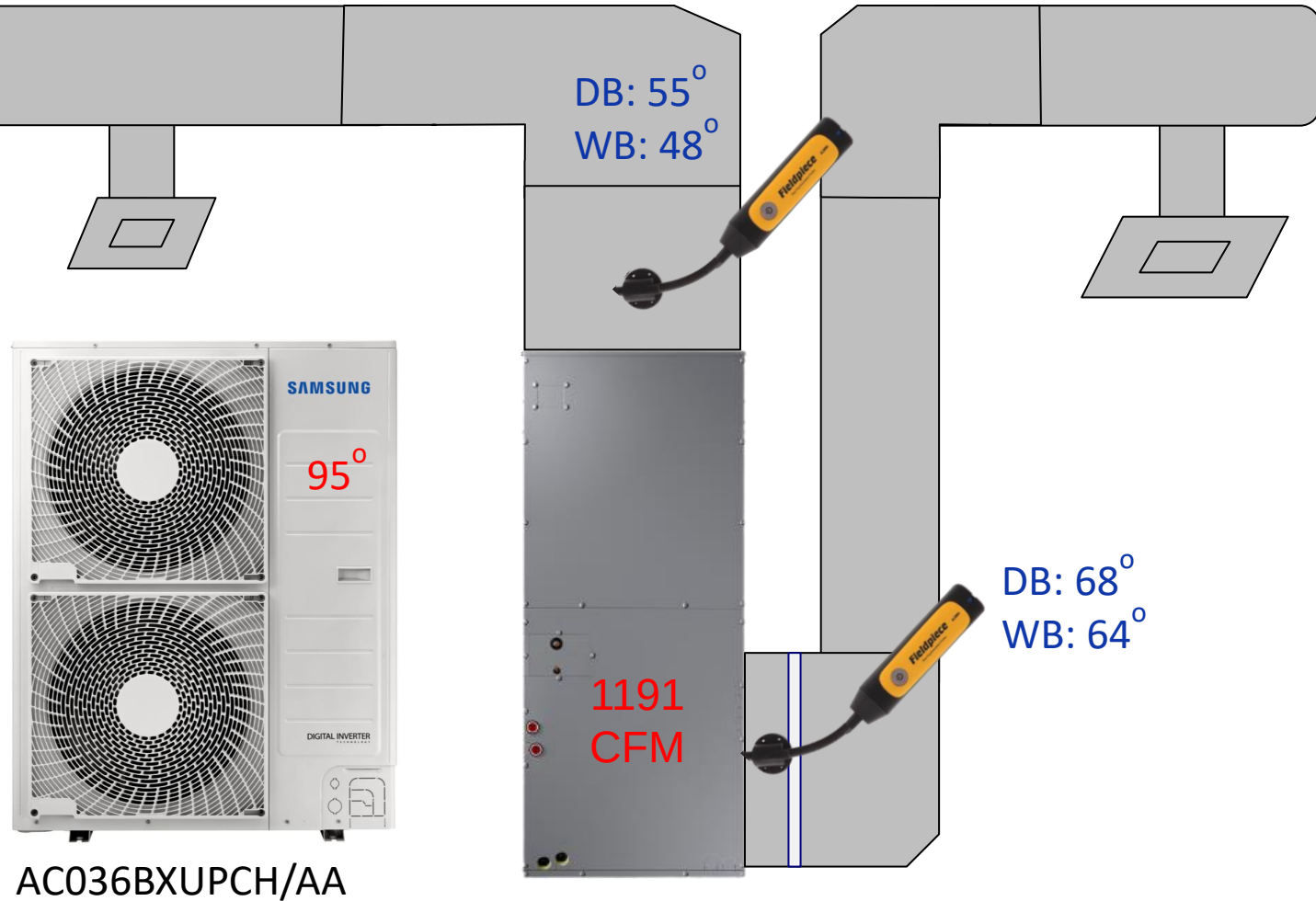


Cooling Capacity Calculations

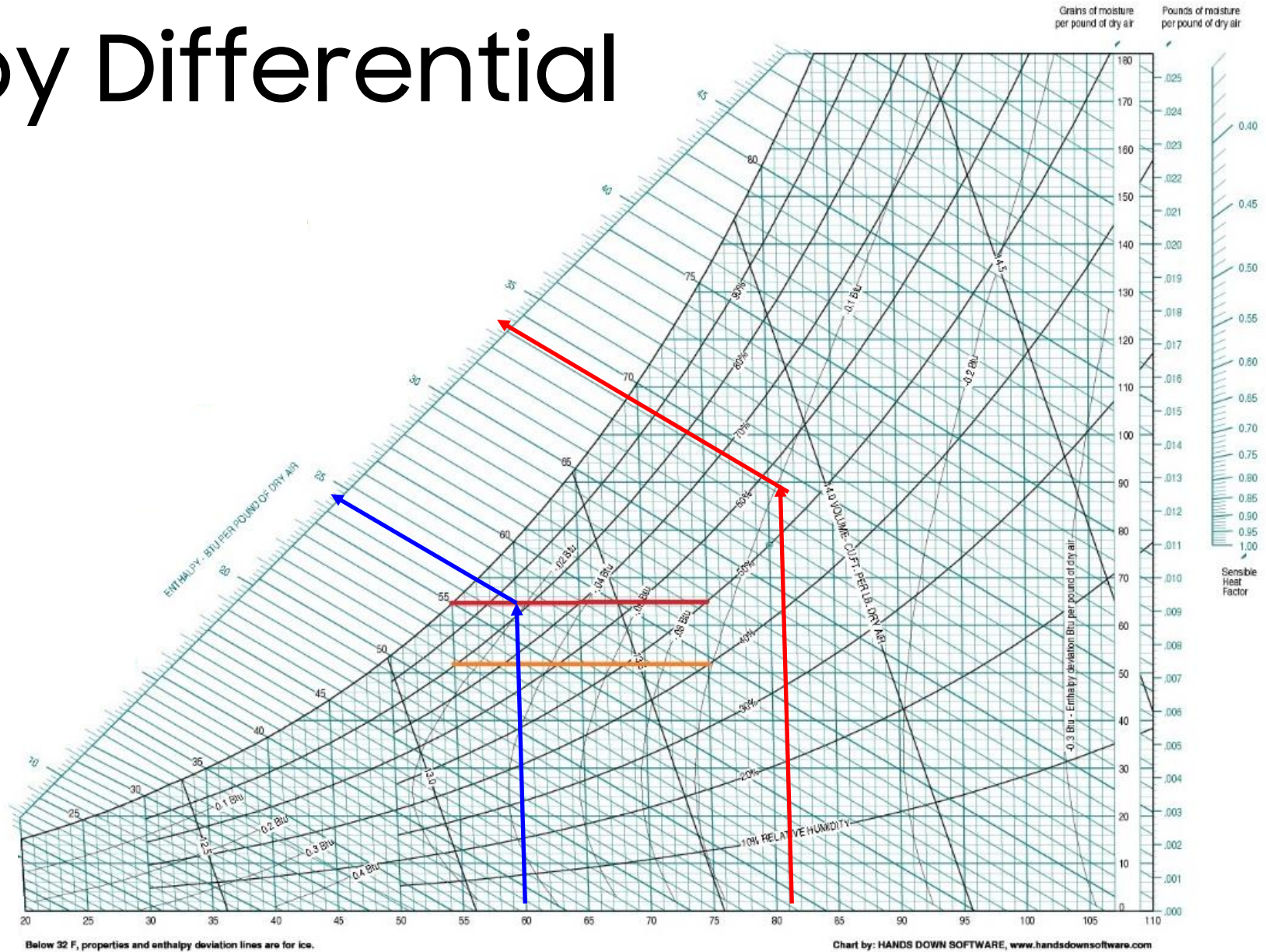
- Cooling BTUs = CFM x Δh x 4.5
- CFM = The fan CFM plotted on the fan tables, or measured at several points in the system
- Δh = the enthalpy change through the system (we get this number by converting wet bulb readings.)
- 4.5 = the cooling BTU multiplier at sea level, or .075-lbs. of air per CFM x 60 (minutes in an hour.) This factor will also change at higher altitudes. BTU multipliers will also change due to extreme air temperatures

Calculating Cooling Capacity

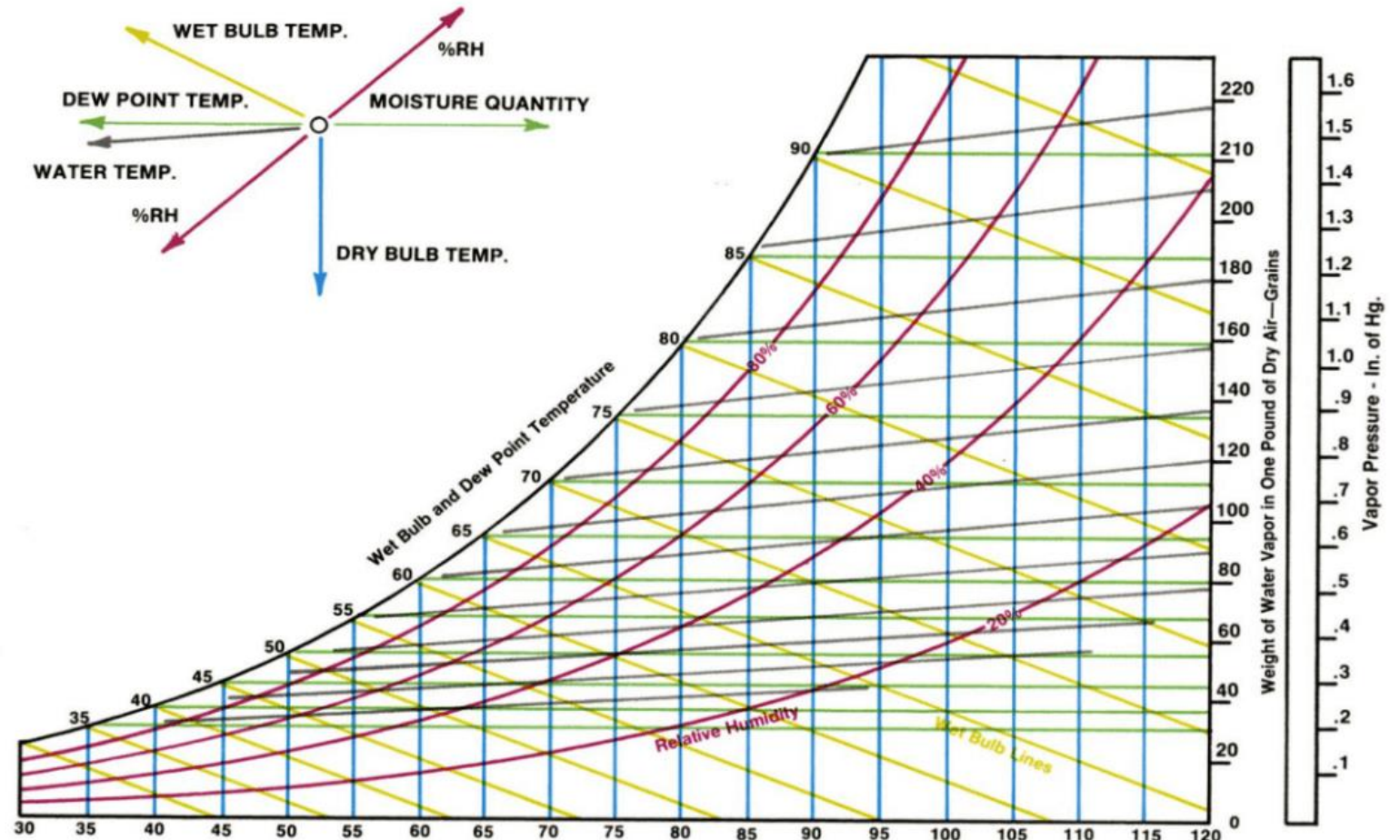
Verify that the filter and system is clean
Set Fan to High Verify ESP Setting
Set controller to cool
Measure The ESP of the system
Measure the Enthalpy
Differential across the coil
Use the Fieldpiece App to
calculate BTU's



Enthalpy Differential



Psychrometric Charts



Calculate Enthalpy

- Using a psychrometric calculator determine the enthalpy of the return and supply air then calculate the difference
 - Return Air: 68 BD – 63.9 WB – Enthalpy 26.7
 - Supply Air: 55 BD – 48 WB – Enthalpy 22.3
 - Enthalpy 4.4

Free online Psychrometric Calculator [Print](#)

☒ IP ☐ SI altitude ft b press in.Hg

T (°F) dry bulb	rH (%) rel.humidity	TDP (°F) dew point	W (GPP) abs. humidity	h (btu/lb) enthalpy	Tw (°F) wet bulb	sp. vol (cft/lb)		
68	65	59.2	75	26.7	63.9	13.46	↺ Calculate	✕ Remove
55	91	48	55	22.3	48	13.14	↺ Calculate	✕ Remove

[+ Add line](#)

Cooling Capacity Formula

- $BTUH = CFM \times 4.5 \times \text{Delta Enthalpy}$
- As an example, measurements inside the main ducts of the air handler:
 - Before the evaporator (return): DB: 68° F, WB: 64° F, Enthalpy = 26.7 BTU/lb.
 - After evaporator (supply): DB: 55° F, WB: 48° F, Enthalpy = 22.3 BTU/lb.
- System Enthalpy: $26.7 - 22.3 = 4.4$ BTU/lb.
- System CFM = 1191CFM
- $BTUH = 1191 \times 4.5 \times 4.4$
- $BTUH = 24,00$ BTUH delivered
- System is a 3-ton system?
- Is this good or bad?

Model	Model Number		AC036BXUPCH/AA
	US Code		CXH36UPB
Capacity	Nominal Capacity	Cooling (Btu/h)	36,000
		Heating (Btu/h)	34,600
	Capacity Range ¹	Cooling (Btu/h)	16,500 - 45,000
		Heating (Btu/h)	13,000 - 50,000

Effect of operating conditions on system performance

- In this example the outside temperature is 95 degrees.
- The system is performing as it should under these conditions.

• CXH36UPB (AC036BXUPCH/AA)

Cooling

TC : Total Capacity, SHC : Sensible Heat Capacity, PI : Power Input

Outdoor Temp. (°F, DB)	Indoor Temperature (°F, DB / WB)																				
	68 / 57			72 / 61			77 / 64			80 / 67			82 / 70			86 / 72			90 / 75		
	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI
	MBH	MBH	kW	MBH	MBH	kW	MBH	MBH	kW	MBH	MBH	kW	MBH	MBH	kW	MBH	MBH	kW	MBH	MBH	kW
-4	19.60	18.27	1.91	25.55	20.16	2.50	30.10	23.53	3.05	35.00	25.10	3.70	37.80	24.67	3.93	39.20	26.14	3.99	41.30	26.95	4.06
70	26.04	22.28	1.87	33.95	24.57	2.43	39.99	28.68	2.97	46.50	30.60	3.60	50.22	30.07	3.82	52.08	31.87	3.88	54.87	32.86	3.95
95	24.01	21.04	2.09	30.35	23.14	2.60	38.05	28.00	3.33	36.00	27.00	3.43	50.74	30.20	4.64	52.10	32.06	4.71	54.81	33.00	4.78
115	23.73	20.19	3.48	28.93	21.62	4.03	30.88	23.76	4.06	32.50	24.05	4.10	34.45	22.94	4.14	35.75	24.87	4.17	36.73	25.55	4.14
126	17.96	16.11	2.71	21.89	17.25	3.12	23.37	18.96	3.14	24.60	19.19	3.17	26.08	18.31	3.20	27.06	19.84	3.22	27.80	20.38	3.20

Make Life Simple



11:40 AM

Fieldpiece Save

SDP2

Return Air	Supply Air
64,589.1 BTU Capacity	62.4 Target Dry Bulb
85.0 °F Dry Bulb	65.0 °F Dry Bulb
65.0 °F Wet Bulb	60.0 °F Wet Bulb

SMAN4

Suction Line	Liquid Line
60.0 PSIG Pressure	130.0 PSIG Pressure
33.9 °F Vapor Saturation	74.0 °F Liquid Saturation
40.0 °F Temperature	70.0 °F Temperature
8.0 °F Actual Superheat	16.0 °F Actual Subcooling
10.0 °F	10.0 °F

13:35

AC + Refrigeration

testo 549i, ... P_{ref} R134a

List Trending Table

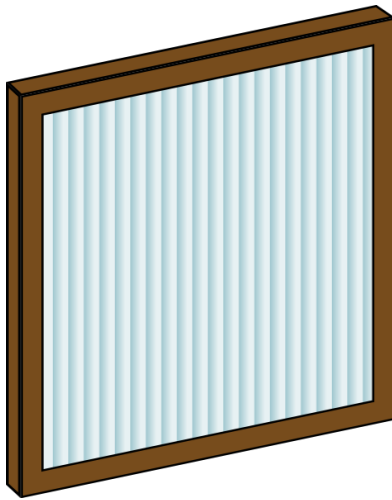
021	LP	254	HP
10,14	bar	27,39	bar
10,0	°C	45,6	°C
020	toh	294	tcu
16,2	°C	43,3	°C
6,2	Δtoh	2,3	Δtcu
	K		K



Things that affect Air Flow Filter Selection

Know what's in your system!

Air Flow: 1000 CFM
300 FPM



16 x 25 Air Filters @ 1,000 CFM

290	430	575	720	970*	*)
0.09	0.15	0.22	0.31	0.46	nc

Air Flow: 1400 CFM
500 FPM

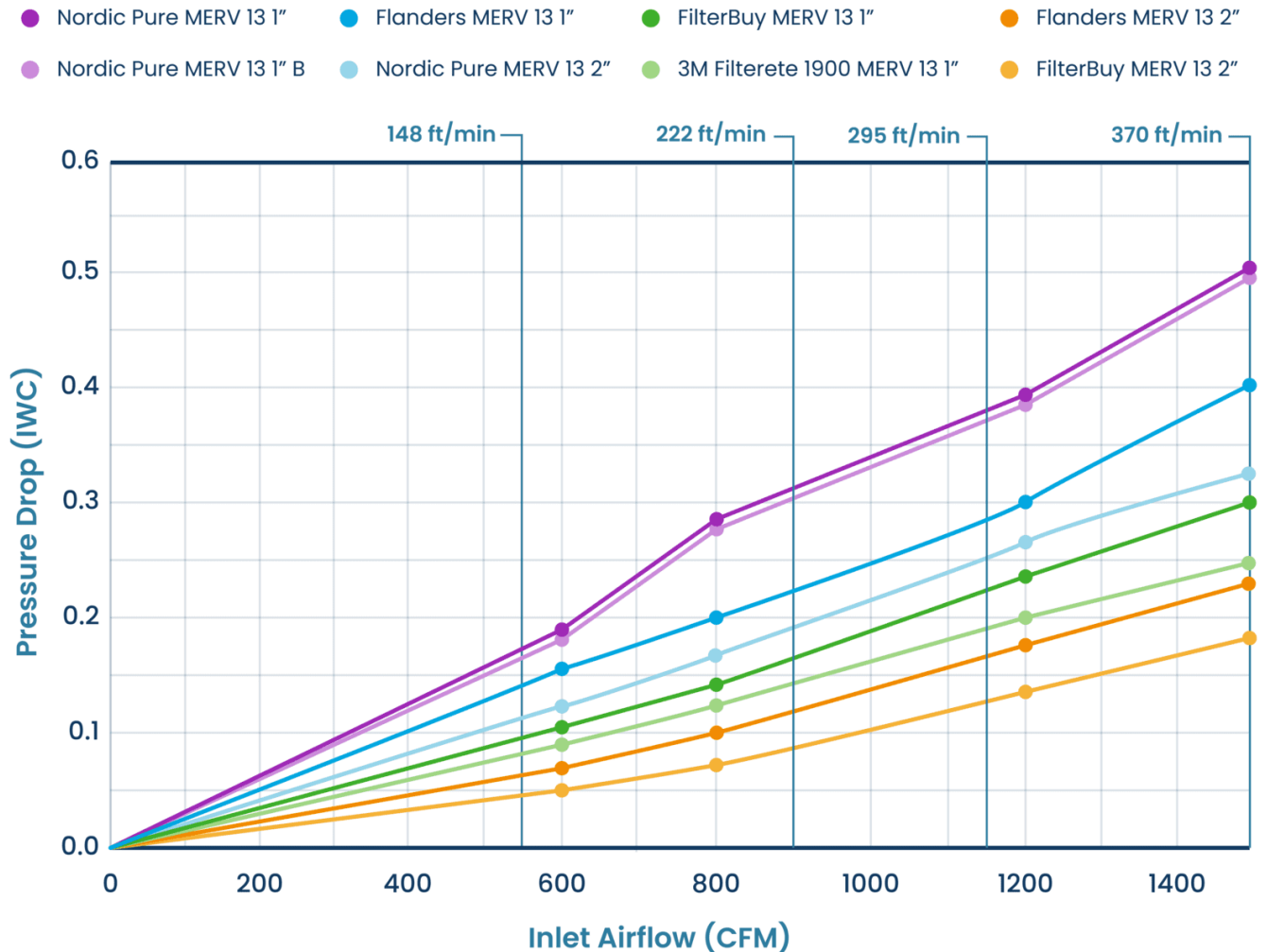


Resistance : .2" wc

All filters are not created equally

- Know the characteristics of the filter you're selecting for a job.
- Misapplying a filter can cause performance issues with the system

MERV 13 Pressure Drop Measurements

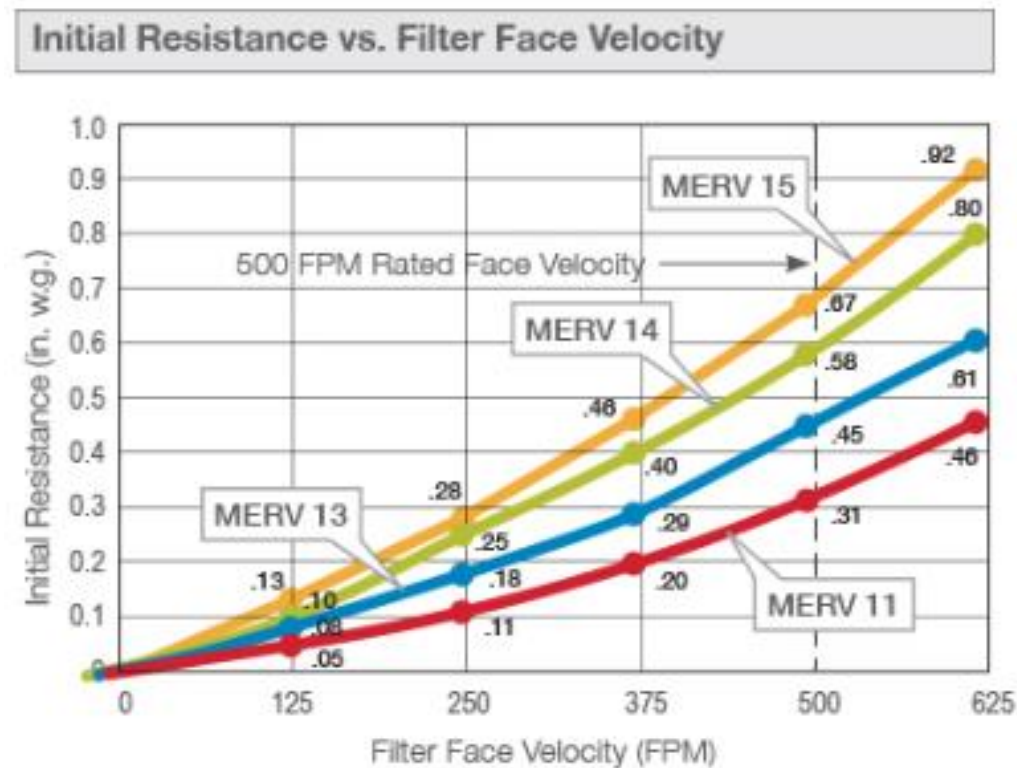


*Testing was conducted by the **California Energy Commission**. MERV Filters tested included various models from 3M, FilterBuy, Flanders, Nordic Pure, Honeywell, & US Home Filter.

sanalife
sanalifewellness.com

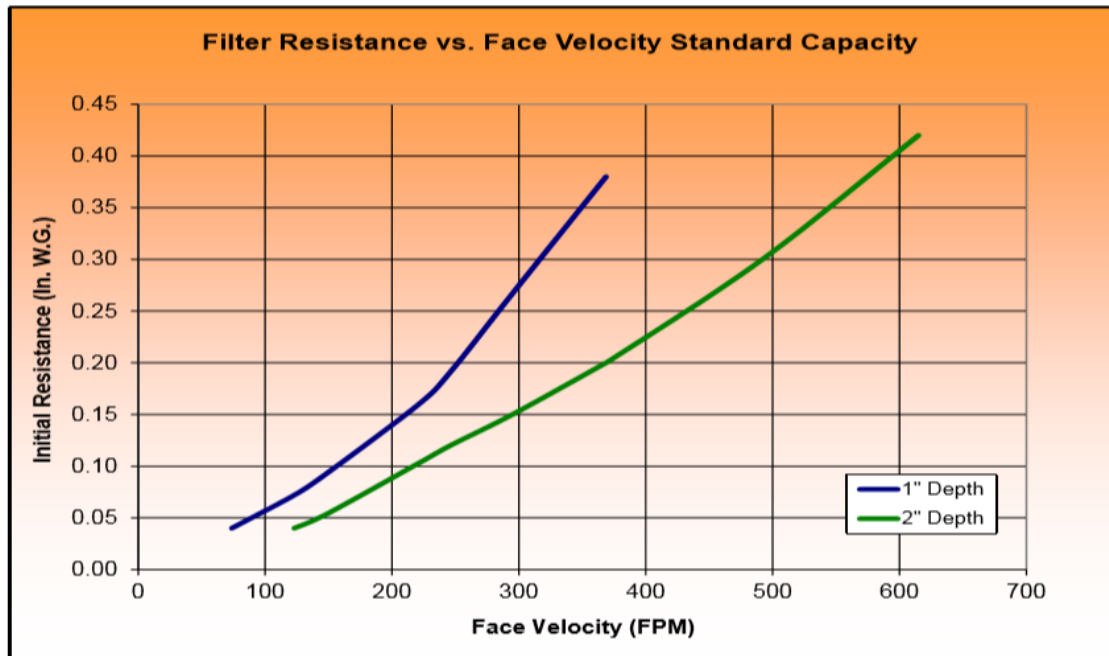
Filter Selection

- If the homeowner purchases a filter with a higher MERV rating, can it effect the system performance?

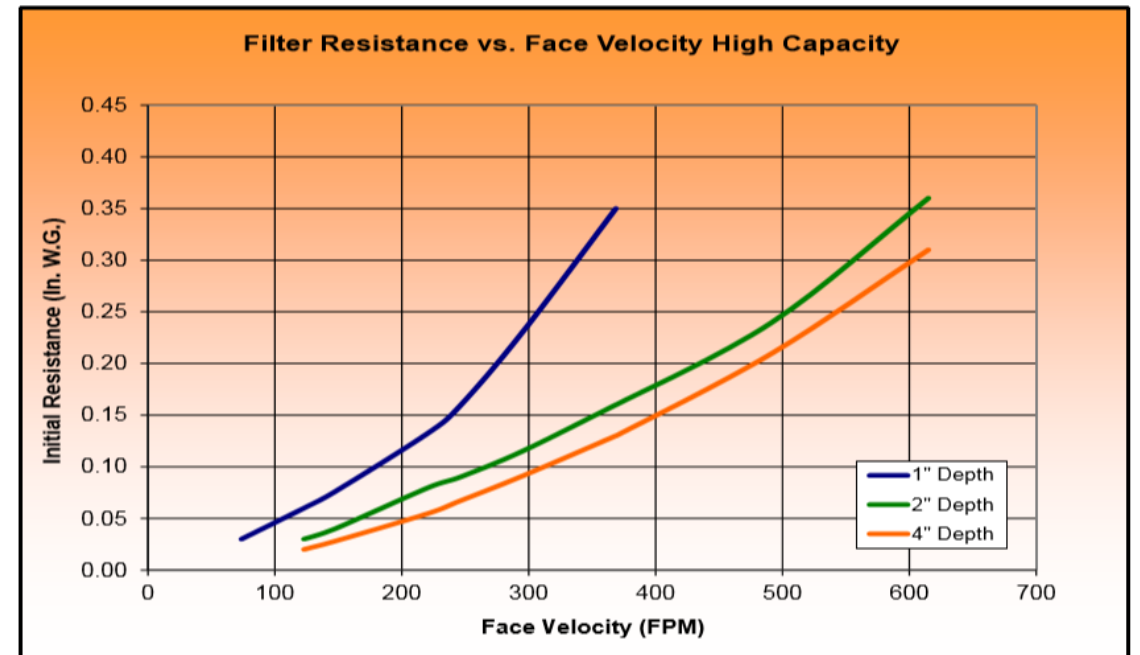


Filter Pressure Drop vs Face Velocity

- Understanding the velocity and pressure drop relationship.



Standard-Capacity (KP)

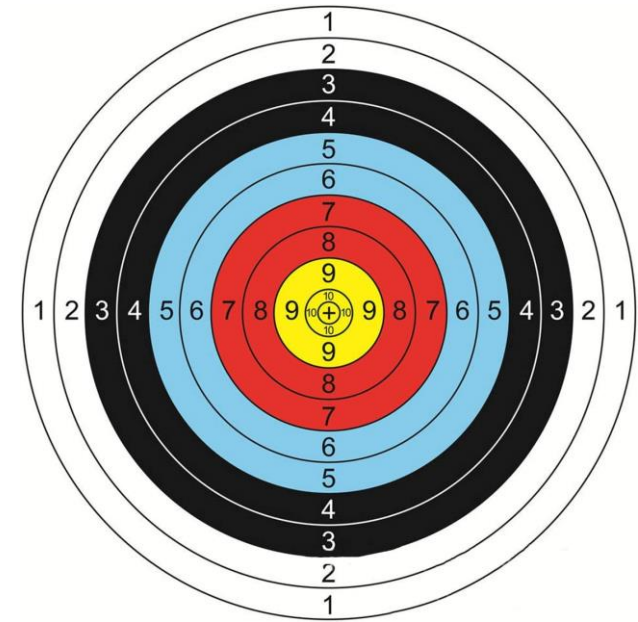


High-Capacity (KPH)

Purolator Key Pleat Merv 8 Filter Performance

Putting It All Together

- The operating characteristics are determined during the testing process using PE diagrams.
- From these tables engineers can calculate the operating characteristics at any given condition for each mode.
- They use this information to establish targeted operating values of the system for the conditions and mode it is running at.
- The system components will adjust to varying conditions to maintain the “Targeted” operation.



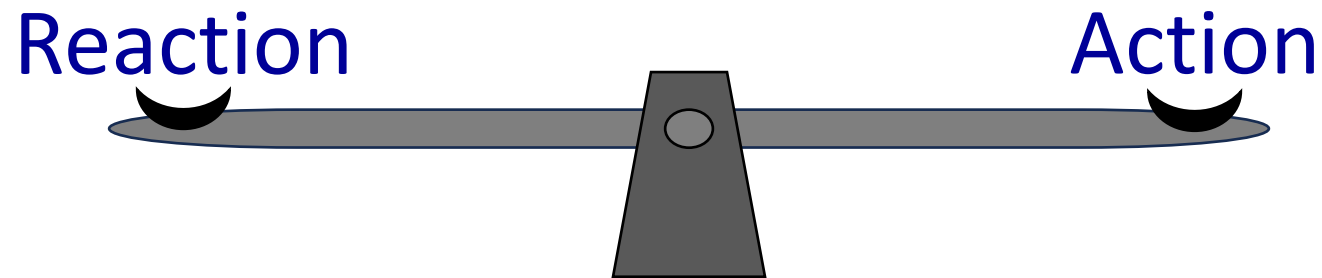
What to look for.

- Here is the caveat
 - The system are dynamic and constantly adjusting
 - We can not precisely identify these targets.
 - There for when troubleshooting we look for measurements to be with in a range, Not a specific value
- Sometimes it feels like you're hitting a moving target.
- We can use the controls sequence of operation to get an operating range for the heating and cooling cycles.



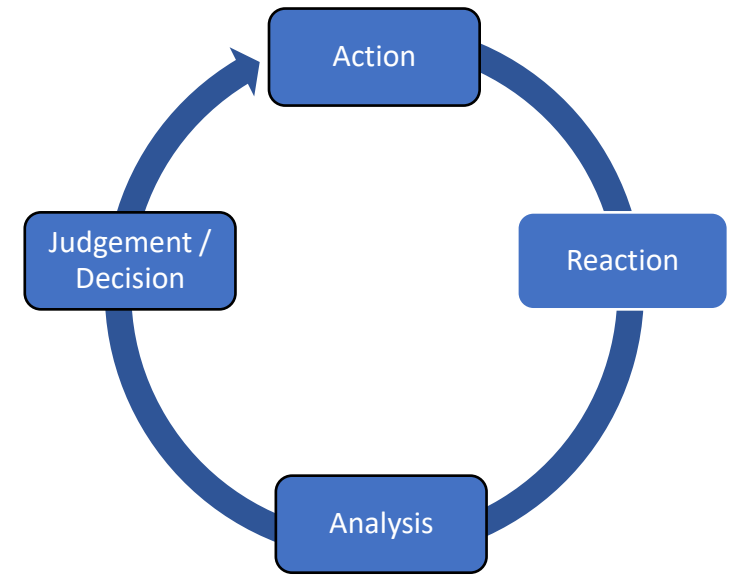
VRF System Operation

- The operation of a VRF system is a constant balancing act between changes in the system and the operation of components.
- For every change within the system there is a reaction by the components to maintain their target operation.
- One change within the system can cause a domino effect of component reactions.
- It's important to understand the cause and effects that take place within the VRF system to understand the system operation.



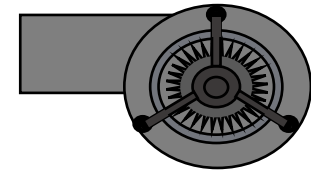
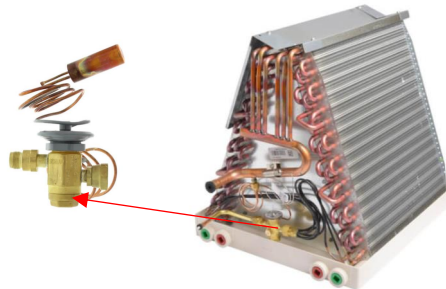
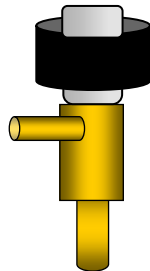
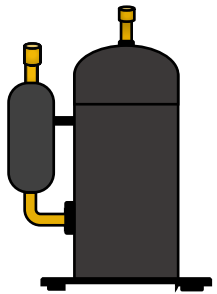
System Cycle Logic

- The operation of a Variable system is a constant balancing act between changes in the system and the operation of components.
- For every change within the system there is a reaction by the components to maintain their target operation.
- One change within the system can cause a domino effect of component reactions.
- It's important to understand the cause and effects that take place within the VRF system to understand the system operation.

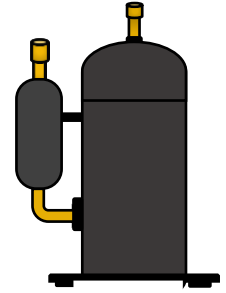


Interrelationship between components

- What effect does the operation of one component have on the others?
- How does the system react to changes in the system?
- What is the domino effect of actions within the system that occur when there is a change within the system
- For example, what happens effect does being under charged have on the system operation?



The Compressor

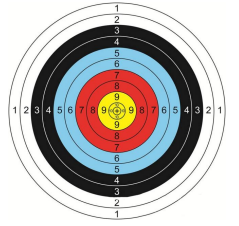
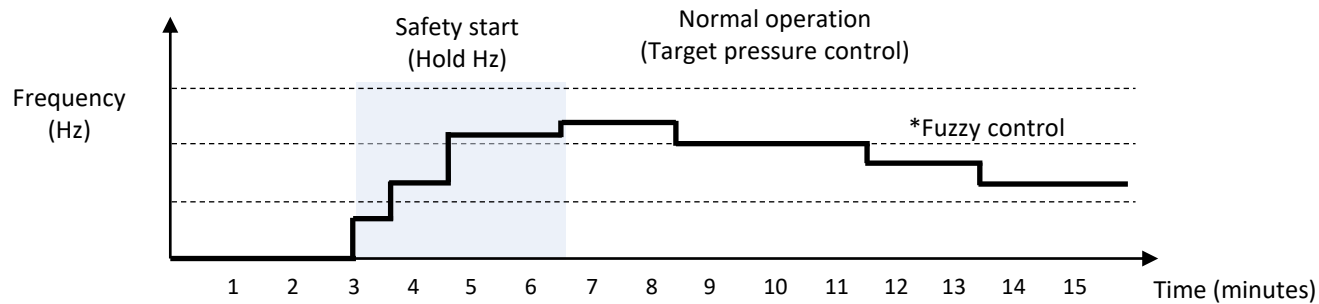


- The compressor's primary function is to maintain pressure in the heating and cooling modes.
- The speed at which the refrigerant flows through the system is directly related to the speed of the compressor.
- The speed of the compressor influences the system pressures and temperatures.

Control logic

■ Compressor frequency control

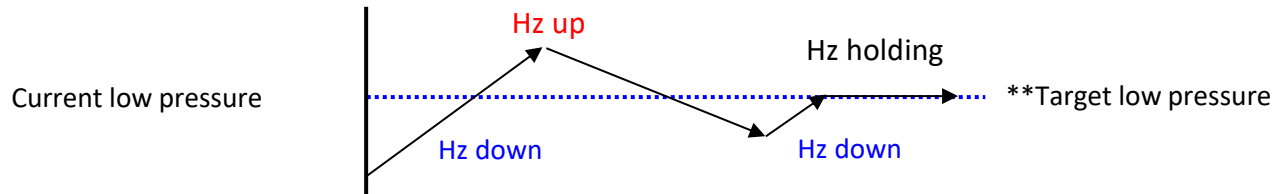
- The compressor frequency is determined by both the **Current pressure & Target pressure**.



*The compressor frequency is determined by fuzzy logic which consists of 2 variables.

- . ΔP (Current pressure/Target pressure)
- . Current ΔP – Previous ΔP

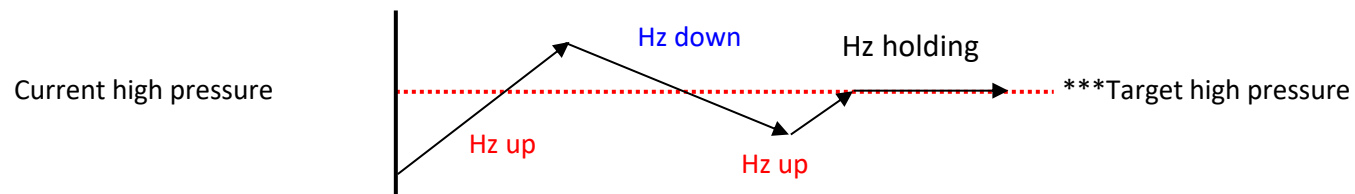
< Cooling target low pressure >



**The target low pressure is different according to the model.

The average target low pressure is approximately 128 psi (9kg/cm²) at the Y2 stage (High speed) and 142.2 psi (10 kg/cm²) at the Y1 stage (Medium speed).

< Heating target high pressure >

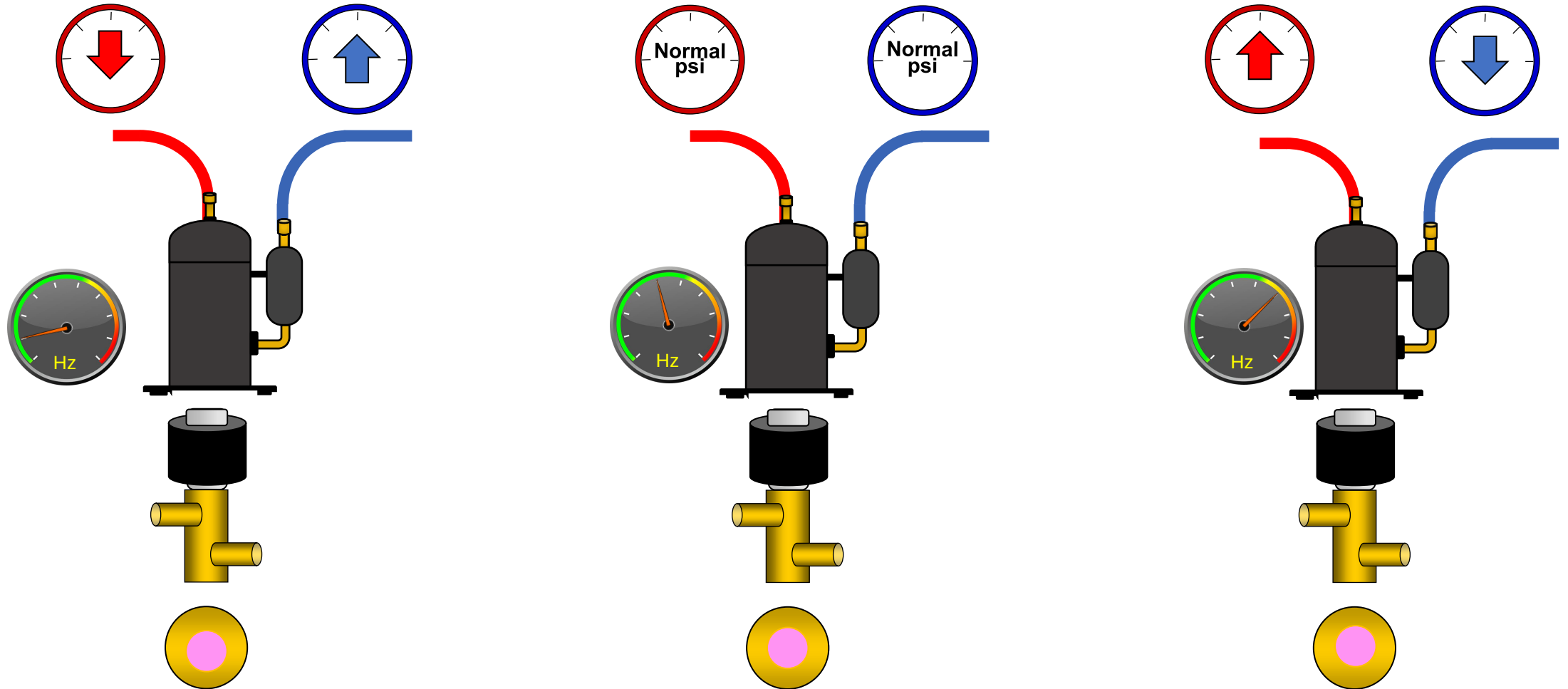


***The target high pressure is different according to the model.

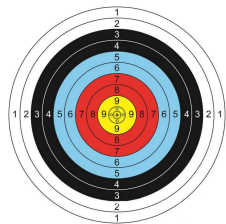
The average target high pressure is approximately 370 psi (26 kg/cm²) at the Y2 stage and 355.6 psi (25 kg/cm²) at the Y1 stage.

Relation between Hz and Pressure

(With no other changes in the system)



Control logic



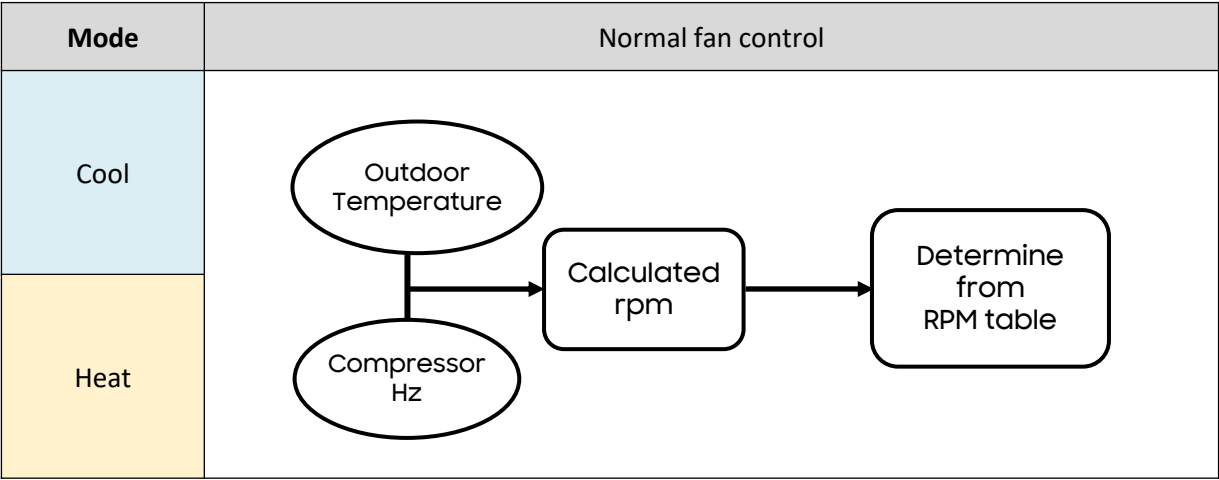
The initial rpm of the outdoor fan is determined by the outdoor temperature.

Cooling operation	
Conditions	*Fan RPM
Outdoor temperature $\geq 104\text{ }^{\circ}\text{F}$ ($40\text{ }^{\circ}\text{C}$)	800
$104\text{ }^{\circ}\text{F}$ ($20\text{ }^{\circ}\text{C}$) < Outdoor temperature < $104\text{ }^{\circ}\text{F}$ ($40\text{ }^{\circ}\text{C}$)	650
Outdoor temperature $\leq 104\text{ }^{\circ}\text{F}$ ($20\text{ }^{\circ}\text{C}$)	350

Heating operation	
Conditions	*Fan RPM
Outdoor temperature $\geq 41\text{ }^{\circ}\text{F}$ ($5\text{ }^{\circ}\text{C}$)	500
$23\text{ }^{\circ}\text{F}$ ($-5\text{ }^{\circ}\text{C}$) < Outdoor temperature < $41\text{ }^{\circ}\text{F}$ ($5\text{ }^{\circ}\text{C}$)	650
Outdoor temperature $\leq 23\text{ }^{\circ}\text{F}$ ($-5\text{ }^{\circ}\text{C}$)	800

* Example of 3 ton model.

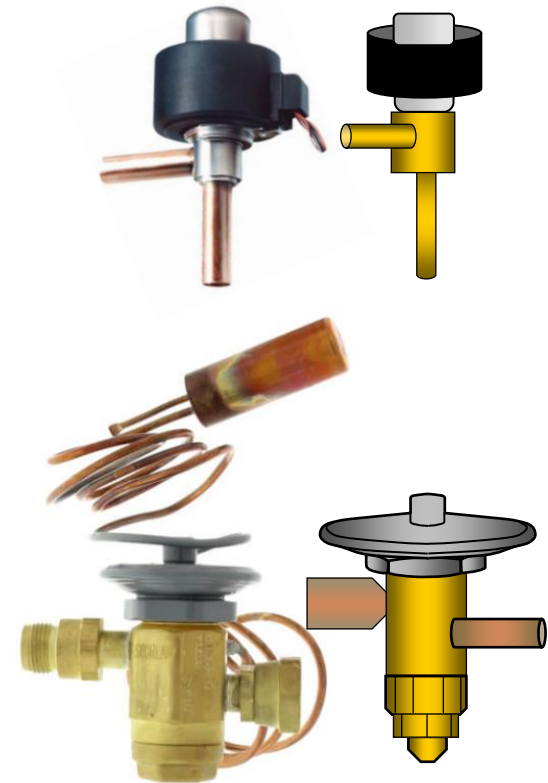
After the initial RPM, next outdoor fan RPM is determined by outdoor temperature and compressor Hz.



- Outdoor fan control

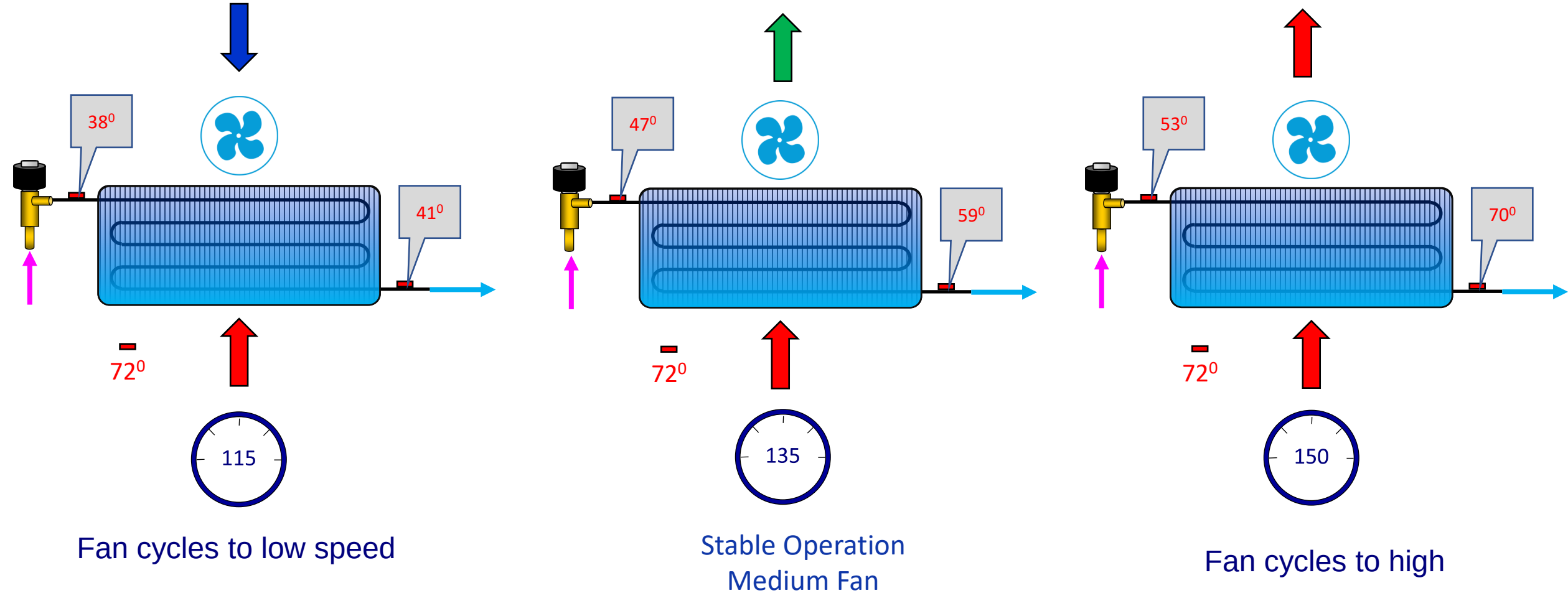
Metering Device Control Logic

- The purpose of the metering device is to create a pressure drop and regulate refrigerant flow.
- The ODU uses a EEV in the heating mode to maintain evaporator super heat of 5.4 degrees
- The IDU uses a TEV in the cooling mode to maintain superheat of approximately 12 degrees .
- No twwhat device is being used their function and operation are basically the same and in



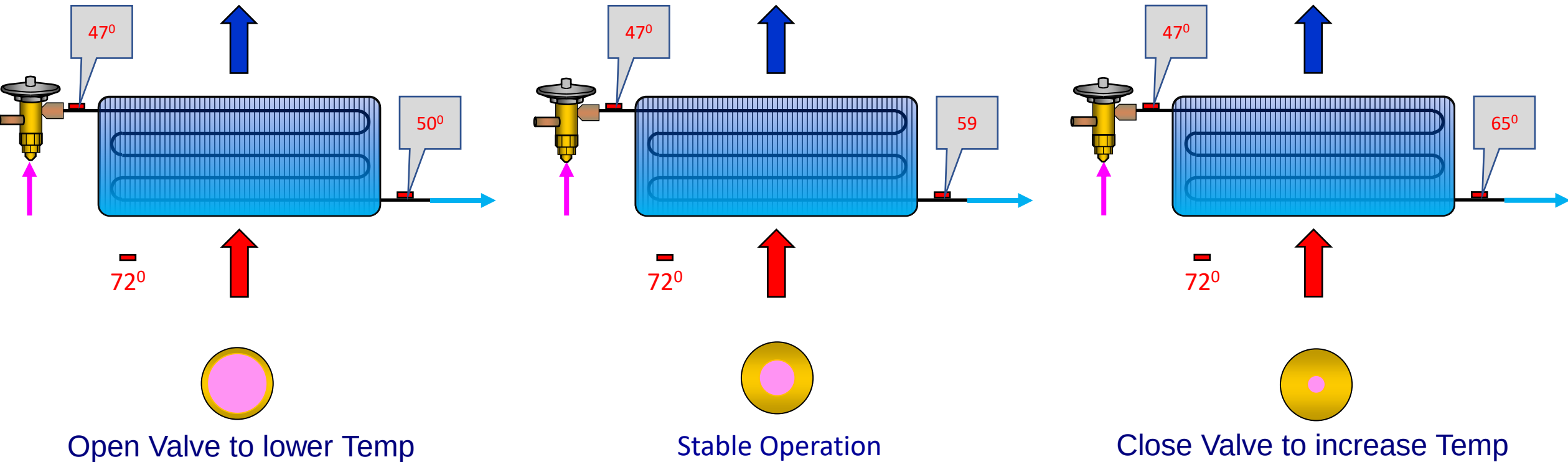
Air flow effects on Evaporator Temperature

(With no other changes in the system)



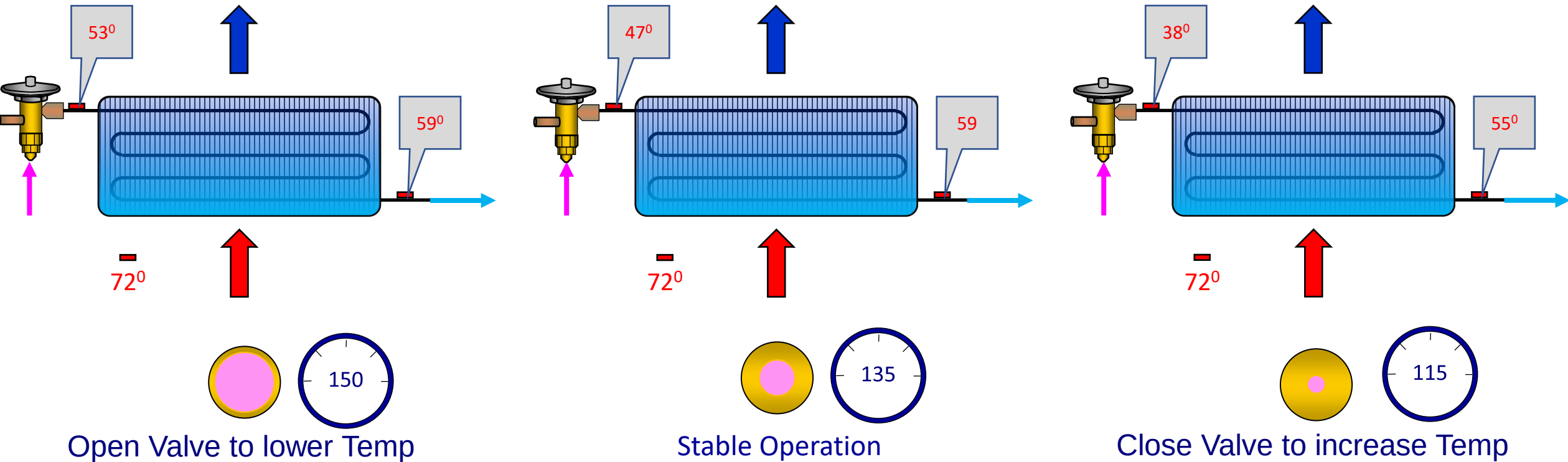
LEV / TEV opening Effects on Super Heat

(With no other changes in the system)



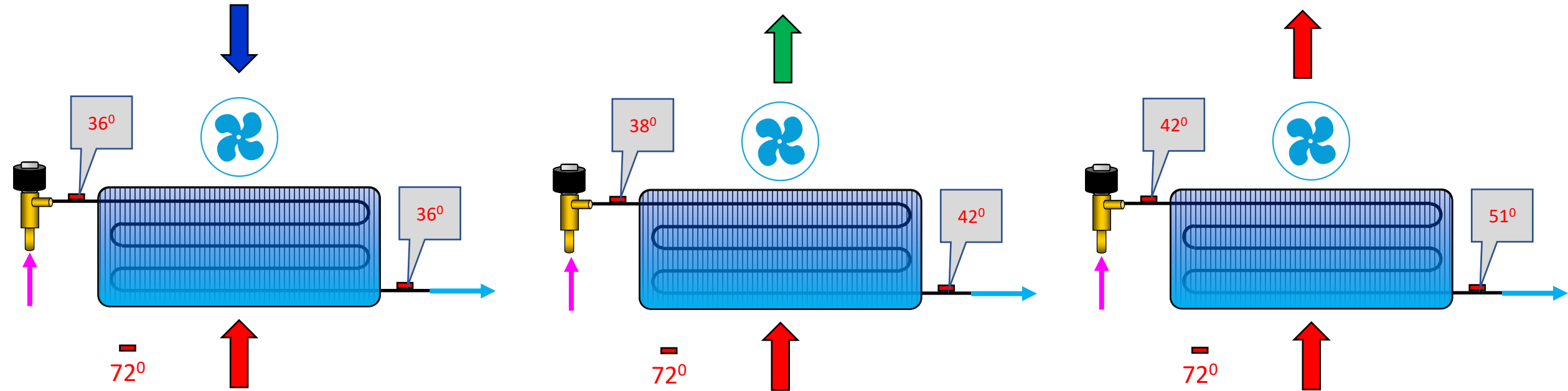
LEV / TEV opening Effects on Pressure

(With no other changes in the system)



Air flow effects on superheat heat

(With no other changes in the system)



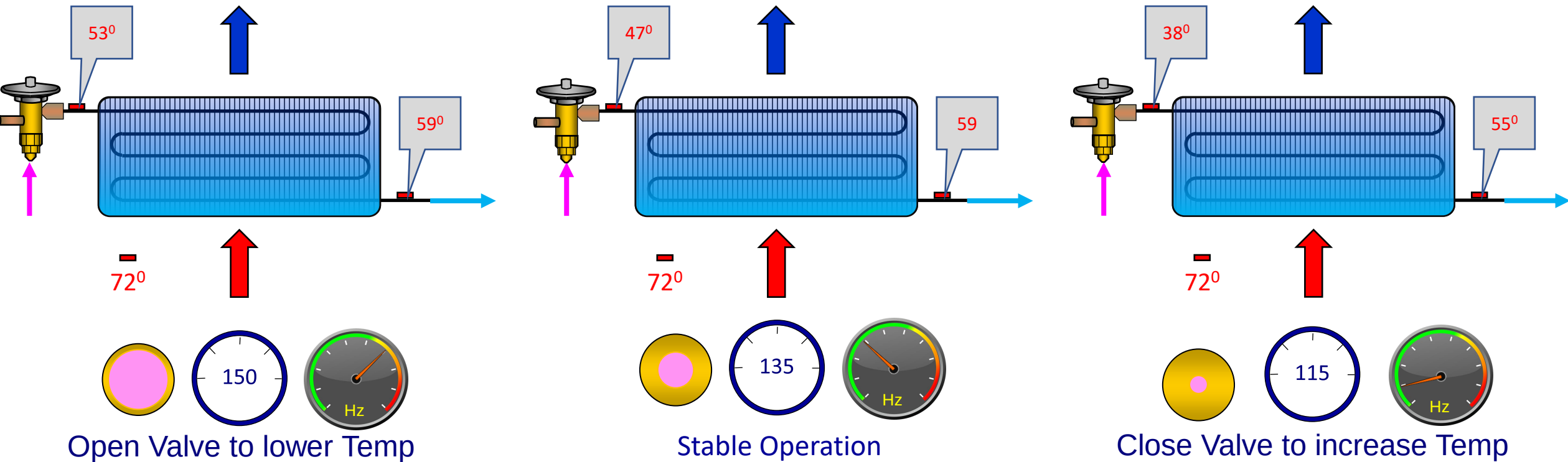
Fan cycles to low speed

Stable Operation
Medium Fan

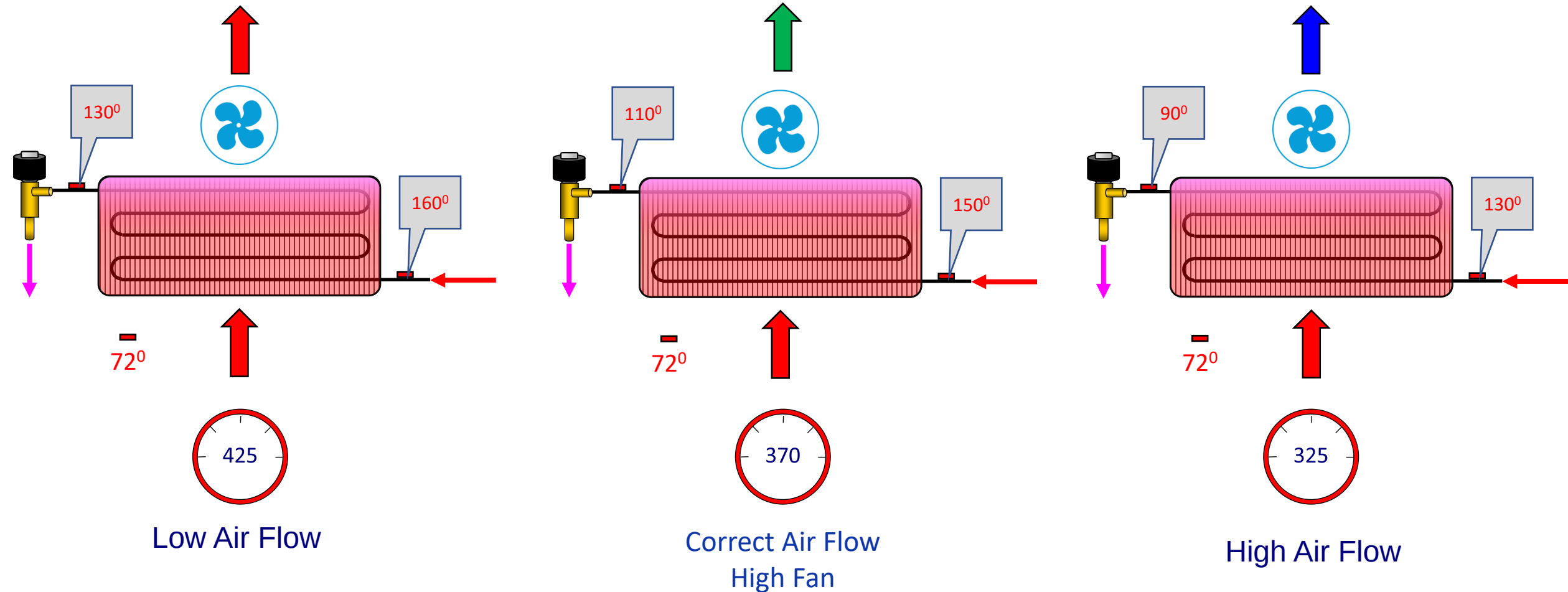
Fan cycles to high

LEV / TEV opening Effects EntireSystem

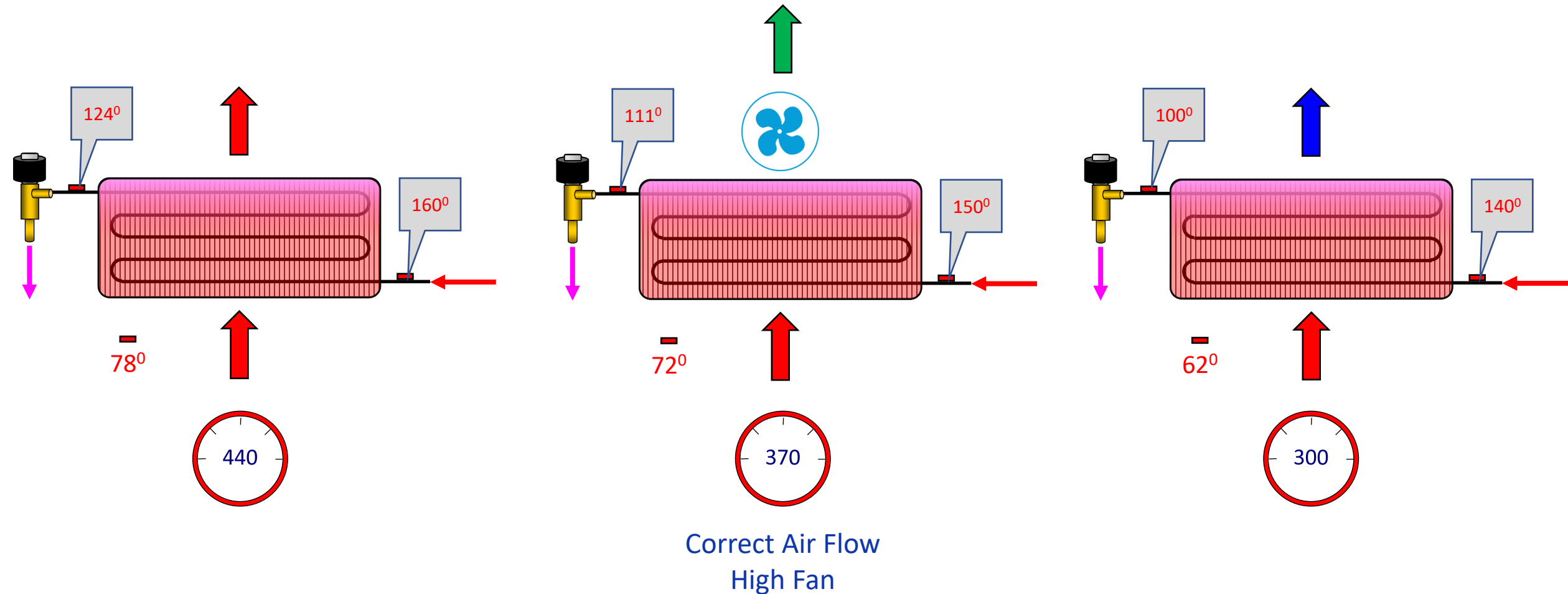
(With no other changes in the system)



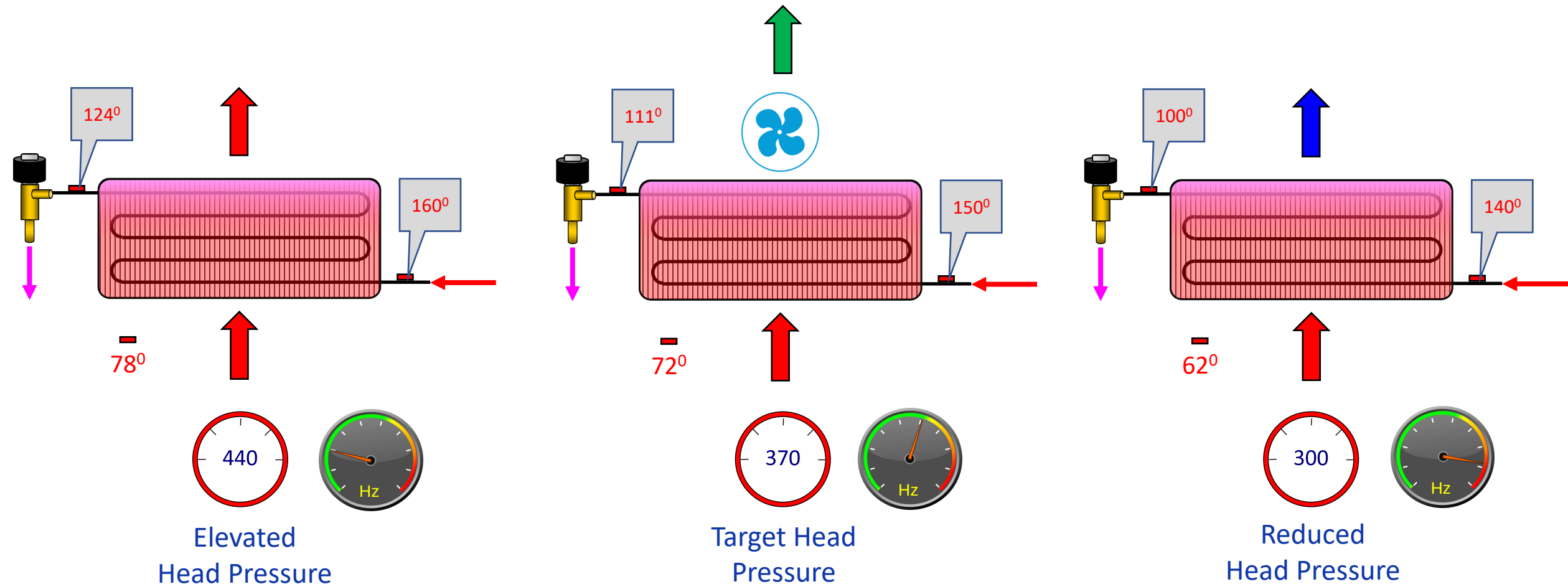
Air flow effects on head PSI



Space Temperature effects on heat PSI



How does the compressor respond to changes in head pressure

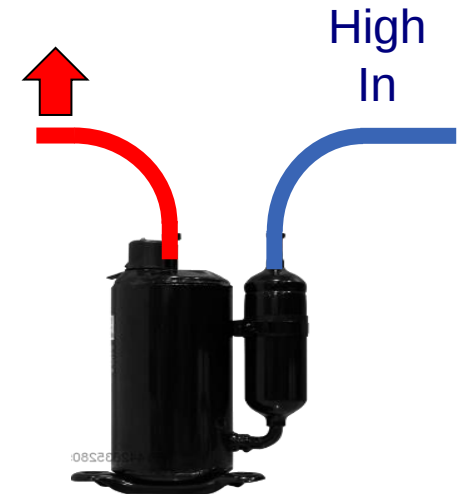
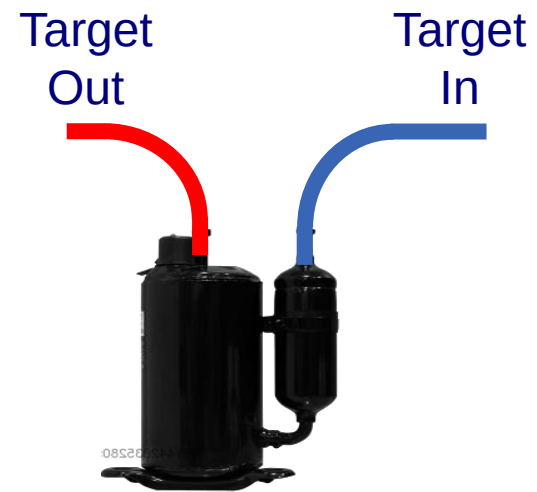
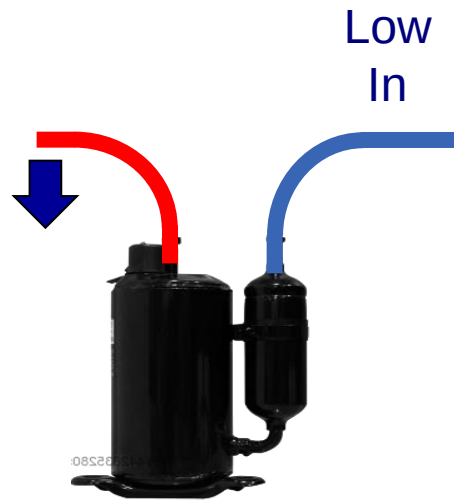


Discharge Temperature Control

- The discharge temperature is directly related to the suction super heat and the compression ratio.
- If the discharge temperature is low and the system wants to increase the discharge temperature, then it will slow down the compressor allowing the refrigerant to sit in the evaporator longer to absorb more heat
- If the system wants to lower the discharge temperature it will increase the speed of the compressor.
- Discharge temperature control is a key factor in the operation of a VRF system.



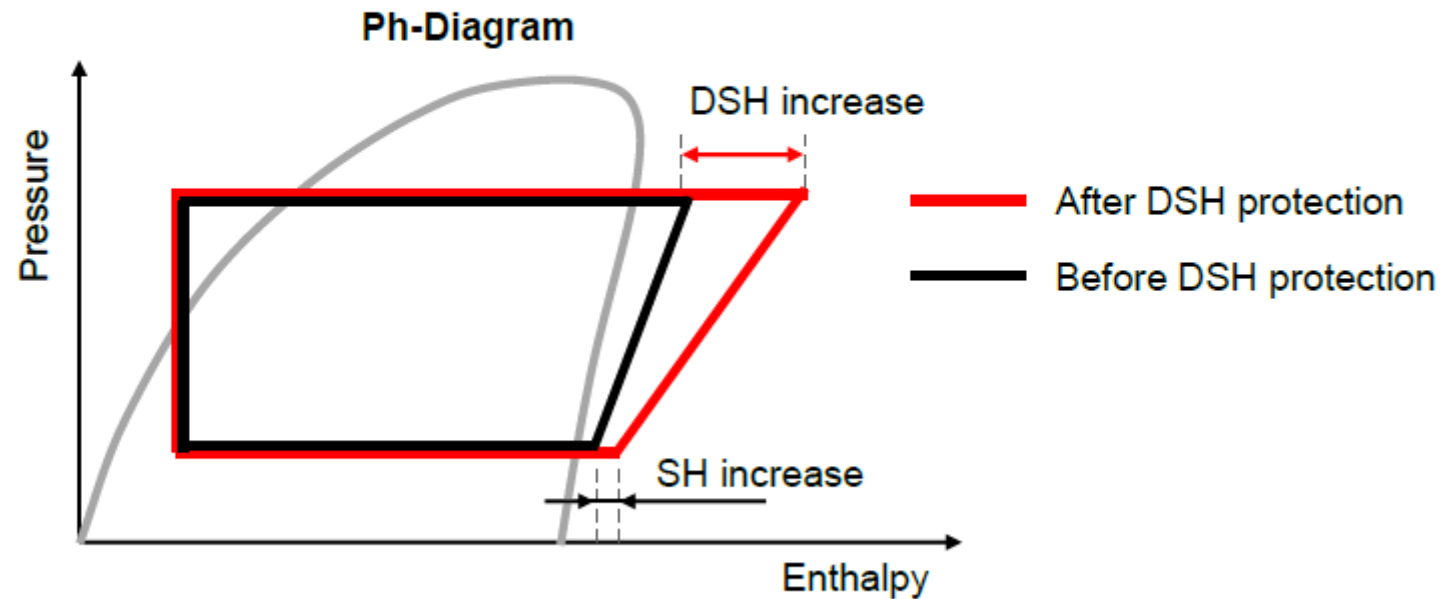
Compressor Temperature In Vs Out



Superheat-Discharge Temperature Relationship

Concept

- The compressor Suction Super Heat (SH) is raised to increase the DSH.



- $DSH = \text{Compressor discharge temp.} - \text{High pressure saturation temp.}$

The Compressor

Speed Vs Compression Ratio

(With no other changes in the system)

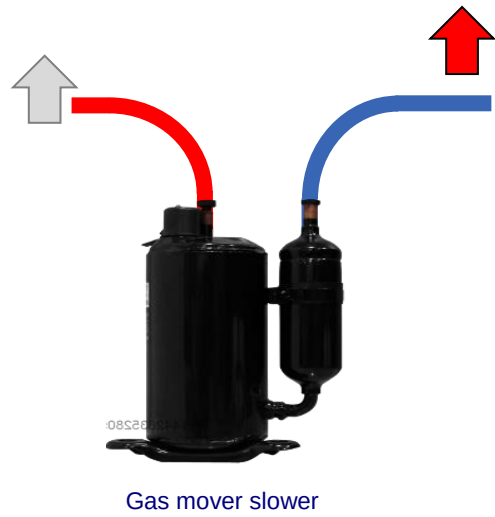
Normal Compression Ratio



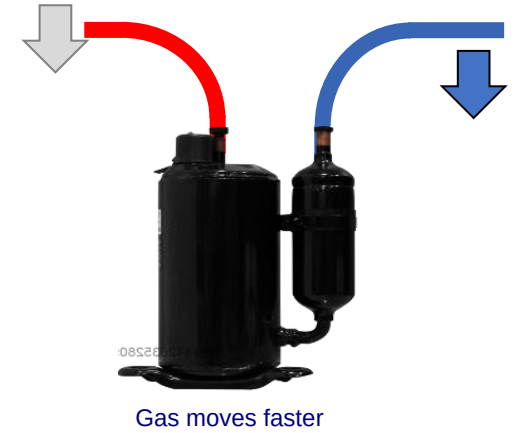
The Compressor

Speed Vs Velocity of the refrigerant (SH)

(With no other changes in the system)

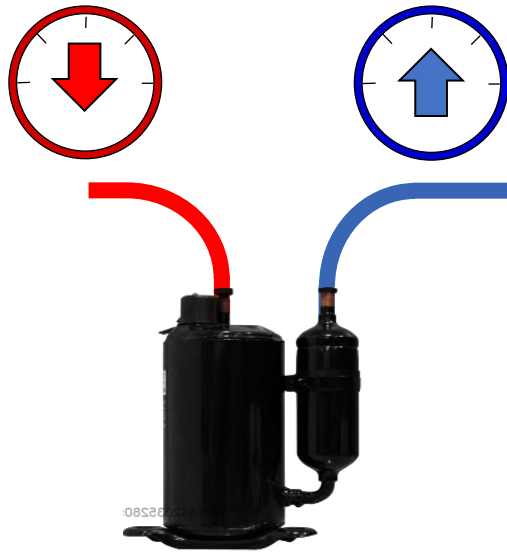


Normal Superheat



The Compressor Speed Vs Pressure Relationships

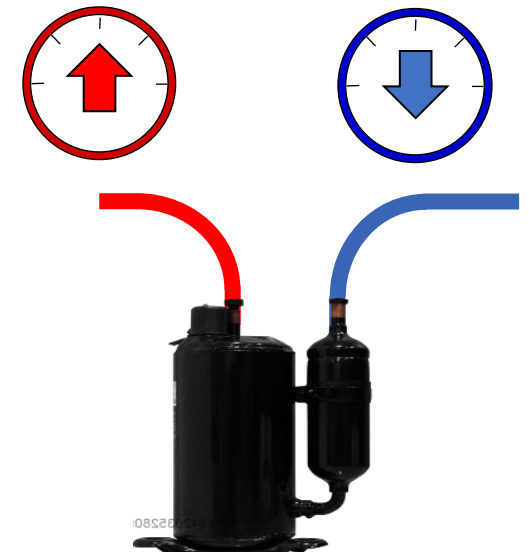
(With no other changes in the system)



Low Compression
Ratio



Stable Operation

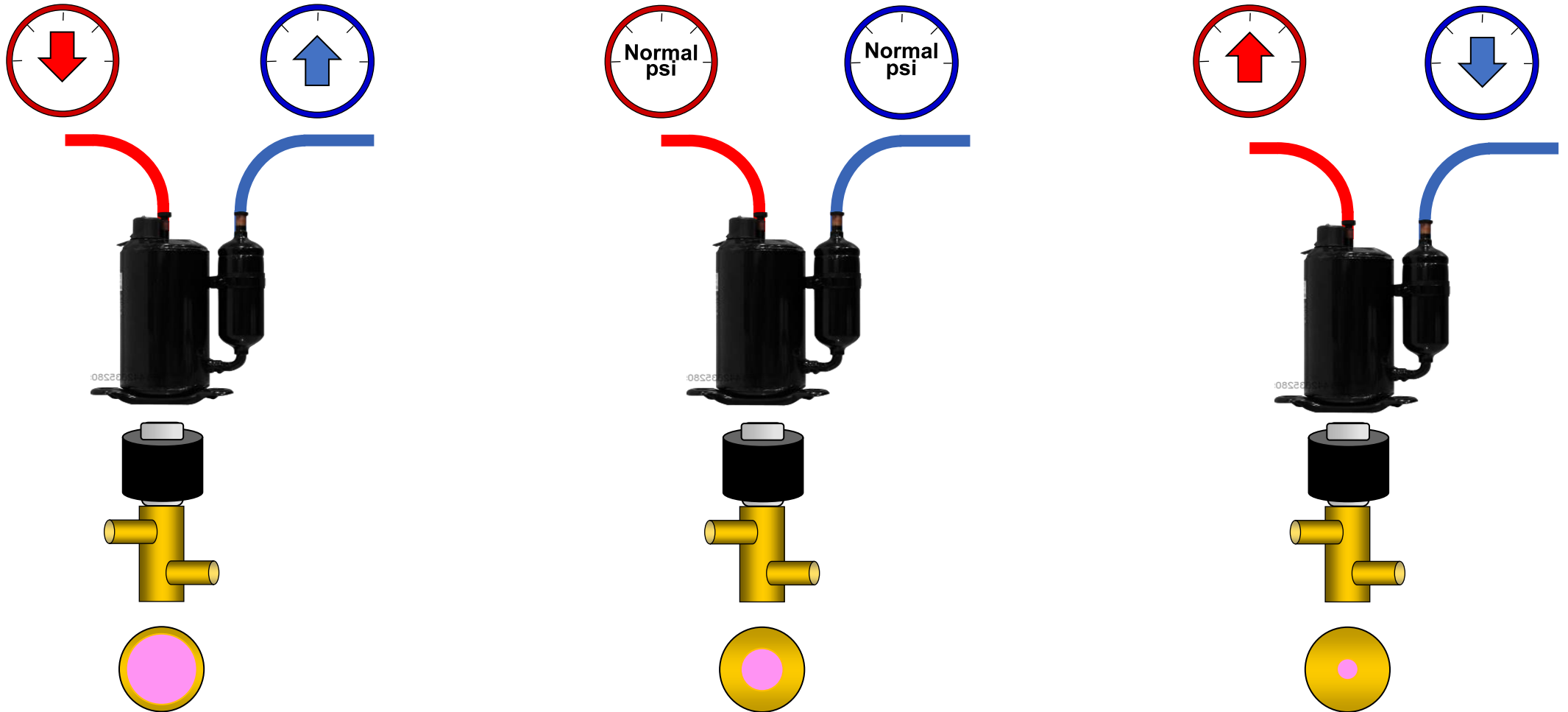


High Compression
Ratio



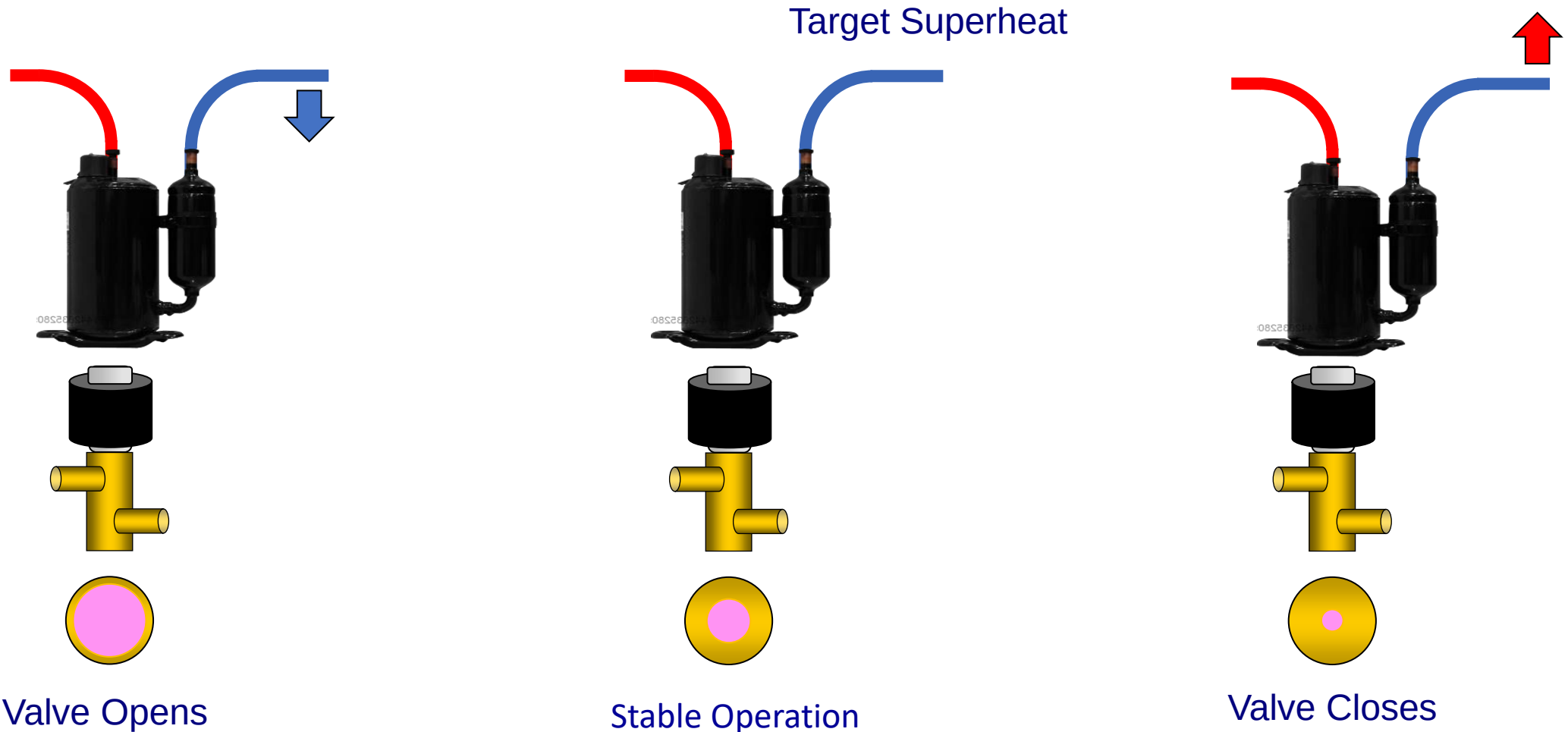
LEV Opening Vs Pressure Relationships

(With no other changes in the system)



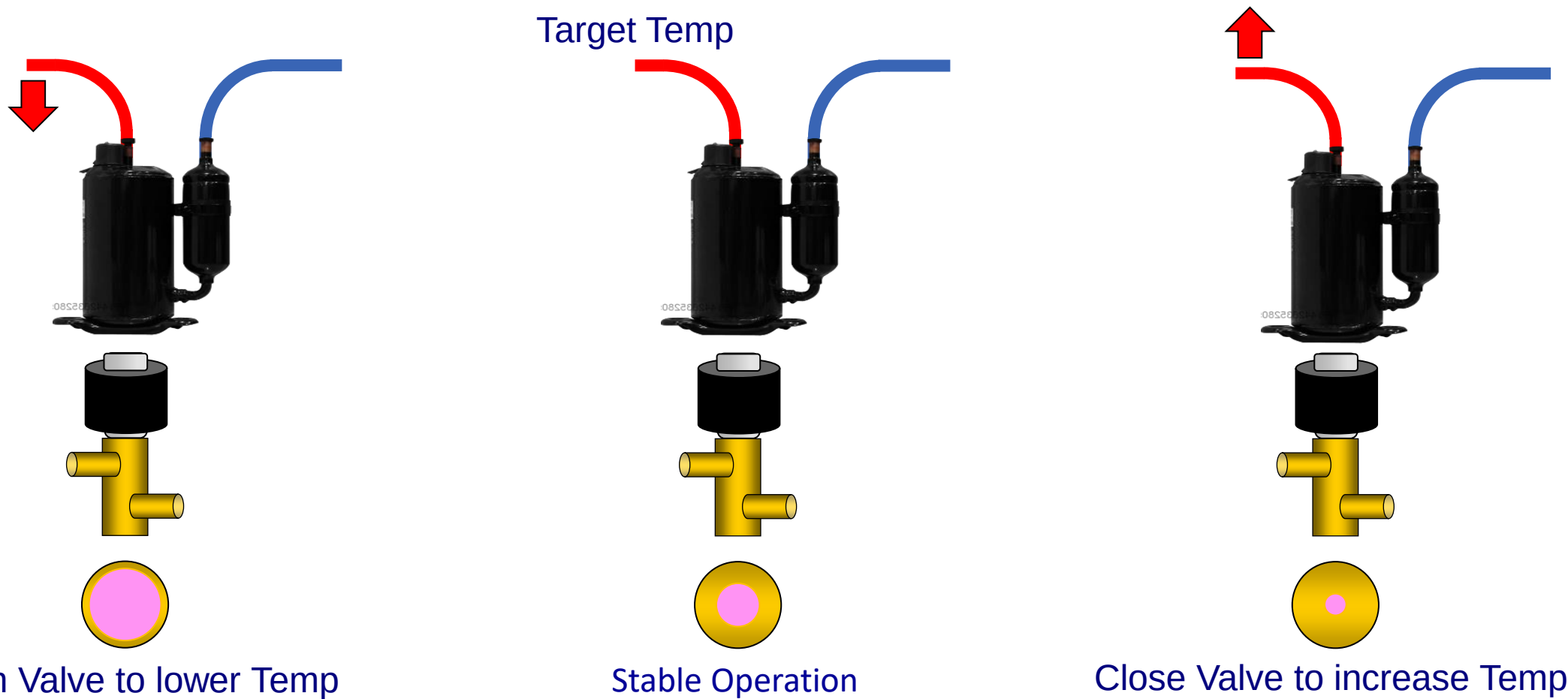
LEV Opening Vs Superheat

(With no other changes in the system)



LEV Opening Vs Discharge Temperature Relationships

(With no other changes in the system)



Discharge Temperature protections Vs Compressor Frequency

(With no other changes in the system)

Flood Back Condition



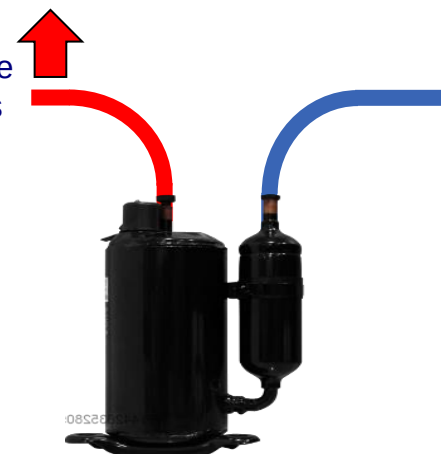
Compressor Slow Down

Target Temp



Stable Operation

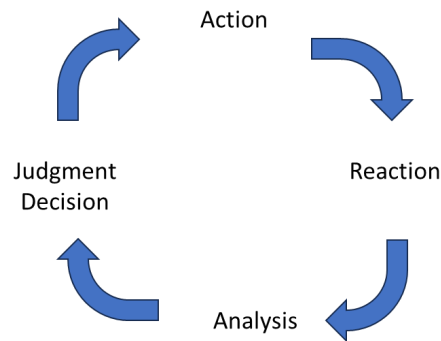
High Discharge Temperatures



Compressor Slows Down

Putting It All Together

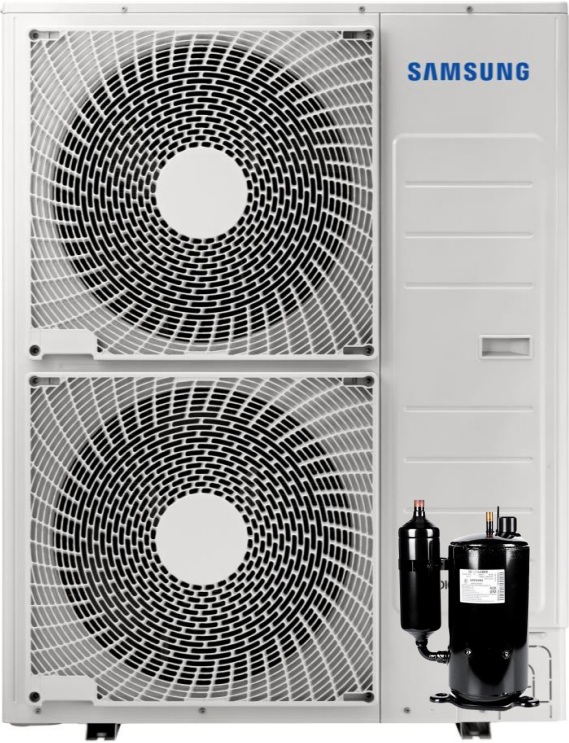
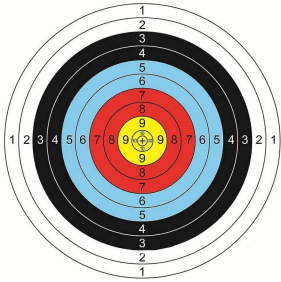
- Let's apply what we used to explain the system operation as a whole.
- We will discuss the control process and how the system respond changes within the system.
- We will review the cause and affect of component operation.



Compressor Targets

Example:

AC036BXUPCH/AA



Target Pressure		Saturation Temperature	
Stage2	Stage1	Stage2	Stage1
9	10	43.9	49.6
128	140		



Target Pressure		Saturation Temperature	
Stage2	Stage1	Stage2	Stage1
26	25	116.1	113.4
370	256		



Putting it all together

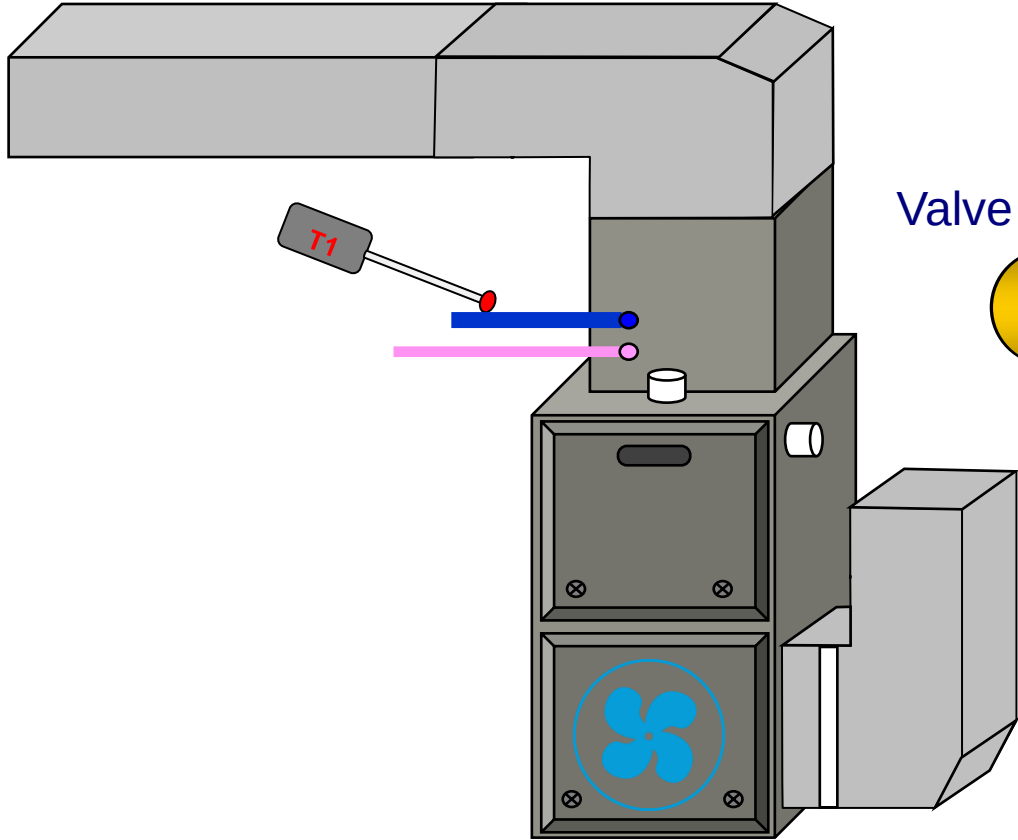
System Operation

Target Pressure	
Stage2	Stage1
9	10
128	140



Stage 1
Stage 2

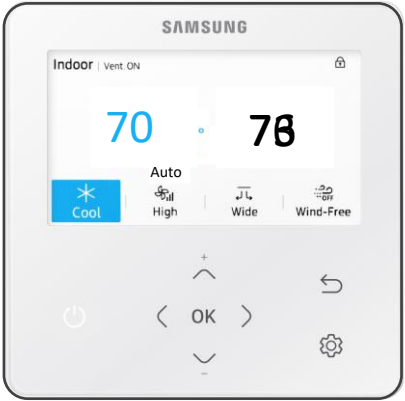
Hz.



Valve Opening

Saturation Temperature	
Stage2	Stage1
43.9	49.6

Stage 1 Fan Speed
Stage 2 Fan Speed

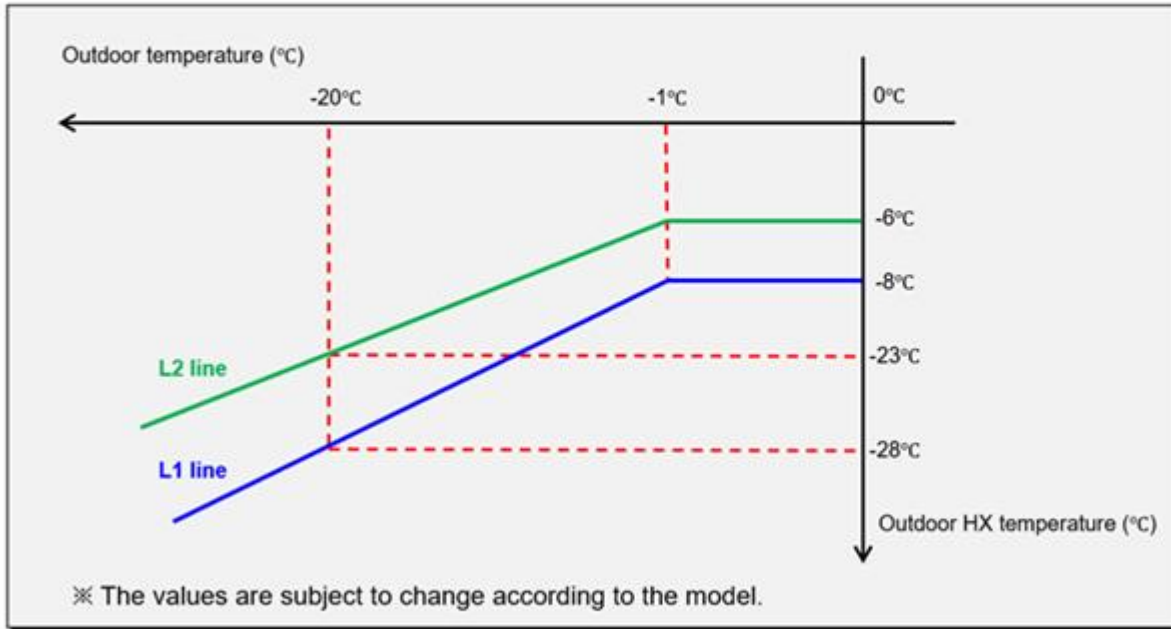


Integration Solutions

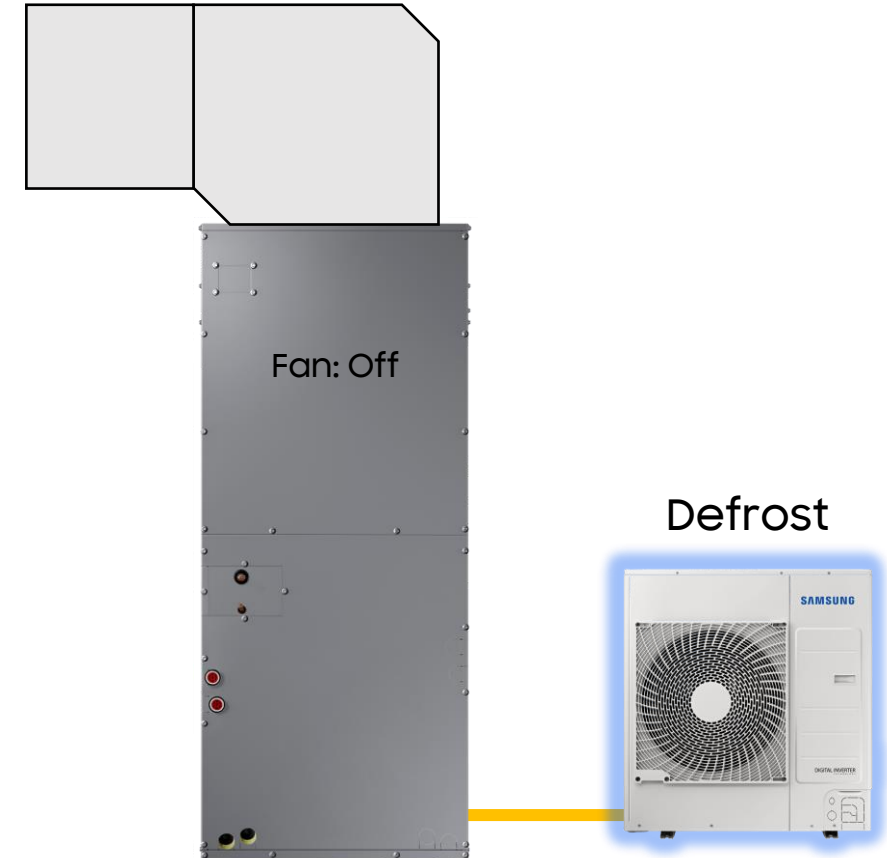
- Defrost Operation

- Entry conditions of defrost operation are as follows. (① or ② or ③)

- ① Outdoor HX temp. is under L2 line during 120min(cumulated)
- ② Comp. operation time is upper 35min & Outdoor HX temp. is under L1 line during 3min(continued)
- ③ Outdoor HX temp. is under -4°C during 360min(cumulated)



- If Outdoor HX temp. is reached 15°C , defrost operation is finished.
- Max defrost operation time = 12min

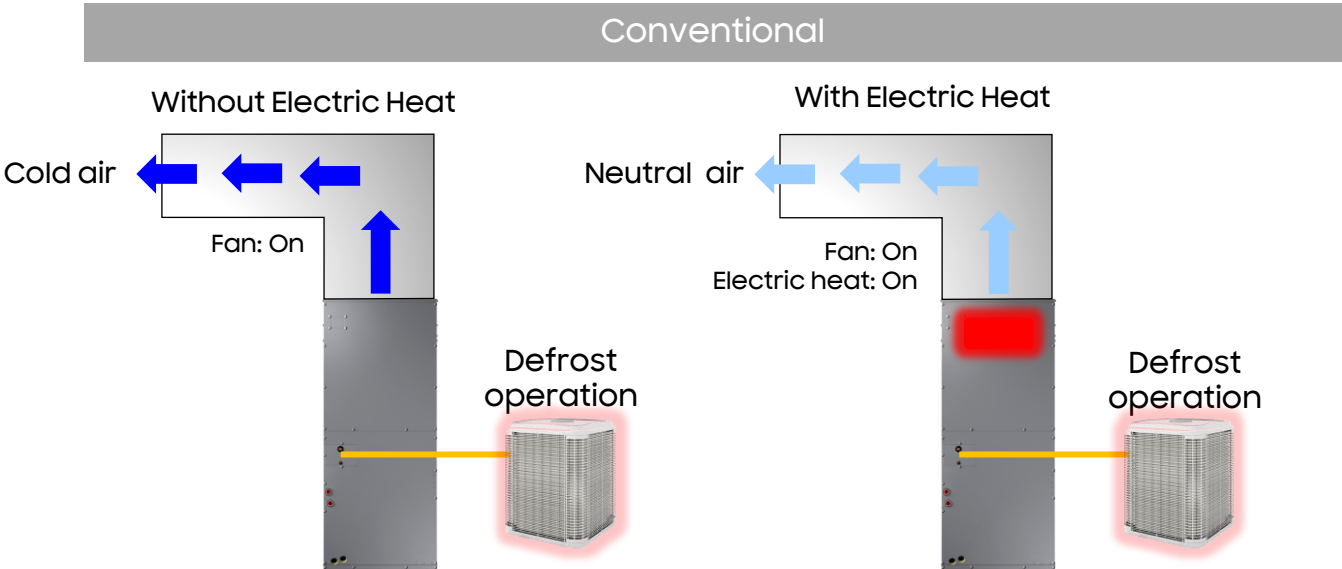


Defrost

- Cold Wind Prevention

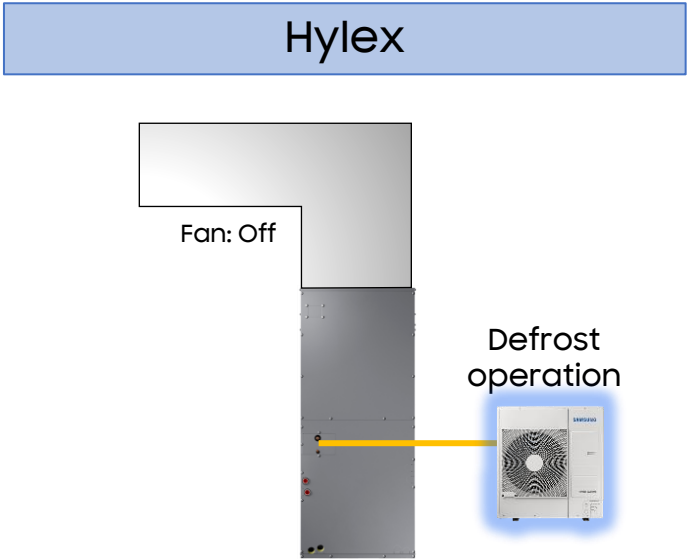
The ACU will stop the indoor fan:

- ✓ During defrost cycles
- ✓ When the coil is first warming up (cold air prevention)



ACU Option
Settings

SW No.	Function
SW 2-12	Fan stop control during defrost operation
On	Disable
Off	Enable(Default)



Defrost Enhancements

ODU Setting

- 11
 - Forced Defrost interval
- 12
 - Reduce Noise During Defrost
- 13
 - Defrost Termination offset.

Cumulative defrost hour	Main	1	1	0	0	unused	default
				0	1	90 Mins	
				0	2	60 Mins	
				0	3	40 Mins	
Defrost Hz Offset	Main	1	2	0	0	unused	default
				0	1	-5Hz	
				0	2	-10Hz	
				0	3	-15Hz	
Exiting defrost Offset	Main	1	3	0	0	unused	default
				0	1	+3°C (5.4°F)	
				0	2	+6°C (10.8°F)	
				0	3	+9°C (16.2°F)	

Service



- All service starts with verifying the application and installation
- Is there a call from the thermostat?
- Is that call being sent to the Air Handler/Furnace?
- If the compressor is running, is it hitting the desired pressure for the mode its running in?
- If not, why?
- Is the pressure above or below target?
- What can cause this condition?

ODU service modes.

- *Key button of the main board



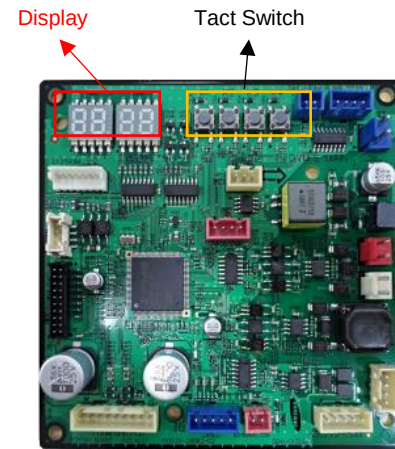
K1 button (Press)	Key function	7 segment display	Remark
1 time	Heating test mode	“K” “1” “Blank” “Blank”	System operates in heating mode [set 104°F (40°C)] and checks for any error.
2 times	Defrost test mode	“K” “3” “Blank” “Blank”	When outdoor temperature is under 50°F(10°C) for 1 minute after pressing K1 for defrost test mode.
3 times	End key operation	-	

K2 button (Press time)	Key function	7 segment display	Remark
1 time	Cooling test mode	“K” “2” “Blank” “Blank”	System operates in cooling mode [set 37.4°F (3°C)] and checks for any error.
2 times	Inverter check	“K” “4” “Blank” “Blank”	The inverter PBA check function by a universal inverter checker.
3 times	Pump down	“K” “6” “Blank” “Blank”	Cooling operation with a maximum 10 minutes operating time.
4 times	Reserved	-	-
5 times	On device inverter checker	“K” “A” “Blank” “Blank”	Refer to the next page
6 times	Auto test mode	“K” “C” “Blank” “Blank”	System determines test mode (cool or heat) based on ambient temperature -Test mode will continue to operate for 10 hours ; any key button press pressing will terminate the test mode. - If an error occurs during the test operation, the test mode will terminate and display the associated error codes
7 times	Auto check (Installation commissioning mode)	“K” “E” “Blank” “Blank”	System will determine test mode (cool or heat) based on ambient temperature - Test mode will continue to operate for 10~20mins and self-terminate test mode if no errors are found. Any key button press will terminate the test mode. - If an error occurs during the test operation, the test mode will terminate and display the associated error codes
8 times	End key operation	-	-

K3 button (Press)	Key function	7 segment display	Remark
1 time	Reset		

* The K4 button is the view mode. For more information, refer to the installation manual.

Gathering Information.



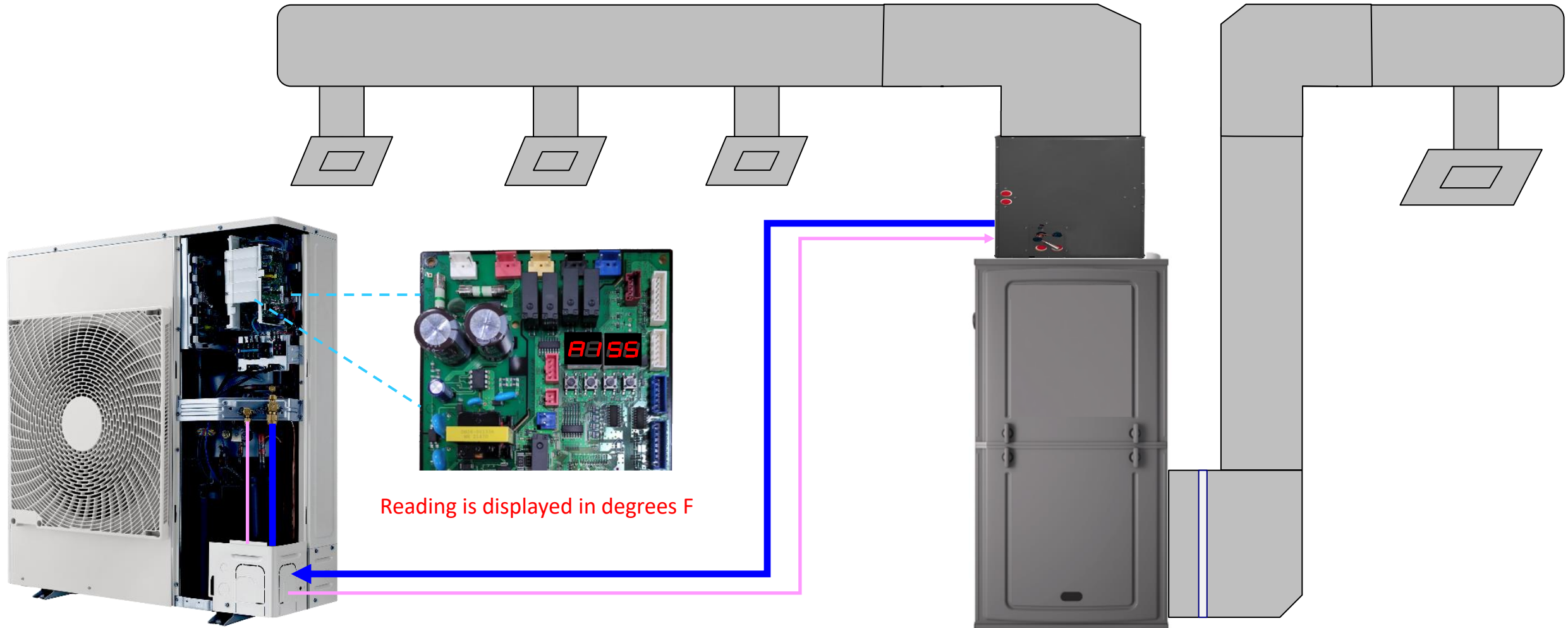
Service Information

- You will need to use the K 4 button to gather additional service information.
- You will also need to use your meters and gauges to verify operating data.
- The following slide review the most commonly used K 4 button options.

K4 short push	Display contents	SEG1	SEG2	SEG3	SEG4	Unit
1	Order frequency	1	Hundreds digit	Tens digit	Units digit	Hz
2	Current frequency	2	Hundreds digit	Tens digit	Units digit	Hz
3	The number of preset indoor units	3	Hundreds digit	Tens digit	Units digit	EA
4	Ambient temperature sensor	4	Hundreds digit or - ¹⁾	Tens digit	Units digit	°F or °C ²⁾
5	Compressor discharge sensor	5	Hundreds digit	Tens digit	Units digit	°F or °C ²⁾
6	Eva-Mid sensor	6	Hundreds digit or - ¹⁾	Tens digit	Units digit	°F or °C ²⁾
7	Condenser sensor	7	Hundreds digit or - ¹⁾	Tens digit	Units digit	°F or °C ²⁾
8	Current	8	Tens digit	Units digit	The first place of decimals	A
9	Outdoor fan RPM	9	Thousands digit	Hundreds digit	Tens digit	rpm
10	Target discharge temperature	A	Hundreds digit or - ¹⁾	Tens digit	Units digit	°F or °C ²⁾
11	EEV	B	Hundreds digit	Tens digit	Units digit	step
12	The capacity sum of indoor units	C	Tens digit	Unit digit	The first place of decimals	kBtu/h or kW ³⁾
13	Protective control	D	0: Cooling 1: Heating	<Protective control> 0: No Protective control 1: Freezing 2: Non-stop defrosting 3: Over-load 4: Discharge 5: Total electric current	<Frequency status> 0: Normal 1: Hold 2: Down 3: Up_limit 4: Down_limit	-
14	IPM temperature	E	Hundreds digit or - ¹⁾	Tens digit	Units digit	°F or °C ²⁾
15	The number of connected indoor units	F	0	Tens digit	Units digit	EA
16	ESC EEV(CAM)	G	Hundreds digit	Tens digit	Units digit	step
17	ESC IN sensor	H	Hundreds digit or - ¹⁾	Tens digit	Units digit	°F or °C ²⁾
18	ESC OUT sensor	I	Hundreds digit or - ¹⁾	Tens digit	Units digit	°F or °C ²⁾
19	Suction sensor	J	+ / - Hundreds for Fahrenheit	Tens digit	Units digit	°F or °C ²⁾
20	Pressure sensor	K	+ / - Hundreds for Fahrenheit	Tens digit	Units digit	psi or kg/cm ² ⁴⁾

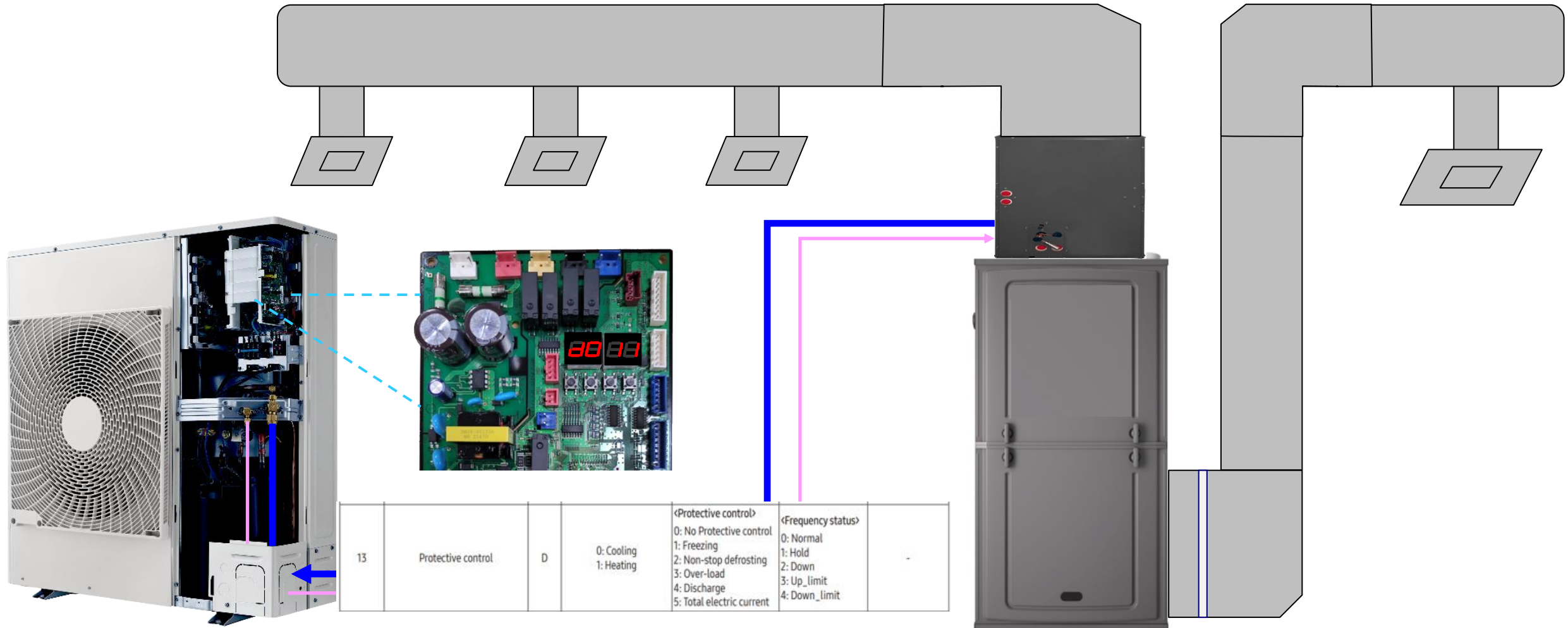
K4 10 times ^(A) Target Discharge Temp.

88.88



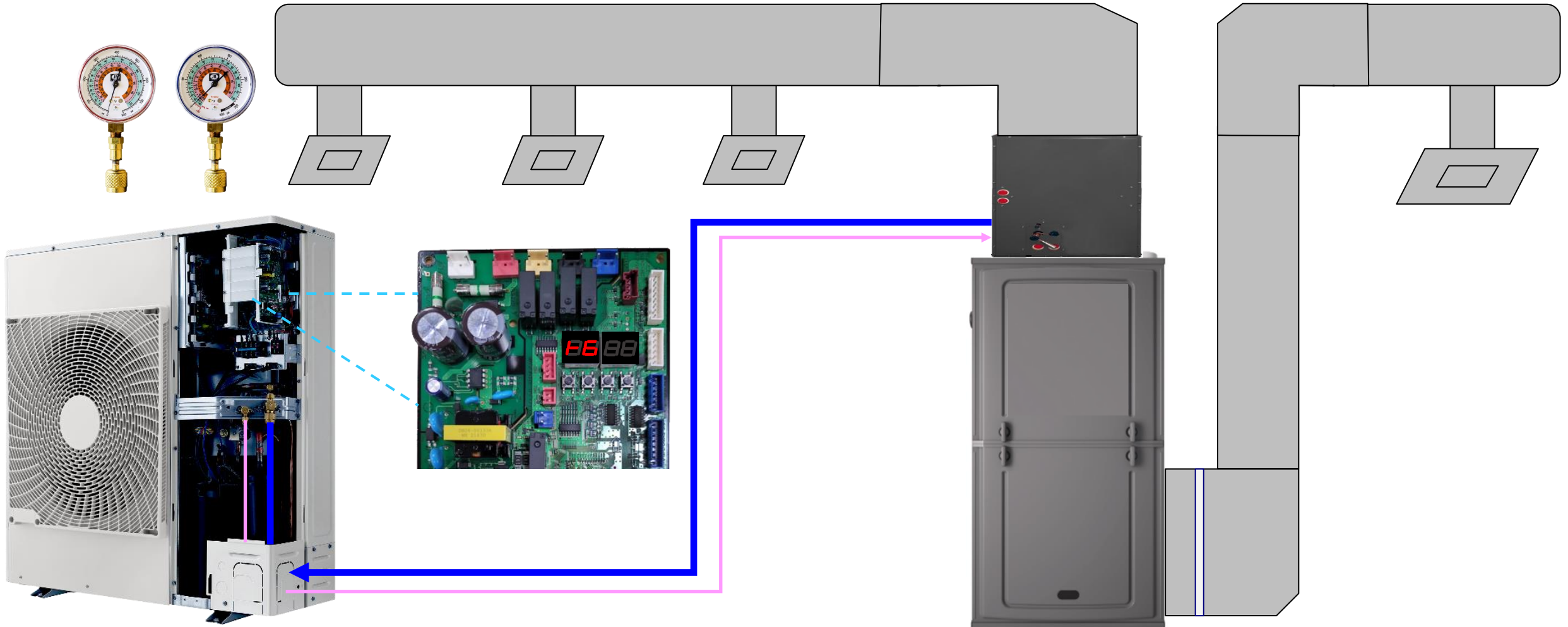
K4 13 times ^(D) Protection Mode

88 88



K4 20 times (K) Pressure

88 88



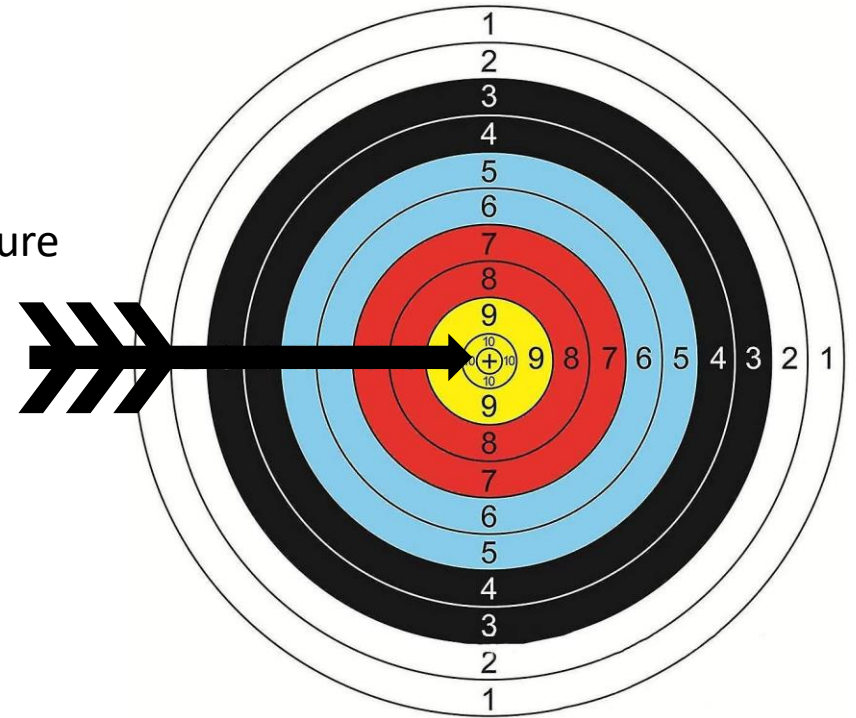
Refrigerant trouble shooting

- Troubleshooting refrigerant issues with variable speed equipment is no different than troubleshooting single stage systems.
- The difference is that the components within the variable speed system will adjust or modulate based on the issue.
- For example, if a system has low air flow in the cooling mode. This causes low suction pressures.
- On a single stage system, the compressor speed does not respond to this drop in pressure.
- With Variable speed equipment they see this drop in pressure and commands the compressor to slow down to maintain the target suction pressure.
- Both systems experience low suction pressure, low superheat and reduced capacity.

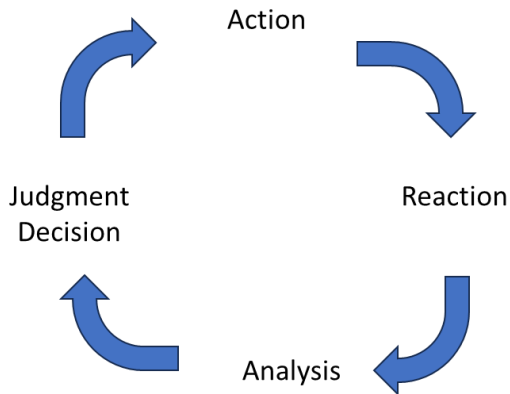
Refrigerant Troubleshooting

- Is the system operating and maintaining its targets.
- If not is the actual value above or below the target?
- Why?
- How will the system adjust to this value?

System Targets:
Discharge Temperature
Suction Pressure
Discharge Pressure
IDU Superheat
ODU Superheat
Sub-Cooling



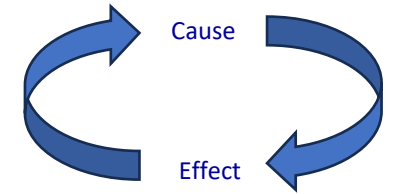
Refrigerant Trouble Shooting Variable Speed Systems is the same as unitary except the system adjust accordingly to the system deficiencies.



Air Conditioning Troubleshooting Chart				
System Problem	INDICATORS			
	Discharge Pressure	Suction Pressure	Superheat	Sub-Cooling
Over Charge	High	High	Low	High
Undercharge	Low	Low	High	Low
Liquid Line Restriction (in outdoor)	High	Low	High	High
Liquid Line Restriction (at indoor)	Low	Low	High	High
Low Evaporator Air Flow	Low	Low	Low	Low
Dirty Condenser	High	High	Low	Low
Low Outdoor Temperature	Low	Low	High	High
Inefficient Compressor	Low	High	High	High
Metering Device Not Opening	Low	Low	High	High
Un-insulated Sensing Bulb	High	High	Low	Low
Low Load	Low	Low	Low	Low

Under Charge

- In the cooling mode the suction pressure will drop due to a shortage of gas in the system.
- Head pressure will be low
- The compressor will slow down to try to maintain target suction pressure.
- Evaporator super heat will be high
- The discharge temperature will be high
- There will be a reduced amount of Sub-Cooling.
- There will be a lack of cooling performance



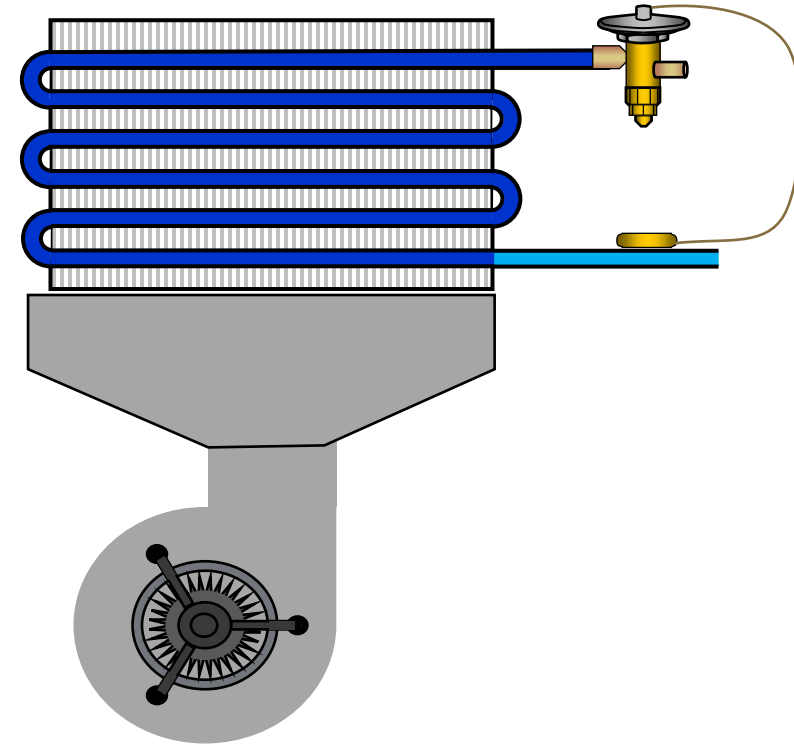
Over Charge

- In the heating mode the head pressure will be high due to an excess refrigerant in the system.
- The compressor will achieve target head pressure at lower than need Hz.
- The discharge temperature will be low
- The ODU superheat will be lower - normal and the EEV's will be driven closed to its lower limit.
- There will be a large amount of Sub-Cooling.
- There will be a lack of heating performance because of the reduced compressor speed.

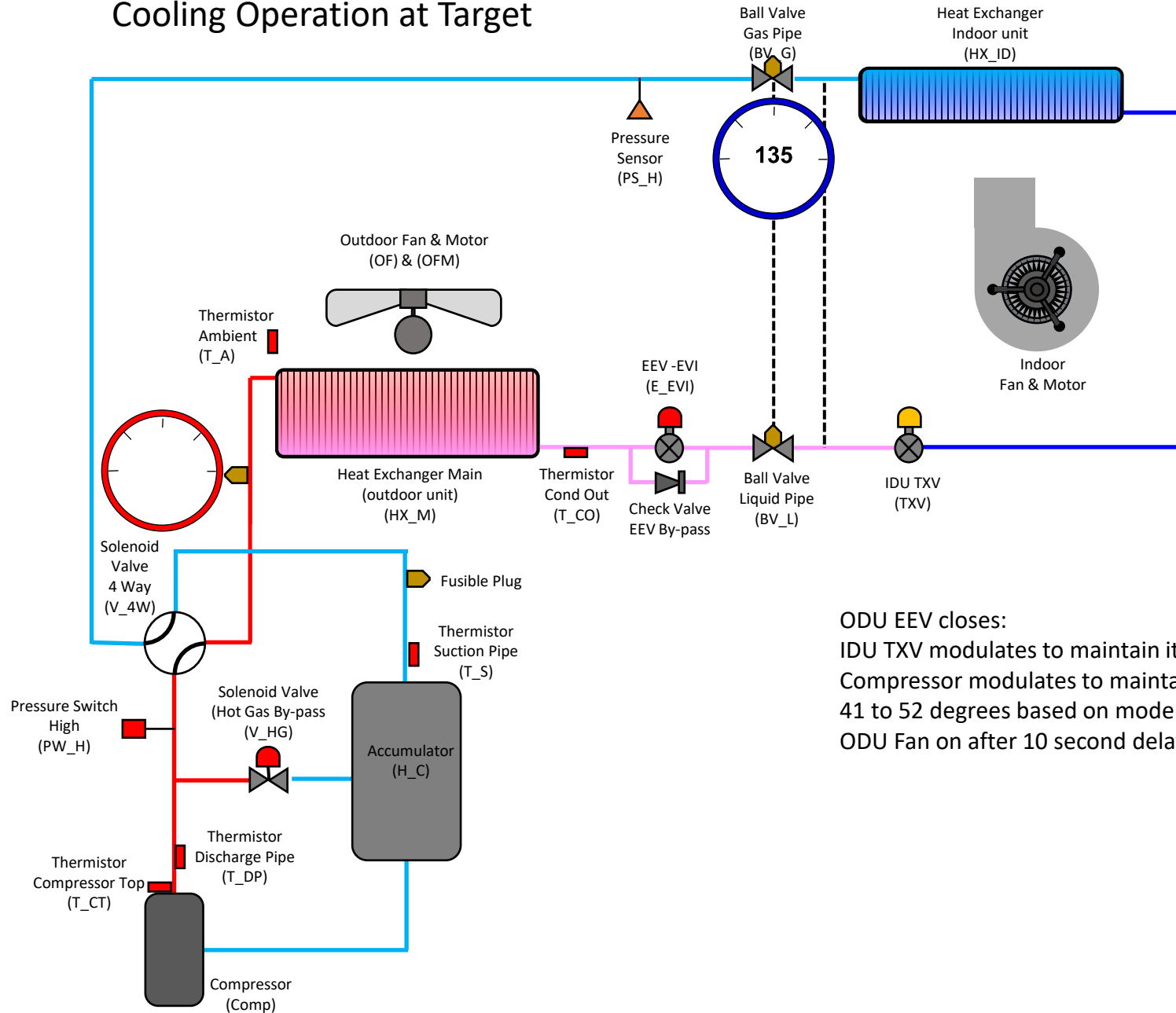


Low Air Flow

- In the cooling mode the suction pressure will drop due to a low air flow
- Head pressure will also be low
- The compressor will slow down to try to maintain target suction pressure.
- Evaporator super heat normal or slightly lower.
- The discharge temperature will be low
- There will be an increase in the amount of Sub-Cooling.
- There will be a lack of cooling performance



Cooling Operation at Target



ODU EEV closes:

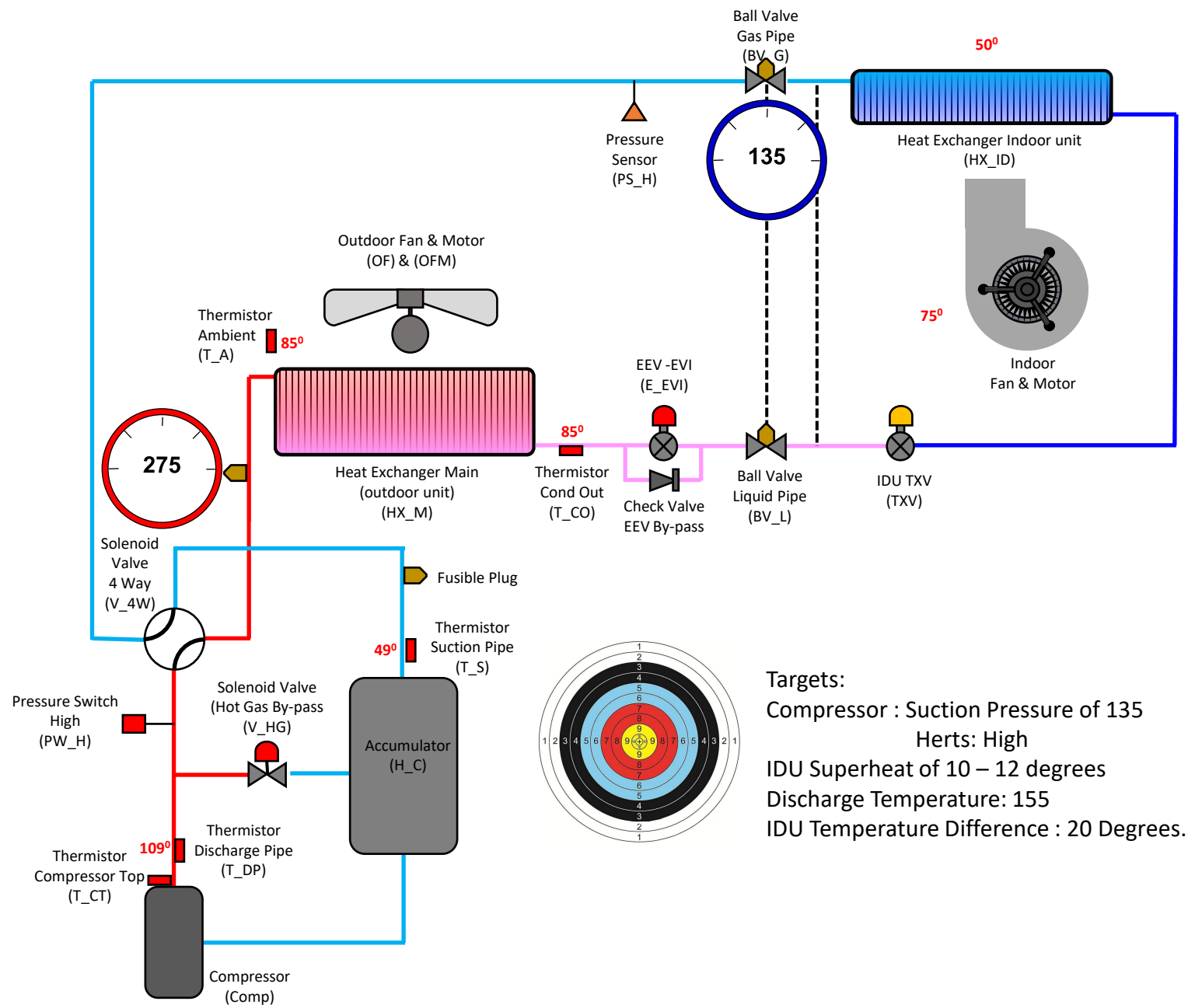
IDU TXV modulates to maintain it's design superheat.

Compressor modulates to maintain set low side pressure in the range of 41 to 52 degrees based on model.

ODU Fan on after 10 second delay.

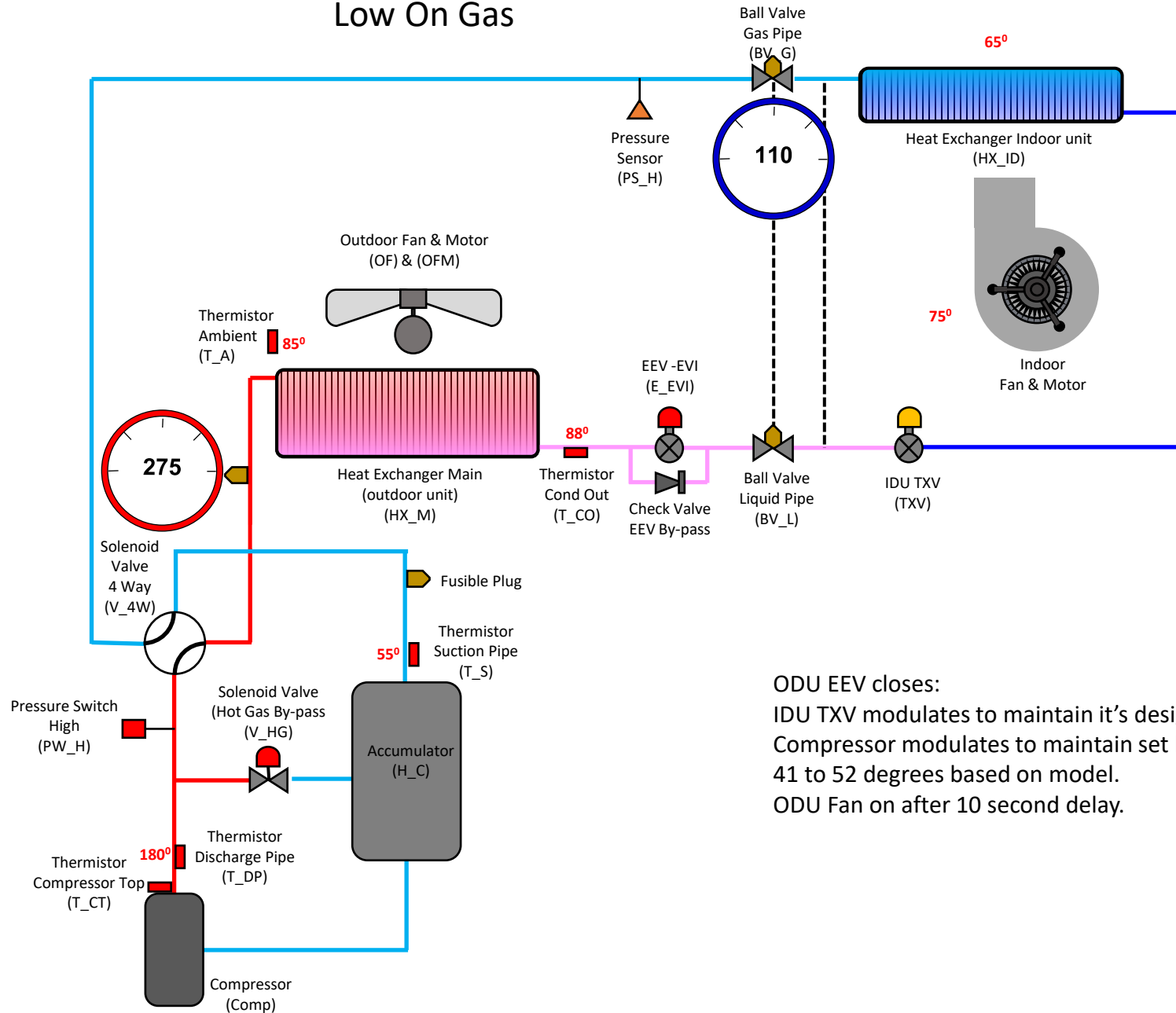


Troubleshooting Exercise



- Targets:
- Compressor : Suction Pressure of 135
 - Herts: High
 - IDU Superheat of 10 – 12 degrees
 - Discharge Temperature: 155
 - IDU Temperature Difference : 20 Degrees.

Low On Gas



ODU EEV closes:

IDU TXV modulates to maintain it's design superheat.

Compressor modulates to maintain set low side pressure in the range of 41 to 52 degrees based on model.

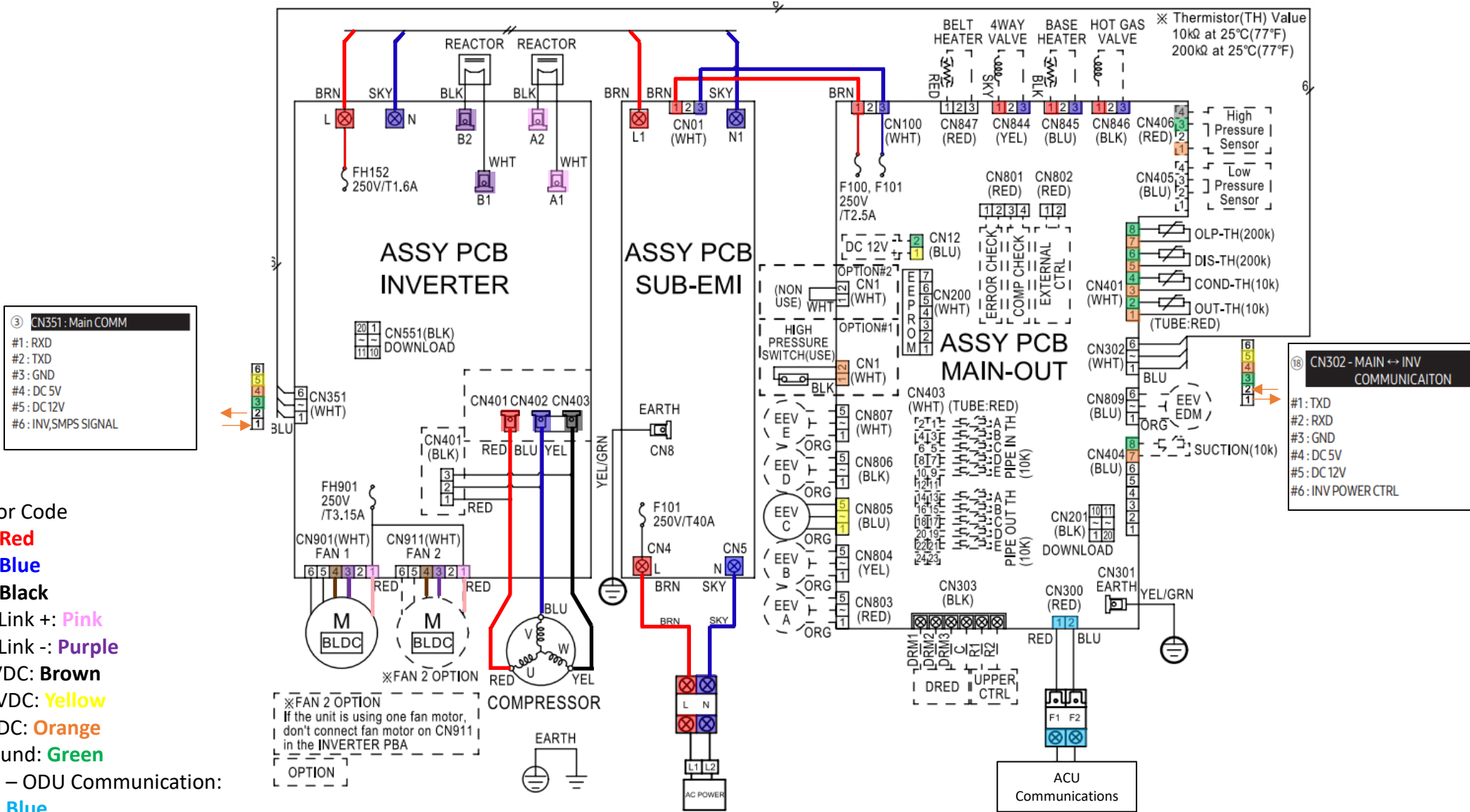
ODU Fan on after 10 second delay.



Electrical Troubleshooting

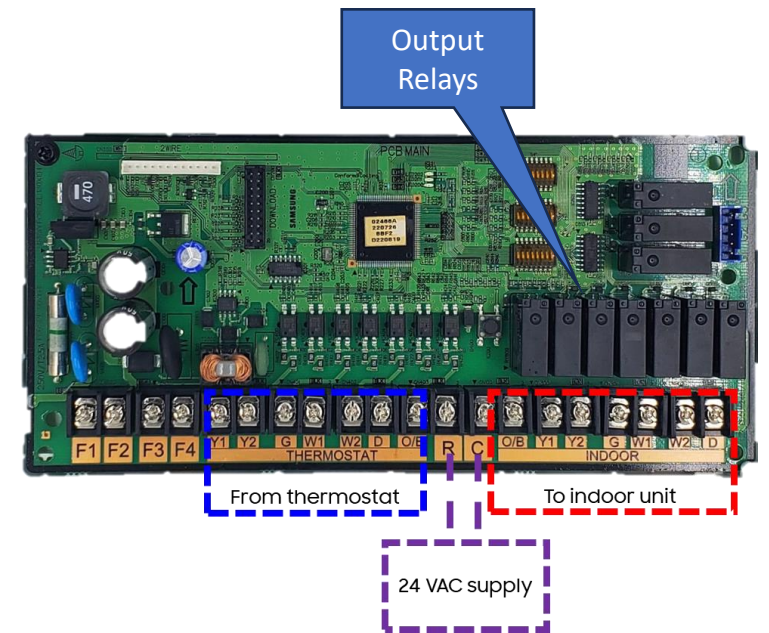
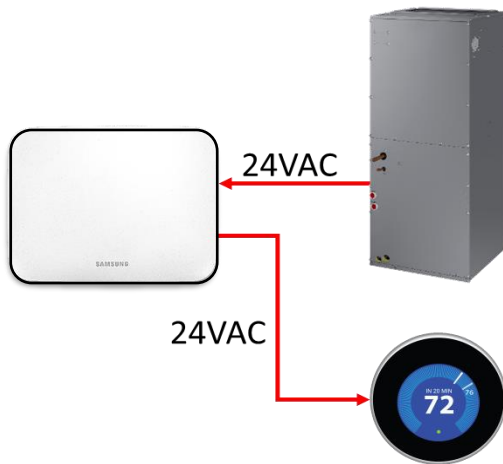
- This section will focus on the ODU and ACU electrical systems.
- Understanding the power flow through the ODU is key in effectively troubleshooting ODU Component.
- Troubleshooting individual components are covered in other Samsung Trainings.
- Properly using an electrical meter will make troubleshooting the ODU and ACU

ODU Power Flow

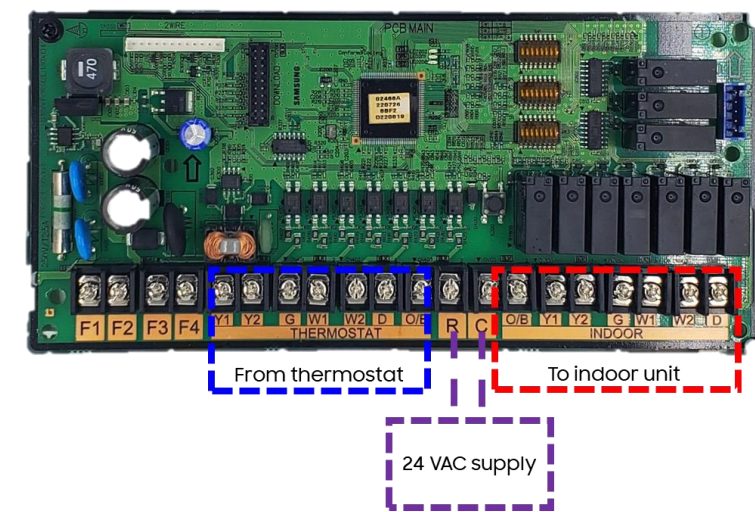
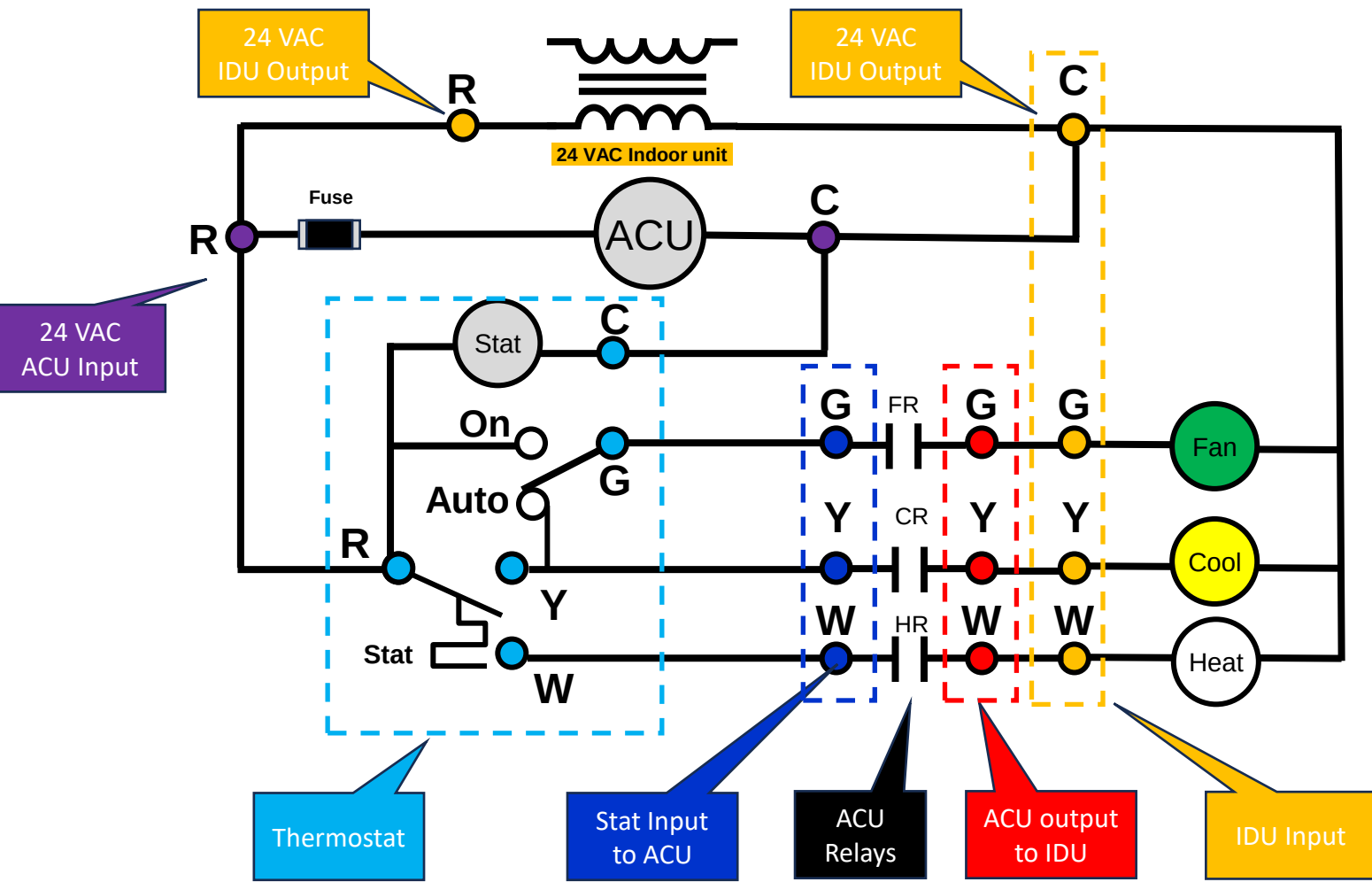


Indoor Control Power Flow Review

- 24 VAC is supplied to the ACU from the IDU
- 24 VAC for the thermostat is supplied from the ACU
- The ACU monitors the inputs from the thermostat and..
- Controls the output from the ACU to the IDU using relays on the ACU

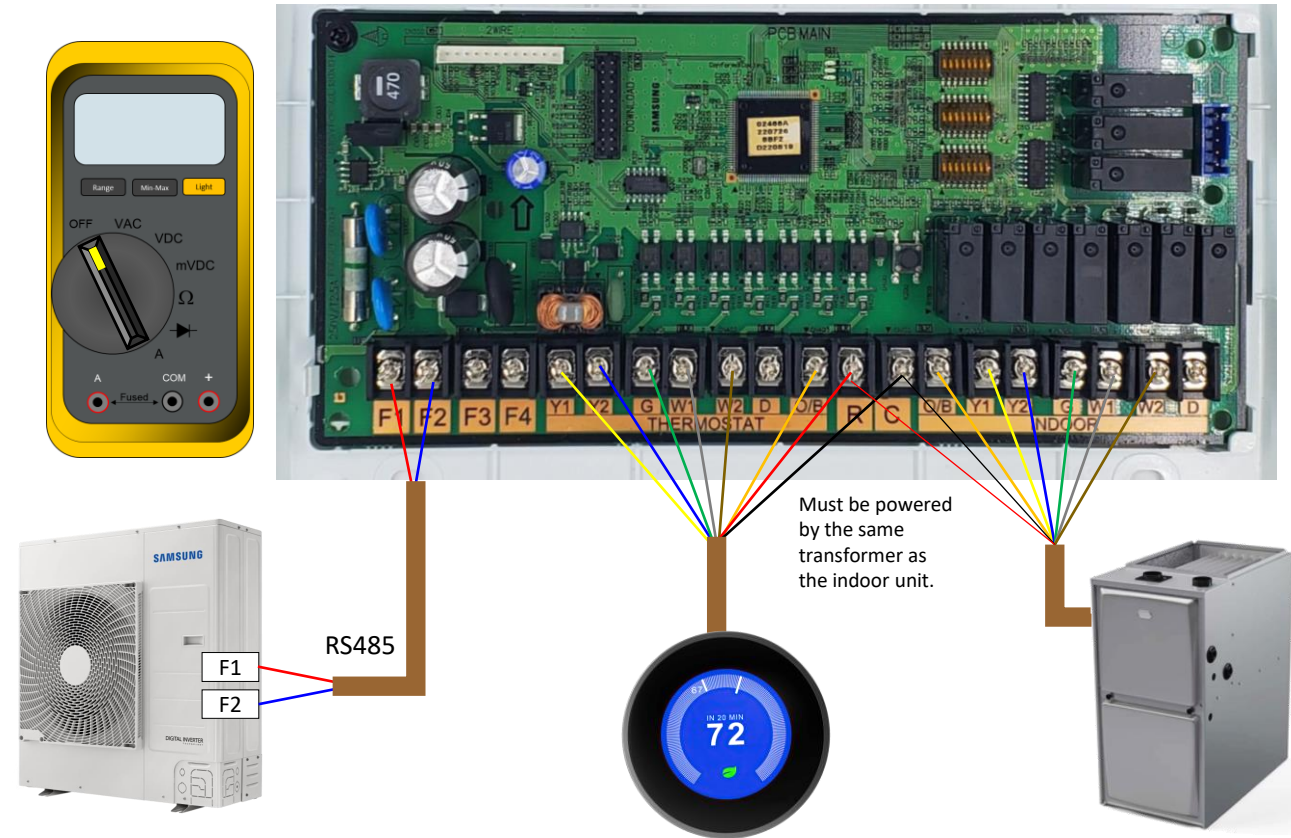


Simplified 24 VAC Power Flow



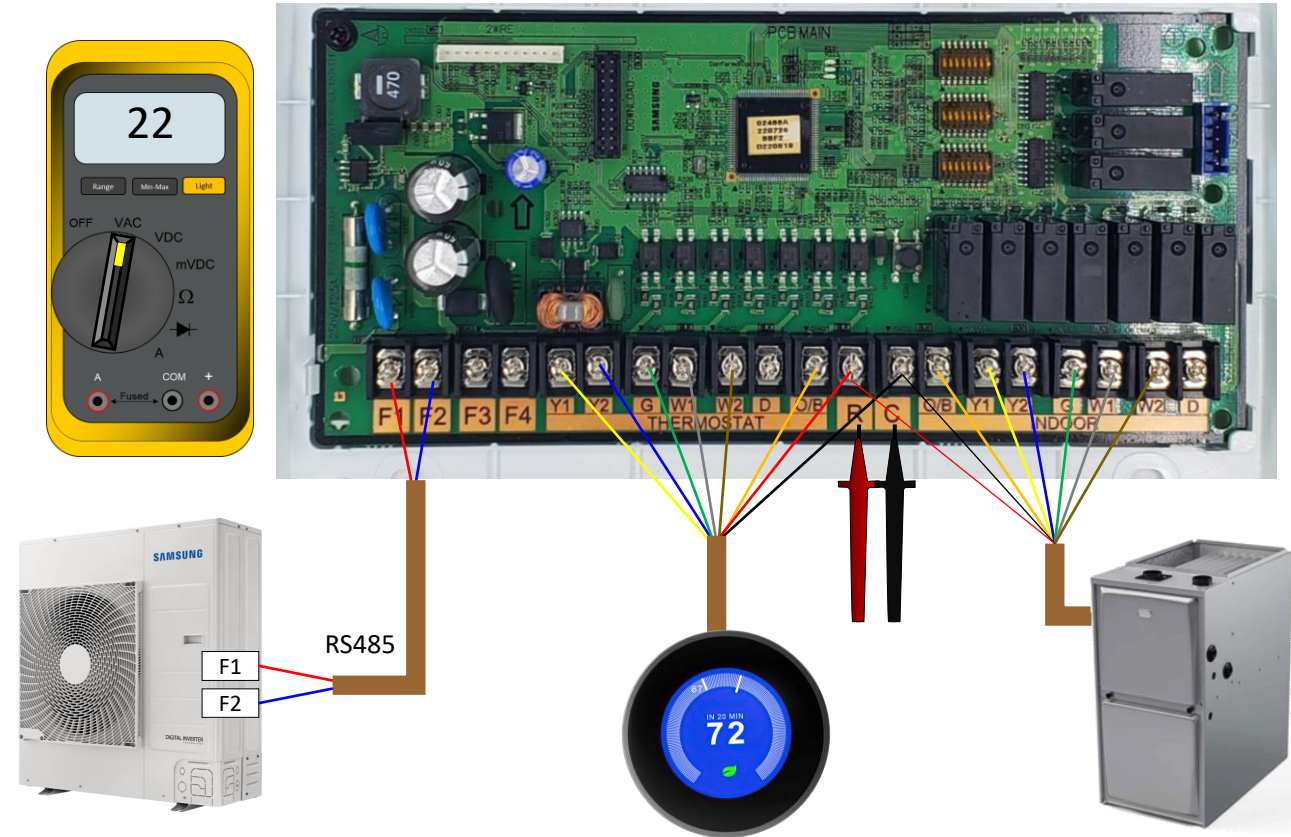
How to verify a call from the thermostat

- There are many methods that contractors use to test a call for heating and cooling.
- The most accurate method is to measure the power between R & C to verify your source voltage.
- Then measure voltage between the terminal(Y,1, Y2, G W1, W2 ...)and common.
- If you have the source voltage, then there is a call and there are no issue with the controller or wiring.
- If the measured voltage is different from source voltage this can indicate an issue within the system that needs to be checked further.
- IF the voltage is 0 there is no call



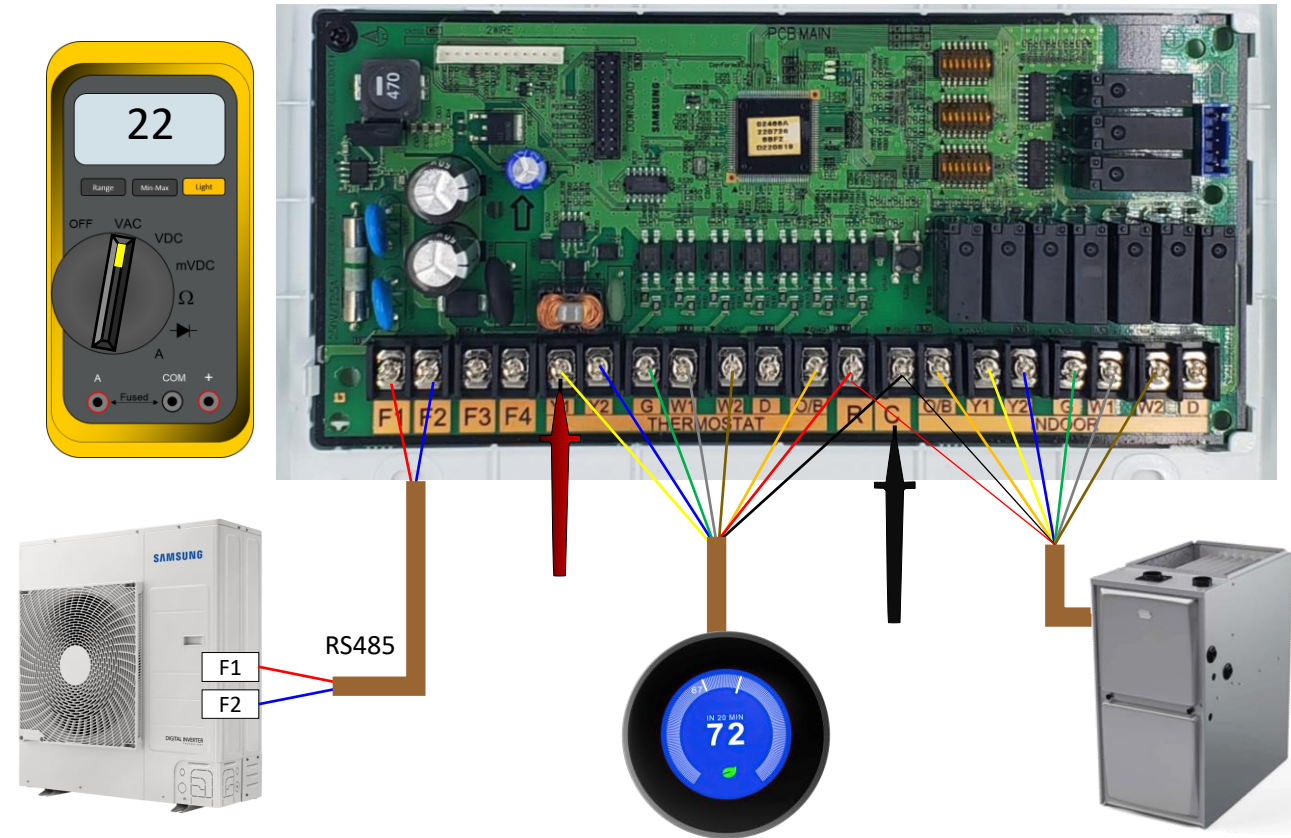
Measuring Source Voltage

- Set you meter to measure AC Volts
- Place your black lead on the C (common) terminal and the Red lead on the R terminal and record the voltage .
- If the voltage is low measure the voltage at the indoor unit.
- If there is a difference between the voltage measured at the indoor unit and ACU there is a problem that needs further investigating.



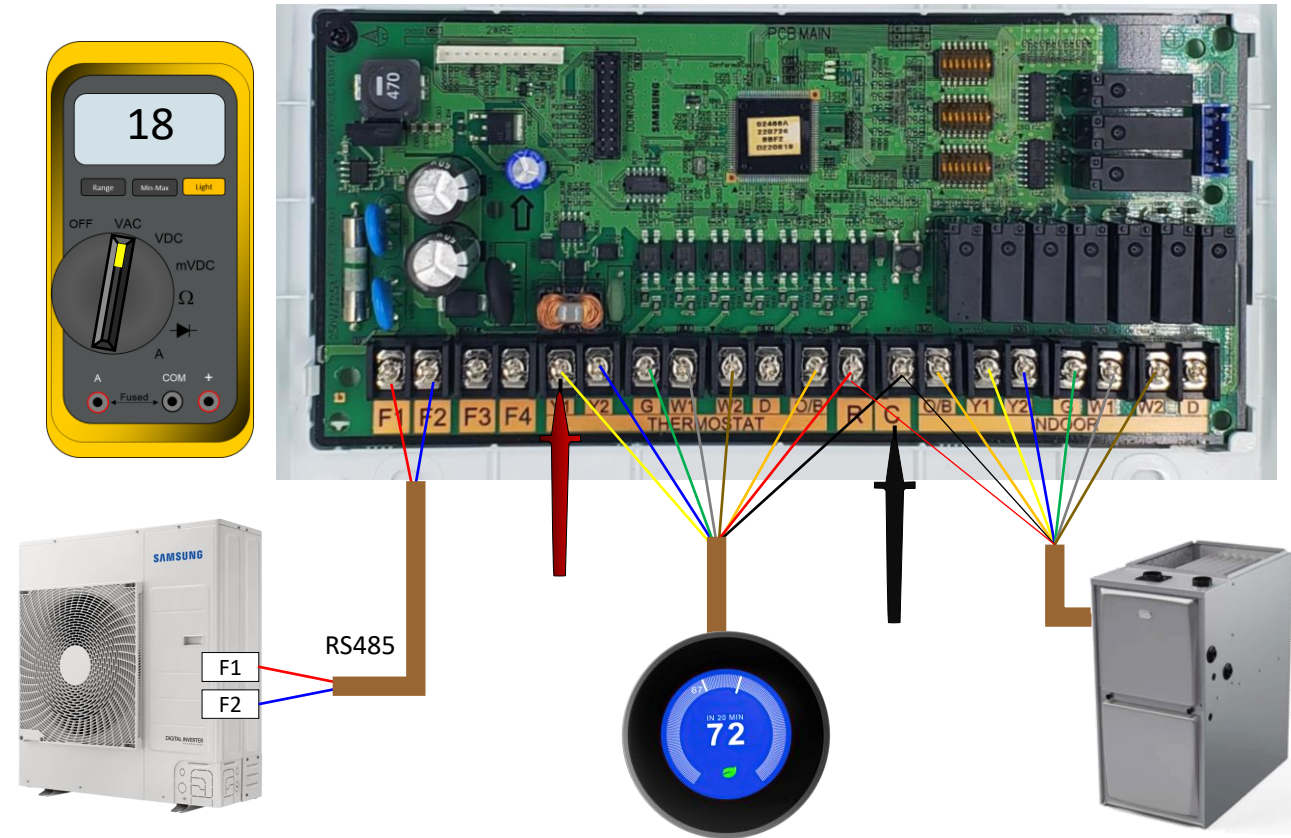
Testing a Call for Y1

- Set you meter to measure AC Volts
- Leave the black lead on C
- Place the read lead on Y1.
- If your measured voltage equals source voltage you have a call.
- If it is 0 then there is no call.



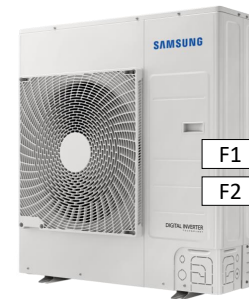
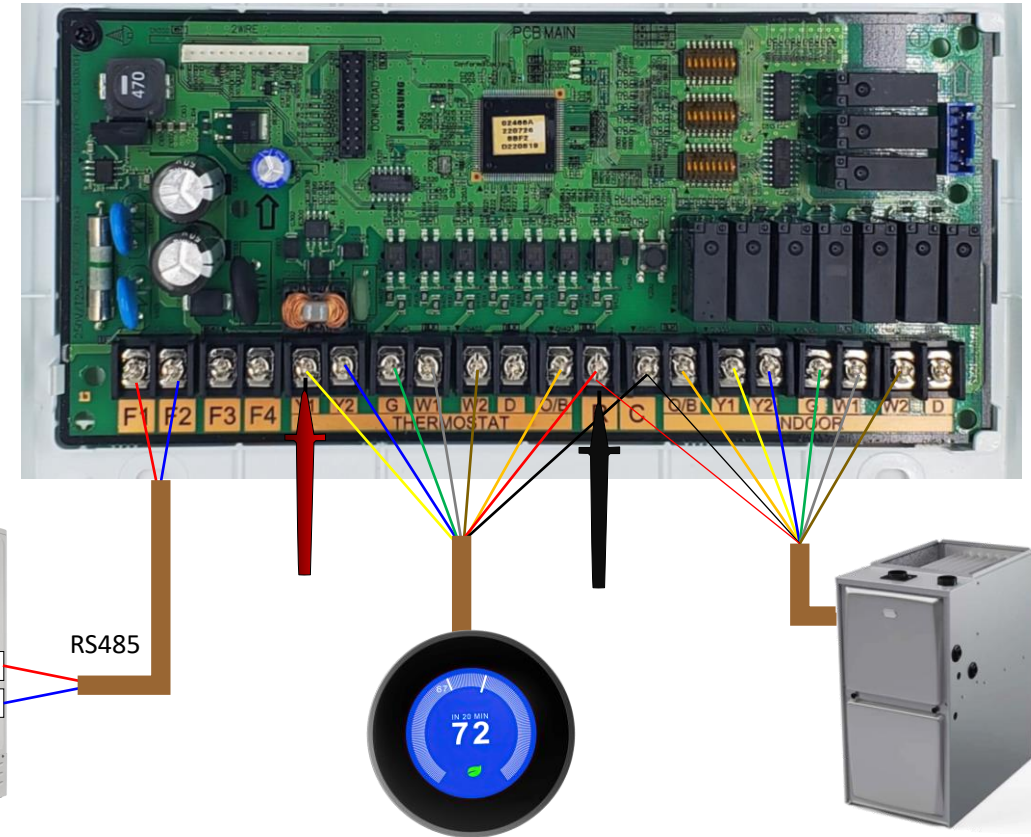
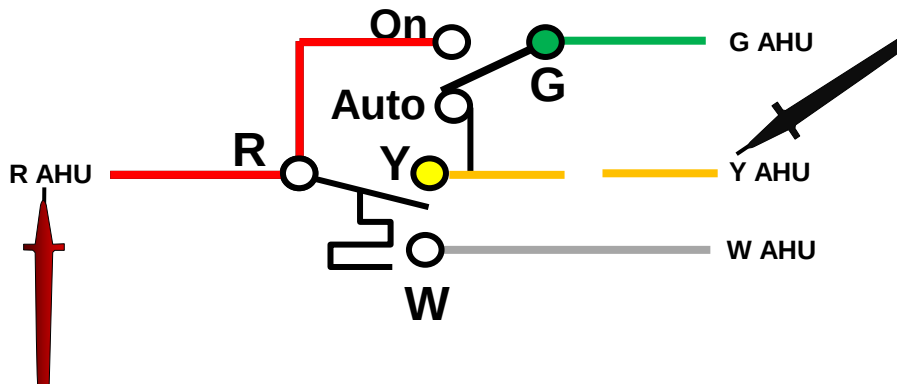
Testing a Call for Y1

- If your measured voltage differs from source voltage this indicates there is an issue with the thermostat or wiring.
- Further investigation is required to determine why.



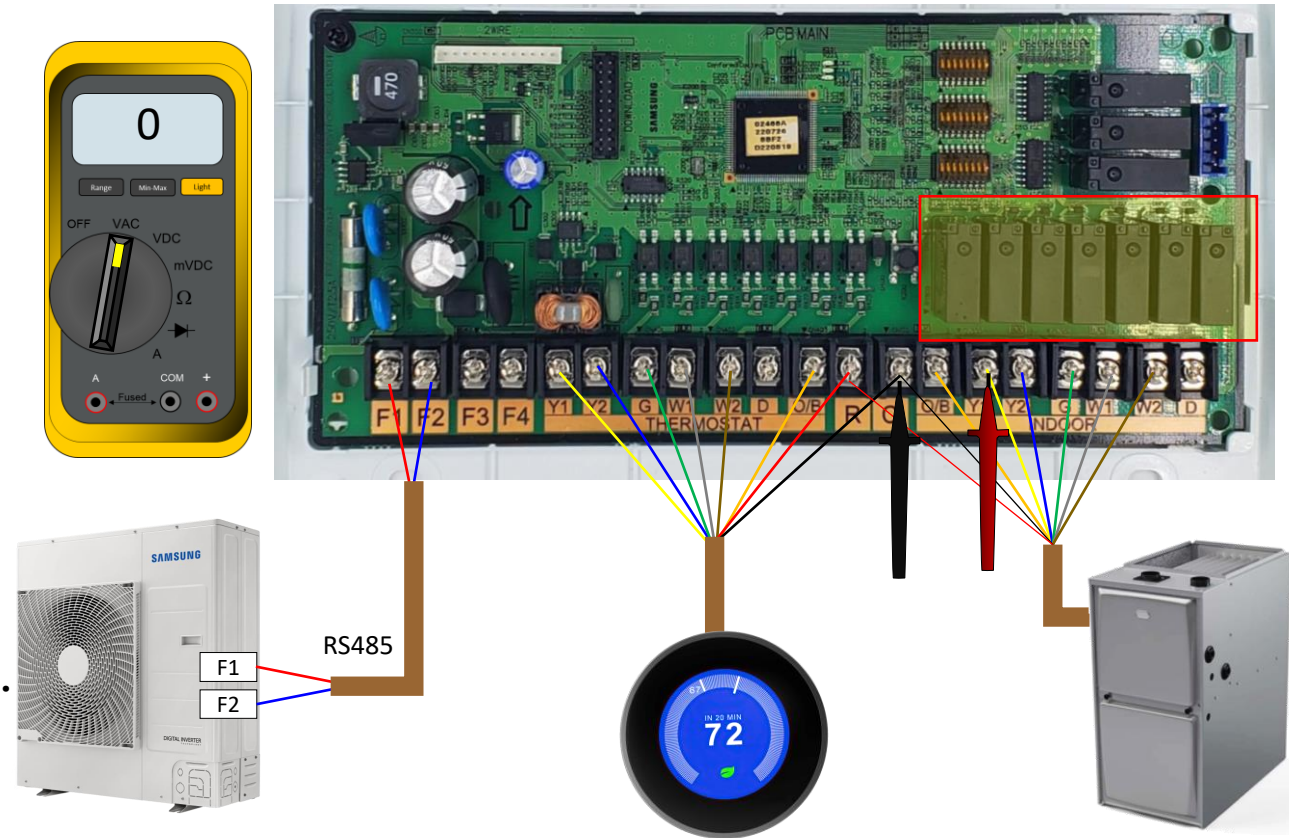
Testing a Call for Y1

- What does this reading tell you?
- It can be a misleading or inconclusive reading.
- Its always better to look for voltage rather than no voltage.



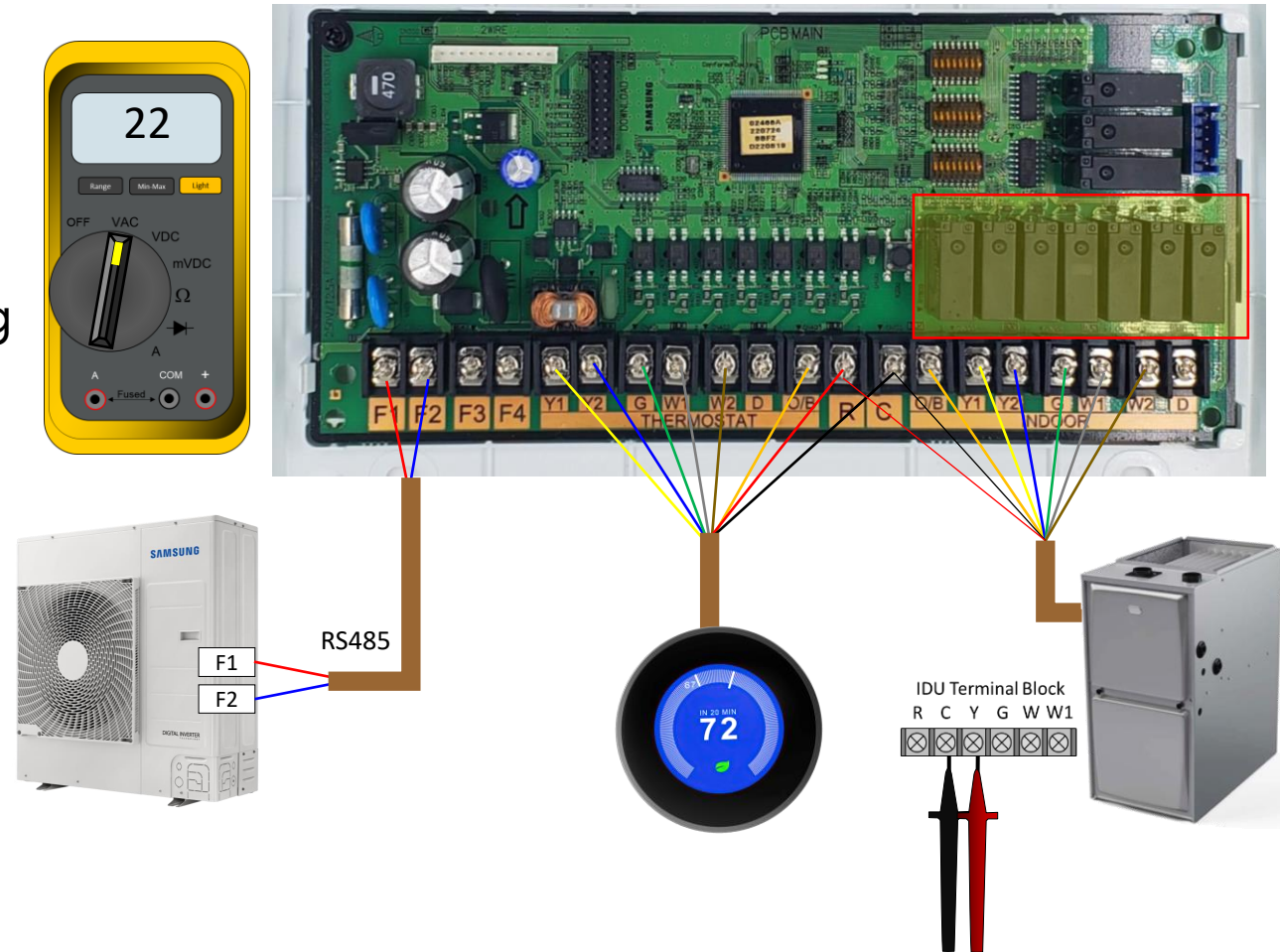
Testing a Call for Y1 to the Indoor unit

- Once you verified a call from the thermostat, the next step is to test the ACU output to the indoor unit?
- If you measure 0 volts there is no call for Y1 from the ACU to the IDU
- The signal from the thermostat to the IDU is controlled by a relay in the ACU.
- The dip switch settings will determine when the relay contacts are opened and closed.
- Before condemning the ACU check the system configuration settings to determine if the relay is supposed to be open or closed.



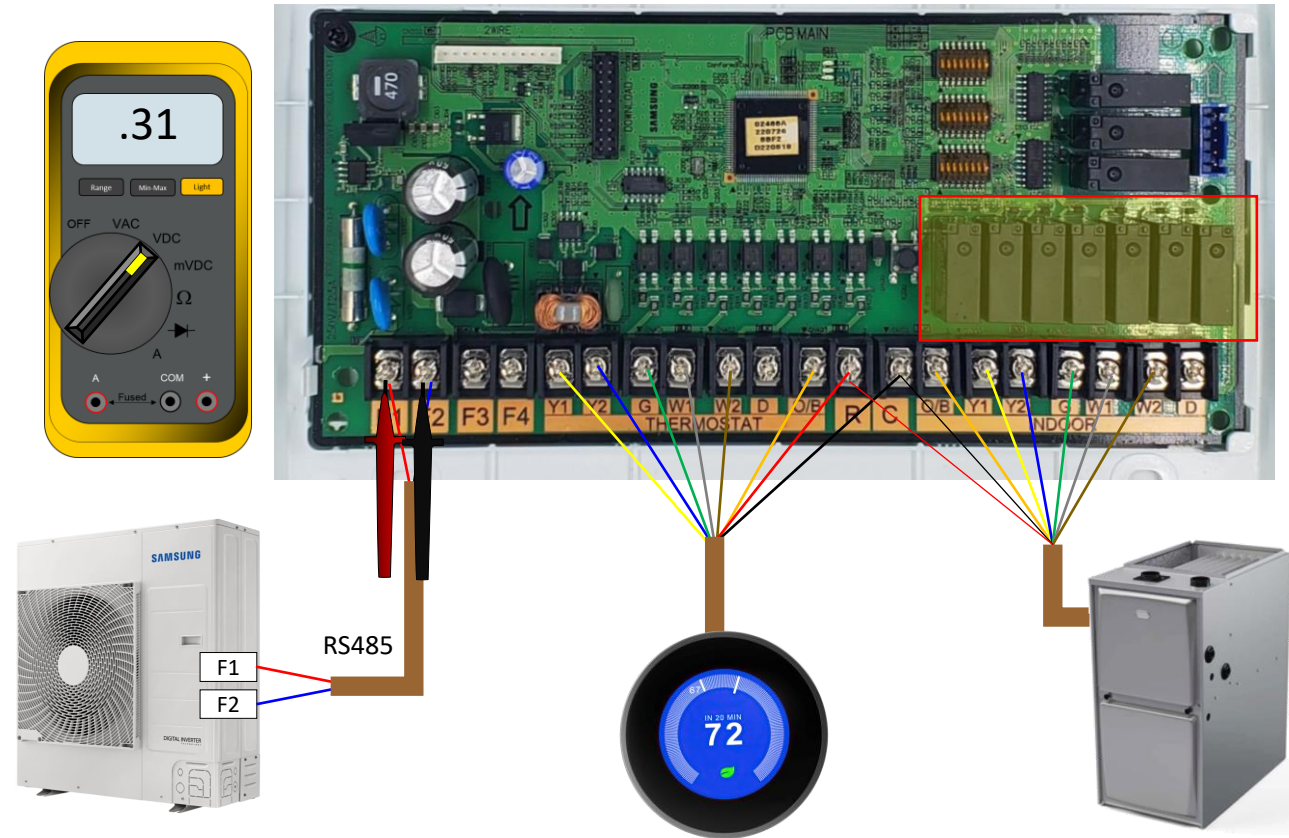
Testing a Call for Y1 to the Indoor unit

- Once you verified a call from the ACU to the indoor unit, the next step is to test the voltage at between Y1 and C at the indoor unit?
- Never assume that just because the thermostat is "calling" that the signal is being received at the indoor unit.
- This is why testing the voltage through out the system is so critical.



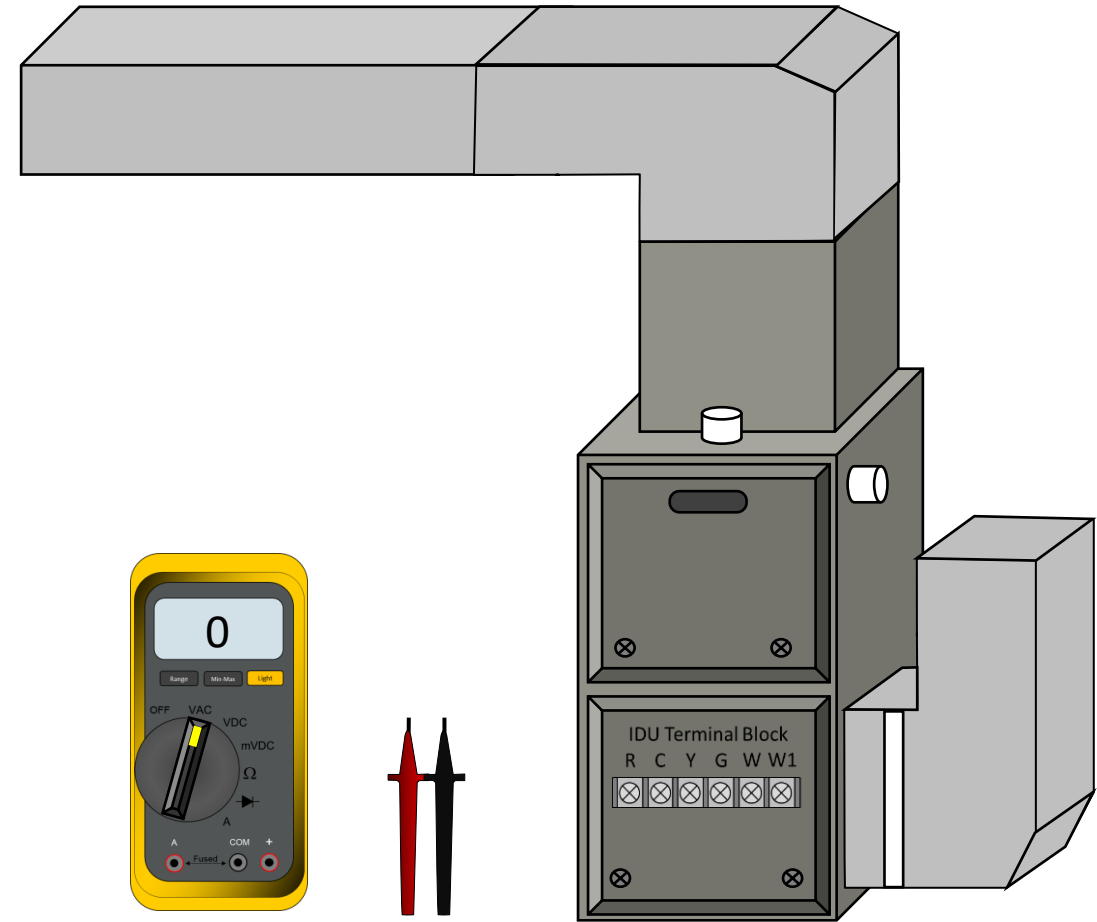
Testing communication between the ODU and ACU

- The ODU communicates with the ACU on the Samsung Set layer (F1 and F2).
- This is an RS 485 communication network.
- The measured voltage between F1 and F2 should be between .2 and .5 VDC



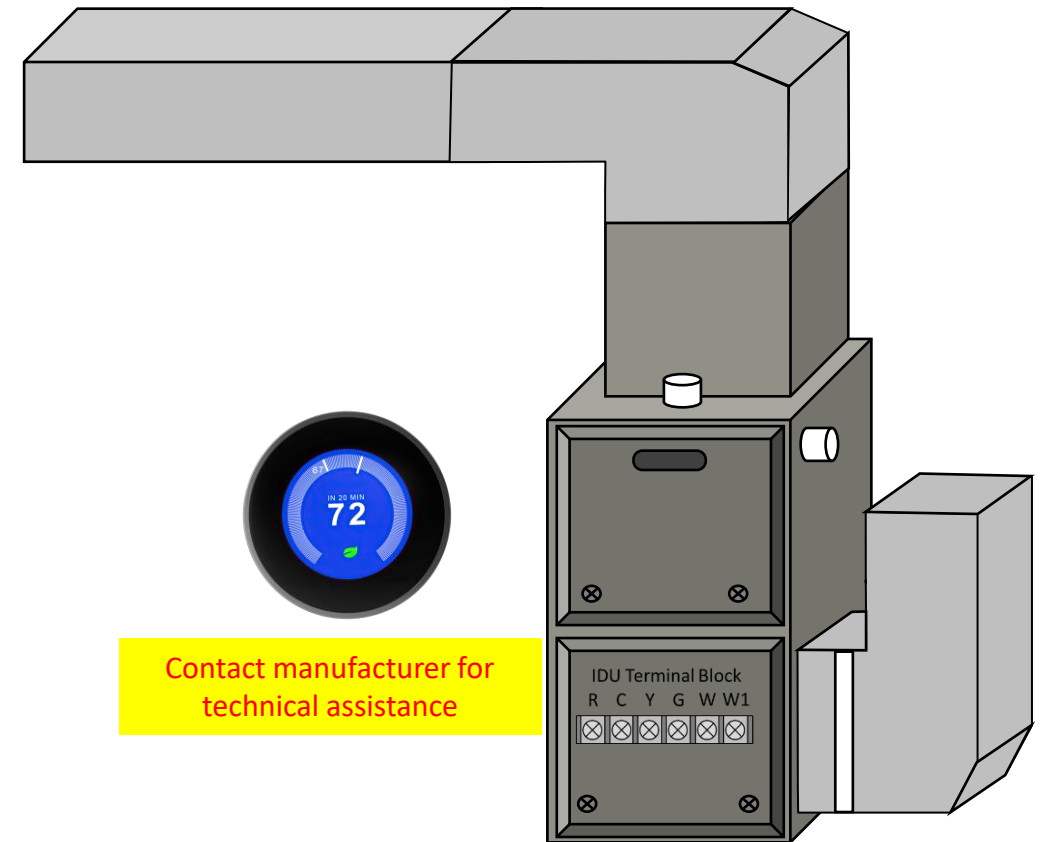
Jumper Wires

- Do not Jumper the terminal at the Indoor unit.
- The ACU will not register this.
- The ODU will not run.
- Measure the voltage to verify a call from the ACU
 - 24 VAC from R to C = Control Voltage Present
 - 24 VAC from Y to C = Call for Cooling
 - 24 VAC from G to C = Call for Fan
 - 24 VAC from W to C = Call for Heat 1st stage
 - 24 VAC from w1 to C = Call for Heat 2nd Stage



IDU or Thermostat Issues

- If you have a call to the IDU and the unit does not operate as expected. You need to contact the IDU manufacturer for technical assistance.
- Samsung is not responsible for troubleshooting 3rd party devices
- This includes thermostats, humidifiers or any other devices.

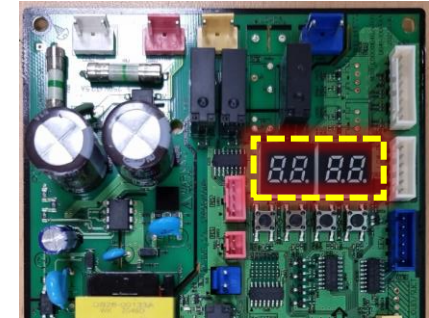


Component Trouble shooting

- Not sure how in-depth I want to go here using the ODU function to test the system like test modes and the inverter checker function.
- Do we really want to get into component checking in this class example compressor, fan or circuit boards? Thoughts ??

Service Features

- On-Board Inverter checker
 - Verify inverter board status using the built-in inverter checker
 - Enable test using "K" buttons on the PCB and the table below

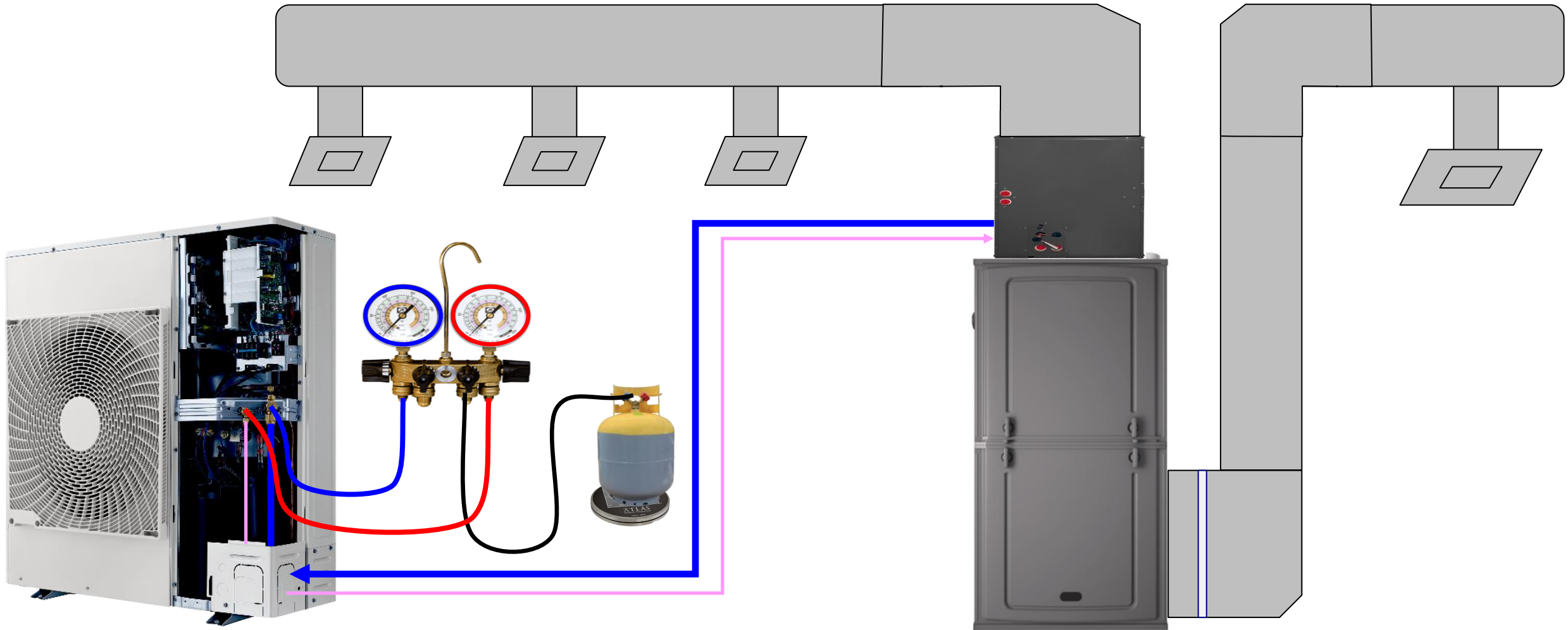


Test result will appear on display

K1 (Number of presses)	Key operation	Display on segment				
		Under progress	OK (PBA OK)	NG (PBA failure)	Check (Manual check needed)	Fail to the check (Try again)
3 times	Inverter fault detection	"I" "1" "-" "-"	"I" "1" "O" "K"	"I" "1" "N" "G"	"I" "1" "C" "h"	"I" "1" "F" "L"

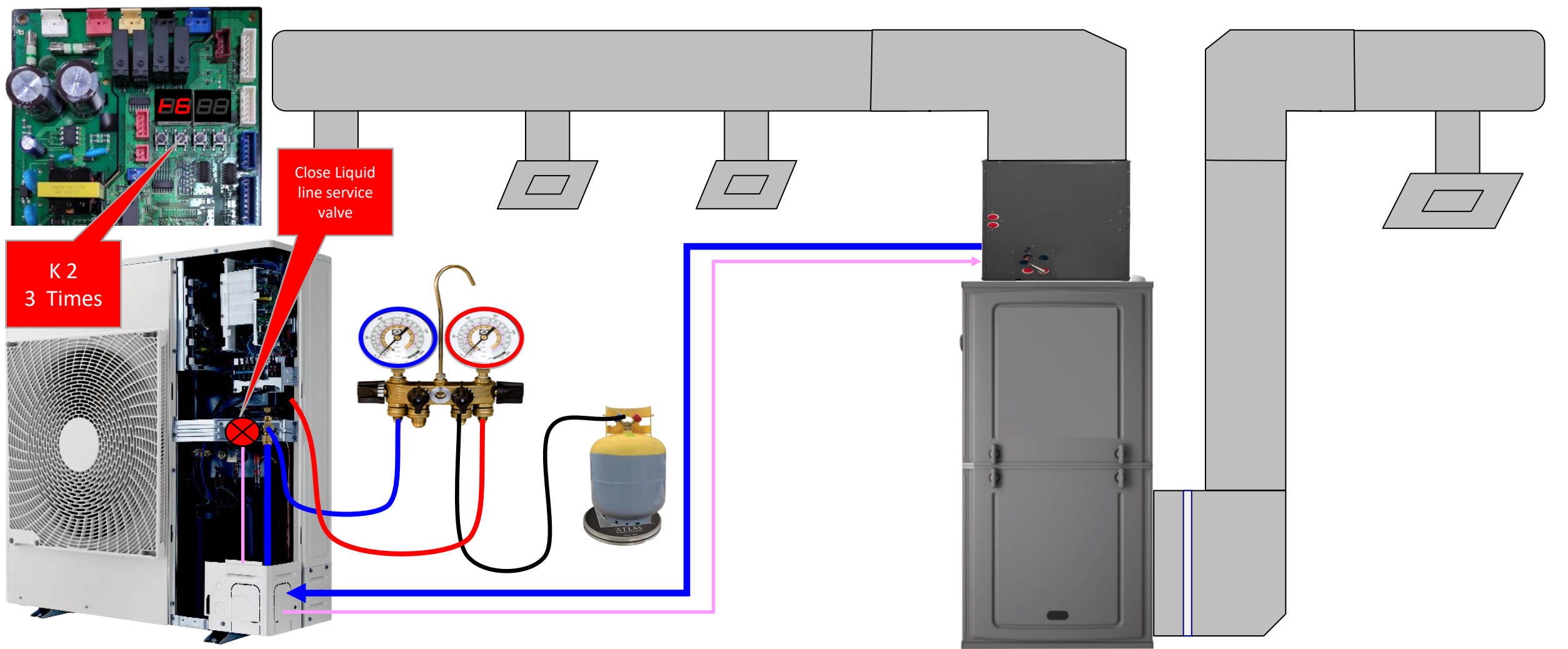


System Pump Down



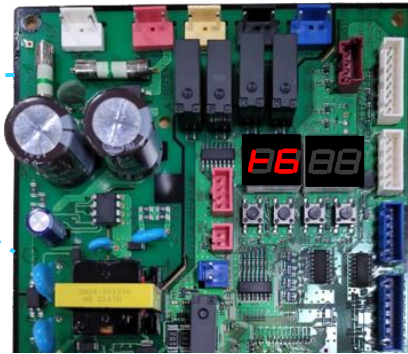
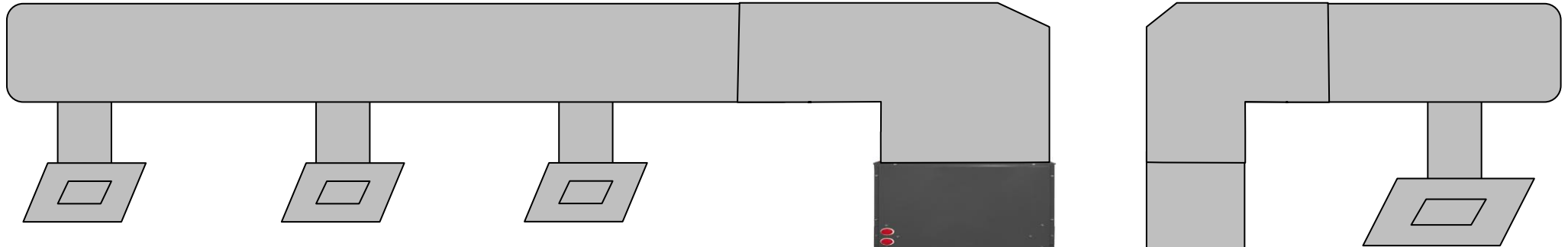
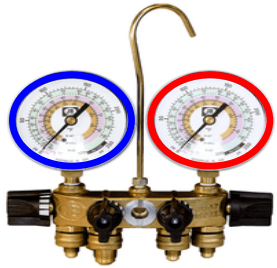
System Pump Down

EE EE

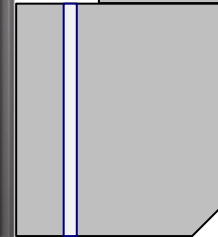


System Pump Down

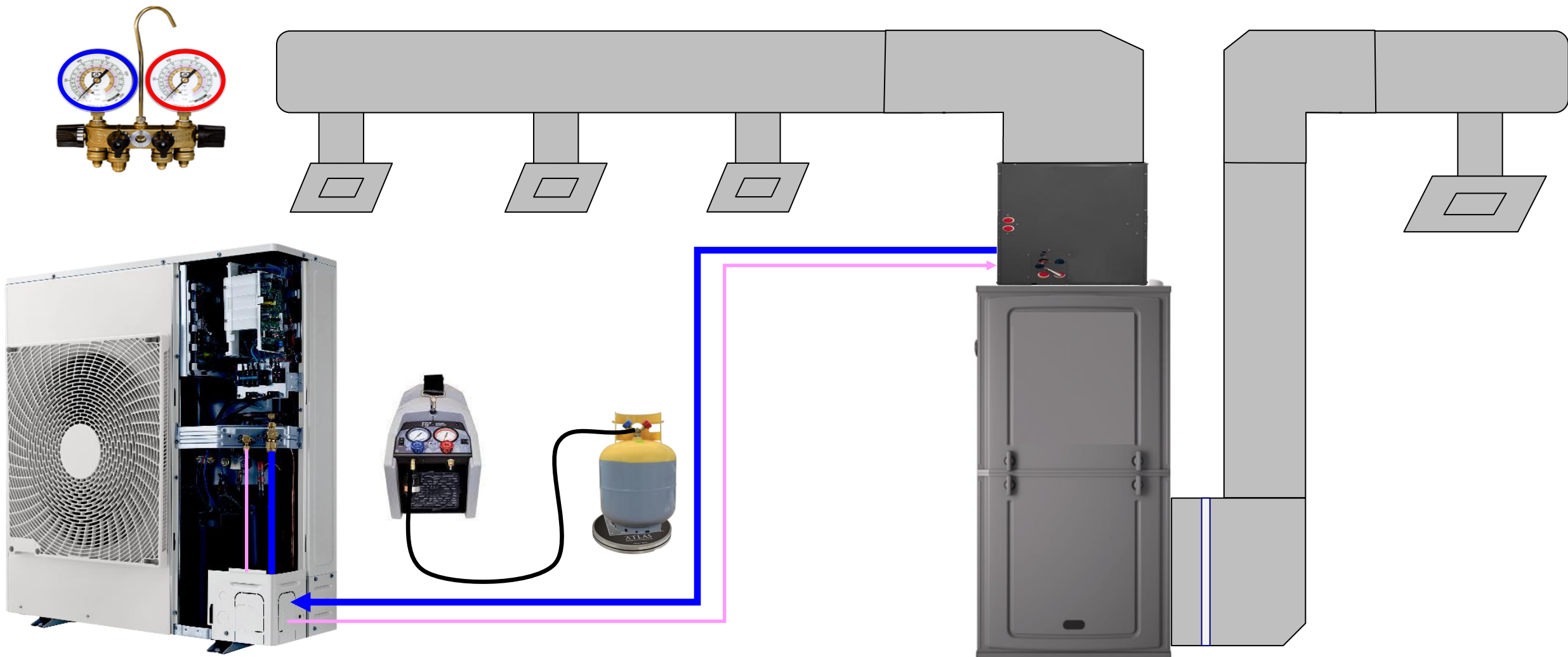
AA AA



K2 3 times

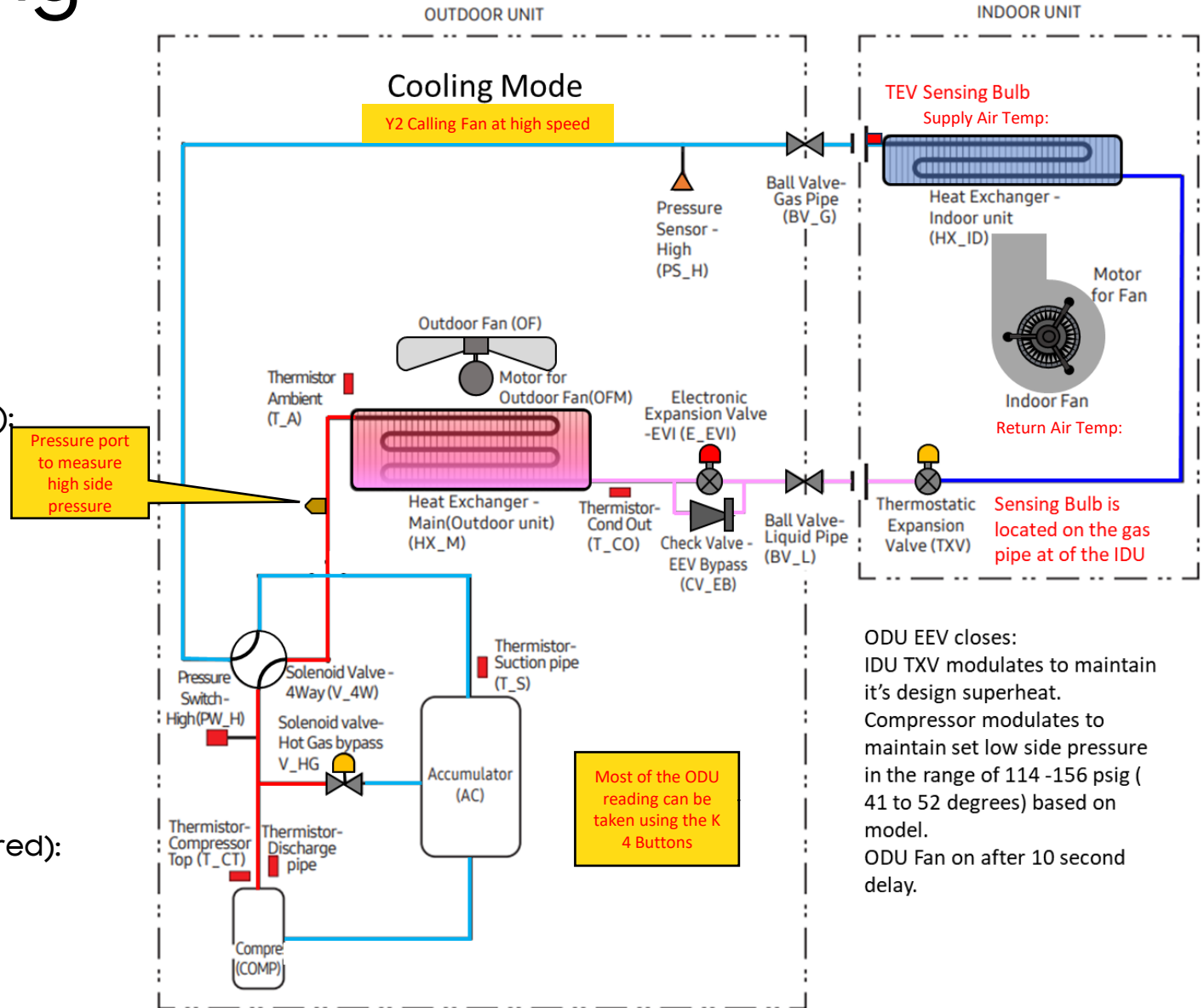


System Pump Down



Hylex Troubleshooting Information

- has CFM been Verified? (Y - N)
- Outdoor Unit information
 - Suction Pressure(K -4 20x):
 - Suction Pipe Thermistor Reading(K -4 19x):
 - Actual Suction Pipe Temperature (measured):
 - High Side Pressure (measured):
 - Condenser Out Thermistor Reading(K -4 7x):
 - Discharge Temperature (K -4 5x):
 - Target Discharge Temperature(K -4 10x):
 - Protection Mode Reading (k - 4 13x)
- Evaporator
 - Return air Temperature and RH (measured):
 - Supply air temperature and RH (measured):
 - Liquid line Temperature (measured):
 - Evaporator Saturation Temperature (measured):
 - Evaporator Superheat (calculated)



Discharge Temperature (K -4 5x): _____
Target Discharge Temperature (K -4 10x): _____
Protection Mode Reading (k - 4 13x): _____
Suction Pressure(K -4 20x): _____
Suction Pipe Thermistor (K -4 19x): _____
Ambient Air Thermistor (K -4 4x): _____

Size _____
Length _____

ODU
L1 – L2 _____

Size _____
Length _____

R to C terminal _____
G to C terminal _____
Y1 to C terminal _____
Y2 to C terminal _____

Directions:

Complete the following form and return to your RTSM or STS. Some information can be obtained using the K 4 Buttons on the outdoor unit and others will need to be taken manually. Air Flow MUST BE VERIFIED for proper operation.

Suction Pipe Temperature _____
Minus
Low Side Saturation Temperature _____

High Side Saturation Temperature _____
Minus
Liquid pipe 1 Temperature _____

Return Air Temperature _____
Minus
Supply Air Temperature _____

Suction Pressure High Side Pressure

Wet Bulb: _____
Dry Bulb: _____

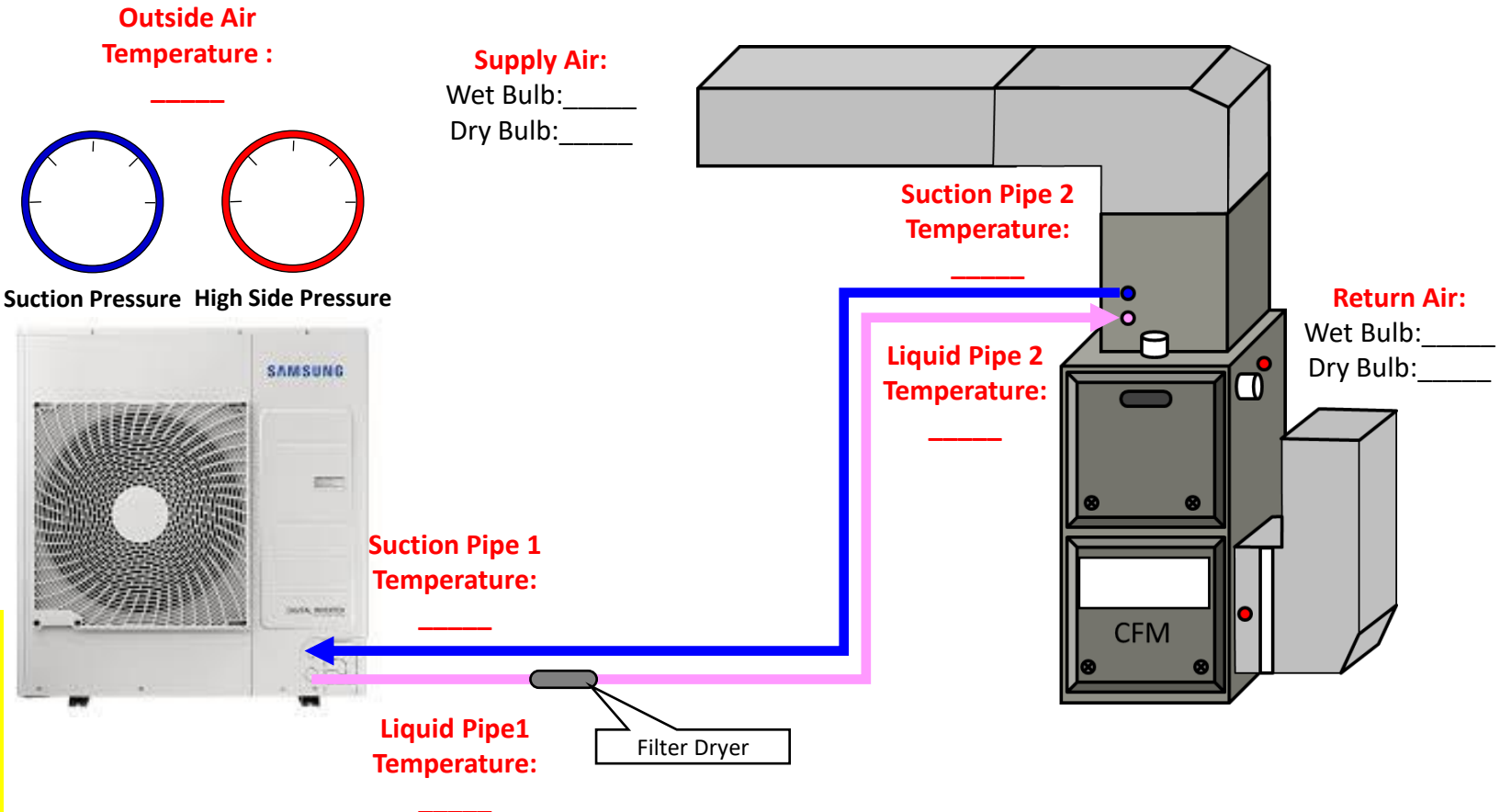
**Suction Pipe 2
Temperature:**

Liquid Pipe 2 Temperature:

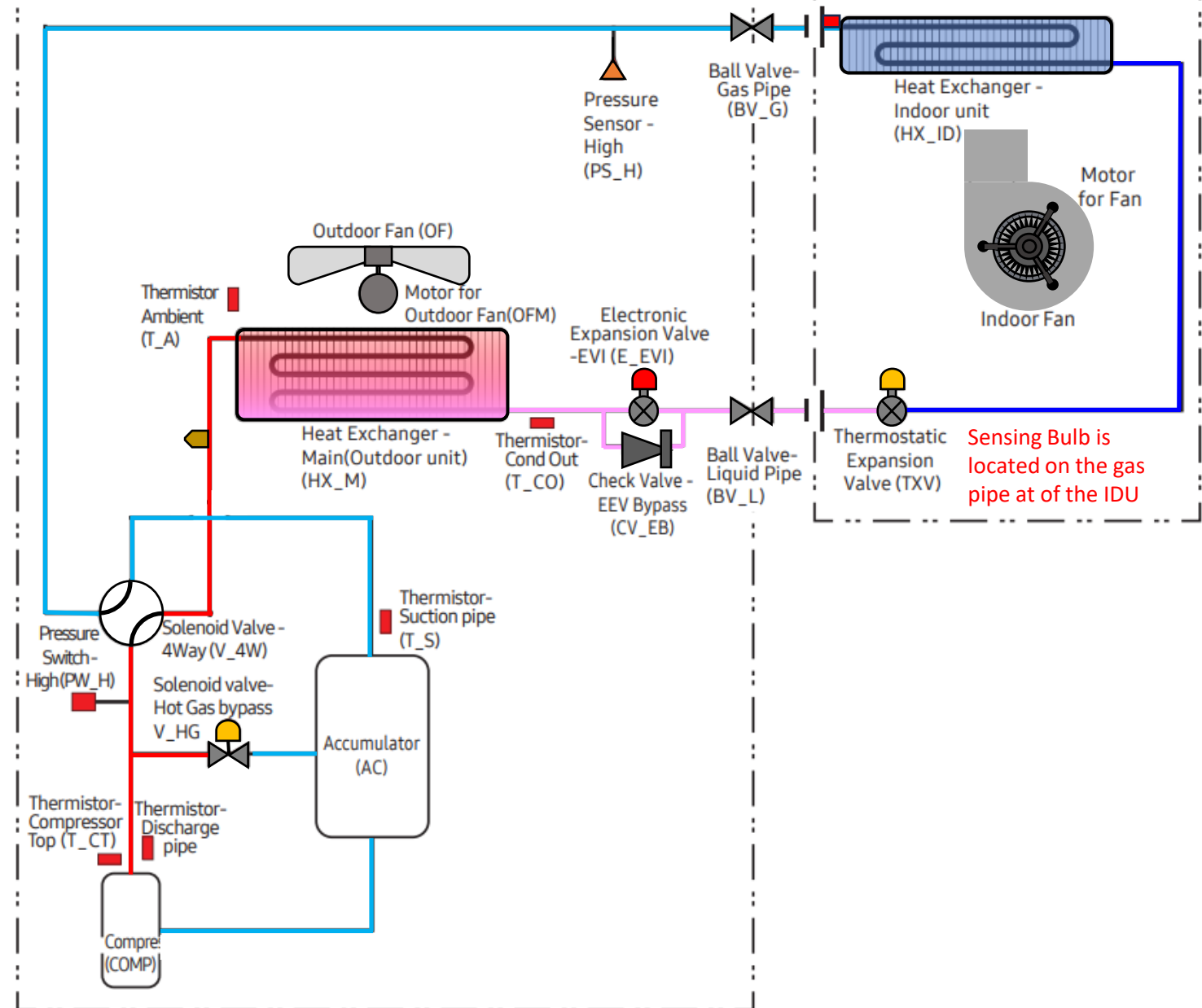
Wet Bulb: _____
Dry Bulb: _____

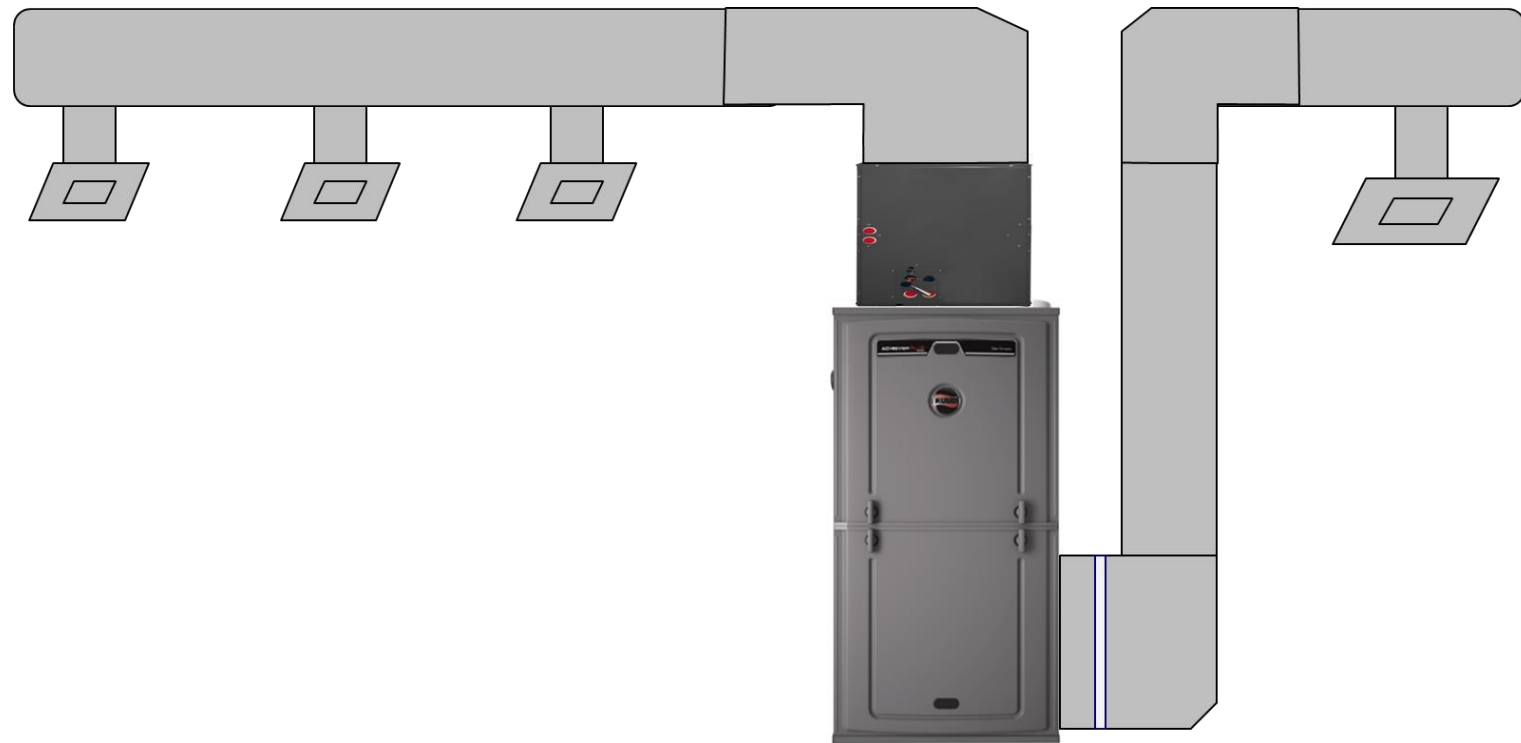
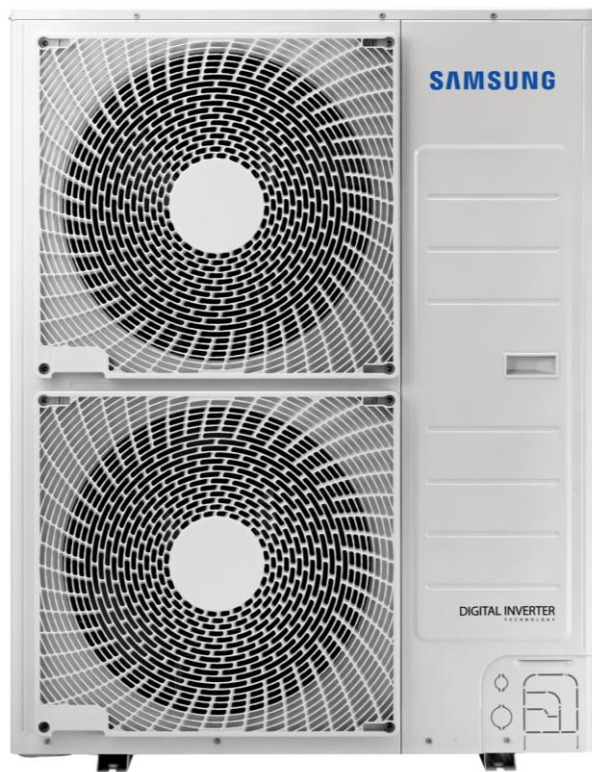
**Suction Pipe 1
Temperature:**

Liquid Pipe1
Temperature:

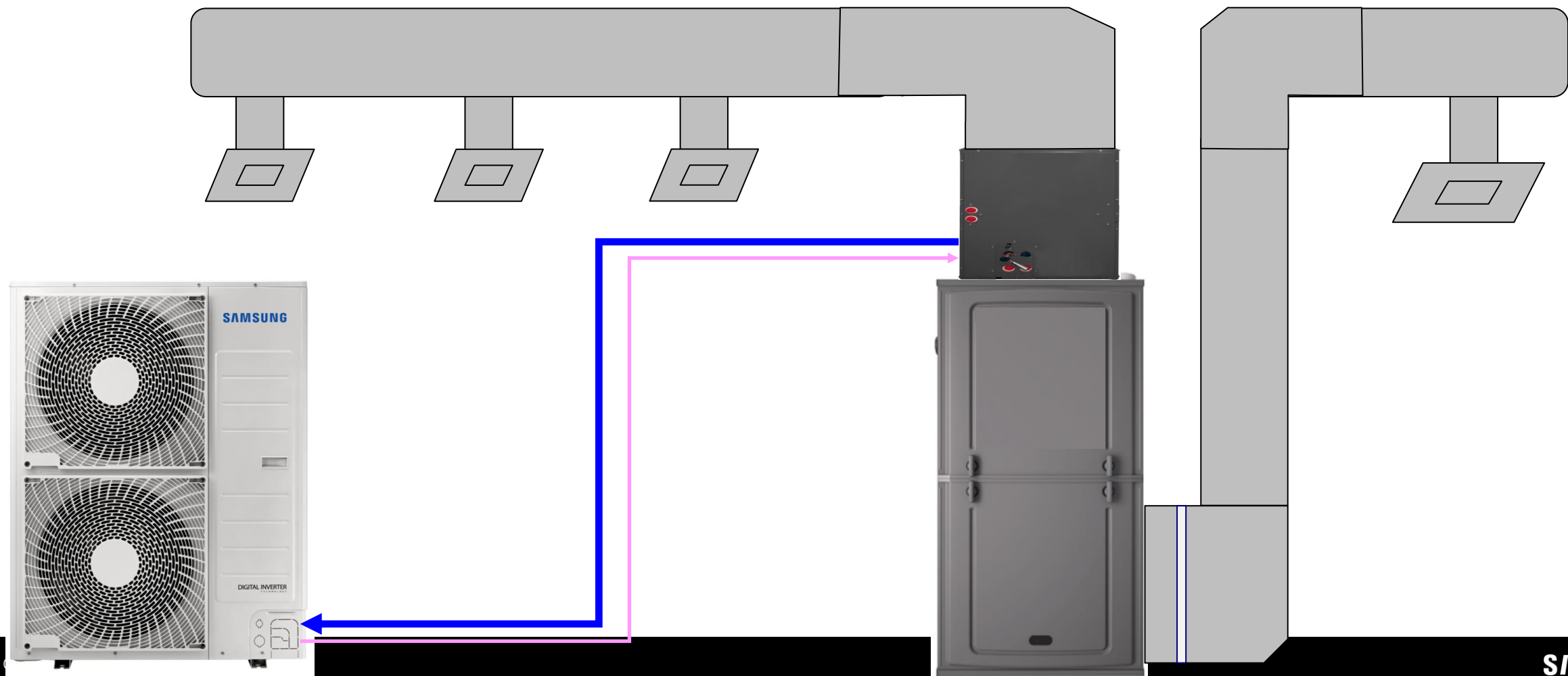


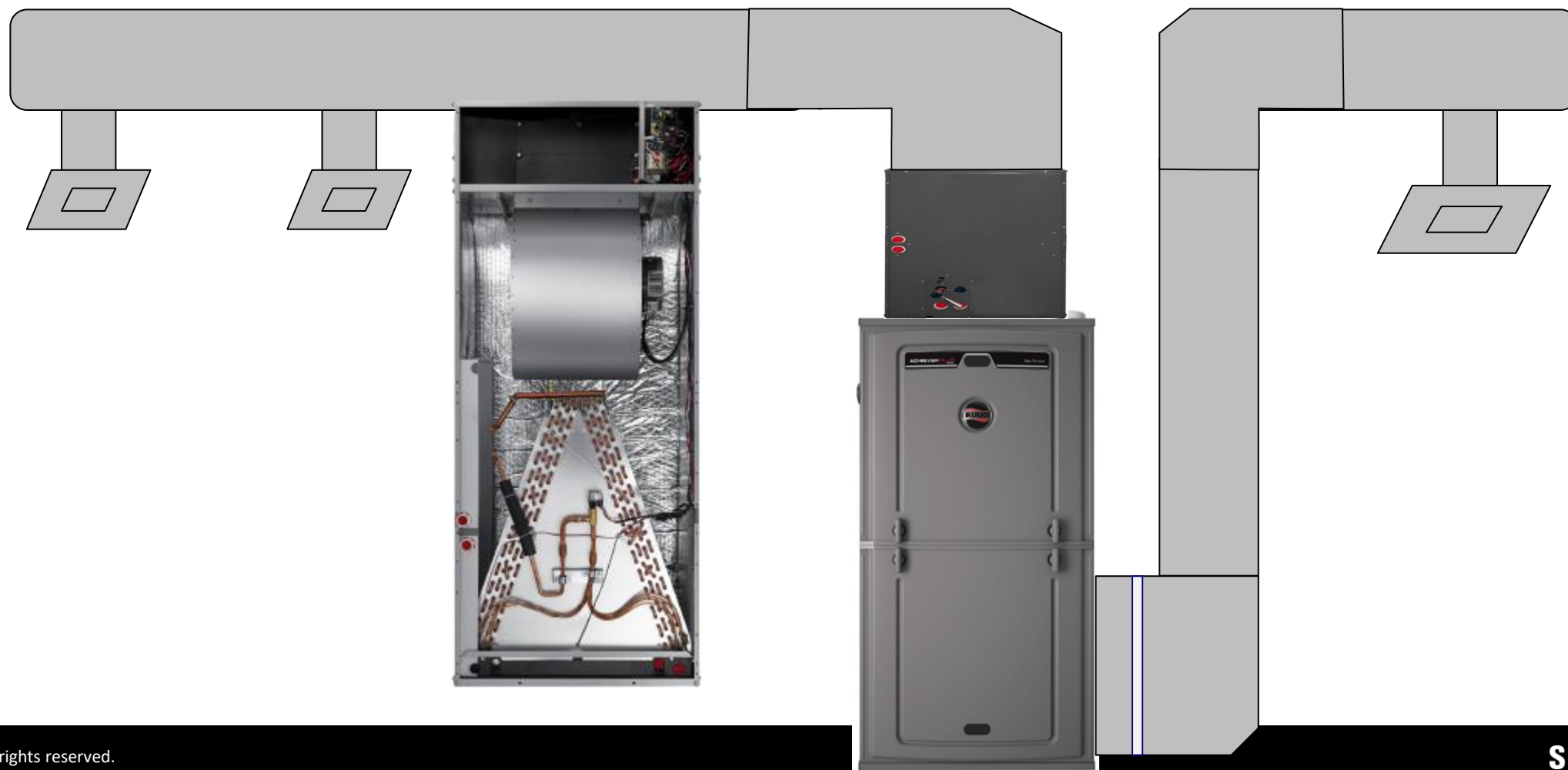
Cooling Mode





AA



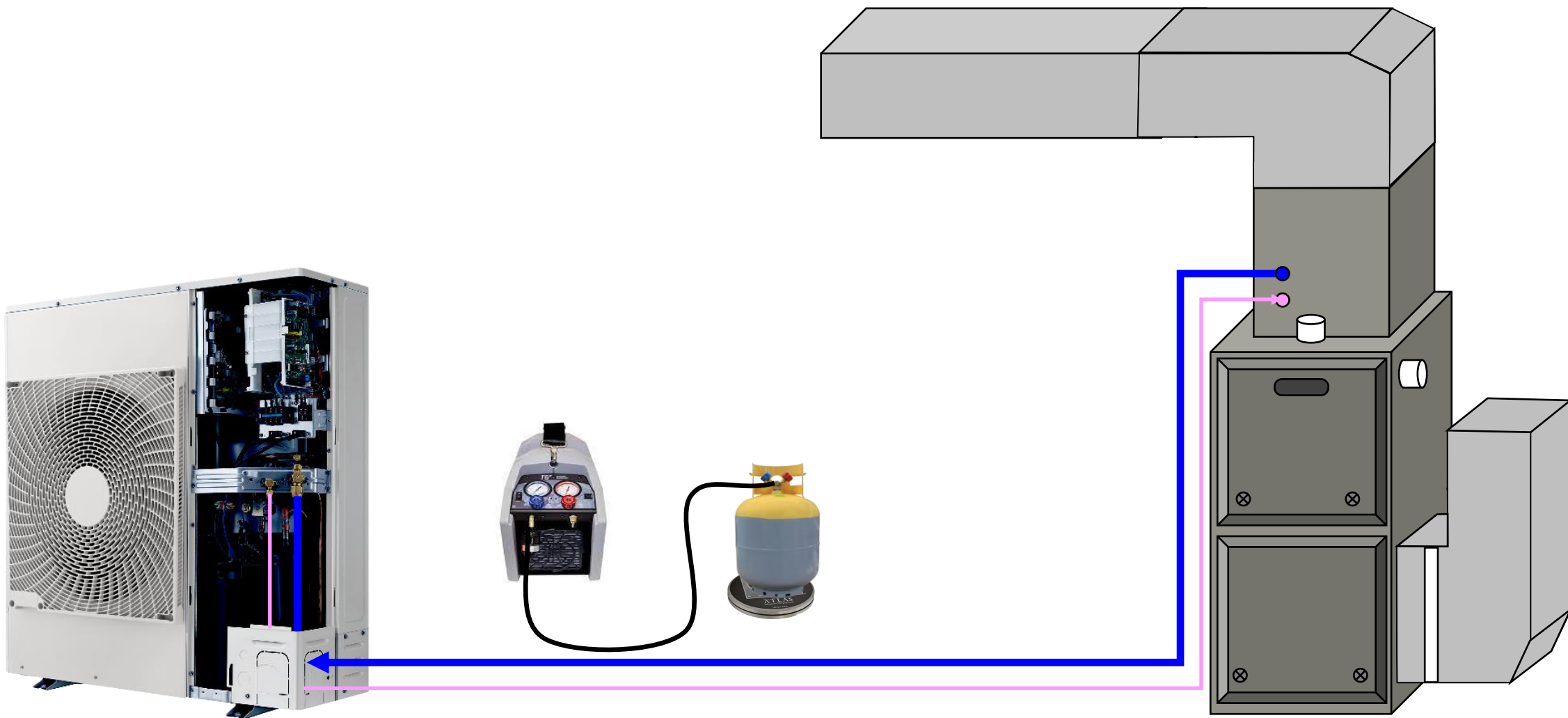


- 1) Performances are based on the following test conditions.
- - Cooling : Indoor temperature : 80°F(26.7°C) DB, 67°F(19.4°C) WB, Outdoor temperature : 95°F(35°C) DB, 75°F(23.9°C) WB
- - Heating : Indoor temperature : 70°F(21.1°C) DB, 60°F(15.6°C) WB, Outdoor temperature : 47°F(8.3°C) DB, 43°F(6.1°C) WB
- - Equivalent refrigerant pipe length 7.5m (24.6ft), Level differences 0m (0ft)

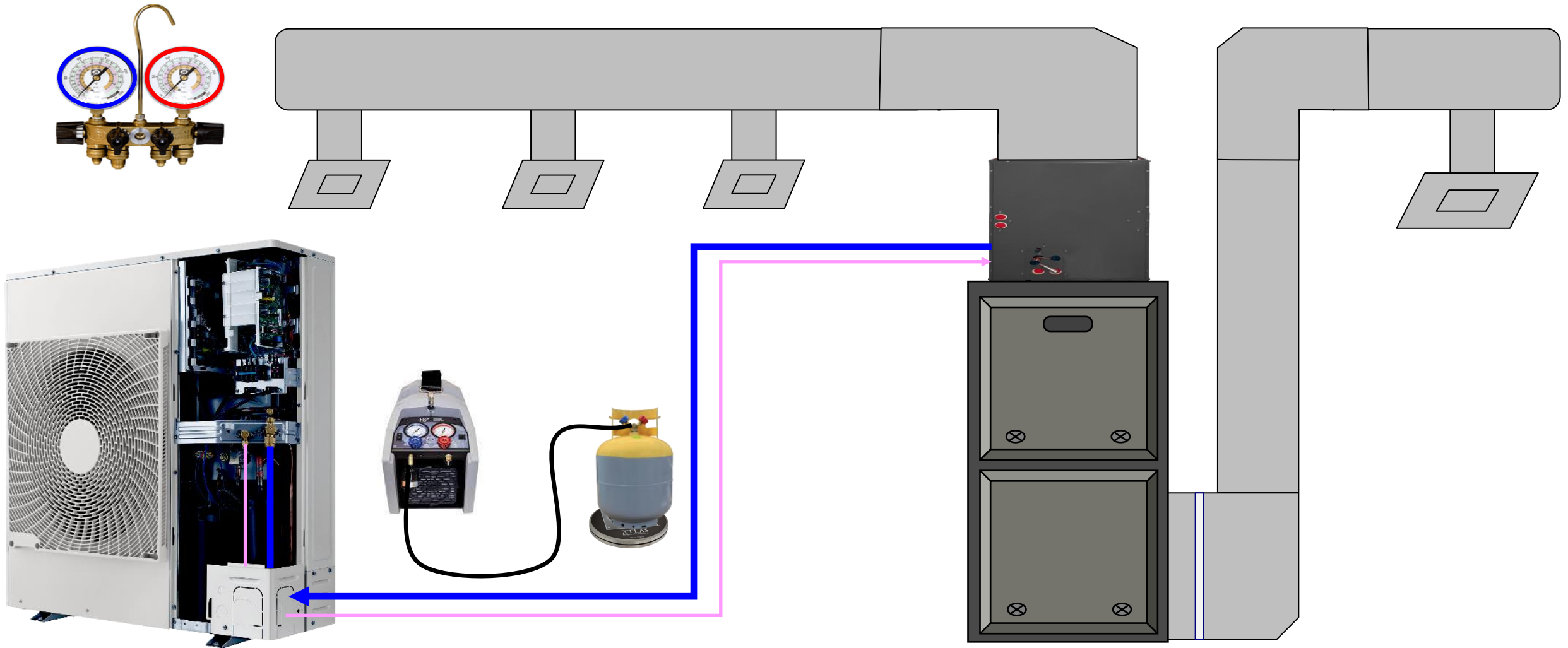
Cooling

TC : Total Capacity, SHC : Sensible Heat Capacity, PI : Power Input

Outdoor Temp. (°F, DB)	Indoor Temperature (°F, DB / WB)																				
	68 / 57			72 / 61			77 / 64			80 / 67			82 / 70			86 / 72			90 / 75		
	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI
	MBH	MBH	kW	MBH	MBH	kW	MBH	MBH	kW	MBH	MBH	kW	MBH	MBH	kW	MBH	MBH	kW	MBH	MBH	kW
-4	16.69	15.40	1.09	21.75	16.99	1.34	25.63	19.83	1.59	29.80	21.16	1.87	32.18	20.79	1.97	33.38	22.04	2.00	35.16	22.72	2.03
70	16.35	14.56	1.14	21.32	16.06	1.41	25.11	18.75	1.67	29.20	20.00	1.98	31.54	19.66	2.09	32.70	20.83	2.12	34.46	21.48	2.15
95	16.38	14.44	1.66	20.70	15.89	2.03	25.96	19.22	2.57	24.00	18.00	2.38	34.61	20.73	3.53	35.54	22.01	3.59	37.39	22.65	3.64
115	18.10	16.03	2.96	22.07	17.17	3.41	23.56	18.87	3.44	24.80	19.10	3.47	26.29	18.22	3.50	27.28	19.75	3.53	28.02	20.29	3.50
126	13.80	12.85	2.72	16.82	13.76	3.14	17.96	15.13	3.16	18.90	15.31	3.19	20.03	14.60	3.22	20.79	15.83	3.24	21.36	16.26	3.22

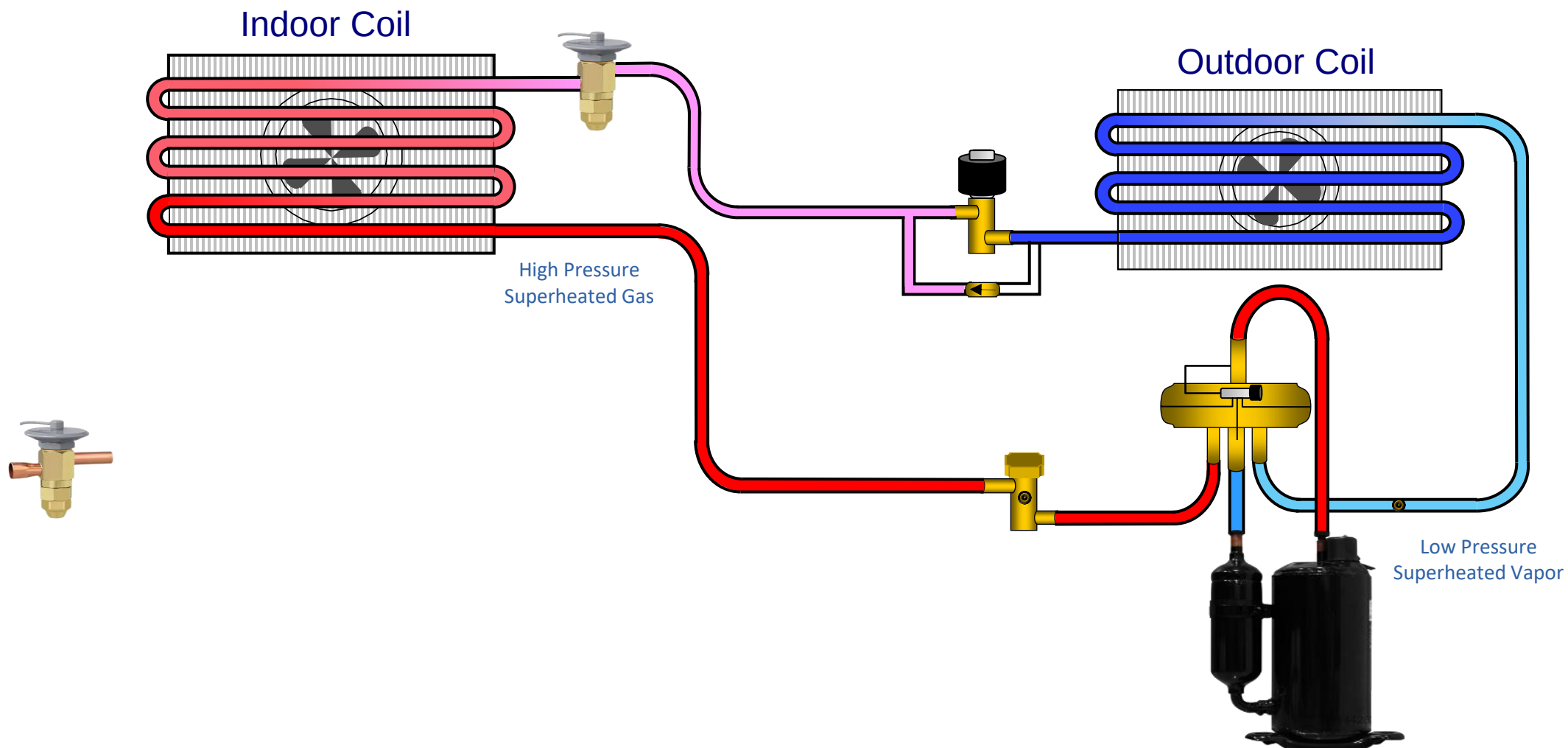


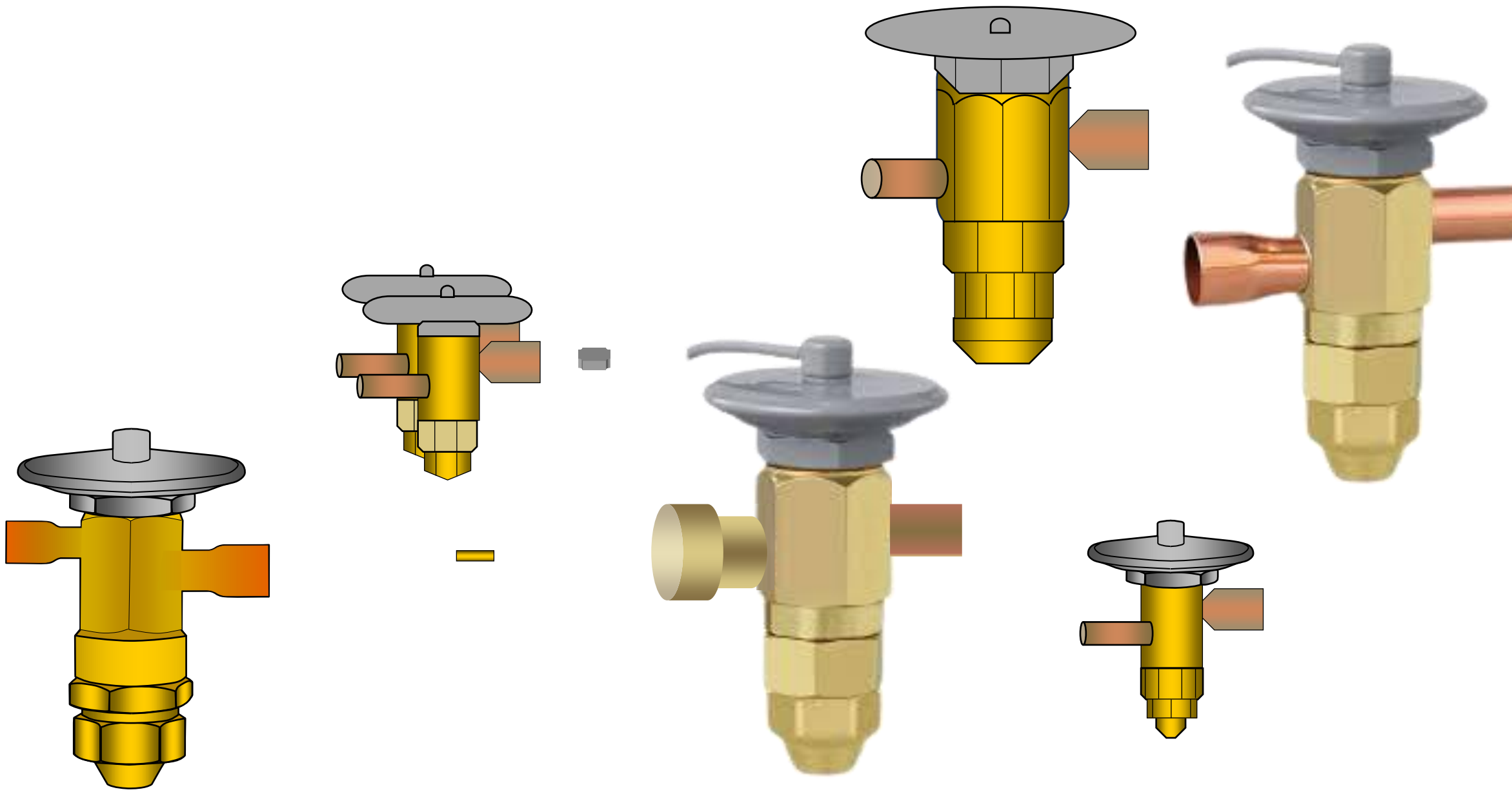
System Pump Down



States of refrigerant within a system

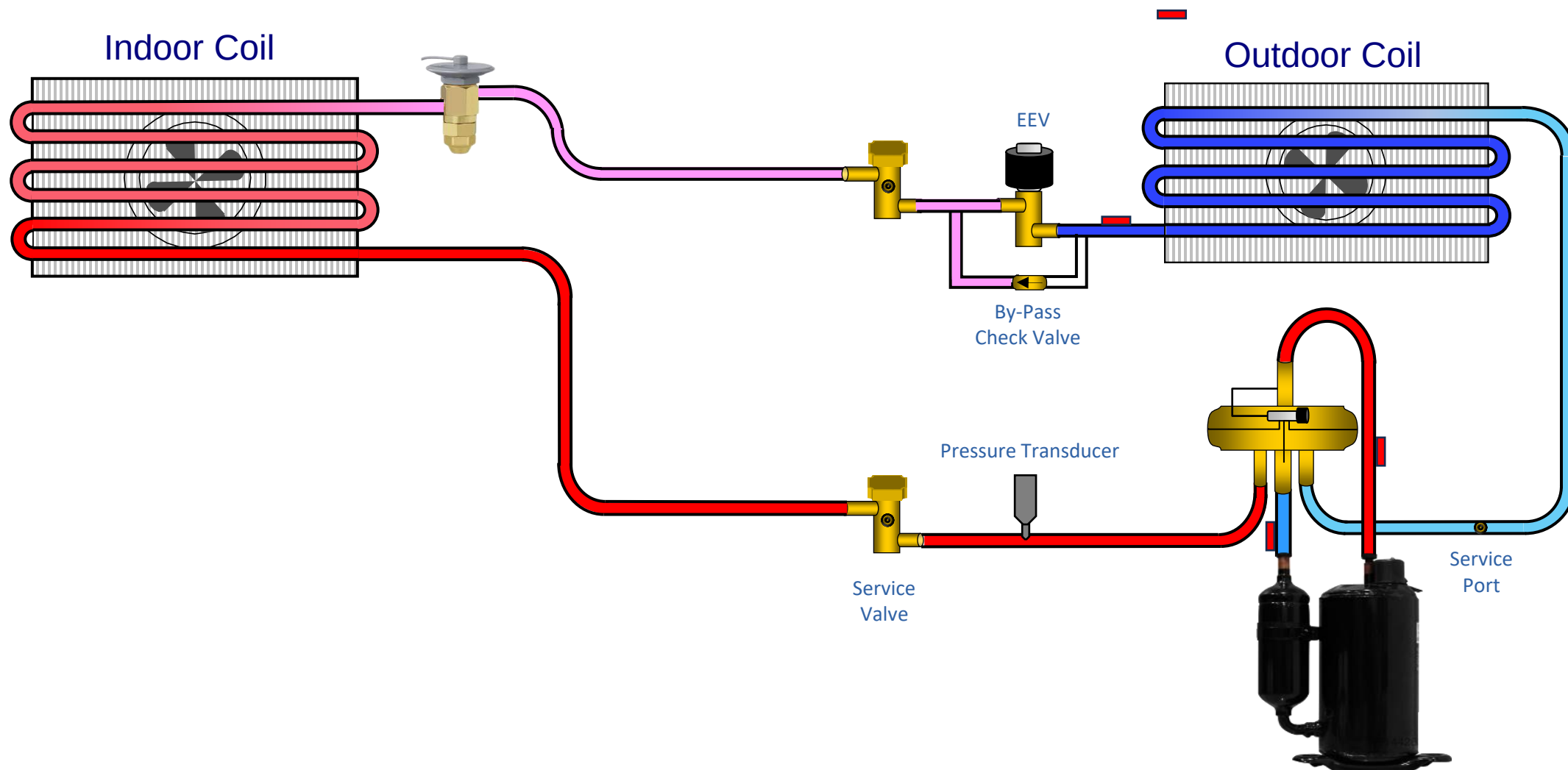
Heating Mode



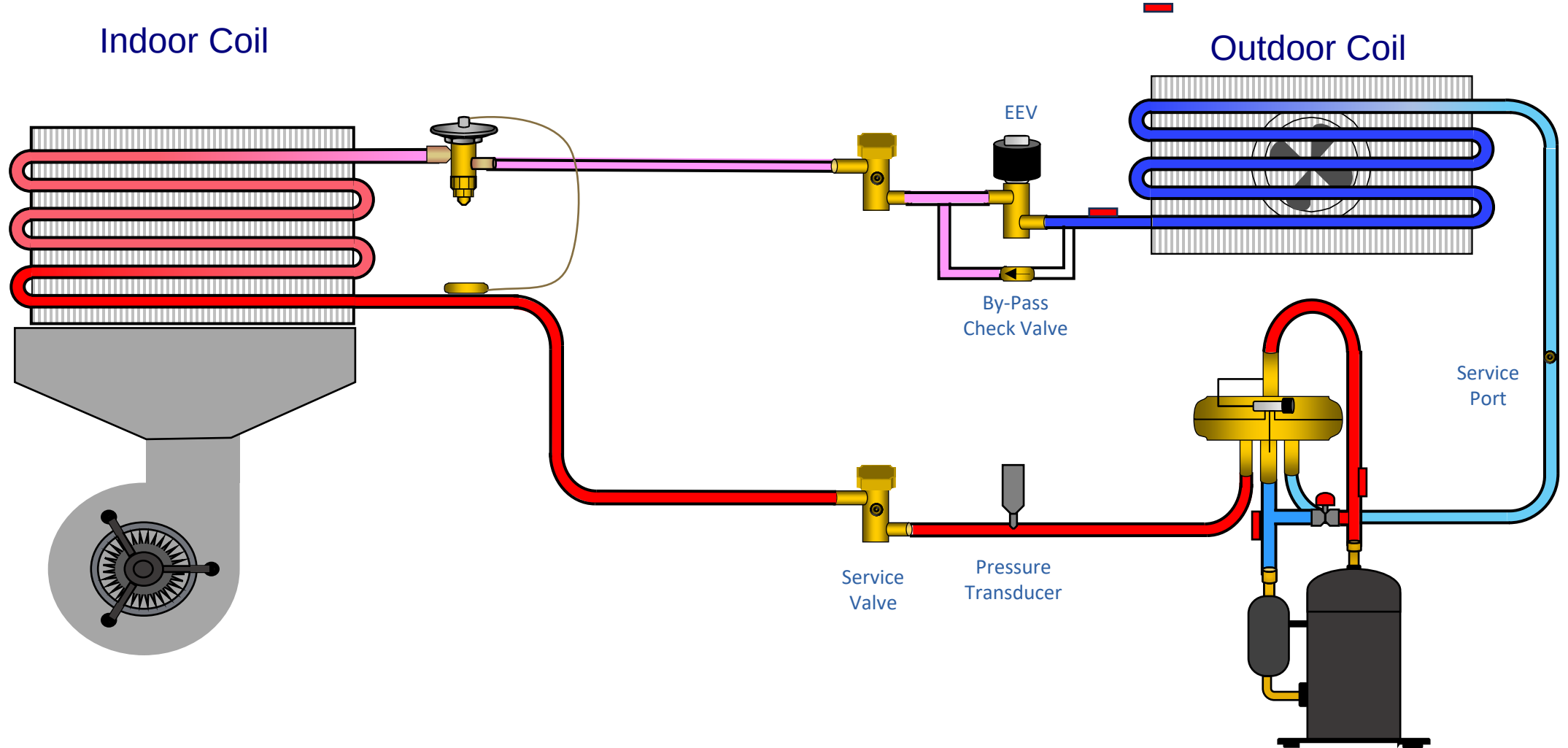


States of refrigerant within a system

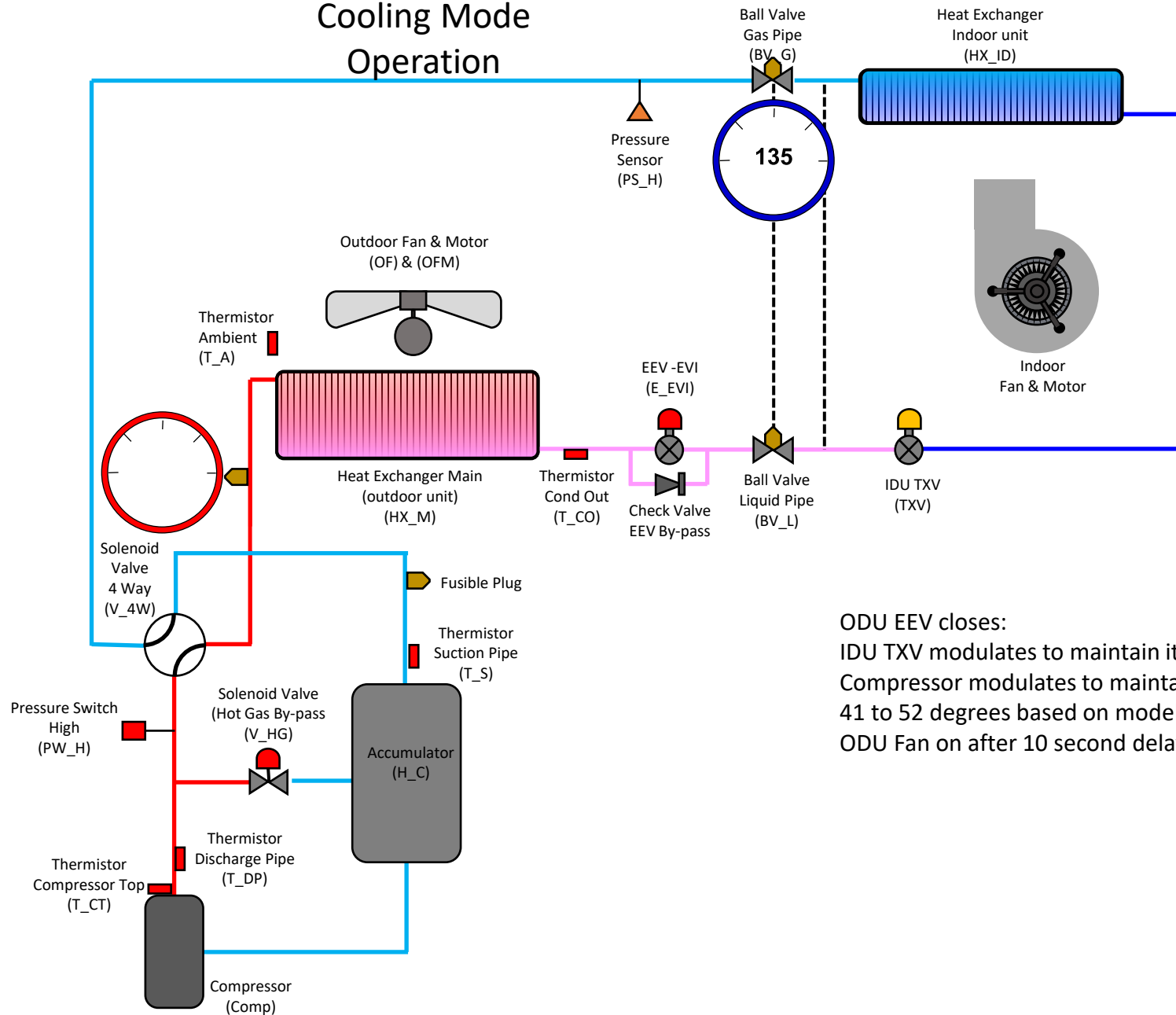
Heating Mode



Hylex Refrigeration System



Cooling Mode Operation



ODU EEV closes:

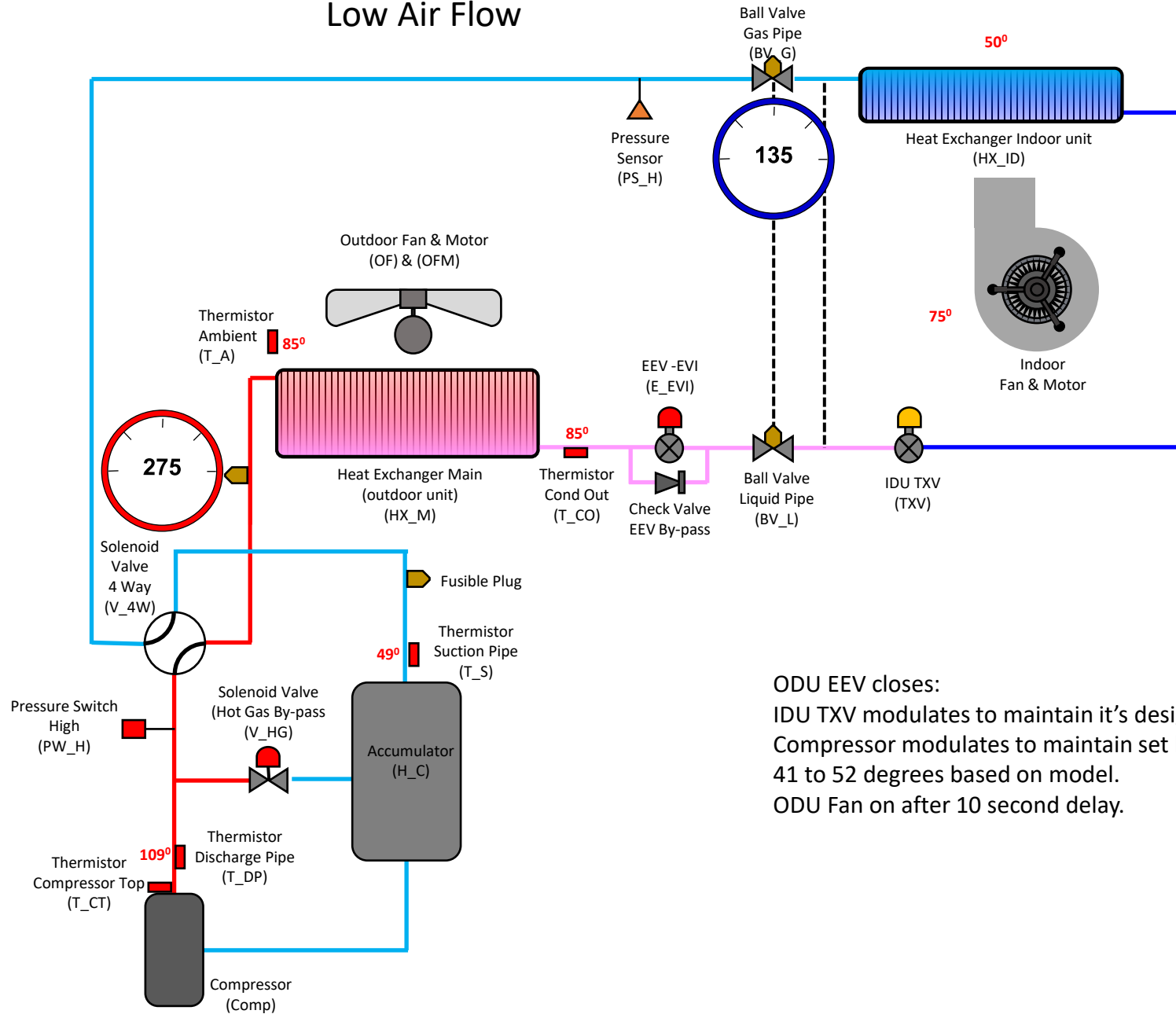
IDU TXV modulates to maintain it's design superheat.

Compressor modulates to maintain set low side pressure in the range of 41 to 52 degrees based on model.

ODU Fan on after 10 second delay.



Low Air Flow



ODU EEV closes:

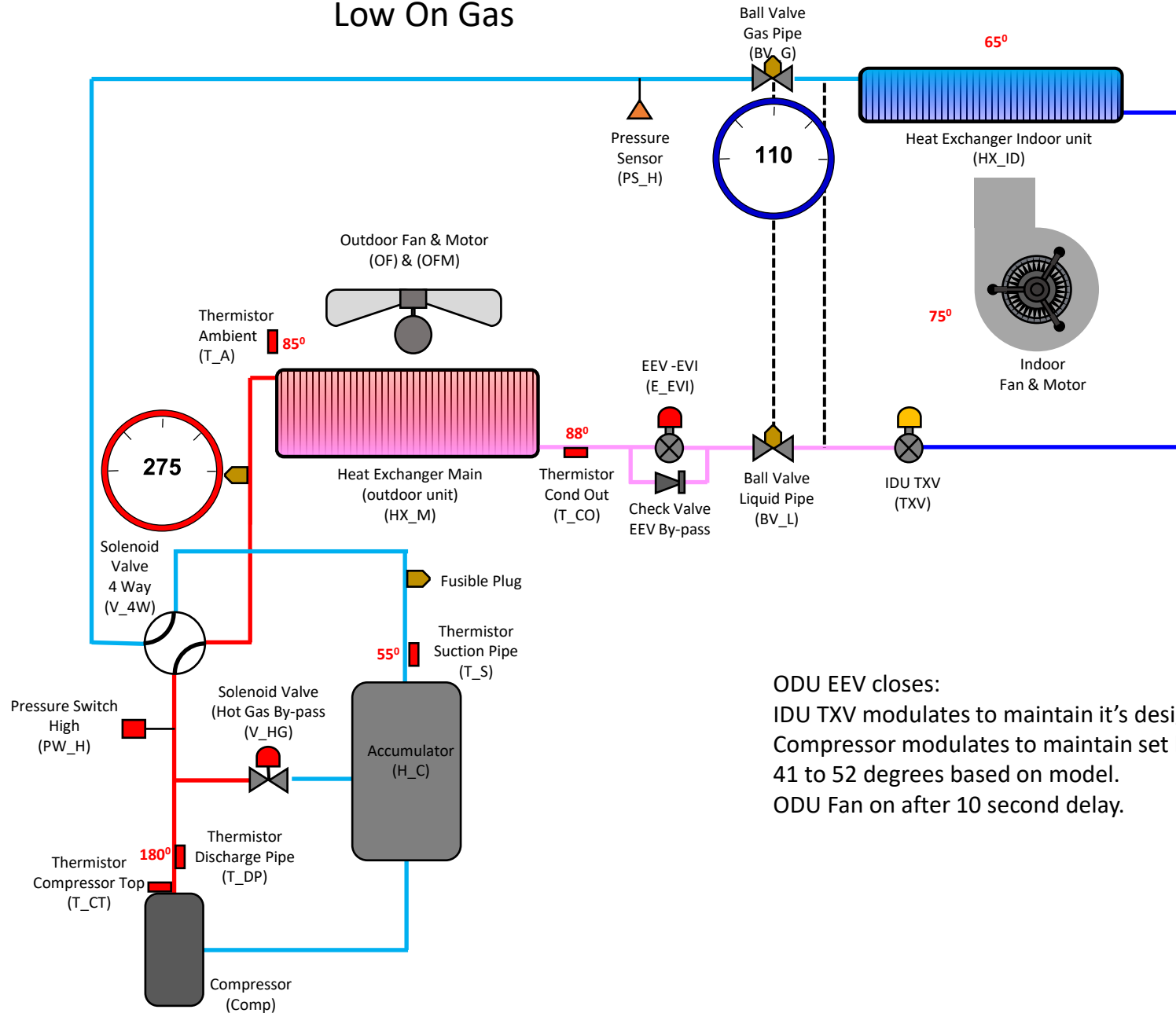
IDU TXV modulates to maintain it's design superheat.

Compressor modulates to maintain set low side pressure in the range of 41 to 52 degrees based on model.

ODU Fan on after 10 second delay.



Low On Gas



ODU EEV closes:

IDU TXV modulates to maintain it's design superheat.

Compressor modulates to maintain set low side pressure in the range of 41 to 52 degrees based on model.

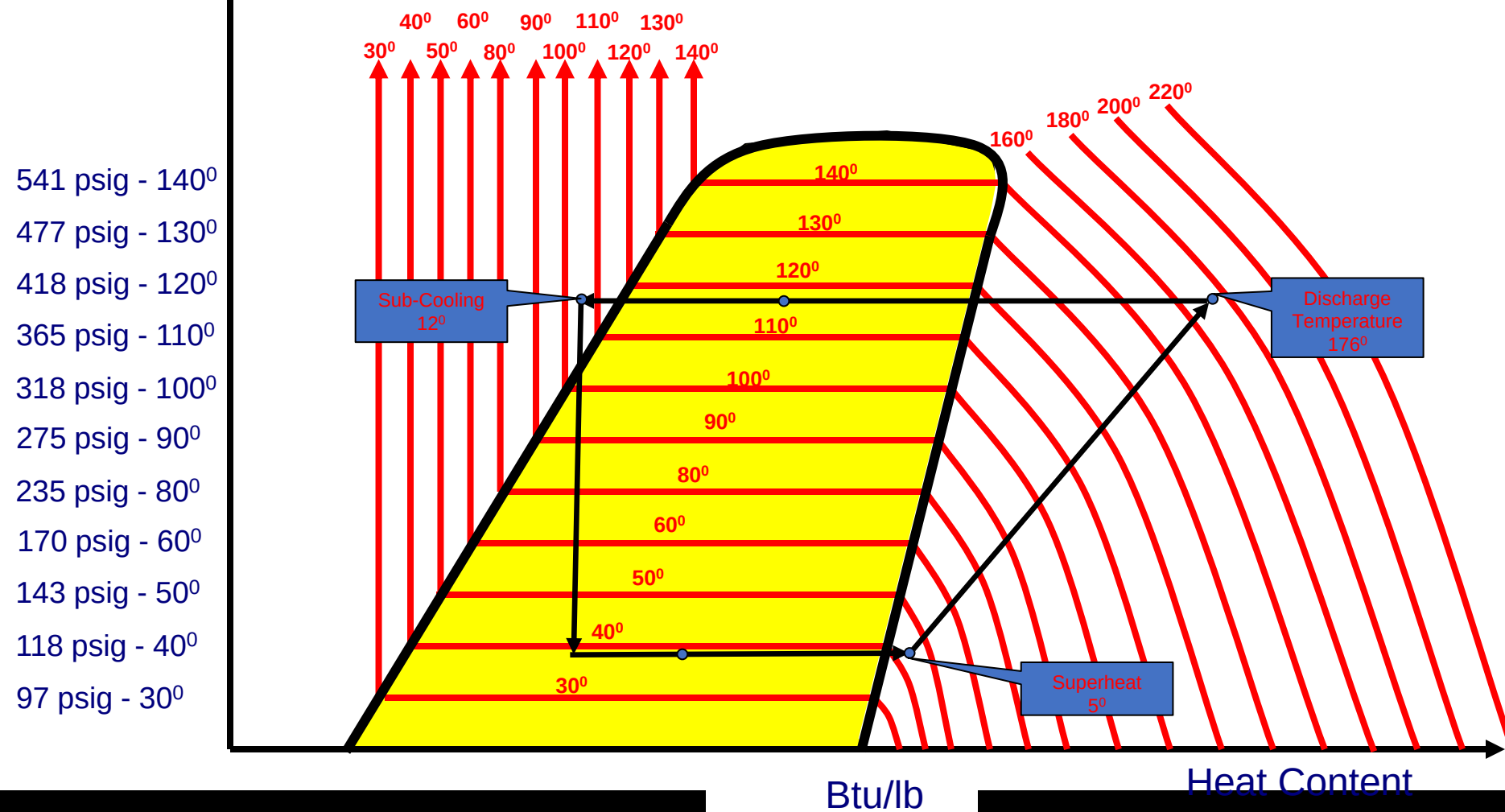
ODU Fan on after 10 second delay.

Pressure (psig)

Target operation 1:1:
Normal

Design Conditions

364 psig - 116°
116 psig - 39°
Discharge Temperature 176°
Super Heat 6°
Sub Cooling 12°
Ambient 95°
Indoor Temp. 80°



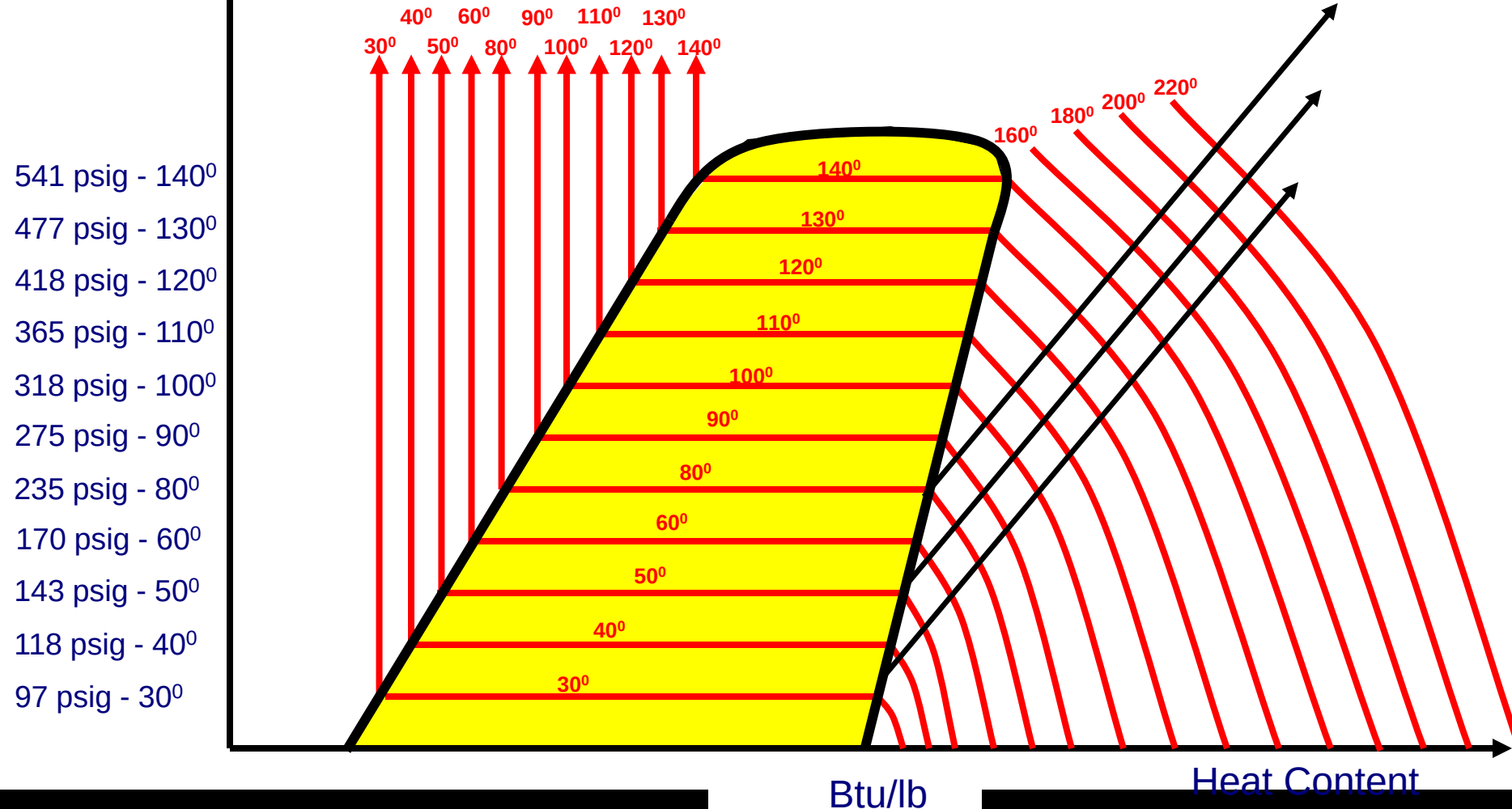
Pressure (psig)

Target operation 1:1:

Normal

Design Conditions

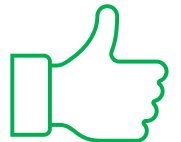
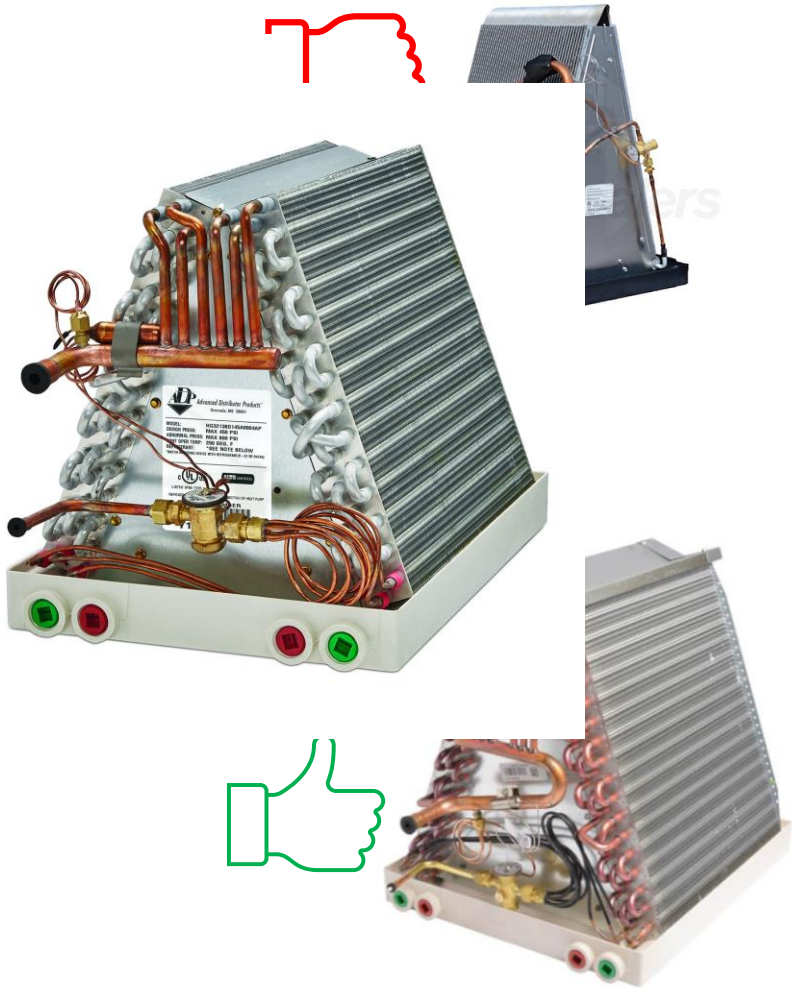
364 psig - 116°
116 psig - 39°
Discharge Temperature 176°
Super Heat 6°
Sub Cooling 12°
Ambient 95°
Indoor Temp. 80°



R-410A Refrigerant

Liquid Line		Line
Line Size (OD)	Charge per Foot	
1/4"	0.19 oz/ft	
5/16"	0.33 oz/ft	
3/8"	0.51 oz/ft	
1/2"	1.01 oz/ft	
5/8"	1.64 oz/ft	
3/4"	2.46 oz/ft	
7/8"	3.27 oz/ft	
1-1/8"	5.58 oz/ft	

Amount of R-410A refrigerant needed per 40° suction temp and 105° liquid temp.



Submittals

SAMSUNG

SUBMITTAL AC024KNZDCHIAA / AC024BXADCHIAA (CNH24ZDK / CXH24ADB) Page 1 of 10
Samsung Multi-Position Air Handler, Single Zone Duct, Split System

Job Name _____
 Purchaser _____
 Submitted to _____
 Unit Designation _____

Location _____
 Engineer _____
 Reference _____ Approval _____ Construction _____
 Schedule # _____

	Specifications	
Model	Indoor Unit Model Number (US Code)	AC024KNZDCHIAA (CNH24ZDK)
	Indoor Unit Model Number (UK Code)	AC024BXADCHIAA (CXH24ADB)
Performance ¹	Nominal Capacity (Cooling / Heating (Btu/h))	24,000 / 27,000
	Capacity Range (Cooling (Btu/h))	8,000 - 30,000
	Heating (Btu/h)	7,000 - 30,000
	SEER / EER	18.7 / 11.0
	COP (nominal heating)	2.80
Power	SEER	10.9
	AHRF Certification Number	207898008
	Voltage (V / V / Hz)	1 / 208-230 / 60
	Operating Voltage Range (VAC)	176-230 V (50/60 Hz)
	Operating Current (min. / std. / max. A)	Cooling (A) 4.3 / 9.9 / 14.9
Dimensions	Heating (A)	3.9 / 14.8 / 24.0
	Max. Breaker (Amps)	30
	Min. Circuit Ampacity (A)	24.1
	R X H X D (in.)	Indoor Unit 17.1/22.4 X 3 X 21
	Outdoor Unit 17.8 X 20.2 X 13.3	
Sound Pressure Level	Purt Connections (in. x in.)	Supply (in.) 12.1/4 X 1/2
	Return (in. x in.)	12.1/2 X 2.1
	Weight (lbs.)	Indoor Unit 36.1
	Outdoor Unit (lbs.)	118.4
	Indoor Unit (dB(A))	20 / 24.1
Operating Temperatures	Cooling / Heating (deg.)	60 / 52
	Indoor	Cooling 23 ~ 122°F (5 ~ 50°C)
	Heating	0 ~ 122°F (1.8 ~ 50°C) (w/ buffer)
	Outdoor	Cooling 53 ~ 122°F (12 ~ 49°C)
	Heating	81 ~ 90°F (18 ~ 32°C)
Pipe Connections	Indoor & Outdoor	High-side (Brass / Flare) 3/8"
	Low-side (Brass / Flare)	5/8"
	Maximum I.D.	1/4"
	Maximum Vertical Separation (ft.)	164
	Condensate Connection (with included strainer)	PPF 3/4" PVC
Refrigerant	Type	R410A
	Charge Method	Electronic Evacuation Valve
	Factory Charge (lbs.)	5.73
	Charge for Additional Refrigerant	24.4 lb.
	Additional Refrigerant	0.11 (lb./ft. over 24.4 lb.)
Compressor	Manufacturer	Samsung
	Model	Inverter Driven, Twin BLDG Rotary
	RSLA	15.9
	Type	Double-inlet, forward curve, centrifugal (with ECM motor)
	AP Volume (CFM @ 0.1 in. H ₂ O)	547 (500 ~ 750 at standard ESP)
Evaporator Fan	Output (W) (FLA @ 115V)	200 W / 2.2 A
	Static Pressure (Standard PWC)	0.20
	Min. / Max. PWC	0.01 ~ 0.29
Condenser Fan	Motor	BLDC Motor Type Fan 111
	FLAT Watts (CFM @ max.)	1.25 AT 125 W / 7.884 CFM
	Configurations	ETL (UL 1995)
	Accessories	PCB base, indoor unit terminal block, Remote fan, control transformer, over-voltage protection, overcurrent thermal protection, fan protection, compressor overcurrent sensor
	Safety	

General Information

- The outdoor unit shall supply power to indoor unit via 14 AWG X 3 power wire
- High-voltage terminal block temperature sensor to disable unit in the event of power connection overheating
- Auto-reset after power loss
- Soft-start compressor minimizing current inrush
- All heat exchangers shall be mechanically bonded aluminum fin to copper tube
- The condensing unit heat exchanger shall spray paint method ISO-80277, the heat exchanger should not unusual risk or corrosion development to 3,000 hours
- Base pan heater equipped as standard
- System shall provide 100% heating capacity at 4°F (-20°C)

Option settings

- The outdoor unit shall have snow accumulation prevention option setting to prevent snow drifting against its side outdoor unit.
- Nighttime Quiet Mode: reduction of operational sound during automatic (manual activation)
- System can be set up as heating/cooling, cooling only, or heating only via outdoor unit option setting
- Maximum Current Control configurable from 50% - 100% via outdoor unit, wired controller, or central controller

Indoor Fan

- Indoor fan is a double inlet, forward curve, centrifugal type
- Three fan speed settings and auto setting
- Field configurable for downflow operation with optional conversion kit

Construction

- Outdoor unit shall be galvanized steel with a baked-on powder coating finish for durability
- Indoor Unit: Insulated, galvanized steel

Controls

- Control wiring shall be 2 X 16 AWG
- No additional interface modules/adapters are required when connecting to Samsung central control system
- The unit shall be operated via a wireless or wired remote control with DC type signal
- Unit shall be temperature support when connected to MMW-WQ050JLN Advanced Wire Control or central control options.
- Signal or wireless controllers must be purchased separately

Refrigerant System

- The compressor shall be hermetically sealed, inverter-controlled BLDG rotary type.
- Refrigerant flow shall be controlled by an electronic expansion valve at outdoor unit

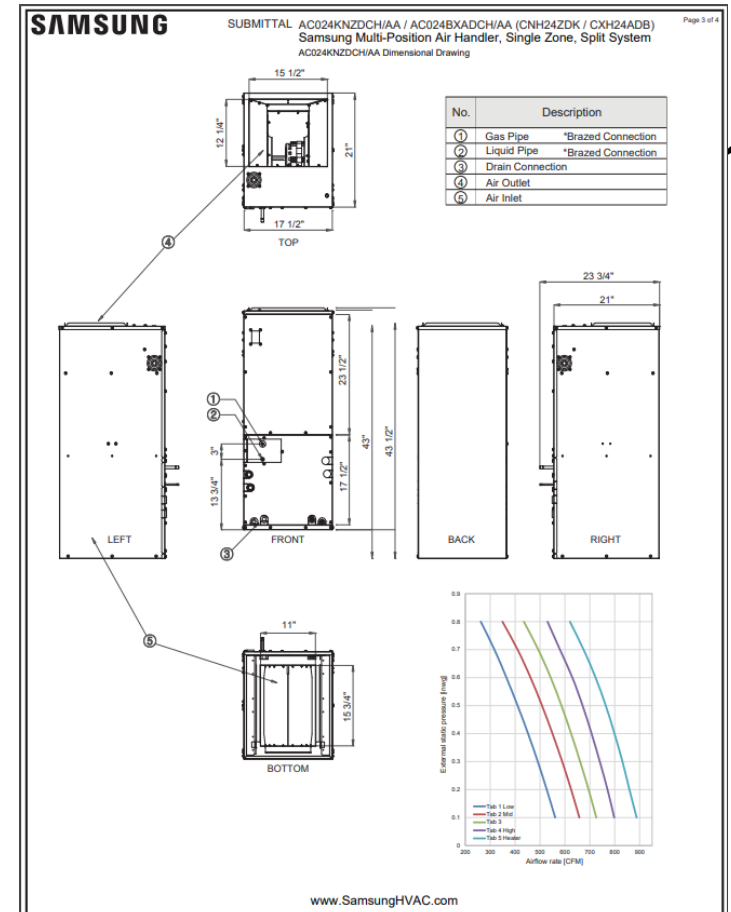
Warranty
 10 years compressor, 10 years parts, 1 year limited labor when registered

¹ Certified in accordance with the AHRI Unitary Small Air-Source Heat Pump (USHP) Certification Program which is based on the AHRI standard of AHRI Standard 1300G.

Samsung HVAC maintains a policy of ongoing development, specifications are subject to change without notice. Refer to www.AHRIdirectory.org for current reference numbers.

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What Service Value Do They Hold?

SAMSUNG

SUBMITTAL AC024KNZDCH/AA / AC024BXADCH/AA (CNH24ZDK / CXH24ADB)
Samsung Multi-Position Air Handler, Single Zone Duct, Split System

Page 1 of 4

Job Name _____
Purchaser _____
Submitted to _____
Unit Designation _____

Location _____
Engineer _____
Reference ☐ Approval ☐ Construction ☐
Schedule # _____

Specifications

Model	Indoor Unit Model Number (US Code)	AC024KNZDCH/AA (CNH24ZDK)
	Outdoor Unit Model Number (US Code)	AC024BXADCH/AA (CXH24ADB)
Performance ¹	Nominal Capacity	Cooling / Heating (Btu/h) 24,000 / 27,000
	Capacity Range	Cooling (Btu/h) 8,000 – 30,000
		Heating (Btu/h) 7,000 – 35,000
	SEER / EER	18.7 / 11.0
	COP (nominal heating)	2.80
	HSPF	10.9
	AHRI Certification Number	207598068
Power	Voltage	ø / V / Hz 1 / 208-230 / 60
	Working Voltage Range (VAC)	176 - 254 (max. 3% deviation from each)
	Operating Current	Cooling (A) 4.3 / 9.9 / 14.5
	(min. / std. / max.)	Heating (A) 3.5 / 12.8 / 24.0
	Max. Breaker	Amps 30
	Min. Circuit Ampacity (A)	24.1
Dimensions	W X H X D	Indoor Unit 17 1/2 X 43 X 21
	(in.)	Outdoor Unit 37 X 39 5/16 X 13
	Duct Connections	Supply (in.) 12 1/4 X 15 1/2
	(W X H)	Return (ID, in.) 17 1/2 X 21
	Weight	Indoor Unit 98.1
	(lbs.)	Outdoor Unit 118.4
Sound Pressure Level	Indoor Unit dB(A)	L / M / H 35 / 38 / 41
	Outdoor Unit dB(A)	Cooling / Heating (high) 50 / 52



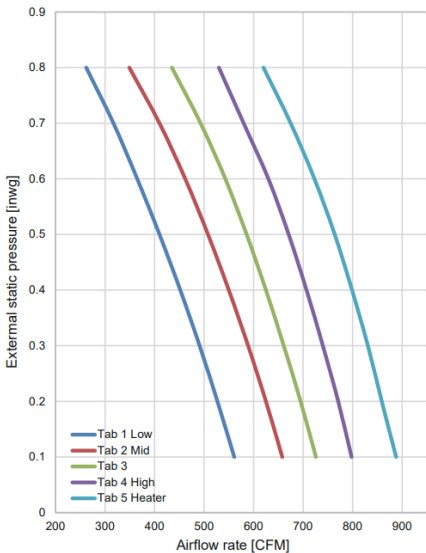
General Information

- The outdoor unit shall supply power to indoor unit via 14 AWG X 3 power wire
- High-voltage terminal block temperature sensor to disable unit in the event of power connection overheating
- Auto-restart after power loss
- Soft-start compressor minimizing current inrush
- All heat exchangers shall be mechanically bonded aluminum fin to copper tube
- The condensing unit heat exchanger salt spray test method: ISO-9227- the heat exchanger showed no unusual rust or corrosion development to 3,000 hours.
- Base pan heater equipped as standard
- System shall provide 100% heating capacity at -4°F(-20°C)

Option settings

Operating Temperatures	Outdoor	Cooling	23 ~ 122°F (-5 ~ 50°C)
		Heating	0 ~ 122°F (-18 ~ 50°C) w/ baffle
	Indoor	Cooling	-13 ~ 75°F (-25 ~ 24°C)
		Heating	61 ~ 90°F (16 ~ 32°C) T ≤ 80°F (27°C)

Evaporator Fan	Type	Double-inlet, forward curve, centrifugal (with ECM motor)	
	Air Volume	CFM (L/M/H)	547/ 636 / 759 (at standard ESP)
	Output (W) / FLA (A)		290 W / .72 A
	Static Pressure	Standard ("WC)	0.20
		Min. / Max. ("WC)	0.01 - 0.79



S-Net Information

EEPROM Version	
Inverter1 Micom	
Main Micom	
Serial Number	
Location	Parkside ODU
ESC OUT (°F)	-
ESC IN (°F)	-
ESC EEV	0
Phase Current	0
Compressor Current1 (°F)	0
Main EEV2	0
Main EEV1	480
CondOut Temp. (°F)	46
Fan RPM	0
Counting high OD Temp.	-
High OD Temp.	-
Target Discharge (°F)	79
IPM1 Temp (°F)	47
Comp Top1 (°F)	62
Discharge1 (°F)	54
Outdoor Temp. (°F)	46
Current Frequency1	0
Order Frequency1	0
Target Frequency1	0
DC Link1 (°F)	169
Error Code	0
Operation Mode	Stop
4Way Valve	Off
Capacity Sum(Indoors) (Btu)	-
Total Indoor	-
Time	15:43:33

- S-net does not show system pressure or suction thermistor.
- This information will need to be gathered manually.

e203

Replace the outdoor unit inverter or Main PCB.

1. If E203 Error to 7-segment of outdoor unit Main PCB is displayed, replace the Inverter PCB and then outdoor unit Main PCB is normal.
2. If Red - ON, Green - OFF, and Yellow - Blink are displayed in the LED display of outdoor unit Inverter, and there is no indication in 7-segment of outdoor unit Main PCB, replace the Main PCB → The outdoor unit Inverter PCB is normal.

4-1-1. Test operation mode and check mode

■ Display Option key

Key	Push type		Mode	Display			
				SEG 1	SEG 2	SEG 3	SEG 4
K1	Short	1st	Heating test mode ¹⁾	H	1	0	0
		2nd	Defrost test mode ²⁾	H	3	0	0
		3rd	End Key operation	0	0	0	0
K2	Short	1st	Cooling test mode ³⁾	H	2	0	0
		2nd	Inverter check	H	4	0	0
		3rd	Pump down	H	6	0	0
		4th	Reserved	H	0	0	0
		5th	On Device Inverter Checker (Comp#1) ⁴⁾	H	A	0	0
		6th	Auto test mode ⁵⁾	H	C	0	0
		7th	Auto check ⁶⁾ (Install commissioning mode)	H	E	0	0
		8th	End Key operation	0	0	0	0
K3	Short	1st	Reset	0	0	0	0



1) Heating test mode : system operates in heating mode (set 104°F (40°C)) and checks for any error.

2) Defrost test mode operates when outdoor temperature is under 50°F(10°C) for 1 minute after pressing K1 for defrost test mode.

3) Cooling test mode : system operates in cooling mode (set 37.4°F (3°C)) and checks for any error.

4) Indication on the display and action to take when an inverter fault is detected.

5) Auto Test Mode: System will determine test mode (cool or heat) based on ambient temperature

* Test mode will continue to operate for 10 hours; any K button press pressing any K button will terminate the test mode

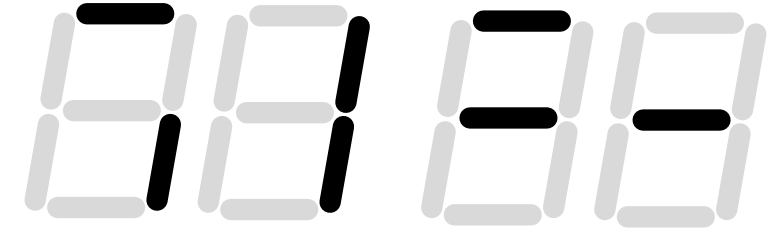
* If an error occurs during the test operation, the test mode will terminate and display the associated error codes

6) Auto Check Mode: System will determine test mode (cool or heat) based on ambient temperature

* Test mode will continue to operate for 10~20mins and self-terminate test mode if no errors are found; pressing any K button press will terminate the test mode

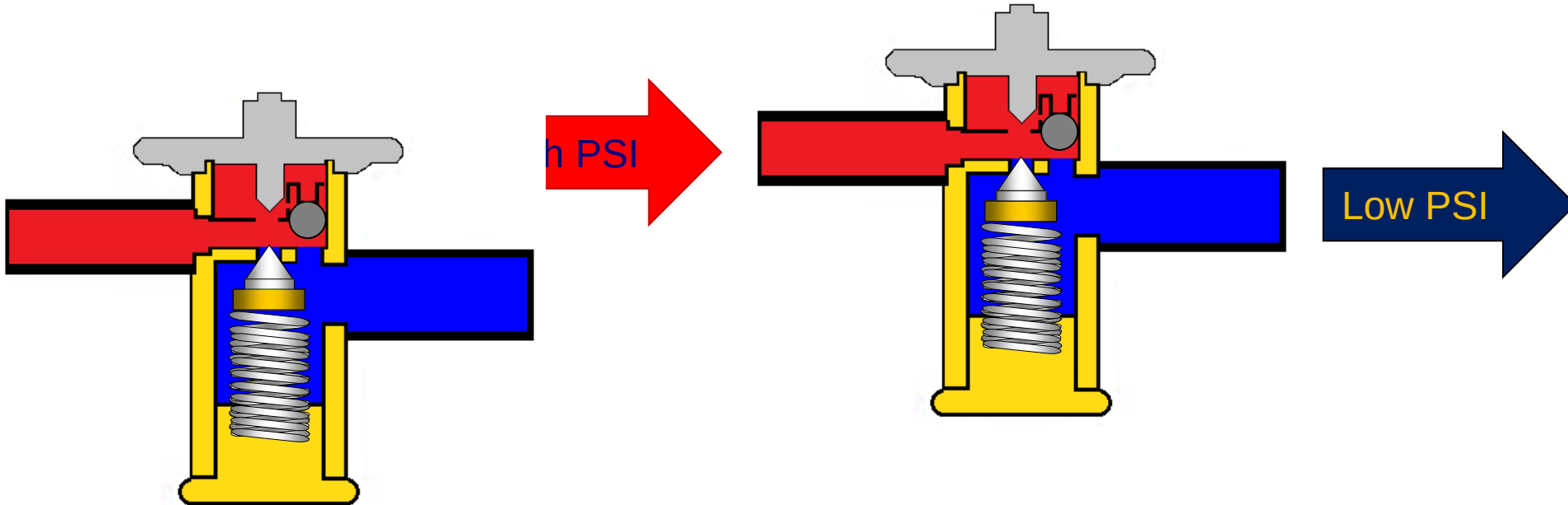
* If an error occurs during the test operation, the test mode will terminate and display the associated error codes

Inverter Check

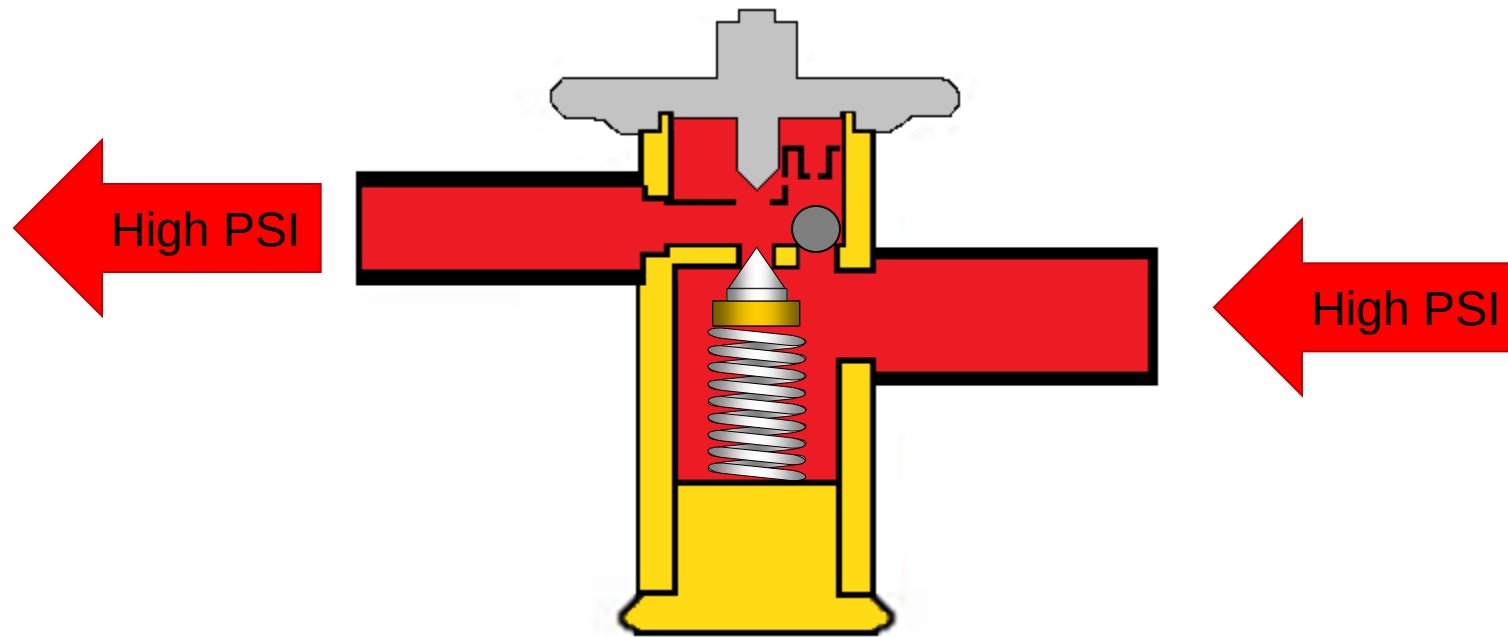


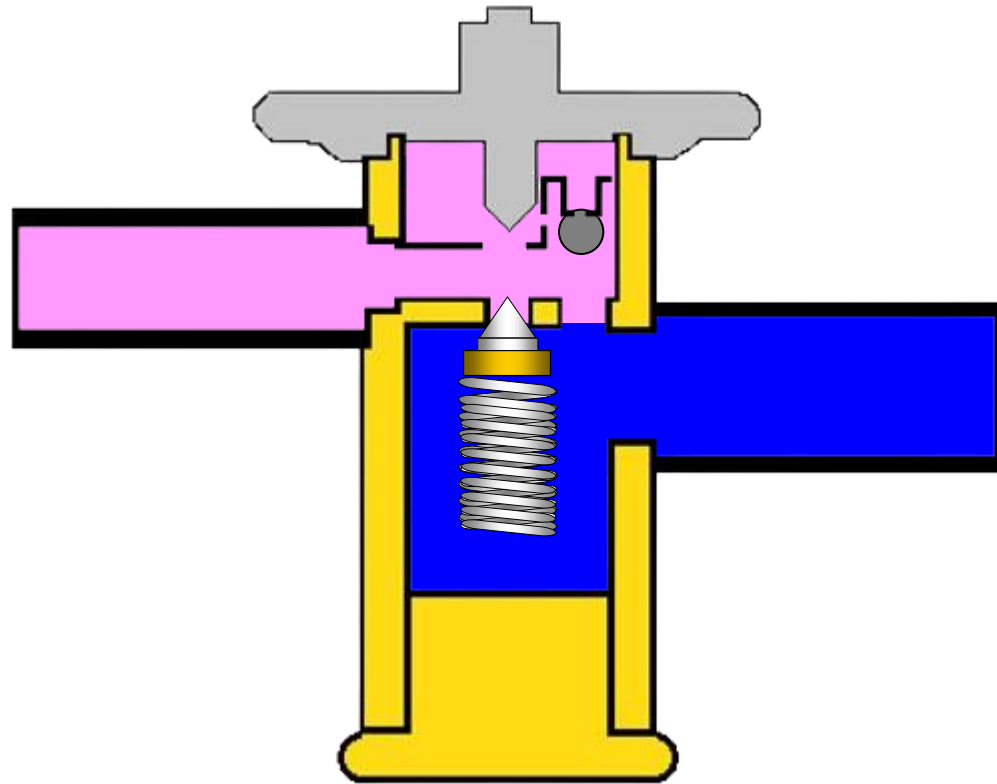
	SEG1	SEG2	SEG3	SEG4	Action to take
Fault detection is in progress	<i>E</i>	<i>E</i>	<i>E</i>	<i>E</i>	-
OK	<i>E</i>	<i>E</i>	<i>E</i>	<i>E</i>	-
NG	<i>E</i>	<i>E</i>	<i>E</i>	<i>E</i>	PBA defect: Replace the PBA
Check	<i>E</i>	<i>E</i>	<i>E</i>	<i>E</i>	Manual inspection is required
Going into fault detection mode failed	<i>E</i>	<i>E</i>	<i>E</i>	<i>E</i>	Try fault detection again

Heat Pump TEV Metering

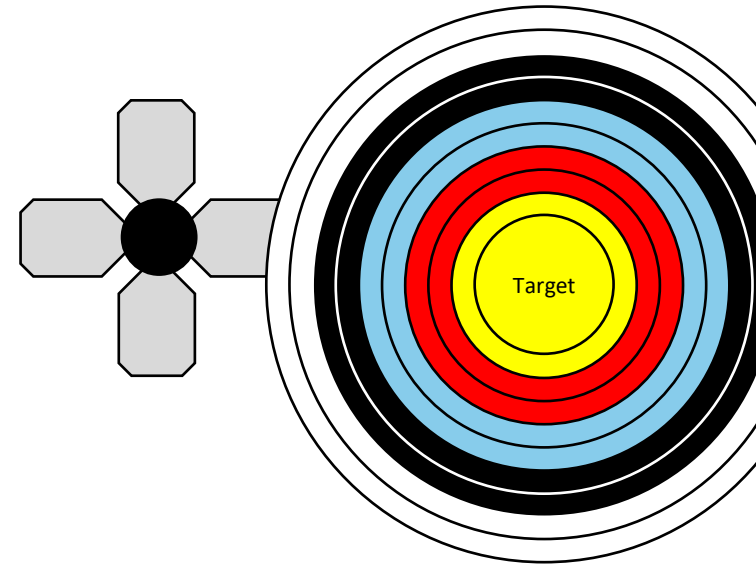
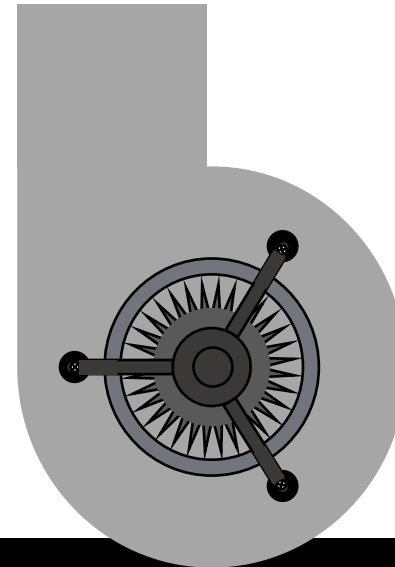
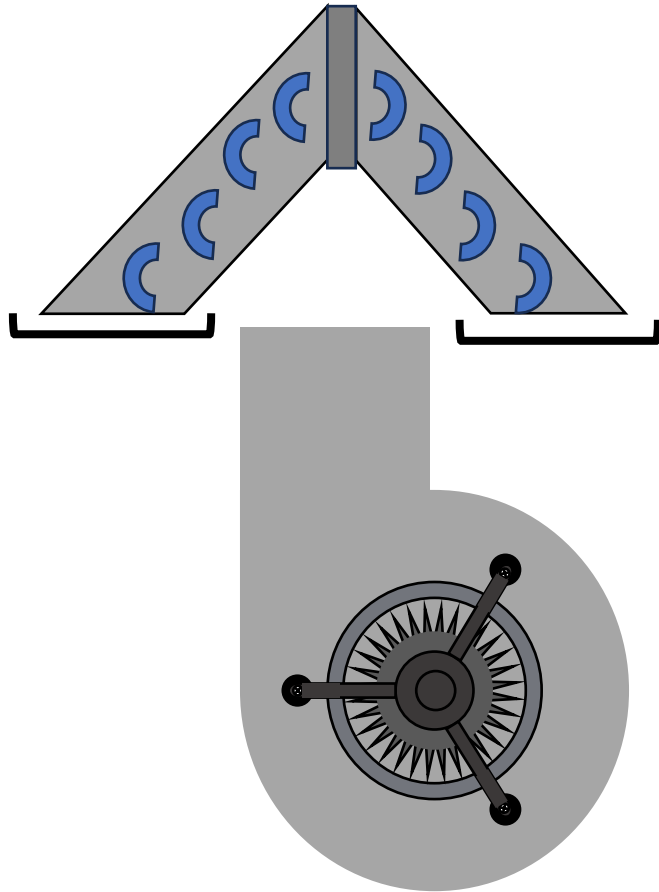


Heat Pump TEV By-Pass

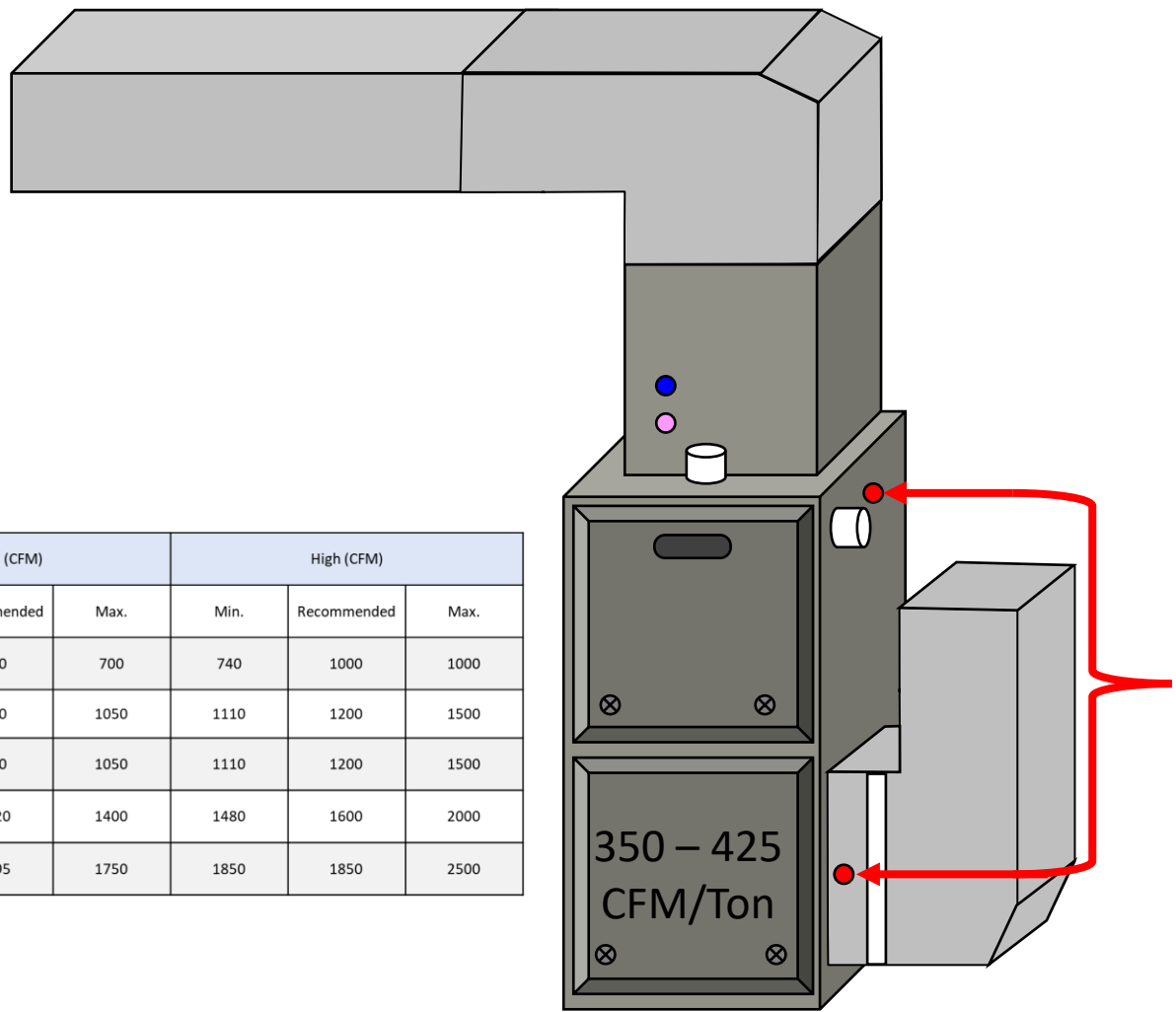




Air Flow



Model Number	Low Speed (CFM)			Medium (CFM)			High (CFM)		
	Min.	Recommended	Max.	Min.	Recommended	Max.	Min.	Recommended	Max.
AC024BXUPCH/AA	360	500	500	518	700	700	740	1000	1000
AC036BXUDCH/AA	540	600	750	777	840	1050	1110	1200	1500
AC036BXUPCH/AA	540	600	750	777	840	1050	1110	1200	1500
AC048BXUPCH/AA	720	800	1000	1036	1120	1400	1480	1600	2000
AC060BXUPCH/AA	900	900	1250	1295	1295	1750	1850	1850	2500

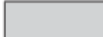


ESP of the fan is the amount of pressure the fan creates to overcome system resistances. ESP measure at the in let of the fan and outlet of the furnace (before the coil).

Fan Performance charts



Motor TapP	(inwg)	CFMM	otor Tap	P(inwg)C	FM
1	0.1	1260	4	0.1	1538
	0.2	1167		0.2	1500
	0.3	1075		0.3	1462
	0.4	981		0.4	1422
	0.5	887		0.5	1379
	0.6	793		0.6	1331
	0.7	699		0.7	1280
	0.8	606		0.8	1224
	0.9	510		0.9	1158
	1	415		1	1078
2	0.1	1379	5	0.1	1739
	0.2	1293		0.2	1705
	0.3	1210		0.3	1670
	0.4	1130		0.4	1633
	0.5	1045		0.5	1597
	0.6	963		0.6	1559
	0.7	878		0.7	1522
	0.8	795		0.8	1485
	0.9	711		0.9	1446
	1	627		1	1406
3	0.1	1475			
	0.2	1410			
	0.3	1348			
	0.4	1282			
	0.5	1216			
	0.6	1148			
	0.7	1076			
	0.8	1003			
	0.9	929			
	1	850			

 = Default Setting

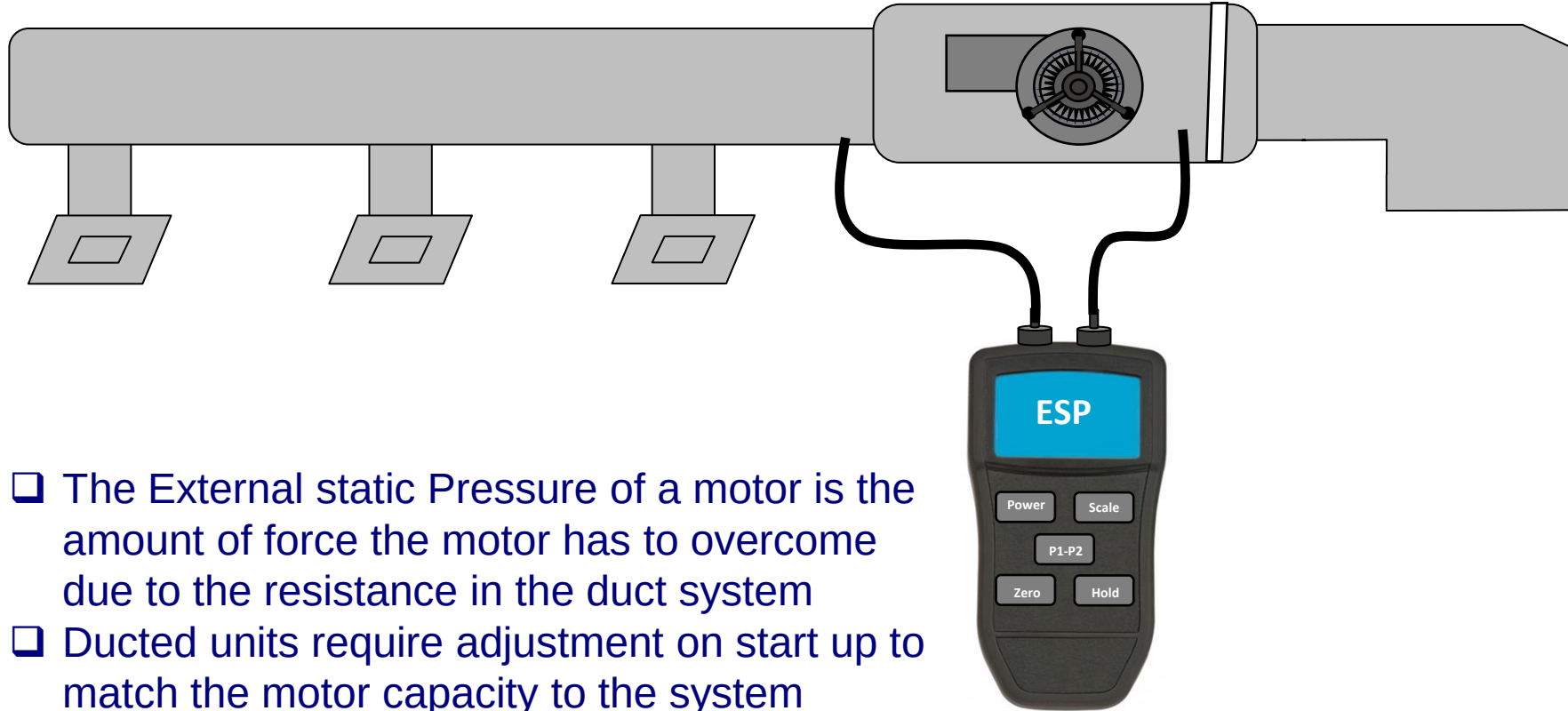
AC048KNZDCH/AA

HP : 3/4

Default motor taps:

High / Mid /Low = 4/2/1

Measuring ESP



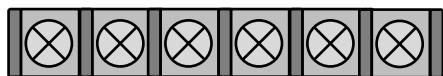
- ❑ The External static Pressure of a motor is the amount of force the motor has to overcome due to the resistance in the duct system
- ❑ Ducted units require adjustment on start up to match the motor capacity to the system resistance.

TABLE 19
AIR FLOW PERFORMANCE – (-)92P & (-)(-)92MSP SERIES MODELS

Model	Motor HP [W] Blower Size IN [MM]	CFM Air Delivery External Static Pressure, "W.C. [kPa]								
		Speed Tap	0.1 [0.02]	0.2 [0.05]	0.3 [0.07]	0.4 [0.10]	0.5 [0.12]	0.6 [0.15]	0.7 [0.17]	0.8 [0.19]
(-)92PA0401317MSA & (-)(-)92MSP040B30A	1/4 [187] 11 x 7 [279 x 178]	Low	732 [345]	695 [328]	669 [316]	639 [302]	605 [286]	547 [258]	512 [242]	487 [230]
		Med. Lo	875 [413]	845 [399]	820 [387]	792 [374]	766 [362]	726 [343]	659 [311]	605 [286]
		Med. Hi*	1073 [506]	1049 [495]	1004 [474]	975 [460]	943 [445]	900 [425]	864 [408]	798 [377]
		High	1279 [604]	1239 [585]	1213 [572]	1174 [554]	1121 [529]	1069 [505]	1033 [488]	980 [463]
(-)92PA0601317MSA & (-)(-)92MSP060B30A	1/2 [373] 11 x 8 [279 x 203]	Low	830 [392]	793 [374]	755 [356]	705 [333]	654 [309]	599 [283]	544 [257]	496 [234]
		Med. Lo	940 [444]	908 [429]	875 [413]	821 [387]	766 [362]	710 [335]	653 [308]	584 [276]
		Med. Hi*	1168 [551]	1139 [538]	1109 [523]	1060 [500]	1010 [477]	942 [445]	874 [412]	802 [379]
		High	1421 [671]	1385 [654]	1349 [637]	1291 [609]	1232 [581]	1147 [541]	1062 [501]	976 [461]
(-)92PA0701317MSA & (-)(-)92MSP070B30A	1/2 [373] 11 x 8 [279 x 203]	Low	852 [402]	827 [390]	802 [379]	752 [355]	701 [331]	662 [312]	623 [294]	580 [274]
		Med. Lo	986 [465]	955 [451]	923 [436]	883 [417]	843 [398]	790 [373]	737 [348]	686 [324]
		Med. Hi*	1238 [584]	1211 [372]	1183 [558]	1139 [538]	1094 [516]	1036 [489]	978 [462]	897 [423]
		High	1493 [705]	1452 [1685]	1410 [665]	1364 [644]	1318 [622]	1257 [593]	1195 [564]	1084 [512]
(-)92PA0851521MSA & (-)(-)92MSP085C50A	3/4 [559] 11 x 10 [279 x 254]	Low	1313 [620]	1275 [602]	1236 [583]	1221 [576]	1184 [559]	1141 [538]	1098 [518]	1053 [497]
		Med. Lo	1488 [702]	1452 [685]	1415 [668]	1384 [653]	1328 [627]	1279 [604]	1229 [580]	1170 [552]
		Med. Hi*	1732 [817]	1704 [804]	1676 [791]	1630 [769]	1579 [745]	1498 [707]	1417 [669]	1343 [634]
		High	1983 [936]	1929 [910]	1874 [884]	1810 [854]	1742 [822]	1663 [785]	1583 [747]	1477 [697]
(-)92PA1001521MSA & (-)(-)92MSP100C50A	3/4 [559] 11 x 10 [279 x 254]	Low	1329 [627]	1313 [620]	1296 [612]	1260 [595]	1224 [578]	1171 [553]	1117 [527]	1049 [495]
		Med. Lo	1535 [724]	1498 [707]	1461 [690]	1422 [671]	1382 [652]	1325 [625]	1268 [598]	1203 [568]
		Med. Hi*	1796 [848]	1748 [825]	1700 [802]	1656 [782]	1612 [761]	1536 [725]	1460 [689]	1370 [647]
		High	2019 [953]	1961 [925]	1903 [898]	1843 [870]	1782 [841]	1687 [796]	1592 [751]	1489 [703]
(-)92PA1151524MSA & (-)(-)92MSP115D50A	3/4 [559] 11 x 11 [279 x 279]	Low	1249 [589]	1234 [582]	1219 [575]	1183 [558]	1146 [541]	1112 [525]	1078 [509]	1017 [580]
		Med. Lo	1440 [680]	1429 [674]	1417 [669]	1381 [652]	1344 [634]	1313 [620]	1281 [605]	1221 [576]
		Med. Hi*	1770 [835]	1750 [826]	1730 [816]	1696 [800]	1662 [784]	1611 [760]	1559 [736]	1463 [690]
		High	2123 [1002]	2077 [980]	2031 [959]	1983 [936]	1934 [913]	1856 [876]	1777 [839]	1665 [786]

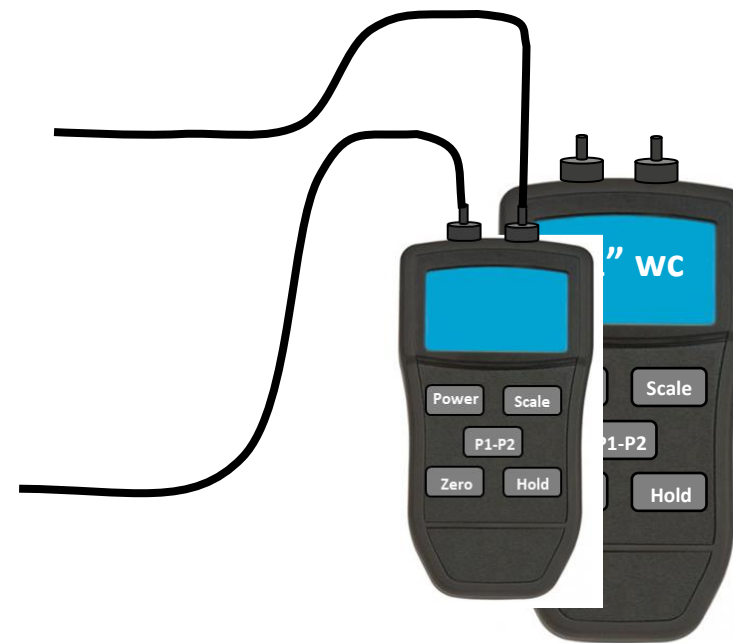
IDU Terminal Block

R C Y G W W1

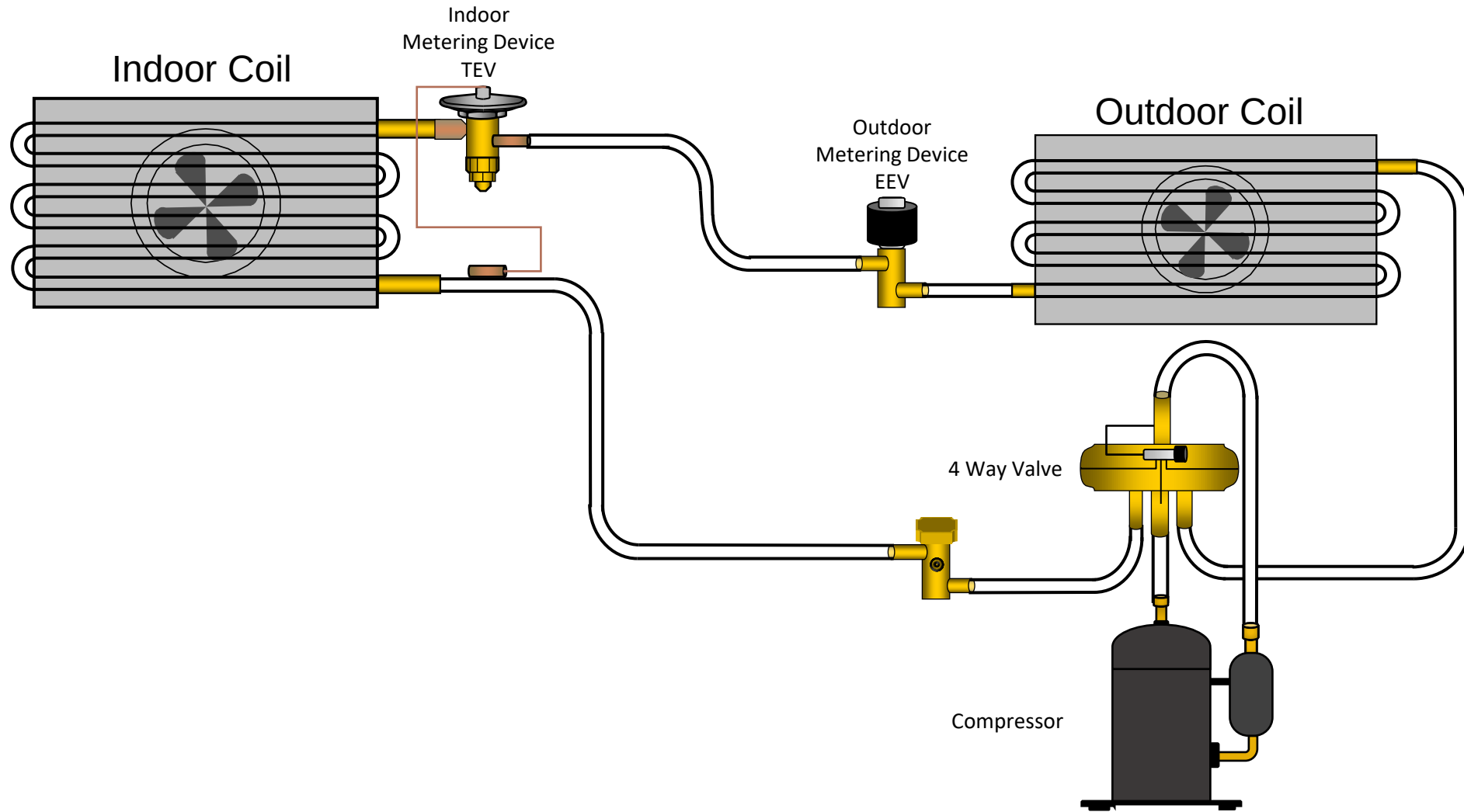


T1

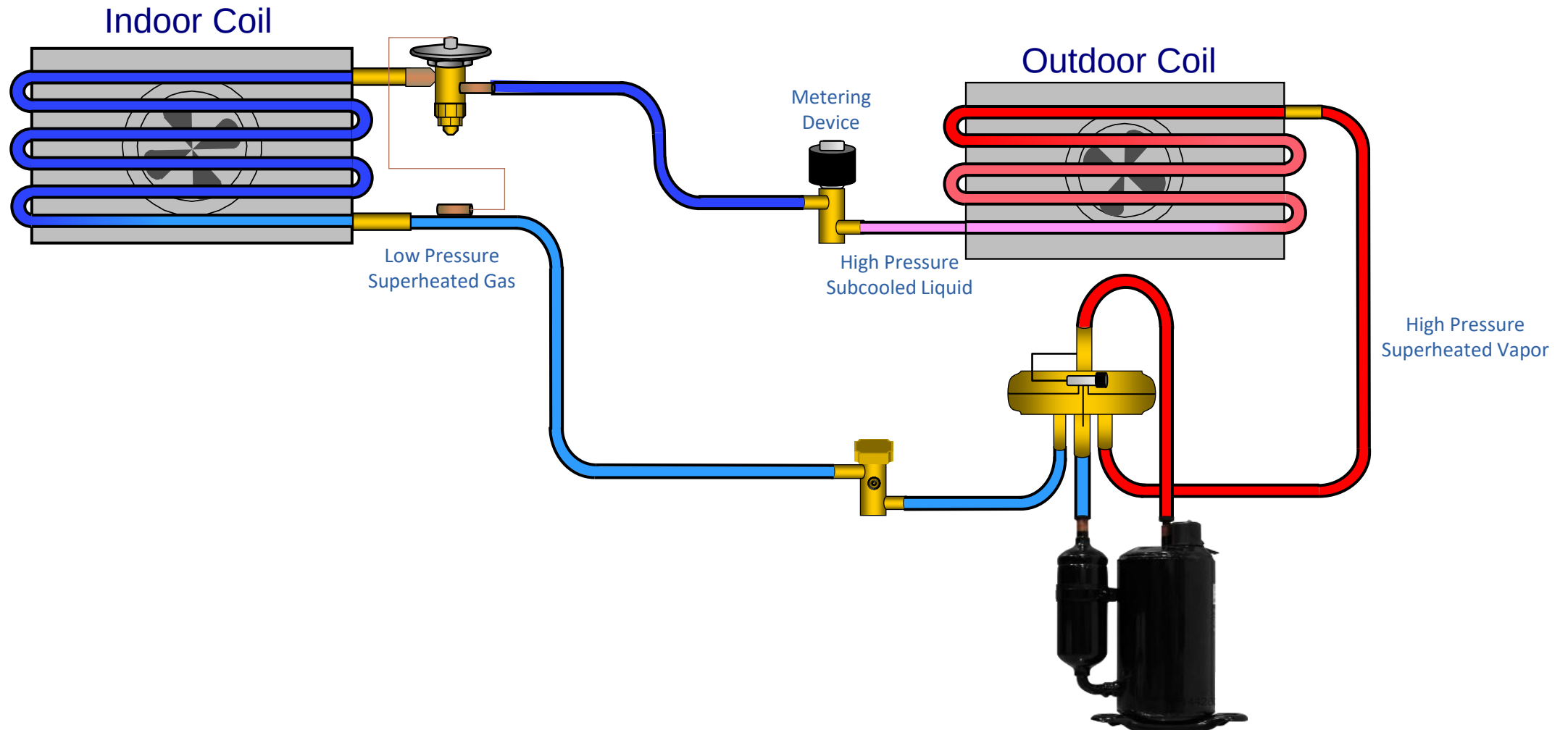
T2



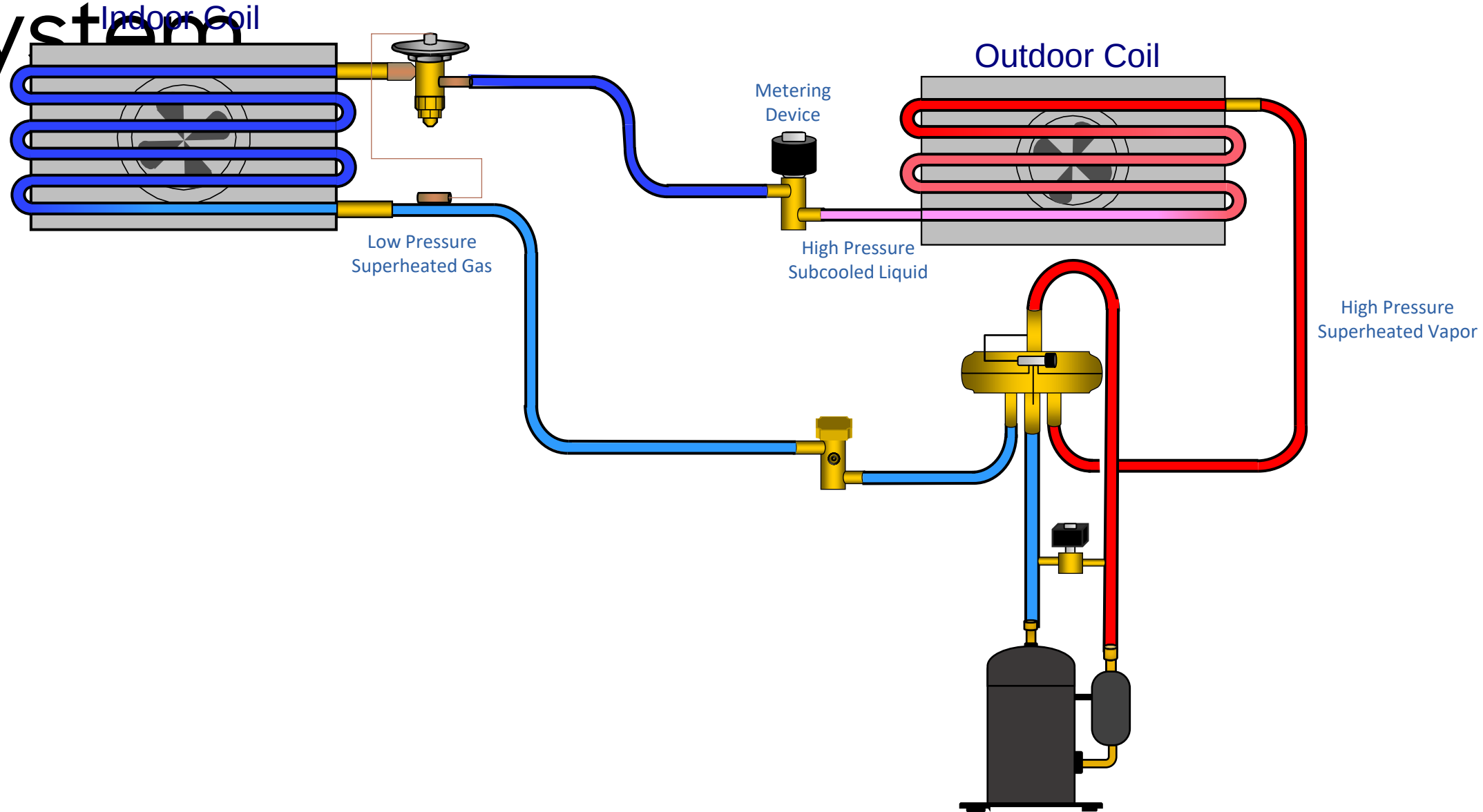
States of refrigerant within a system



States of refrigerant within a system



States of refrigerant within a system



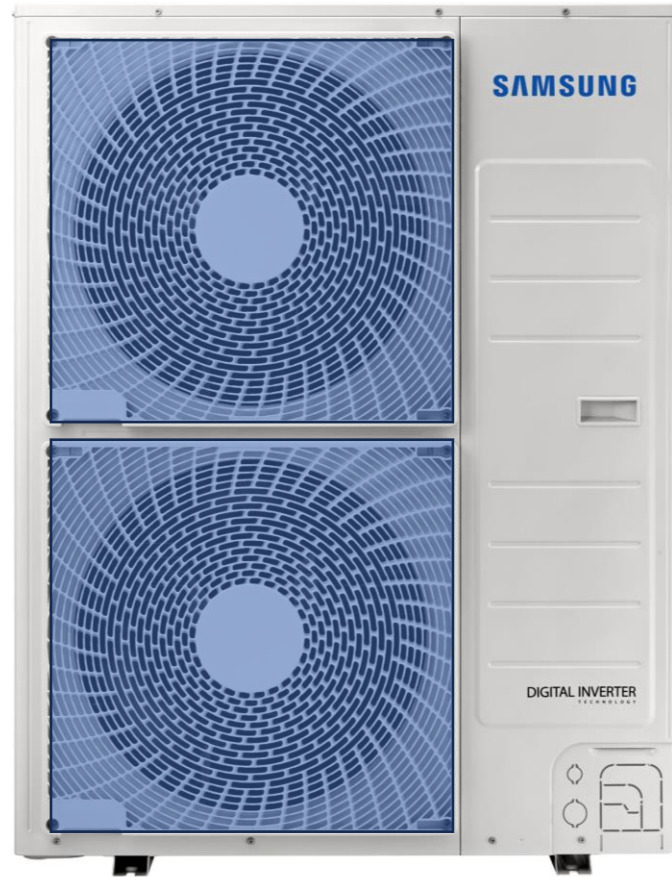
Function of System Components in heating mode



EEV



Pressure Transducer



Thermistor



4 Way Valve



Solenoid Valve

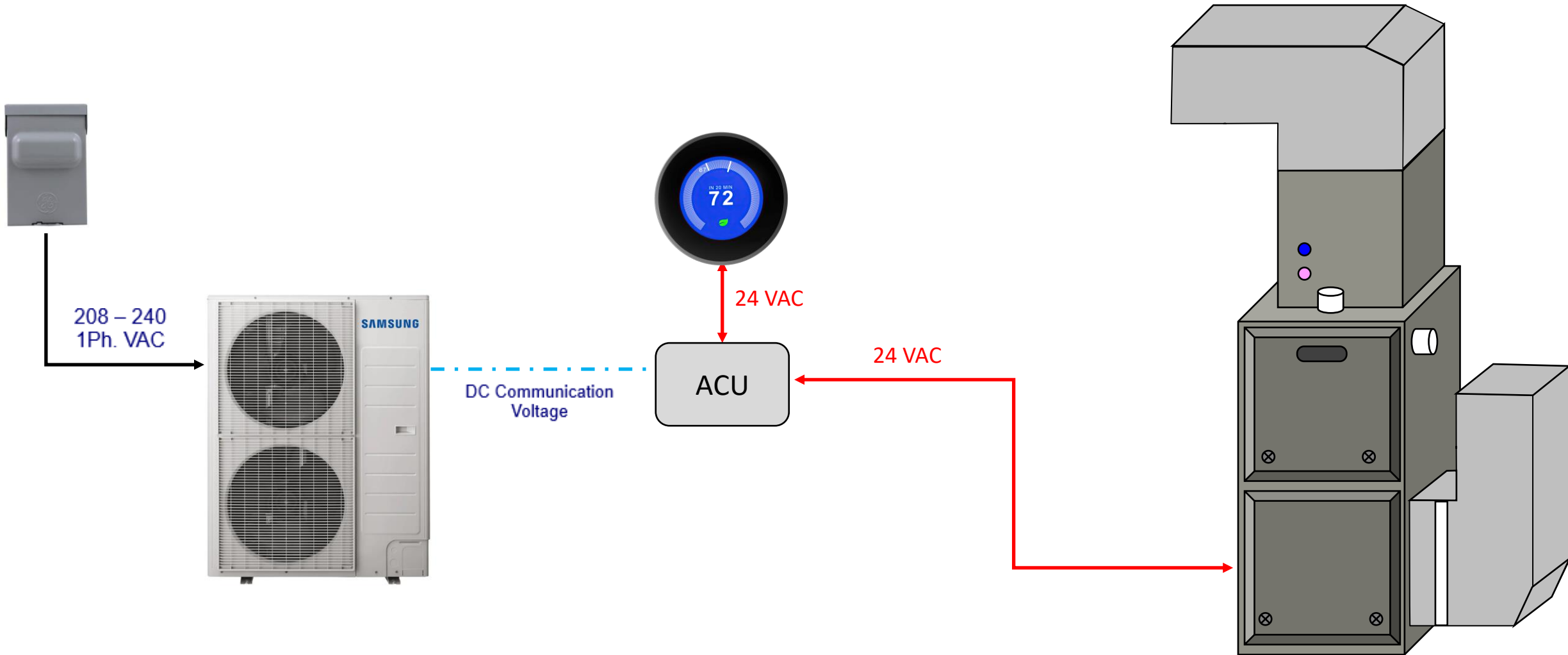


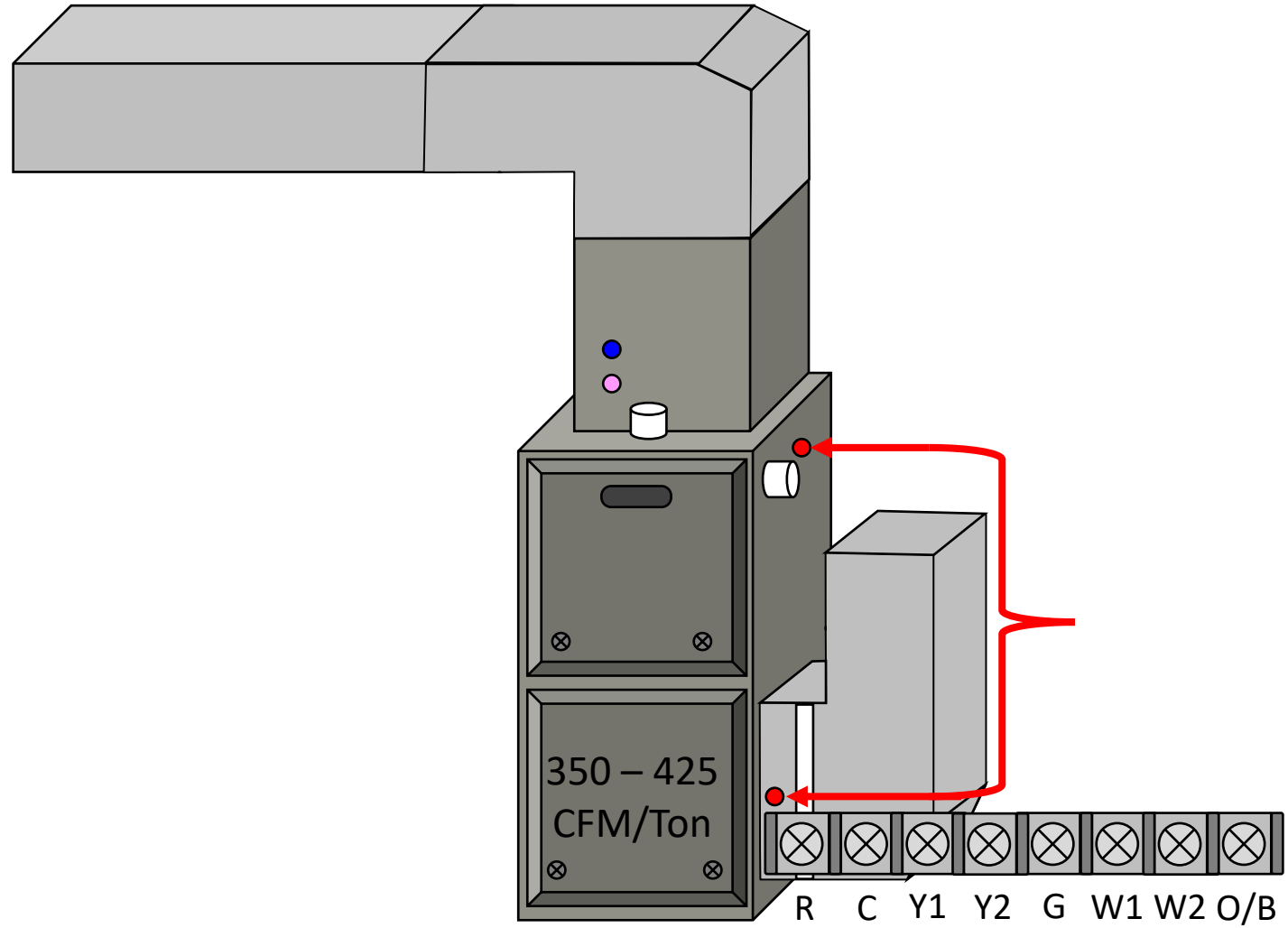
Fan Motor



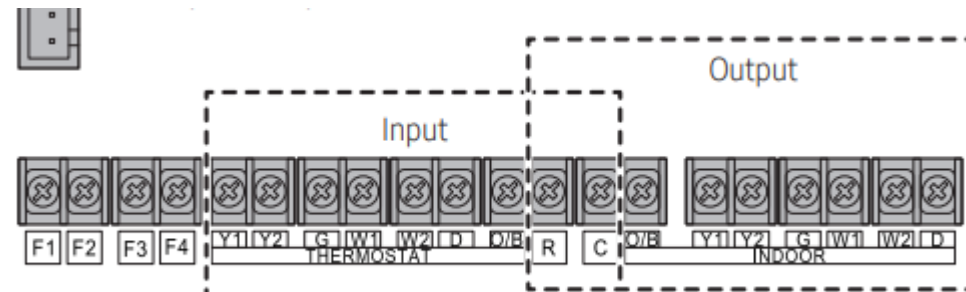
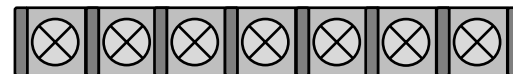
Compressor

Electrical Voltages





R C Y G W W1 O/B



Thermostat Terminal Block

